PsyRAT User Manual

Version 0.1.0-beta

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Contents

This manual documents PsyRAT 0.1.0-beta, the Psychophysiologist's Reliability Analysis Toolbox: a MATLAB toolbox that uses generalizability theory to evaluate the psychometric reliability of event-related potential (ERP) scores, with Bayesian variance-component estimation via CmdStan.

The version above is the only version number printed in this manual. Your installed version appears in the startup banner when you run `psyrat_start`, and in the header of every table the toolbox exports.

How to read this manual

- New to the toolbox? Start with Chapter 2, Quick start, then work through the tutorials (Chapters 10 to 13).
- New to generalizability theory? Read Chapter 1 and Chapter 4 first.
- Hit an error message? Go straight to Chapter 18, Troubleshooting.
- Migrating from the ERA Toolbox? Chapter 1 notes what changed; the workflow chapters (6 to 9) are the new reference.

Conventions used in this manual

- GUI control names appear with their exact on-screen capitalization: click Analyze, select the Preferences button.
- Dialog titles are quoted as the toolbox sets them. macOS file panels do not display a title, so a dialog the manual calls "titled Data" opens there as an untitled panel.
- File names and MATLAB commands appear in `code font`.
- The tutorial datasets ship with the toolbox in `test_data/`, and the expected results printed in the tutorials were produced from those exact files at the settings each tutorial states.
- Reliability is the umbrella term. Dependability refers to the absolute-error coefficient and generalizability to the relative-error coefficient; the two are not interchangeable, and the manual always names the one it means.

1. Why analyze the reliability of ERP scores?

Reliability is a property of scores (the data in hand), not a property of measures. Everything this toolbox does follows from that one sentence, and most of this chapter is spent defending it against a habit the event-related potential (ERP) literature has held for decades. If you are already convinced, Chapter 2 puts a number on screen in one sitting.

1.1 Reliability is a property of scores, not measures

Most published ERP work cites reliability rather than estimating it. A methods section reports that internal consistency for error-related negativity (ERN) scores has been established elsewhere, supplies a citation, and moves on. Clayson and Miller (2017b) named this the common misunderstanding that reliability is a fundamental property of a measure. It is not. A reliability estimate describes one particular set of scores, obtained from one sample, with one task and one processing pipeline.

The evidence against borrowing is not subtle. A preregistered reliability generalization study pooled 189 internal-consistency estimates of ERN scores drawn from 68 samples nested in 43 studies (n = 4,499 participants). The pooled estimate was .68, 95% CI [.63, .72], and about 90% of the variation among those estimates was not attributable to sampling error (Clayson, 2020). An average that heterogeneous is not a number anyone can inherit.

Population is one reason. Baldwin, Larson, and Clayson (2015) estimated the dependability of ERN and error positivity (Pe) scores in 34 participants meeting criteria for major depression, 29 meeting criteria for an anxiety disorder, and 319 controls. Within-person variance exceeded between-person variance in every group, and the controls needed fewer trials than the clinical groups to reach the same dependability. Under a fixed eight-trial inclusion rule, internal consistency ran from .52 to .60 in psychopathology samples against .67 to .69 in controls (Clayson, 2025). Same component, same rule, different scores, different reliability.

Task is another. ERN scores recorded during flanker, Stroop, and go/no-go tasks show both shared and unique variance, and their internal consistency differs by task (Clayson & Miller, 2017b). Then there is the processing pipeline, which is where the variability is worst. A multiverse analysis computed 3,456 ERN scores per person across defensible choices about filtering, ocular correction, reference scheme, baseline window, sensor, and scoring approach. Those choices changed both the mean amplitudes and the variability of the scores (Clayson, 2024). Reliability moves with all of it. Your pipeline is not the cited study's pipeline, and there is no reason to expect its estimate to survive the move.

Trial-count rules of thumb are the most common form the borrowing takes. Olvet and Hajcak (2009) reported that six to eight error trials could support internal consistency near .50 in undergraduates, and six to eight became a field-wide inclusion rule (Clayson, 2025). Across the ERN literature as a whole, the reliability generalization study put the requirement at 9 trials for .70 and 16 trials for .80. After influential estimates were dropped it was 10 and 17 (Clayson, 2020). Neither pair is your number. Both were averaged over samples, tasks, and pipelines that are not yours, and the second pair was estimated from the very heterogeneity that makes the first pair unsafe. Computing reliability on the data in hand replaces a borrowed cutoff with one your own data justify, which is the entire reason PsyRAT reports a trial cutoff at all.

One kind of reliability also does not stand in for another. Carbine et al. (2021) recorded four food-related ERP scores from 132 participants twice, two weeks apart. Coefficients of equivalence (internal consistency) exceeded .96 for every score. Coefficients of stability (test-retest) in the same data ranged from .48 to above .65. Scores that separate people cleanly inside one session need not preserve that ordering across two weeks, and a paper reporting only the first coefficient has not addressed the second.

Now the costs. Poor reliability caps the correlations you can observe. The maximum correlation between two measures is the square root of the product of their reliabilities, so nothing else about the hypothesis matters until that ceiling is high enough. If ERN scores are reliable at .50 and the second measure at .90, no observed correlation can exceed .67 in absolute value. Raise the ERN score reliability to .90 and the ceiling rises to .90 (Clayson & Miller, 2017b, Table 1). A null individual-differences result under the first condition is weak evidence about the relationship and strong evidence about the measurement.

Low reliability also costs statistical power. Take a simulated independent-samples t test at alpha = .05 with a true standardized difference of 0.5. Power was .65 when scores were reliable at .70, .74 at .85, and .80 under perfect reliability (reported in Clayson & Miller, 2017b). Holding true-score variance constant, better reliability shrinks error variance, and shrinking error variance is what buys the power.

Group comparisons are where it gets genuinely misleading. Suppose two groups differ by the same true amount on two tasks and one task is measured with less error. The effect size will be larger for the better-measured task. Simple-effects tests following a significant interaction then track the measurement difference rather than the construct (Clayson & Miller, 2017b). High measurement error can also produce sign errors, so a clinical group can appear to show larger scores than controls when the true difference runs the other way (Clayson, 2024). Comparing groups whose scores differ in reliability means comparing measurement quality alongside everything else. The only way to find out whether that is happening is to estimate reliability separately in each group. PsyRAT does that by construction: it estimates separate variance components for every group and event, and every coefficient it reports is labeled with the cell it came from.

None of this requires heroics. Report the reliability threshold you adopted and the estimate you obtained, report how you estimated it, and justify the trial cutoff you used for exclusions (Clayson & Miller, 2017b, Table 2). On the threshold itself, .70 is a floor for exploratory work and .80 is more appropriate in most contexts (Clayson & Miller, 2017b). Decisions about individual people, of the kind a clinical application would require, are usually held to .90 or higher, and ERN scores in clinical samples do not generally reach that yet (Clayson, 2025). Chapter 17 works through what to write down.

1.2 What PsyRAT does and does not do

PsyRAT takes a long-format table of single-trial ERP scores and estimates the reliability of the averaged scores those trials produce. Variance components come from a Bayesian multilevel model, fit in CmdStan whenever it is installed, so every coefficient arrives as a point estimate with an interval rather than as a bare number (a native fallback engine covers a subset of designs with approximate intervals; Chapter 5 explains). The models tolerate unequal trial counts across participants and cells, which is the ordinary situation after artifact rejection and which coefficient alpha cannot handle without truncating the data (Clayson, 2024).

From a single session, PsyRAT reports internal consistency for each group and event, the number of trials needed to reach a reliability threshold you set, and which participants clear it. Asking for subject-specific error variances adds per-participant (subject-level) estimates, one coefficient per person rather than one for the sample. It handles two-event difference scores and the four-event difference-of-differences contrast. Supply an occasion column and it estimates test-retest reliability. Split means with an items-per-split count are analyzed as splits rather than as trials. Supply a continuous between-person dimension and it estimates reliability as a function of that dimension. Reliability can then be reported conditionally rather than as one number for a whole sample (Rast & Clayson, in press). Chapter 5 helps you choose, and Chapter 16 lists every design.

PsyRAT also imports epoched ERP files (EEGLAB `.set` and EP Toolkit exports; ERPLAB `.erp` files are averaged and import as reference only) and scores single-trial component amplitudes into the table the analysis needs. That path is Chapter 8.

The toolbox is GUI-first, and this manual is organized around its screens. Every analysis the Process New Data workflow runs is also available from `psyrat_run`, which uses the same estimation pipeline and the same seed contract, so the same inputs at the same seed on the same CmdStan build give the same variance components. Neither route is the lesser one, and I use both. Chapter 14 is the scripting reference.

There are four things PsyRAT does not do. It does not record or preprocess EEG: filtering, re-referencing, ocular correction, and epoching happen in your existing pipeline before PsyRAT sees anything. It does not decide what counts as adequate reliability, because you type the threshold and the toolbox applies it. It does not establish validity, since reliability limits validity without supplying it, and a dependable score can still index something other than the construct it was named after. And it does not deliver a biomarker: these coefficients say how precisely you measured, which is a precondition for a clinical claim rather than a substitute for one (Clayson, 2025).

Migrating from the ERA Toolbox

PsyRAT descends from the ERP Reliability Analysis (ERA) Toolbox, which is documented in Clayson and Miller (2017a). That paper describes the predecessor, not this toolbox, so cite it for ERA. For PsyRAT, cite Rocha et al. (2026) for the estimators together with the release you ran; `README.md` carries the current citation guidance.

If you used ERA, expect renaming rather than a port. Toolbox function names now begin with `psyrat_` where they began with `era_`, so the launcher is `psyrat_start`. Processed results are written with the `.psyrat` extension, and the View Results file dialog lists only those. Read Chapter 6 through Chapter 9 rather than assuming ERA habits carry over: the input contract and every output are documented there from scratch. I make no compatibility promise in either direction, and ERA result files and ERA scripts are not supported here.

1.3 The three coefficient terms used everywhere

Three words do a great deal of work in this manual, and they are not interchangeable.

Reliability is the umbrella term. Use it for the general idea that scores consistently distinguish among people, or within a person over time. The manual uses it that way and never as a stand-in for either coefficient below.

Dependability (the absolute-error coefficient, written phi) asks how well a participant's observed score reflects their absolute level on the construct across the conditions being generalized over. It charges systematic differences among trials or occasions to error (Rocha et al., 2026). If the value of the score matters (a cut score, a change score, a comparison against a norm), dependability is the coefficient you want. It is the toolbox's default.

Generalizability (the relative-error coefficient) asks how well the scores preserve relative differences among people, and it leaves facet main effects out of the error term (Rocha et al., 2026). If a systematic shift between occasions is expected and irrelevant (habituation, say), and all that matters is whether the same participants stay high, generalizability is the coefficient that ignores the shift.

The two coincide when facet means do not differ. If scores do not systematically differ across trials or occasions, absolute and relative error variance are the same quantity, and the two coefficients are equal (Rocha et al., 2026). Otherwise dependability is the smaller of the two, because absolute error variance contains every term relative error variance contains and adds the facet main effects on top.

Both coefficients are ratios of universe-score variance to universe-score variance plus error, which is why both live on the same 0-to-1 scale as a classical reliability coefficient and are read the same way. What differs between them is only what got counted as error. Chapter 4 gives the formulas and restates each of them in words. Everywhere else, the label tells you which one you are looking at. The G-Theory Coefficient control on the viewer selects between Dependability and Generalizability, and the exported overall and cutoff tables name the coefficient that produced them.

2. Quick start: your first analysis

This chapter runs one analysis end to end on data that ship with the toolbox. You will click through five screens, wait for CmdStan, and read one coefficient off a table. Nothing here is a scientific decision. Chapter 10 makes the same run properly, with a defensible choice at every step; this one is the shortest honest path to a number.

2.1 Before you start

You need PsyRAT installed and on the MATLAB path, and you need CmdStan installed and discoverable. Chapter 3 covers both, including the guided installer.

The dataset is `test_data/manual_ern_singlesession.csv`, which ships with the toolbox. It is a simulated flanker study: 80 participants in two independent groups (GroupA and GroupB), one session, and four events crossing flanker congruency with response accuracy (`inc_err`, `inc_cor`, `con_err`, `con_cor`). There is one row per single trial, 8,908 rows in all, and the measurement column holds ERN-like mean amplitudes in microvolts. Trial counts vary on purpose: errors are scarce (12 to 36 per participant) and correct trials are plentiful (24 to 40), because unequal trial counts are the normal case and PsyRAT is built for them. No participant data are involved. The file is fully simulated, and every generating parameter is documented in the generator that produces it, which Chapter 10 names.

2.2 Start the toolbox

At the MATLAB prompt, type:

```
psyrat_start
```

A short banner prints before any window appears. It confirms that the toolbox files are on the path, that CmdStan, MatlabProcessManager, and MatlabStan were found, and that each dependency is at a version the toolbox has been tested against. It prints the PsyRAT version you are running, which is the version to quote in a bug report. PsyRAT also checks for a newer release at startup, and if it finds one it raises a window you have to answer before you can go on. It reads ”You are using an old version of the PsyRAT Toolbox. It is recommended that you update the toolbox. Would you like to do so now?” and offers Yes and No.

If the dependency check fails, PsyRAT does not open the home screen at all. It prints what is missing and offers to install it, which is Chapter 3's territory. Come back once the banner reports that CmdStan was found.

The home screen asks ”What would you like to do?” and offers three buttons: Process New Data, Import + Score ERP, and View Results. The first two labels are drawn on two lines.

2.3 Load the data

Selecting the Process New Data button opens a file dialog titled Data, filtered to the formats PsyRAT reads (`.xlsx`, `.xls`, `.csv`, `.dat`, `.txt`, `.ods`). Choose `manual_ern_singlesession.csv` from `test_data/`. The console prints:

```
Loading Data...
This may take awhile depending on the amount of data...
```

```
Ensuring dependents are found in the Matlab path
PsyRAT Toolbox files found
CmdStan, MatlabProcessManager, and MatlabStan files found
CmdStan version is current
MatlabProcessManager version is current
MatlabStan version is current



PsyRAT Version 0.1.0-beta
```

Figure 1: The startup banner and dependency check printed by psyrat_start.

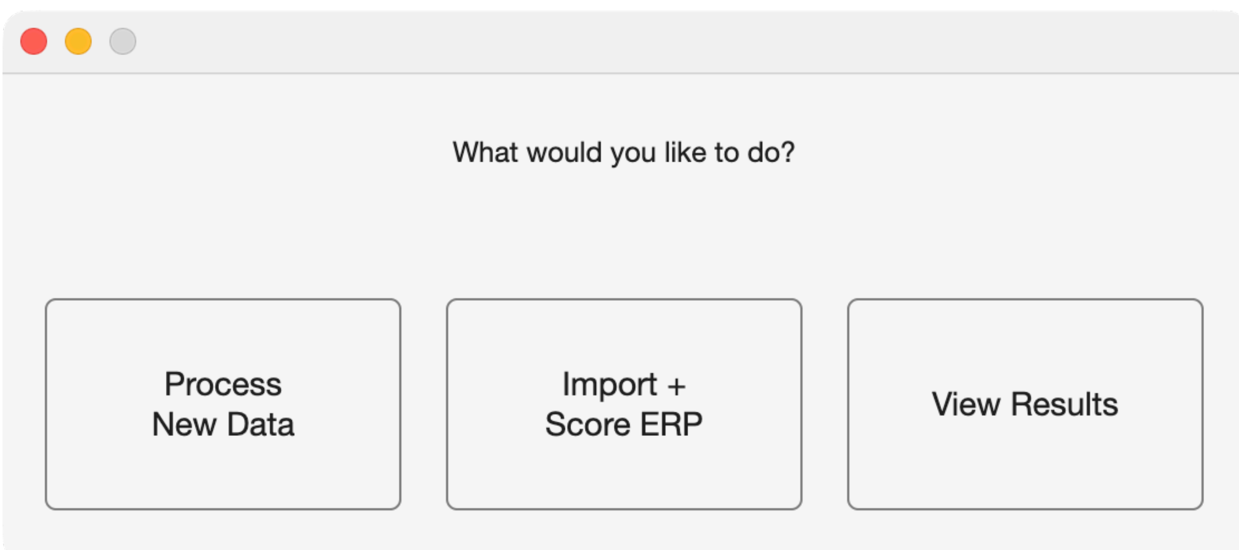


Figure 2: The PsyRAT home screen.

For a file this size it does not. If you cancel the dialog instead of choosing a file, PsyRAT returns to the home screen and raises a File Error dialog reading ”No data file was selected.”

2.4 Specify Inputs

The Specify Inputs window names the loaded dataset across the top and lists each PsyRAT input beside a popup of your file's column headers. PsyRAT preselects what it can by matching header names, and this file uses the canonical names, so the four rows that matter should already read:

- Participant ID: id
- Measurement: meas
- Group ID: group
- Event Type: event

Leave everything below them alone. Occasion ID, Dimension 1 (dynamic), Dimension 2 (dynamic), and Items per split (n_i) all read (none), which is how PsyRAT encodes a facet that is not in play. Split type and Residual correlation are inert without their facets. Estimation engine reads Automatic (recommended), which uses CmdStan whenever CmdStan is installed.

VERIFY those four preselections before you click anything else. The auto-detect matches header text first, and when no header matches it falls back to the last numeric column no other role claims. In a long-format export that is usually the score, but if your score column is not last (say `subjid, ern, trialnum`), the fallback lands on the wrong numeric column and the run reports its reliability. Chapter 6 covers the naming conventions that keep the auto-detect honest.

2.5 Process one event

Selecting the Select Events button opens Specify Which Events to Process. It asks you to ”Select which events you would like to be processed:” and lists the four event labels alphabetically: `con_cor`, `con_err`, `inc_cor`, `inc_err`. All four start selected. Click `inc_err` so that it is the only one highlighted, then click Save.

One event is the simpler first run. PsyRAT fits a separate model for every event, with group-specific variance components inside each one, so four events across two groups means eight cells to read. That is a longer wait for no extra insight the first time through. The incongruent-error cell is the ERN cell and the one the rest of the manual keeps returning to. Chapter 10 adds the incongruent-correct event and processes the two incongruent events.

Set the Event Type popup FIRST. If Event Type still reads (none) when you press Select Events, PsyRAT sends you back to Specify Inputs and raises an Event Column Not Defined dialog reading ”No event column is selected.”

Leave Select Groups alone. Both groups are processed by default, each with its own variance components, which is what makes a later group comparison interpretable at all.

2.6 Preferences

Selecting the Preferences button opens Specify Processing Preferences. Accept every default on that screen and click Save. For the record, the settings that matter for this run are, as control name and value:

- Number of Chains: 4

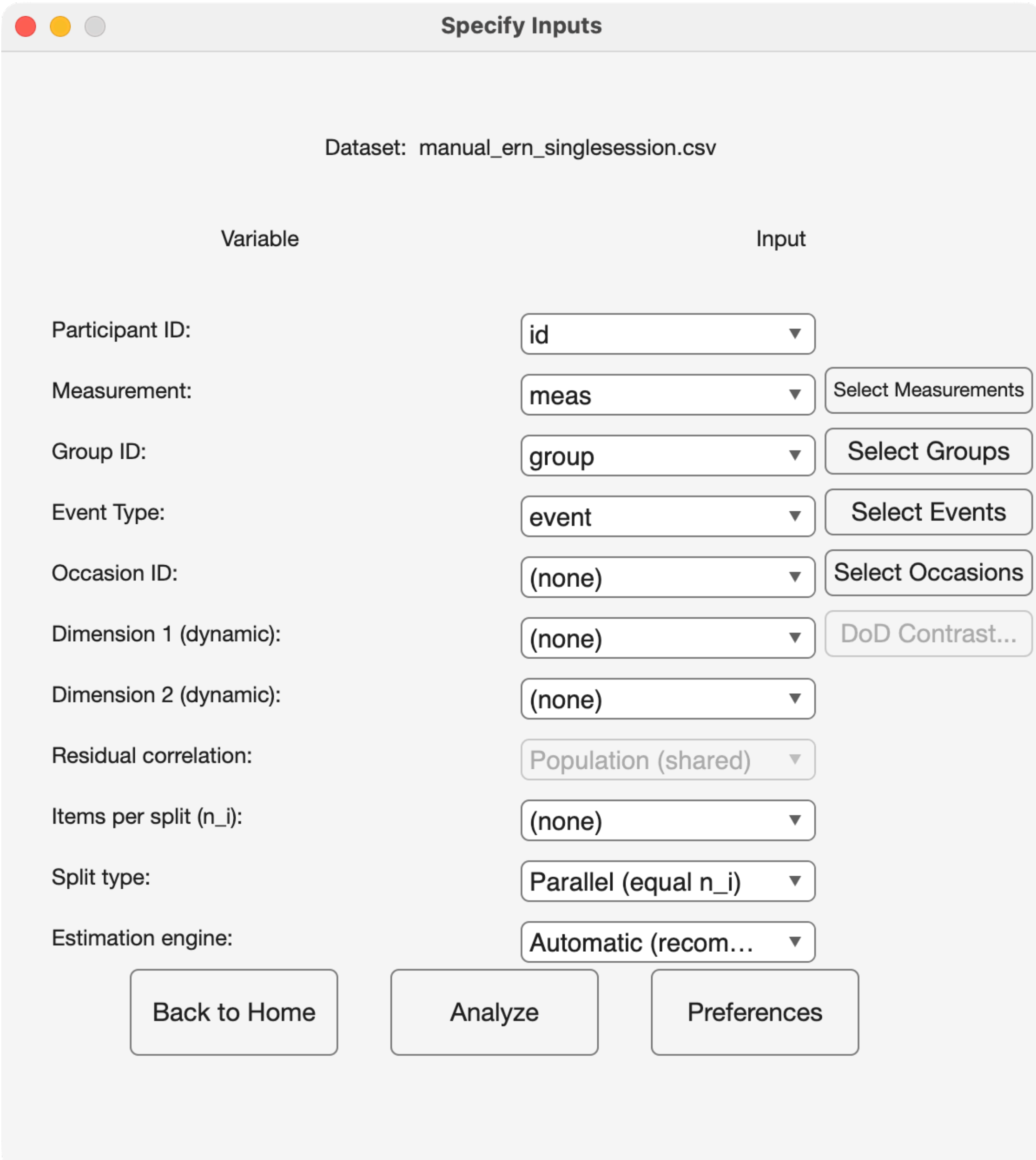


Figure 3: The Specify Inputs screen, with the four canonical columns auto-detected.

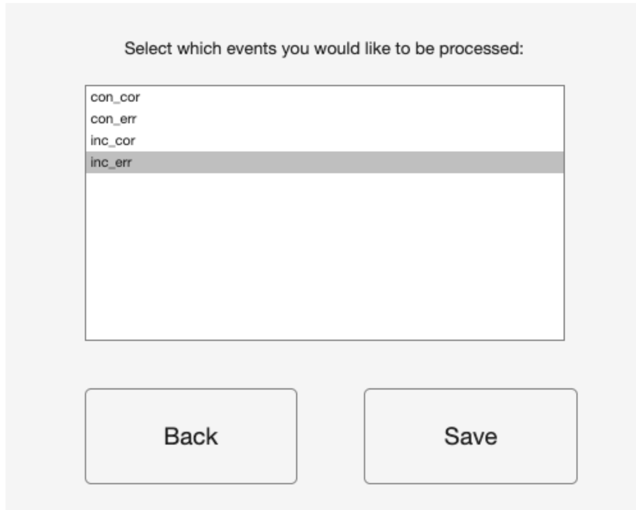


Figure 4: Selecting the incongruent-error event only.

- Warmup iterations: 5000
- Sampling iterations: 5000
- Random seed: 12345
- Observation likelihood family: Gaussian
- Estimate subject-specific error variances: No
- Estimate internal consistency of difference scores: No

Those defaults are generous rather than minimal, and that is deliberate. Four chains is one more than the fewest the toolbox will accept (it refuses fewer than three). The 10,000 iterations per chain (5,000 warmup plus 5,000 sampling) are set high enough that a one-facet fit like this one should not make you think about sampling at all. If a model does not converge, PsyRAT offers to rerun with warmup and sampling doubled. Starting generous mostly buys back that second run. Test-retest designs are the ones that genuinely need long chains, and the toolbox says so on screen when an occasion column is selected and the total iterations sit below 10,000.

The seed is the reproducibility contract. Leave it at 12345 and this analysis repeats exactly, for you and for anyone else holding the file. Change it and the draws change, and so does the third decimal place of everything downstream.

2.7 Run it

Selecting the Analyze button checks your inputs and then asks where to save the results, in a dialog titled Save output file as. It opens in the folder your data file came from, and the output takes the `.psyrat` extension. Name it something plain, such as `quickstart`, and type only that base name: the dialog supplies the extension, and typing `quickstart.psyrat` over the preselected name saves the file as `quickstart.psyrat.psyrat` on macOS.

> CAUTION. Choose a save location whose FULL path contains no spaces. CmdStan does not accept whitespace in filenames or paths, and PsyRAT checks both before it starts. If either contains a space you get a dialog titled Whitespace detected, reading ”The selected filename/path contains whitespace.” and ”CmdStan does not support whitespace in filenames or paths.”, and the save dialog reopens until you choose a path

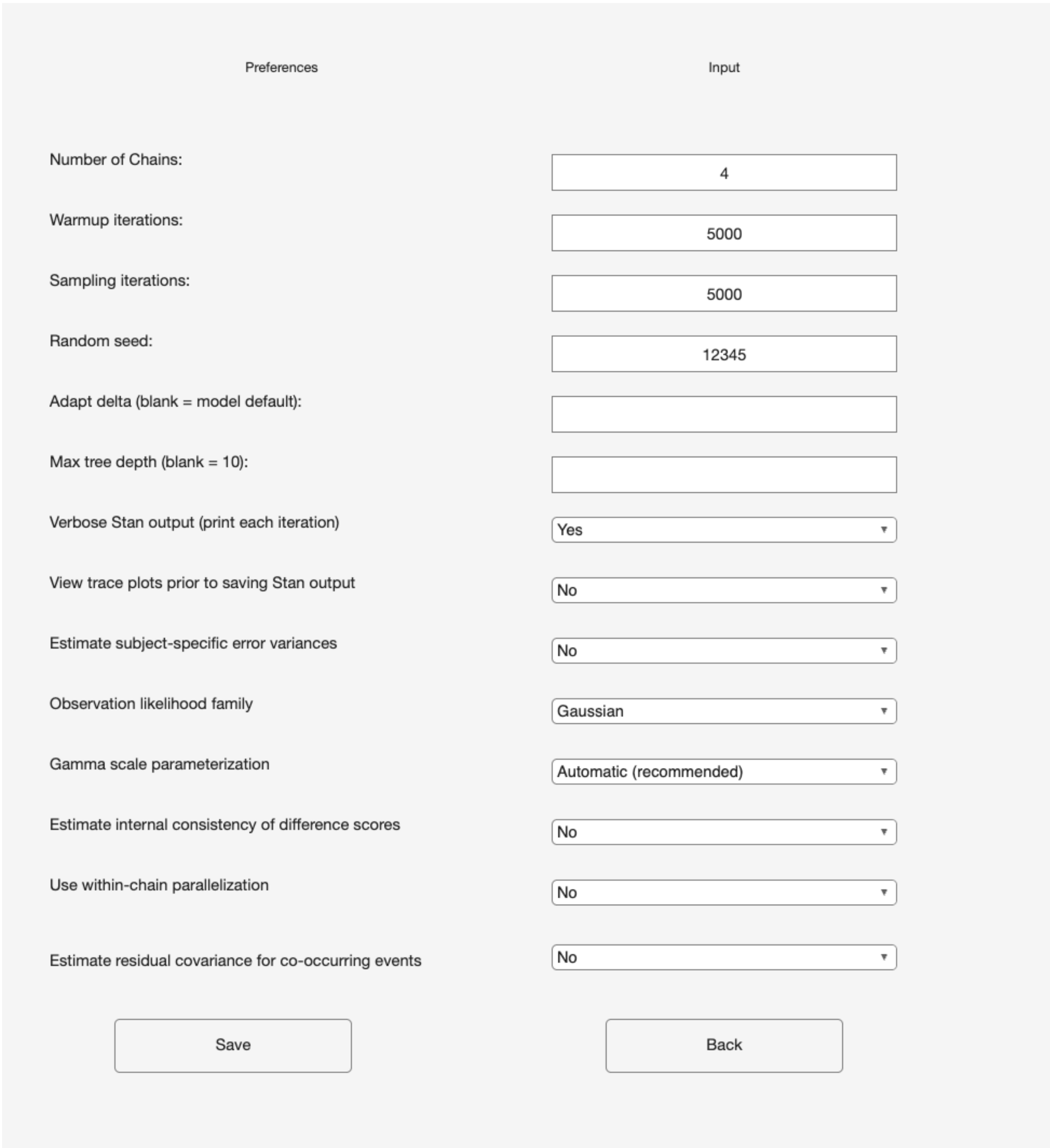


Figure 5: Specify Processing Preferences at the shipped defaults.

> without one. A folder named My Data is enough to trigger it, and so is an account name with a space in it.

Once the location is accepted, PsyRAT closes the setup window and the console takes over:

```
Processing measurement 1/1: meas

Preparing data for analysis...

Working on event 1 of 1

This may take a while depending on the amount of data

Model is being run in cmdstan
We have to compile the model first...
compile: --- Translating Stan model to C++ code ---
compile: bin/stanc  --o=...
compile: --- Compiling C++ code ---
compile: clang++ -std=c++17 ...
compile: --- Linking model ---
Stan is sampling with 4 chains...
```

CmdStan compiles a C++ program for this design before sampling starts, and the `compile:` lines are that step. The compile happens on every run, not only the first, because each run compiles under a fresh model name. It takes several seconds per model, the `clang++` lines are the only output while it works, and a pause after them is normal.

Sampling follows, and the shipped preferences show it. Verbose Stan output (print each iteration) defaults to Yes, so each chain prints a progress line, prefixed with its output file (`output-1.csv:` through `output-4.csv:`), as it counts through 5,000 warmup and 5,000 sampling iterations. A line appears every 500 iterations, since PsyRAT sets the refresh interval to a tenth of the sampling count. Two kinds of message appear before the iteration lines and neither is a failure. Each chain first prints a CmdStan warning that its data file is being read as an RDump file, a deprecated format; that is how PsyRAT hands the data to CmdStan, and the warning has no bearing on the fit. Some chains then print ”Informational Message: The current Metropolis proposal is about to be rejected” with an exception naming a location parameter that is not finite. In this run two of the four chains did so, several times each, and every one of those messages came before that chain's ”Iteration: 1 / 10000” line: they are the sampler trying random starting values during initialization, not sampling going wrong. A message that appears with an iteration number attached is the one to read, and messages from early warmup are adaptation finding its footing rather than a failure.

When the chains have converged the console says so, and the results are written:

```
Model converged

Saving Processed Data...
```

> CAUTION. When estimation finishes, PsyRAT may raise a dialog titled Estimation diagnostics before the results window opens. It appears when the fit failed to converge, when sampling produced divergent transitions, or both, and the toolbox WAITS for you to dismiss it. It is not always drawn in front of the MATLAB desktop. If the cursor

looks busy and nothing further happens, MATLAB is not hung: the dialog is sitting behind another window. Bring it forward and click it. Until you do, the results window will not open.

If the chains do not converge you get a different window instead, headed ”Chains did not converge” and asking whether to rerun with more iterations. Clicking Rerun doubles warmup and sampling and starts over. Clicking Do Not Rerun saves the non-converged fit, which is occasionally useful for diagnosis and never useful as a result.

2.8 Read the result

The View Results Setup window opens by itself with the new file already loaded. It names the dataset and the measurement column, shows a Cell/Event row reading `inc_err` because one event was processed, and offers a Reliability Cutoff box (0.8), a G-Theory Coefficient popup (Dependability), and an Output Selection column of checkboxes, all of them checked.

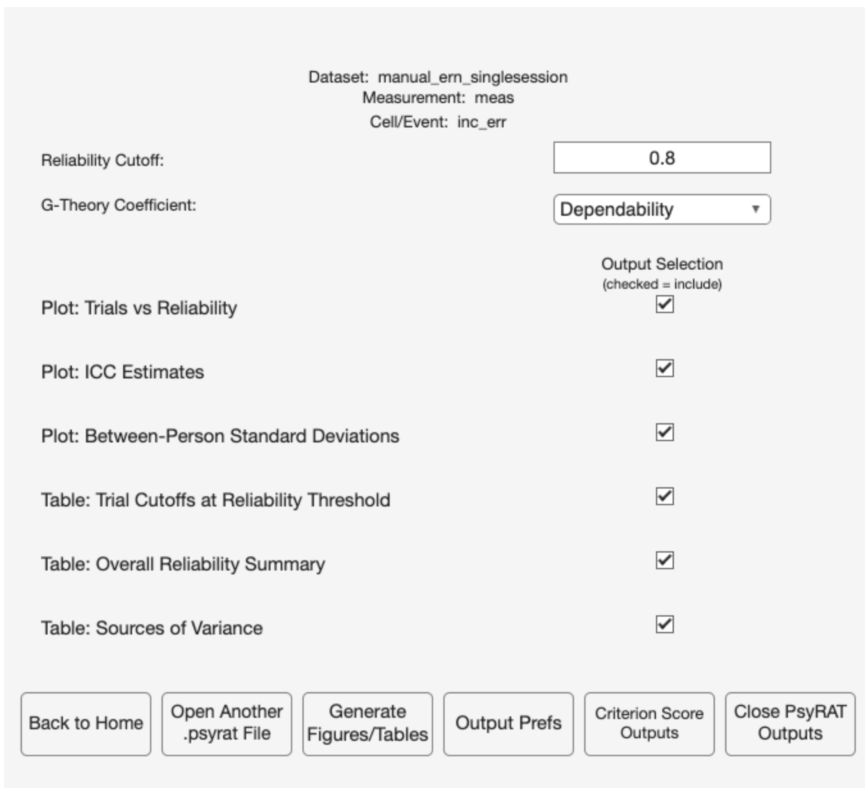


Figure 6: View Results Setup for the single-session analysis.

Confirm that Table: Overall Reliability Summary is checked, leave the rest as they are, and click Generate Figures/Tables.

Several windows open at once. The one to read now is titled Dependability Analyses and headed Overall Dependability. It carries one row per group, because only one event was processed. The columns give the label, the numbers of participants included and excluded at your cutoff, the coefficient, and a summary of trial counts.

The Dependability cell for GroupA reads .72, with a 95% credible interval of [.56, .83]. Read it this

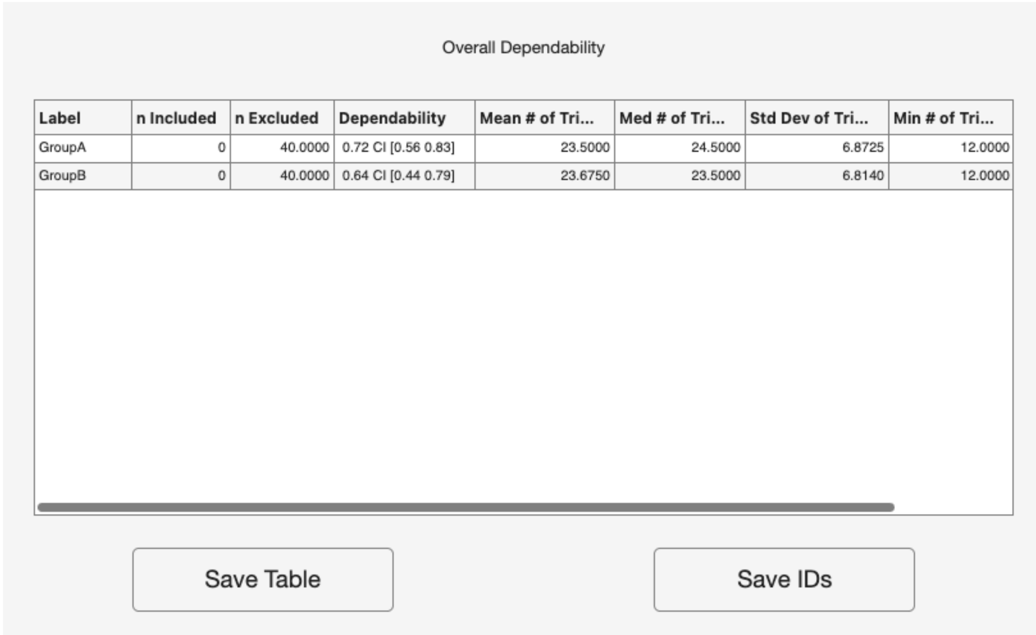


Figure 7: The Overall Dependability table, opened by the Table: Overall Reliability Summary checkbox.

way. Of the variance in GroupA's mean incongruent-error scores, computed at the average number of error trials its participants actually have, roughly that proportion reflects stable differences between participants rather than measurement error. The interval says how precisely the proportion is known. GroupB has its own row and its own answer, which is the whole point of estimating the groups separately.

Notice the n Included column: it reads 0 for GroupA, because the trial cutoff the next paragraph reports lies beyond every participant's error-trial count, so nobody is retained. When a group retains nobody at the cutoff, the coefficient is evaluated at the mean trial count of ALL its participants, so it describes the file as collected rather than a retained sample.

The trial cutoff is where this dataset earns its keep. PsyRAT reports the smallest number of trials at which the dependability point estimate reaches the 0.8 you left in the Reliability Cutoff box, and error trials here are scarce. In this file the cutoffs come back as 38 trials for GroupA and 55 for GroupB, and no participant supplied more than 36, so PsyRAT says so in a dialog titled Extrapolation beyond data. It reads "Not enough trials are present in the current data" and "Cutoffs represent an extrapolation beyond the data", and the trial-cutoff table (the window titled Results of Increasing the Number of Trials on Dependability) carries -1 where the coefficient at the cutoff would go. A value of -1 indicates that an appropriate cutoff could not be realistically found given the threshold you specified. That is a finding about your data rather than a malfunction, and Chapter 10 takes it seriously: at the .70 threshold that chapter argues for, the cutoffs land inside the data.

Selecting Save Table opens a dialog headed "Where would you like to save table?" offering Excel and comma-separated formats. The saved file carries a header block above the table with the toolbox version, the dataset name, the cutoff to four decimal places, and the run's provenance (seed, estimation engine, convergence status). Keep it. Chapter 17 says which of those lines belong in a manuscript.

Selecting Close PsyRAT Outputs closes the figures and tables PsyRAT drew and leaves every other MATLAB figure alone. Your results are already on disk in the `.psyrat` file, and the View Results button on the home screen reopens them without refitting anything.

Success! You have a dependability estimate with a credible interval, a trial cutoff, and a saved analysis you can reopen.

2.9 What just happened

PsyRAT read your table, split it by event, and fit one Bayesian multilevel model per event in CmdStan, with a separate set of variance components for each group inside it. That fit is the G-study: it partitioned the variance in single-trial scores into a between-person component, a trial component, and residual error, and returned posterior draws for each. The D-study then took those draws and projected them onto the score a researcher actually analyzes, which is an average over n trials, producing a coefficient for every value of n. That projection is the curve in the trials plot. It also produces the trial cutoff, by walking n upward until the point estimate reaches your threshold, and the single number in the summary table, evaluated at the mean trial count of the retained participants (or, as in this run, of all participants when a group retains nobody at the cutoff). Because the projection runs on draws rather than on point estimates, every coefficient carries a credible interval instead of a standard error borrowed from an asymptotic argument. Chapter 4 does all of this properly, with the formulas.

2.10 Where to go next

Chapter 10 reruns this analysis carefully, with the incongruent-correct event added, and interprets what comes out. After that, Chapter 11 covers test-retest, Chapter 12 covers difference scores, and Chapter 13 covers reliability from data splits. If you would rather have the theory before the practice, read Chapter 4. If anything above failed, Chapter 18 lists the errors by their literal text.

3. Installation and dependencies

PsyRAT itself is plain MATLAB code, and MATLAB code needs no installing. What needs installing is CmdStan, which does the Bayesian estimation, and the C++ toolchain CmdStan needs to compile each model. That is the whole problem. This chapter is deliberately short on click-by-click steps, because the ones I wrote for the predecessor toolbox ran to pages and then rotted with the operating systems they described. Requirements keep. Dialogs do not.

If the toolbox is already working on your machine, skip to Section 3.4 and confirm it.

3.1 What you need

Component	Version	Where it comes from
MATLAB	Tested on R2025b (Update 5)	You install it
CmdStan	2.26 or newer; 2.38.0 is the targeted and tested release	The guided installer, or you
MatlabStan (PsyRAT fork)	2.15.1.0	Ships with the toolbox
MatlabProcessManager	0.5.1	Ships with the toolbox
C++ toolchain	make and a C++ compiler	Your platform (see below)
Statistics and Machine Learning Toolbox	Conditional, see below	MathWorks
Deep Learning Toolbox	Optional	MathWorks

MATLAB. The tested release is R2025b (Update 5), and every release claim for the toolbox rests on runs made on the author's own machine. No minimum release is enforced anywhere in the code, and the precise minimum has not been pinned. The effective floor is R2022a: the dynamic-reliability surface plot calls `clim`, introduced in that release, and the export code calls `exportgraphics` and `writetable` with `WriteMode`, introduced in R2020a, all without fallback. Beyond those the toolbox relies on `table`, `string`, `categorical`, and the `matlab.unittest` framework, all of which are old features by now. Between R2022a and R2025b, the honest answer is that nobody has checked.

CmdStan. CmdStan is not bundled and must be installed separately. The generated Stan models use the `array[N] T name;` declaration syntax introduced in Stan 2.26, so anything older cannot compile them. Version 2.38.0 is the version the guided installer fetches and the version used for the accuracy and integration runs. If you already have a CmdStan on your machine, it is checked at startup: an installed version below 2.26 produces a warning when the toolbox opens and a hard error before the first model is compiled. The check is deliberately permissive, so a CmdStan whose version cannot be read is allowed through rather than blocked.

The two bundled dependencies. MatlabStan and MatlabProcessManager ship inside the toolbox in `bundled_dependents/`, and there is nothing to install for either. The bundled MatlabStan is a patched fork, not the upstream release, and the patches are load-bearing: they add CmdStan auto-discovery and within-chain threading, they remove three Statistics and Machine Learning Toolbox calls (two in the fit reader, which is the pair that would otherwise have made the toolbox mandatory on the default path, and one in a separately licensed subpackage), and they repair an indexing bug in the draw extraction. Use the bundled copy. If a different MatlabStan is on your MATLAB path ahead of it, the startup check says so.

C++ toolchain. CmdStan compiles every model with make and a C++ compiler, so one has to be present before CmdStan can be built or used. On macOS that is the Xcode Command Line Tools, and the installer offers to trigger the Apple installer for you. On Windows it is GNU make plus g++, both on the PATH; Rtools provides them, and the installer cannot install them for you. On Linux it is make plus g++, from `build-essential` on Debian and Ubuntu or the Development Tools group on Fedora and RHEL.

Statistics and Machine Learning Toolbox. For most analyses you do not need it. Three designs need it even on the default CmdStan path: the gamma (scaled chi-square) family combined with concurrent difference scores, which is analysis 7 (two-event difference), analysis 8 (its subject-level variant), and analysis 10 (two-facet test-retest difference). Those three read their results through a Gaussian copula that calls `normcdf` and `gaminv`, and neither function has a base-MATLAB equivalent. Both dispersion parameterizations are affected. The failure is late and expensive, because extraction runs after sampling has finished: without the toolbox you lose a completed fit to an undefined-function error rather than being stopped before the run. If you plan to use those designs (see Chapter 15), confirm the toolbox is installed first. Whether other designs have a similar dependency has not been settled. The toolbox is also required by both native estimation engines, which are the CmdStan-free fallbacks described in Chapter 5.

Deep Learning Toolbox. Optional. It supplies automatic-differentiation gradients for the native HMC engine, which runs without it on numeric gradients instead.

3.2 Install

FIRST, get the toolbox. Download a release from the PsyRAT repository, or clone it.

Put it somewhere whose full path contains no spaces. CmdStan does not accept whitespace in paths or filenames, and the toolbox enforces that on the files it writes: choose an output folder or an output filename containing a space and you get a Whitespace detected dialog rather than a failed run. Nothing enforces it on the folder you unpack the toolbox into, or on the folder CmdStan is built in, so that one is on you. `~/Documents/MATLAB/PsyRAT` is a safe choice. `~/My Lab Stuff/PsyRAT` is not.

Then run this in MATLAB, once, from inside the toolbox folder:

```
psyrat_start
```

That is also the installation step. `psyrat_start` adds its own directories to the MATLAB path (the toolbox root, `subroutines/`, and `bundled_dependents/`) and saves the path, so later sessions find it from anywhere. Do not `addpath(genpath(...))` the whole repository. Doing so also adds tooling and artifact folders, and a stale copy of a function inside one of those will shadow the real file with no warning at all.

If the Stan dependencies are missing, `psyrat_start` prints a warning that they are not in the MATLAB path, stops before the home screen, and raises a small window offering two buttons: Install Dependents and Exit. Selecting Install Dependents runs the guided installer, which you can also run directly with `psyrat_installdependents`.

The installer asks one question first, in a save dialog headed "Where would you like to save the directory for the dependents?”. Answer it with a folder whose path has no spaces. The installer creates a folder called PsyRATDependents there and puts everything inside it. Choose `~/Documents/MATLAB` if you have no preference, because that is one of the locations the toolbox searches on its own later (Section 3.3), and a PsyRATDependents folder somewhere else is then found only through the saved MATLAB path.

From there the installer works down a checklist and shows its progress in a window named Installation Summary.

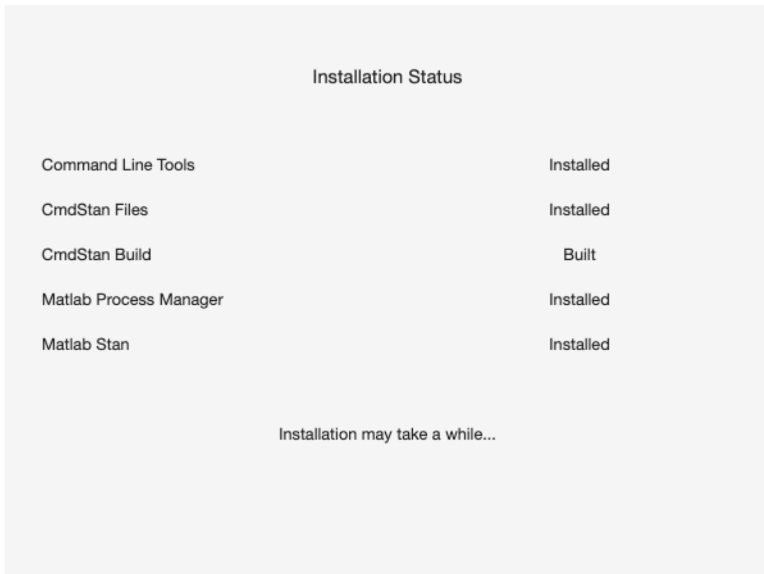


Figure 8: The Installation Summary window, shown while the guided installer works. Each row is one dependency; the status column changes from Not Installed to Installed as each step finishes.

The window lists five rows: Command Line Tools, CmdStan Files, CmdStan Build, Matlab Process Manager, and Matlab Stan. Each carries a status of Installed or Not Installed (Built or Not Built for the build row), and each is redrawn as the step behind it completes. The same status text also goes to the Command Window, which is where to look if something fails, because the failure message is printed there and not in the window. The line at the bottom reading ”Installation may take a while...” is not a figure of speech. Building CmdStan compiles a large amount of C++, takes several minutes on a current laptop, and uses all but one of your cores while it does (all of them on a two-core machine).

What the installer actually does: it checks for a C++ toolchain, downloads the CmdStan release tarball for the targeted version from the Stan project's GitHub releases, unpacks it into PsyRATDependents, builds it with `make`, copies the bundled MatlabStan and MatlabProcessManager in beside it, points MatlabStan at the CmdStan it just built, adds the whole PsyRATDependents tree to the MATLAB path, saves the path, and relaunches `psyrat_start`.

On macOS it inserts one extra step. If the Xcode Command Line Tools are not present, the installer starts Apple's own installer and then waits behind a window reading ”Please indicate when the XCode installation is complete”, with a single button labeled Installation Complete. Click it when Apple's installer has finished, not when it has started. The toolbox rechecks the install location rather than taking the button press as proof, so clicking early produces an error telling you to install the command line tools manually.

On Windows the installer cannot set up the toolchain. If GNU make and g++ are not already on the PATH it stops with a message pointing you at Rtools, and you have to install that yourself before rerunning. Windows is not part of the test matrix: the toolbox follows the general MatlabStan and CmdStan support there, but no Windows run has been verified by the author.

On Linux the installer runs, and it is EXPERIMENTAL. That branch mirrors the macOS path (check for make and g++, download the CmdStan tarball, build it), and it has not been verified on a Linux machine by the author. It announces this with a warning when it starts. Estimation itself is fine on Linux once CmdStan is discoverable. If the automatic install fails, install CmdStan yourself and point the toolbox at it as described next.

3.3 Installing or pointing at CmdStan yourself

You do not have to use the guided installer. A CmdStan you built yourself, or one that came with another Stan interface, works as long as the toolbox can find it and it meets the minimum version.

Discovery happens in a fixed order, and the first candidate that looks like a real CmdStan root (a `makefile` and a `bin` directory) is the one used:

1. The environment variables `STAN_HOME`, `CMDSTAN_HOME`, and `CMDSTAN`, in that order. (One further name, checked ahead of these, is reserved for the test harness.)
2. A `cmdstan-*` folder inside `PsyRATDependents` under your home directory: `Documents/MATLAB/PsyRATDependents`, `Documents/PsyRATDependents`, or `MATLAB/PsyRATDependents`. A `cmdstan-*` folder directly in your home directory is checked too. Where one location holds several versioned folders, the newest version wins; pin an older one with an environment variable if you need it.
3. Whatever is reachable on the MATLAB path.

An environment variable therefore always beats a toolbox-managed install, which is what you want when you are deliberately testing against a different CmdStan. The home-directory install beats a MATLAB path entry, which is also what you want, since it keeps an unrelated CmdStan that happens to be on the path from displacing the one built for the toolbox.

The startup check in `psyrat_start` uses this same discovery order, so any CmdStan the list above can find, including one named only by an environment variable, satisfies it. There is no need to add the CmdStan root to your MATLAB path just to get past the launch check. (Earlier versions checked the MATLAB path alone, and an environment-variable-only setup produced a spurious missing-dependencies report at launch; that behavior is gone.)

For building CmdStan itself, follow the Stan project's own instructions at https://mc-stan.org/users/interfaces/cmdstan rather than anything reproduced here. Those steps are maintained by people who test them on every release, which is more than can be said for a printed manual.

3.4 Verify the installation

Run `psyrat_start` and read the Command Window before you touch the buttons. A working installation prints six lines:

```
Ensuring dependents are found in the Matlab path
PsyRAT Toolbox files found
CmdStan, MatlabProcessManager, and MatlabStan files found
CmdStan version is current
```

```
MatlabProcessManager version is current
MatlabStan version is current
```

Three blank lines later comes the banner, reading ”PsyRAT Version” followed by the version string of your installation. Report that string in any bug report. The second line reads ”Directories for PsyRAT Toolbox files added to Matlab path” instead on the first launch after installing, which is the toolbox putting itself on the path, and it reverts to the line shown above afterward.

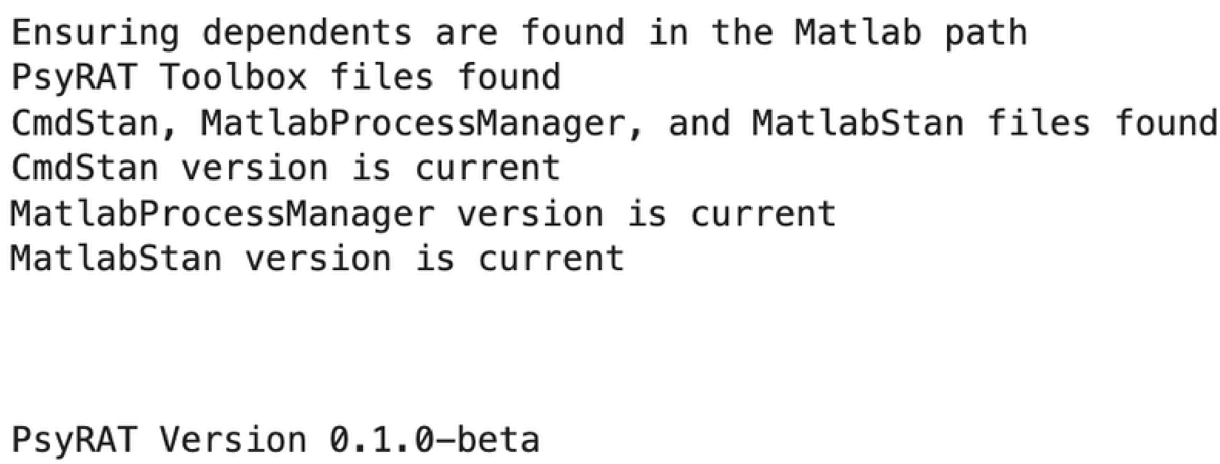


Figure 9: The MATLAB Command Window immediately after psyrat_start, showing the dependency check lines and the version banner.

The three version lines are advisory. For CmdStan and MatlabProcessManager, a dependency behind the version the toolbox targets produces a line saying a new version is available with an offer to update, and one ahead of it produces a warning that it is newer than the version the toolbox was tested with. The offer is a window naming the dependents that are behind, reading ”It is recommended that you update the dependents. Would you like to do so now?” with a warning that updating deletes the old dependents directories, and it offers Yes and No. The MatlabStan line reports something different: whether the copy in use carries the patches PsyRAT ships, with a third outcome when the patch level cannot be determined. When the copy in use is the bundle PsyRAT ships inside its own folder and its stamp does not match this release, the line instead reads ”PsyRAT's bundled pinned copy is in use (updated with the toolbox, not the dependents)”: the bundle cannot be replaced by the dependents updater, so no update is offered for it, and updating the toolbox itself is what refreshes it. Estimation is not blocked in any of these cases.

You may also see a message from the release check, and on a beta build you should expect one. The toolbox asks GitHub for the latest published release and compares it against the version you are running. When there is no published stable release to compare against, or when your version is ahead of the newest one, it says so and recommends caution. That message is informational, and it reports nothing about whether your installation works.

The last check is a real fit. Run the analysis in Chapter 2 and let it finish. Every estimation starts by compiling the generated Stan model to a binary before sampling, which adds about 8 s per generated model on top of the sampling itself (the toolbox names each generated model uniquely, so the compile happens on every run, not just the first). The binaries land in a folder named PsyRAT_CmdStan on your Desktop, or in the current MATLAB directory when there is no Desktop folder, and the folder is safe to empty between sessions. If that first run finishes and produces a results window, everything downstream of this chapter is working.

One thing worth knowing before you start it: the models run in C++, not in MATLAB, so interrupting MATLAB with ctrl+c stops MATLAB waiting for the sampler rather than stopping the sampler.

Success! CmdStan is installed and discoverable, the bundled MatlabStan is talking to it, and the toolbox is on your MATLAB path and will still be there in your next session. If any of the checks above failed instead, the symptom-by-symptom fixes are in Chapter 18.

4. Generalizability theory for ERP scores

An ERP score is an average. A participant's ERN amplitude is the mean of some number of single-trial values, each of which carries the participant's standing on the construct plus whatever else was happening in that epoch. Classical test theory splits such a score into true score and error, which forces every source of inconsistency into one undifferentiated term and makes the resulting coefficient depend on which source the chosen method happens to capture. Generalizability theory replaces that single error term with a set of estimated variance components, one for each source the design can identify, and then builds reliability coefficients out of them (Rocha et al., 2026). PsyRAT implements that framework. This chapter defines the quantities the toolbox estimates, states the formulas it evaluates, and maps each symbol onto the column the toolbox prints.

Readers who want the shortest path to a working analysis should read Chapter 2 first and return here when the output needs interpreting. Readers deciding which analysis their design supports should read this chapter and then Chapter 5.

4.1 Persons crossed with trials

Consider a researcher who records ERN from 80 participants in a flanker task and scores mean amplitude on every error trial. Each participant contributes many trials, and every participant is measured on the same kind of trial, so persons and trials are crossed. A facet is a characteristic of the measurement situation that varies and that could be sampled differently: here, trials. The object of measurement is the participant, and it is not a facet. The universe score is the participant's expected score across all conditions in the universe of generalization, which is the generalizability-theory analog of the true score in classical test theory (Rocha et al., 2026, Table 1, p. 3).

A single-trial score decomposes into a grand mean and three deviations:

$$X_{pi} \;=\; \mu \;+\; (\mu_p - \mu) \;+\; (\mu_i - \mu) \;+\; (X_{pi} - \mu_p - \mu_i + \mu) \qquad (1)$$

In words, the score of person p on trial i equals the mean across persons and trials, plus a person effect, plus a trial effect, plus everything left over. The leftover term is the person-by-trial interaction confounded with residual error, because each person is observed on each trial only once, and the two cannot be separated.

The three deviation terms are uncorrelated in the random-effects model that generalizability theory assumes, so the variance of a single-trial score partitions into three components:

$$\sigma^2_X \;=\; \sigma^2_p \;+\; \sigma^2_i \;+\; \sigma^2_{pi,e} \qquad (2)$$

That is, the total variance of single-trial scores is the sum of variance between persons, variance between trials, and person-by-trial residual variance. The first component is the signal for individual-differences research. The second reflects mean differences among trials, which shift every participant's score in the same direction and therefore do not reorder participants. The third is what averaging over trials reduces.

A facet observed at only one level is hidden: its variance component is not identifiable, and the reliability that results is conditional on the single level observed (Rocha et al., 2026, Table 1, p. 3). A single-session study hides the occasion facet, so its coefficients confound stable person differences with whatever was specific to that session. This is not a defect in the analysis, but it does bound the claim the analysis supports.

Where this is derived: Rocha et al. (2026), Tables 1 and 2 (pp. 3, 7); Baldwin, Larson, and Clayson (2015), Equations 1 and 2 (p. 792).

4.2 Adding occasions

If the same participants return for a second session, occasion becomes an estimable facet and the design becomes persons crossed with trials crossed with occasions. The partition then carries seven components:

$$\sigma_X^2 \;=\; \sigma_p^2 + \sigma_i^2 + \sigma_o^2 + \sigma_{pi}^2 + \sigma_{po}^2 + \sigma_{io}^2 + \sigma_{pio,e}^2 \qquad (3)$$

The three main effects are persons, trials, and occasions. The three two-way interactions are person-by-trial, person-by-occasion, and trial-by-occasion. The final term is the three-way interaction confounded with residual error.

Two of these components carry most of the interpretive weight in retest studies. The person-by-trial component, σ_{pi}^2, is specific-factor error: a systematic source that is not the construct and that reflects idiosyncratic responding across trials. The person-by-occasion component, σ_{po}^2, is transient error: variation unique to the individual that changes across sessions, often reflecting state fluctuations unrelated to the construct (Rocha et al., 2026, pp. 5 to 6). Clayson, Carbine, et al. (2021) report an N2 dataset in which σ_{po}^2 was the largest error component apart from the residual, roughly 40% the size of σ_p^2, which placed a ceiling on retest reliability that no number of additional trials could raise.

Where this is derived: Rocha et al. (2026), Table 3 (p. 8); Clayson, Carbine, et al. (2021), pp. 178 to 183.

4.3 Reading the variance components in PsyRAT output

Selecting Table: Sources of Variance on the View Results Setup screen produces a table of standard deviations, standard errors of measurement, and intraclass correlations, each with a point estimate and a credible interval at the width selected in the viewer (95% by default). The toolbox reports these components on the standard-deviation scale rather than the variance scale, because standard deviations are in microvolts and can be compared directly against the ERP scores themselves. Every formula in this chapter uses variances, so each printed value is squared before it enters a coefficient.

The mapping between the symbols above and the printed columns is the single most useful thing to memorize about the output. For a one-facet analysis:

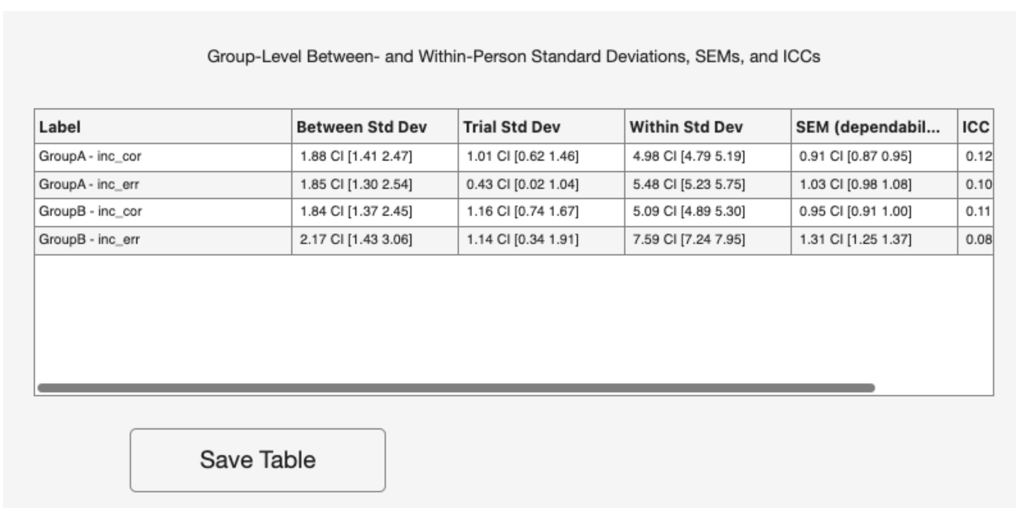
Symbol	Meaning	Printed column
σ_p	between-person standard deviation	Between Std Dev
σ_i	trial main effect	Trial Std Dev
$\sigma_{pi,e}$	person-by-trial residual	Within Std Dev

Symbol	Meaning	Printed column
$\sqrt{\widehat{\sigma}^2(\Delta)}$ or $\sqrt{\widehat{\sigma}^2(\delta)}$	standard error of measurement	SEM (dependability) or SEM (generalizability)
$\widehat{\rho}_{\mathrm{ICC}}$	intraclass correlation	ICC (dependability) or ICC (generalizability)

Two features of that table are easy to misread. The Within Std Dev column holds the residual $\sigma_{pi,e}$ alone, not a combined within-person term, so the trial main effect is reported separately in its own column rather than folded into it. The SEM and ICC headers carry a parenthetical decision type because both quantities change value with the G-Theory Coefficient setting, and an archived export would otherwise be impossible to reconstruct. If the analysis uses data splits rather than single trials, the Trial Std Dev header reads Split Std Dev, and the analysis unit is the split throughout.

Test-retest analyses print a five-column version of the same table, in which Between Std Dev is σ_p and Within Std Dev is the three-way residual $\sigma_{pio,e}$. All seven components are listed in the window reached through the View All Standard Deviations button, two of which repeat the five-column table:

Symbol	Printed column
σ_p	Between Person
σ_o	Between Session
σ_i	Between Trial
σ_{po}	Person x Session
σ_{pi}	Person x Trial
σ_{io}	Session x Trial
$\sigma_{pio,e}$	Within Person (Error)



Label	Between Std Dev	Trial Std Dev	Within Std Dev	SEM (dependabil...	ICC
GroupA - inc_cor	1.88 CI [1.41 2.47]	1.01 CI [0.62 1.46]	4.98 CI [4.79 5.19]	0.91 CI [0.87 0.95]	0.12
GroupA - inc_err	1.85 CI [1.30 2.54]	0.43 CI [0.02 1.04]	5.48 CI [5.23 5.75]	1.03 CI [0.98 1.08]	0.10
GroupB - inc_cor	1.84 CI [1.37 2.45]	1.16 CI [0.74 1.67]	5.09 CI [4.89 5.30]	0.95 CI [0.91 1.00]	0.11
GroupB - inc_err	2.17 CI [1.43 3.06]	1.14 CI [0.34 1.91]	7.59 CI [7.24 7.95]	1.31 CI [1.25 1.37]	0.08

Figure 10: Sources of Variance table for a one-facet analysis, showing point estimates and credible intervals for each component.

Comparing the components against one another is the first diagnostic step in any generalizability study, because their relative sizes indicate which sources of variance dominate the reliability estimate (Clayson, Carbine, et al., 2021, p. 180). Which redesign would help follows from where the dominant component sits in Equation 5, Equation 6, or Equation 9. A large $\sigma_{pi,e}$ relative to σ_p means more trials will help. A large σ_{po} relative to σ_p means more trials will not.

Where this is derived: Rocha et al. (2026), Tables 2 and 3 (pp. 7 to 8); Clayson, Carbine, et al. (2021), p. 180.

4.4 Two coefficients, two decisions

Reliability in generalizability theory is a ratio of universe-score variance to universe-score variance plus error variance, and the two coefficients the toolbox reports differ only in what counts as error. For a one-facet design with scores averaged over n'_i trials:

$$\hat{\sigma}^2(\delta) \;=\; \frac{\sigma^2_{pi,e}}{n'_i} \qquad\qquad \hat{\sigma}^2(\Delta) \;=\; \frac{\sigma^2_{pi,e}}{n'_i} + \frac{\sigma^2_i}{n'_i} \tag{4}$$

Relative error variance, $\hat{\sigma}^2(\delta)$, contains only the component that varies idiosyncratically across persons. Absolute error variance, $\hat{\sigma}^2(\Delta)$, adds the trial main effect, because a shift common to all participants still moves each participant's score away from its universe value even though it leaves the ordering of participants intact.

The generalizability coefficient uses relative error:

$$E\rho^2 \;=\; \frac{\sigma^2_p}{\sigma^2_p + \dfrac{\sigma^2_{pi,e}}{n'_i}} \tag{5}$$

The dependability coefficient uses absolute error:

$$\phi \;=\; \frac{\sigma^2_p}{\sigma^2_p + \dfrac{\sigma^2_{pi,e}}{n'_i} + \dfrac{\sigma^2_i}{n'_i}} \tag{6}$$

In words, generalizability is the proportion of variance in an averaged score that reflects stable person differences once sources that reorder participants are treated as error, and dependability is the same proportion once every measured source is treated as error. For a single-score design, dependability is never larger than generalizability, and the two are equal when $\sigma^2_i = 0$, which happens when trials do not differ systematically in their means. (For difference scores, where estimated trial-effect covariances enter the absolute error, the ordering can reverse in unusual configurations; Section 4.7 returns to this.)

A worked contrast makes the size of the gap concrete. The manual's tutorial dataset is simulated, so its generating values are known: for GroupA incongruent-error trials (the inc_err cell), $\sigma_p = 2.0$, $\sigma_{pi,e} = 5.5$, and $\sigma_i = 1.0$ microvolts, giving $\sigma^2_p = 4.00$, $\sigma^2_{pi,e} = 30.25$, and $\sigma^2_i = 1.00$. Averaged over 17 error trials,

$$E\rho^2 = \frac{4.00}{4.00 + 30.25/17} = \frac{4.00}{5.779} = 0.692$$

The trial main effect enters absolute error only, so dependability sits below that: the absolute error is $(30.25 + 1.00)/17 = 1.838$, and $\phi = 4.00/5.838 = 0.685$. With this dataset's modest trial main effect the gap is small. Now suppose a researcher analyzed real data in which the trial main effect was estimated at $\sigma_i = 2.0$ microvolts, so $\sigma_i^2 = 4.00$. Relative error is unchanged at 1.779, but absolute error becomes $(30.25 + 4.00)/17 = 2.015$, and

$$\phi = \frac{4.00}{4.00 + 2.015} = 0.665$$

Generalizability of 0.692 and dependability of 0.665 describe the same scores. They differ because they quantify different error definitions, and the choice between them is a research decision, not a statistical one. If the inference concerns the relative standing of participants, as it does for correlations, group comparisons, and most individual-differences work, the generalizability coefficient is the relevant one. If the inference concerns a participant's absolute level, as it does for cut scores, clinical thresholds, and any claim about the magnitude of a score, the dependability coefficient is the relevant one. The toolbox exposes this as the G-Theory Coefficient control on the View Results Setup screen, with the options Dependability and Generalizability. Cut scores are always computed with absolute error regardless of that setting, because a threshold decision is by definition an absolute one.

Where this is derived: Rocha et al. (2026), Table 2 (p. 7) and their Section 3.3.1 (p. 5).

4.5 Standard error of measurement

The two error variances have matching standard errors of measurement, which are simply their square roots:

$$\mathrm{SEM}_\delta = \sqrt{\frac{\sigma^2_{pi,e}}{n'_i}} \qquad \mathrm{SEM}_\Delta = \sqrt{\frac{\sigma^2_{pi,e}}{n'_i} + \frac{\sigma^2_i}{n'_i}} \tag{7}$$

The standard error of measurement is expressed in the units of the score, which for ERP amplitude means microvolts, and it quantifies the typical distance between an observed averaged score and the universe score behind it. Continuing the example above, the relative standard error at 17 trials is $\sqrt{1.779} = 1.33$ microvolts and the absolute standard error is $\sqrt{1.838} = 1.36$ microvolts. The between-person standard deviation for that cell is 2.0 microvolts, so an individual averaged score carries roughly 1.4 microvolts of measurement error against 2.0 microvolts of true between-person spread, which is the comparison the standard error of measurement exists to make available, and which is why the dependability there is only .685.

The toolbox evaluates Equation 7 at the trial count actually available rather than at a single trial. That count is the mean number of retained trials across participants who met the trial cutoff, or the median if the median was selected in the viewer preferences. The reported standard error is therefore the standard error of the scores in hand, and it moves with the G-Theory Coefficient setting in step with the coefficient printed beside it.

Where this is derived: Rocha et al. (2026), Table 2 (p. 7).

4.6 The decision study: projecting reliability onto a design

Generalizability theory separates two phases. A generalizability study estimates the variance components. A decision study uses those components to project reliability for a measurement procedure that was not necessarily the one run, most often by varying the number of trials retained for averaging (Baldwin et al., 2015, p. 793). The variance components do not change between the two phases. Only n_i' changes.

Because n_i' appears only in the denominator of Equations 5 and 6, the projection from those two equations is monotonic: more trials yield a higher coefficient, with diminishing returns. (The coefficient of stability of Equation 9 is the exception: when the person-by-trial variance is large relative to the person-by-occasion terms, adding trials can lower it.) For the tutorial values above, dependability rises from 0.390 at 5 trials to 0.561 at 10, 0.685 at 17, 0.793 at 30, and 0.865 at 50. Inverting Equation 6 gives the count a researcher needs for a target coefficient ρ:

$$n_i' = \frac{\rho}{1-\rho} \cdot \frac{\sigma^2_{pi,e} + \sigma^2_i}{\sigma^2_p} \qquad (8)$$

In words, the required number of trials is the target coefficient expressed as odds, multiplied by the ratio of error variance to between-person variance. With $\sigma^2_i = 0$, reaching a dependability of 0.80 requires $4 \times (30.25/4.00) = 30.25$ trials, so 31 trials. Adding the hypothetical trial main effect of $\sigma^2_i = 4.00$ raises the requirement to 34.25, so 35 trials. That is the practical cost of a trial main effect: four additional error trials per participant, in a paradigm where error trials are the scarce resource.

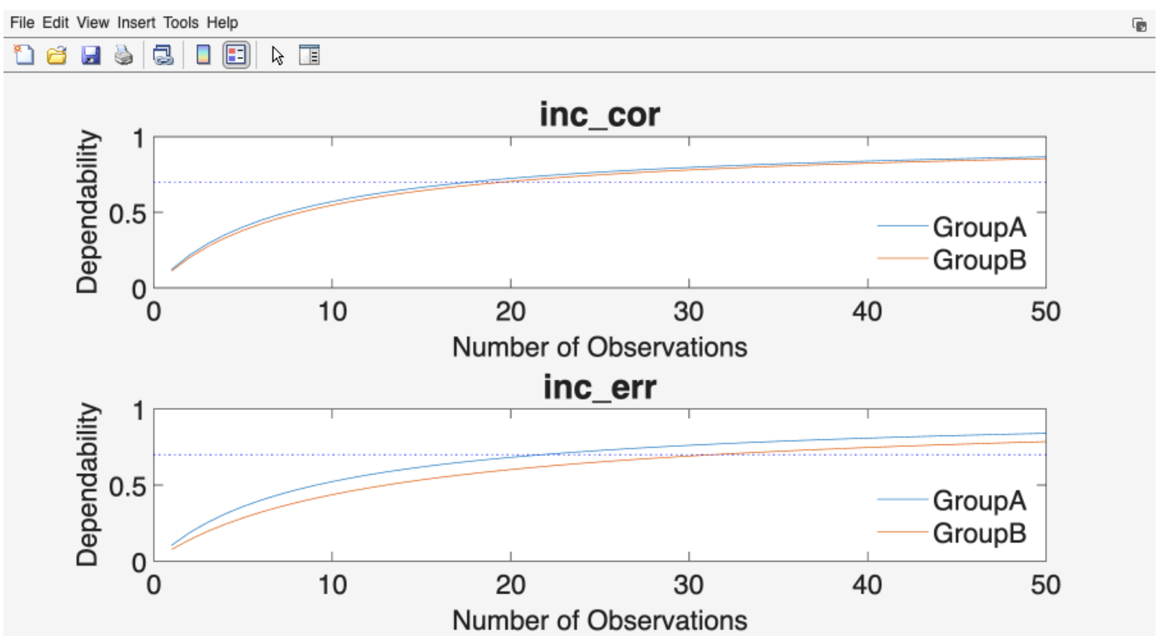


Figure 11: Dependability as a function of the number of trials retained for averaging, one panel per event and one curve per group, with the reliability cutoff drawn as a dotted horizontal line.

The plot produced by the toolbox draws that curve directly, one panel per event, with the number of observations on the horizontal axis and the coefficient on the vertical axis, bounded above at 1.

The dotted horizontal line is the reliability cutoff, and the point where a curve crosses it is the trial count that design requires.

Choosing that cutoff is a judgment the software cannot make. Clayson and Miller (2017b, p. 62) recommend a threshold of at least 0.70 for exploratory ERP research and 0.80 in most contexts. The reasoning behind those two values is that 0.70 buys enough precision to decide whether a measure is worth developing further, while 0.80 is what group-difference research needs before unreliability begins attenuating the effects under test. The toolbox default is 0.80. The exception runs upward, not downward: when a decision is made about an individual participant's score, as in clinical classification, 0.90 is a minimum and 0.95 is preferred. The illegitimate reason to lower the threshold is that the data did not reach it. A threshold chosen after the coefficient is known is not a threshold, and the honest response to a coefficient below the stated cutoff is to report it and to state what the analysis therefore cannot support.

For a two-facet test-retest design the projection has two arguments, n_i' and n_o'. The dependability coefficient of stability, which is the retest analog, takes the form

$$\phi_{CS} = \frac{\sigma_p^2 + \frac{\sigma_{pi}^2}{n_i'}}{\sigma_p^2 + \frac{\sigma_{pi}^2}{n_i'} + \frac{\sigma_{po}^2}{n_o'} + \frac{\sigma_{pio,e}^2}{n_i' n_o'} + \frac{\sigma_o^2}{n_o'} + \frac{\sigma_{io}^2}{n_i' n_o'}} \tag{9}$$

The person-by-trial component sits in the numerator here, because a coefficient of stability treats consistency across trials as part of the universe score and asks only whether scores hold up across occasions. The coefficient of equivalence exchanges those roles, placing σ_{po}^2 in the numerator and σ_{pi}^2 in the error term. The coefficient of trial equivalence and stability keeps σ_p^2 alone in the numerator and treats both interactions as error, which makes it the most demanding of the three.

By the toolbox's convention, every test-retest coefficient defaults to $n_o' = 1$. The estimand is then the reliability of a score collected at one occasion and generalized over the occasion facet, which is the score a researcher actually has in hand after a single visit. Setting $n_o' = k$ instead reports the reliability of a composite averaged over k occasions, a different and generally higher quantity, and the toolbox labels the two distinctly wherever they appear so that a coefficient is never paired with the other estimand's standard error. For the coefficient of stability the $n_o' = 1$ default matches the practice described by Clayson, Carbine, et al. (2021, p. 183). Selecting the multi-occasion composite is an explicit choice, described in Chapter 9.

Where this is derived: Rocha et al. (2026), Table 3 (p. 8); Baldwin et al. (2015), Equation 4 (p. 793); Clayson and Miller (2017b), p. 62; Clayson, Carbine, et al. (2021), p. 183.

4.7 Difference scores

Many ERP measures are contrasts. ERN is frequently scored as error minus correct, and lateralized indices are scored as one hemisphere minus the other. The reliability of such a score is not the reliability of either constituent, and it is usually lower than both.

Write X and Y for the two constituent measures. The universe-score variance of the difference and its two error terms are

$$\sigma_{p(X-Y)}^2 = \sigma_{X(p)}^2 + \sigma_{Y(p)}^2 - 2\sigma_{XY(p)} \tag{10}$$

$$\hat{\sigma}^2(\delta) = \frac{\sigma^2_{X(pi,e)}}{n'_{iX}} + \frac{\sigma^2_{Y(pi,e)}}{n'_{iY}} - \frac{2\sigma_{XY(pi,e)}}{\ddot{n}'_i} \qquad \hat{\sigma}^2(\Delta) = \hat{\sigma}^2(\delta) + \frac{\sigma^2_{X(i)}}{n'_{iX}} + \frac{\sigma^2_{Y(i)}}{n'_{iY}} - \frac{2\sigma_{XY(i)}}{\ddot{n}'_i} \tag{11}$$

Each coefficient places Equation 10 over Equation 10 plus one of the error terms in Equation 11: the relative error term for generalizability, the absolute one for dependability. In words, the numerator is the between-person variance the two conditions do not share, and the denominator adds the measurement error of both conditions minus whatever error they share. Covariance terms are divided by the harmonic mean of the two trial counts, written $\ddot{n}'_i$, the convention Rocha et al. (2026, Table 6) adopt; since $2/\ddot{n}'_i = 1/n'_{iX} + 1/n'_{iY}$, this is exactly the scaling that makes the difference-score error variance vanish when the two conditions' components are identical.

The subtraction in the numerator is why difference scores are typically less dependable than their constituents. Whatever between-person variance the two conditions share is removed from the signal, while the error variances add rather than cancel. The tutorial dataset makes the arithmetic explicit. Error-trial and correct-trial person effects were generated with a correlation of $r = 0.75$, with $\sigma_p = 2.0$ for error trials and $\sigma_p = 1.7$ for correct trials, so $\sigma_{XY(p)} = 0.75 \times 2.0 \times 1.7 = 2.55$. The universe-score variance of the difference is $4.00 + 2.89 - 2(2.55) = 1.79$, which is smaller than either constituent. Error trials are noisier and scarcer than correct trials, so at 17 error trials and 32 correct trials the relative error variance is $30.25/17 + 25.00/32 = 2.561$, and

$$E\rho^2_{X-Y} = \frac{1.79}{1.79 + 2.561} = 0.411$$

The generating trial main effects ($\sigma_i = 1.0$ per cell) are drawn independently across conditions, so their cross-condition covariance is zero and dependability differs from generalizability only by the two σ^2_i/n'_i terms: absolute error is $31.25/17 + 26.00/32 = 2.651$ and $\phi_{X-Y} = 1.79/4.441 = 0.403$. The constituent generalizability coefficients at those same trial counts are 0.692 for error trials and 0.787 for correct trials. The difference score is markedly less reliable than either, and the correlation between conditions is the reason. Had the person effects been uncorrelated, the numerator would have been 6.89 instead of 1.79, and the same error variance would have produced a coefficient of 0.729.

Two further points govern how the toolbox handles these designs. First, the residual covariance $\sigma_{XY(pi,e)}$ is estimable only when the two measures are recorded concurrently, as in an ipsilateral-versus-contralateral contrast. When they come from separate trials, as error and correct trials do, the residual covariance is set to zero (Rocha et al., 2026, p. 15). Second, a low difference-score coefficient does not by itself indicate a flawed measure. When true between-person differences in the contrast are genuinely small, or when participants differ but all to the same degree, the numerator is small or zero and the coefficient is correspondingly low even for a perfect instrument; the toolbox reports a zero numerator as an exact zero with a diagnostic rather than treating it as an error (Rocha et al., 2026, Section 5.1, p. 15). Correlated measurement error runs the other way and can inflate the coefficient. One further property of these estimates deserves stating: the difference-score error terms are differences of estimated components, so a legitimate posterior draw can push a coefficient outside the zero-to-one range, and the toolbox reports the value as evaluated together with a diagnostic rather than silently repairing it.

The difference-score row of the All Standard Deviation Components window, reached from the View All Standard Deviations button, carries an ICC computed at a single observation, labeled

ICC (n'=1). It is a derived quantity for the difference score rather than an entry in the published difference-score formulas; the label records the n'=1 convention, and the derivation is documented here and in Chapter 12.

Where this is derived: Rocha et al. (2026), Table 6 and their Section 5.1 (pp. 13, 15).

4.8 The intraclass correlation

The intraclass correlation is the coefficient evaluated at a single observation. It addresses a different question from a dependability coefficient: not how reliable an averaged score is, but how much of the variance in one trial is between-person variance.

$$\rho_{\text{ICC},\delta} = \frac{\sigma_p^2}{\sigma_p^2 + \sigma_{pi,e}^2} \qquad \rho_{\text{ICC},\Delta} = \frac{\sigma_p^2}{\sigma_p^2 + \sigma_{pi,e}^2 + \sigma_i^2} \tag{12}$$

The absolute version has the plainest reading. An absolute ICC of 0.40 indicates that 40% of the variance in single-trial scores is stable between-person variance and the remaining 60% counts as error for an absolute decision. The relative version removes the trial main effect from the denominator, so its percentage is the share of variance among only those sources that affect the relative standing of participants. The two are equal when $\sigma_i^2 = 0$.

Because the ICC ignores the number of trials, it is a trial-independent index of how much person-level signal each observation carries (Rocha et al., 2026, p. 16). That property is what makes it a useful companion to the dependability coefficient rather than a substitute for it. The tutorial values illustrate the gap. For GroupA error trials the relative ICC is 4.00/34.25 = 0.117 and the absolute ICC, which keeps the trial main effect in the denominator, is 4.00/35.25 = 0.113: fewer than 12% of the variance in a single error trial is between-person variance either way. The dependability of the same scores averaged over 17 trials is 0.685. Averaging is what converts a weak per-trial signal into a usable score, and a design that yields more usable trials can produce more dependable scores from an equally noisy signal.

The toolbox plots these estimates as a forest plot, with the events listed down the left, the intraclass correlation running along the bottom, and one point estimate with its credible interval for each group and event. Reading it means comparing the horizontal positions of the points, which indicate per-trial signal, and the widths of the intervals, which indicate how well that signal was estimated. Overlap between two intervals is not by itself evidence that two events carry the same per-trial signal, since that claim concerns the posterior distribution of their difference and not the two marginal intervals. Events with similar ICCs can still differ substantially in dependability whenever their trial counts differ, which is the point of reporting both.

Where this is derived: Rocha et al. (2026), Table 2 and their Section 8 (pp. 7, 16).

4.9 Why the estimates are Bayesian

Every formula in this chapter takes variance components as input. How those components are estimated determines whether the formulas can be evaluated at all. PsyRAT estimates them with Bayesian multilevel models fitted in CmdStan, for reasons that bear directly on the structure of ERP data.

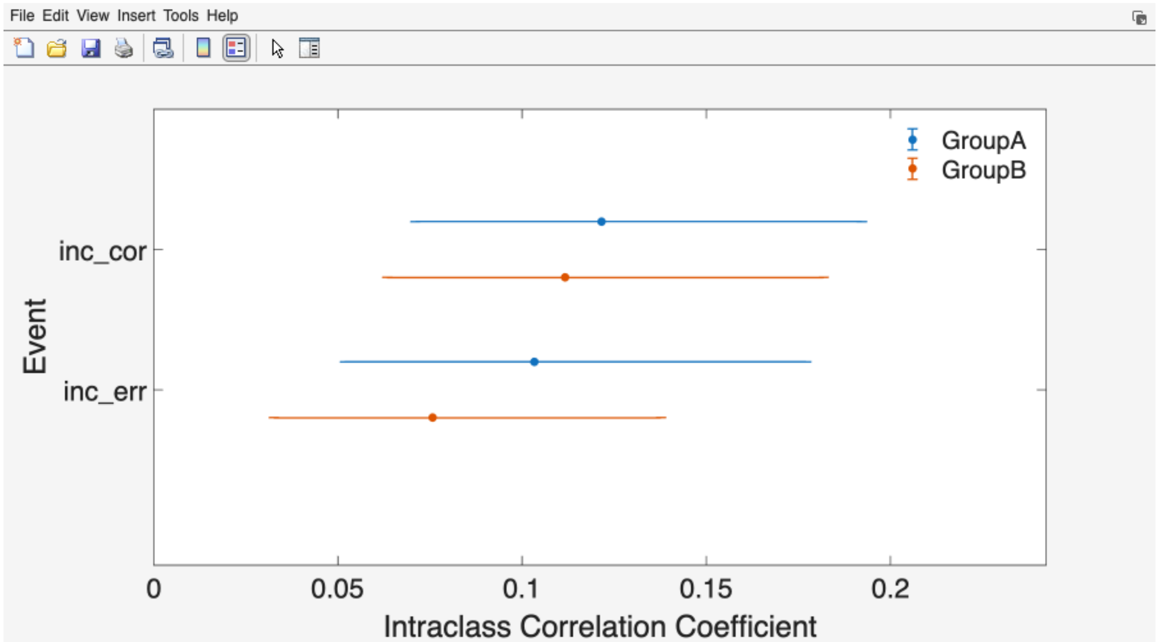


Figure 12: ICC Estimates plot, one point estimate and credible interval per event and group.

Problem with the usual approach	What the Bayesian estimates provide
Random-effects ANOVA on unbalanced data can return negative variance components, which makes the coefficients uninterpretable	Variance components are constrained to be positive by the model's parameter space, so a negative component estimate cannot occur (derived difference-score error terms can still leave the usual range; Section 4.7)
Restricted maximum likelihood sets negative estimates to zero during iteration, which blocks standard errors and intervals	Every component carries a full posterior distribution, so credible intervals accompany the coefficients, standard errors, and intraclass correlations in the main output tables (a few summary windows print point estimates only)
Conventional inference assumes homoscedastic residuals, and departures distort variance-component estimates in the small, unbalanced samples common to psychophysiology	In the subject-level and dynamic designs, residual variance is modeled explicitly rather than assumed constant
Reliability is reported for the group as a whole	The same posterior draws support subject-level coefficients, which the toolbox reports for the designs that estimate them

Trial counts in ERP research are unbalanced by construction, because artifact rejection removes different numbers of epochs from different participants, and error trials are scarce. That is the condition under which the first two problems actually bite (Rocha et al., 2026, Section 7, pp. 15 to 16).

Priors

The toolbox ships weakly informative default priors, and those defaults are calibrated on the ERP microvolt scale. They are not scale-free. A measure recorded on a very different scale, such as standardized scores or a modality whose values are orders of magnitude larger or smaller, can make the same priors informative, which would let the prior rather than the data drive the estimates. Researchers analyzing scores on another scale should set the priors deliberately rather than accepting the defaults. Chapter 15 covers the priors the toolbox exposes and how to override them.

Sampling controls

Four processing preferences govern the fit. Each is set on the Specify Processing Preferences screen, described in Chapter 7.

Control	What it does	Default
Number of Chains	independent sampling runs started from different points; agreement among them is the evidence of convergence	4
Warmup iterations	iterations used to tune the sampler and discarded before estimation	5000
Sampling iterations	iterations retained per chain to form the posterior	5000
Random seed	fixes the random number stream so a rerun reproduces the same draws	12345

The seed is the reproducibility control. Holding the seed, the settings, and the input file fixed reproduces a run's draws exactly on the same machine and CmdStan version, with within-chain parallelization off (Chapter 7), which is what makes an archived analysis checkable. Warmup and sampling iterations are the controls to raise when a model does not converge.

Convergence

The toolbox checks convergence automatically after every fit and applies two rules to every monitored parameter. A run is marked as not converged if any parameter has a potential scale reduction factor (R-hat) of 1.1 or higher, or if any parameter has an effective sample size below ten times the number of chains, which is 40 at the default of four chains. Passing both rules is a necessary condition for trusting the estimates, not a sufficient one.

When a fit fails either rule, the toolbox offers to rerun with the warmup and sampling iterations doubled. Accepting is usually the right response, because non-convergence most often reflects too short a run rather than a misspecified model. The estimates from a non-converged fit should not be reported.

Divergent transitions are counted separately and reported, but they do not trigger a rerun. A nonzero count is a warning that the variance components may be biased even when both convergence rules pass, and the response is more warmup or stronger priors. Chapter 18 covers both cases.

Credible intervals are not confidence intervals

The intervals PsyRAT prints are credible intervals, and the distinction from confidence intervals is not cosmetic. A 95% credible interval is a statement about the parameter given the data, the prior, and the model: the posterior probability that the dependability coefficient lies between the reported bounds is 0.95, up to the Monte Carlo error of a finite sample of draws. A 95% confidence interval is a statement about the procedure: intervals constructed this way would contain the fixed true value in 95% of hypothetical replications, and the particular interval in hand either contains that value or does not. The everyday reading that researchers apply to a confidence interval is the one that is actually licensed for a credible interval. Reporting a credible interval in frequentist language, or the reverse, misstates what was estimated.

Two details of construction belong in a methods section. The toolbox forms each interval from the empirical quantiles of the posterior draws, taking equal tails, so a 95% interval runs from the 2.5th to the 97.5th percentile. The point estimate beside it is the posterior mean of the same draws, not the median or the mode.

Where this is derived: Rocha et al. (2026), Section 7 (pp. 15 to 16). The convergence rules and the priors are the toolbox's own settings.

4.10 Where the framework goes next

The designs in this chapter assume that residual variance is a single quantity shared by all participants and all conditions. Rast and Clayson (in press) integrate generalizability theory with multilevel location-scale models, which relax that assumption and let residual variance vary systematically across persons and conditions. The result is a reliability function rather than a single coefficient, and it supports allocating trials to the participants or conditions where precision is lowest. PsyRAT implements this as its dynamic and conditional reliability analyses, described in Chapter 15.

For readers who want a broader treatment of generalizability theory outside psychophysiology, including its relations to classical test theory and structural equation modeling, Vispoel et al. (2018) is the standard reference. Rocha et al. (2026) is the source of record for the formulas the toolbox evaluates and is the preferred citation for a PsyRAT analysis.

Chapter 5 maps these designs onto the analyses the toolbox offers. Chapter 17 covers what to report. Definitions of the terms used here are collected in Appendix A, and full references in Appendix B.

5. Choosing an analysis

PsyRAT never asks you to name an analysis. You describe your data (which column holds the participant, which holds the score, and which optional columns hold groups, events, occasions, dimensions, or split sizes), you set a few preferences, and you click Analyze. The toolbox reads that description and routes the run to one of 29 numbered analyses. Chapter 16 holds one reference card for each of them.

This chapter is about the description rather than the click sequence. Six choices decide the routing, and every one of them is a question you would answer in a methods section anyway. Chapter 6 covers the file format those choices assume, and Chapter 7 walks the screens they live on.

5.1 The six design axes

Facets: one occasion, or two

A facet is a source of measurement error you deliberately sampled. Every PsyRAT analysis treats trials as a facet. The Occasion ID popup on Specify Inputs decides whether there is a second one. Leave it on none and you get a one-facet design (persons crossed with trials), which is the internal-consistency case: all the data come from one session, and the question is how well a participant's score generalizes over the trials that produced it. Select an occasion column and you get a two-facet crossed design (persons by trials by occasions), which adds the question of how well the score generalizes over occasions. The second facet is expensive in a specific way. Occasion-level variance components are estimated from as many levels as you have occasions, which is usually two, so those components stay wide no matter how many trials or participants you collected.

Score type: single, difference, or difference of differences

Set this with Estimate internal consistency of difference scores, in Preferences. No treats the measurement column as the score, one row per trial. Yes: 2-event difference models both conditions jointly and reports the reliability of their difference (error minus correct, incongruent minus congruent, and so on), which requires exactly two event types in the data. Yes: 4-event DoD reports the reliability of a contrast between two differences (a difference of differences, DoD), (ERP_1 - ERP_2) - (ERP_3 - ERP_4), which requires exactly four. The four-event option also requires you to say which event plays which role, on the Configure DoD Contrast screen, before Analyze will run. Difference scores are not a post-hoc subtraction of two separately estimated coefficients. The constituent scores are modeled together so that their covariance enters the universe-score variance, which is the whole reason a difference can be far less reliable than either score it is built from.

Error level: one error variance, or one per participant

Estimate subject-specific error variances, in Preferences, decides whether the residual is a single population quantity or a separate quantity for each participant. No gives the group-level estimand: one dependability coefficient describing the sample, which is what most published generalizability-theory work reports. Yes fits a location-scale model in which each participant carries their own residual standard deviation, so the output is a coefficient per participant alongside the group summary. Use it when the question is who is measured well rather than how well the measure performs, and expect a longer fit: the model adds one scale parameter per participant, and per-participant scale is harder to identify than a pooled one.

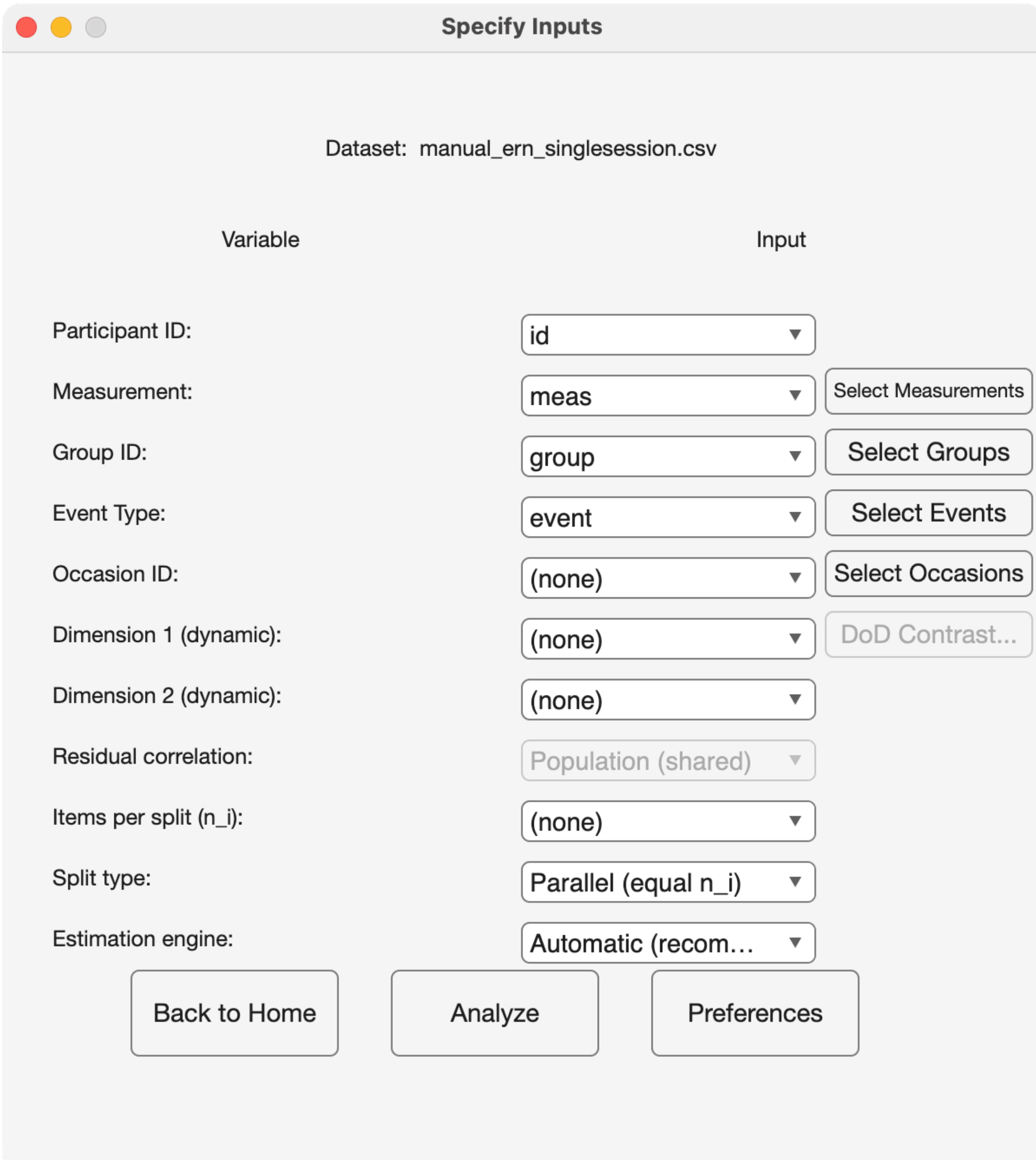


Figure 13: The Specify Inputs screen. Four of the six design axes are set by popups here; the other two live behind the Preferences button.

Moderation: static, or reliability as a function of a dimension

The Dimension 1 (dynamic) popup turns on dynamic reliability, the conditional-reliability model of Rast and Clayson (in press). Selecting a continuous, between-person column (one value per participant, such as a questionnaire total) makes both the mean and the residual scale functions of that standardized dimension, so the reported coefficient becomes a curve rather than a number: reliability at low, average, and high values of the dimension. Dimension 2 (dynamic) adds a second predictor and its interaction with the first. This is the axis to reach for when you suspect the measure works better in one part of the sample than another, which is a common worry when a clinical range and a comparison range are pooled in one dataset. Leave both popups on none for a static analysis, which is the default.

Residual coupling: independent, shared, or per-participant

When the two events of a difference score are recorded on the same trial (ipsilateral and contralateral windows, or two electrodes scored from one epoch), their residuals are not independent, and pretending otherwise distorts the difference-score error variance. Estimate residual covariance for co-occurring events, in Preferences, turns that covariance on. No fixes it to zero, which is right when the two conditions come from separate trials. Yes estimates it. For dynamic subject-level designs a third choice opens up: the Residual correlation popup on Specify Inputs offers Population (shared), one correlation for the sample, or Per-subject, a separate correlation for each participant. That popup is greyed out until difference scores, subject-specific error variances, and residual covariance are all switched on, because it means nothing anywhere else. The four-event contrast never takes a residual covariance in any configuration; all six of its pairwise residual covariances are fixed at zero by the estimand it reports.

Splits: single trials, parallel splits, or nonparallel splits

The toolbox normally expects one row per single trial. If your rows are split means instead (odd trials against even trials, or four random subsets of the trials), select the column holding the number of items averaged into each split in the Items per split (n_i) popup, and declare the splits with Split type. Parallel (equal n_i) says every split holds the same number of items. That case reuses the ordinary one-facet and test-retest machinery unchanged, with the split mean as the score and the number of splits in place of the trial count, so the output looks exactly like a single-trial run with the word trial replaced by split. Nonparallel (unequal n_i) fits the weighted observed-design model instead, and it is identified only by variation in n_i across splits. Equal counts under that setting are rejected before the run starts. Nonparallel splits report the design you actually observed, with no projection to other split counts and no trial-count cutoff, because split means have already lost the item-level partition a projection would need.

5.2 Two more choices, crossed with all six

The observation likelihood is set by Observation likelihood family in Preferences. Gaussian is the default and is what ERP amplitudes want, negative values included. Gamma fits a scaled chi-square (Gamma with a log link) and suits strictly positive, right-skewed scores such as single-trial time-frequency power; non-positive values are rejected under it. The gamma family covers most, not all, of the design grid, and it always runs on CmdStan. Gamma scale parameterization decides what the dispersion submodel is built on, and Automatic (recommended) picks correctly for the design you asked for. The two settings under it are different estimands rather than two views of

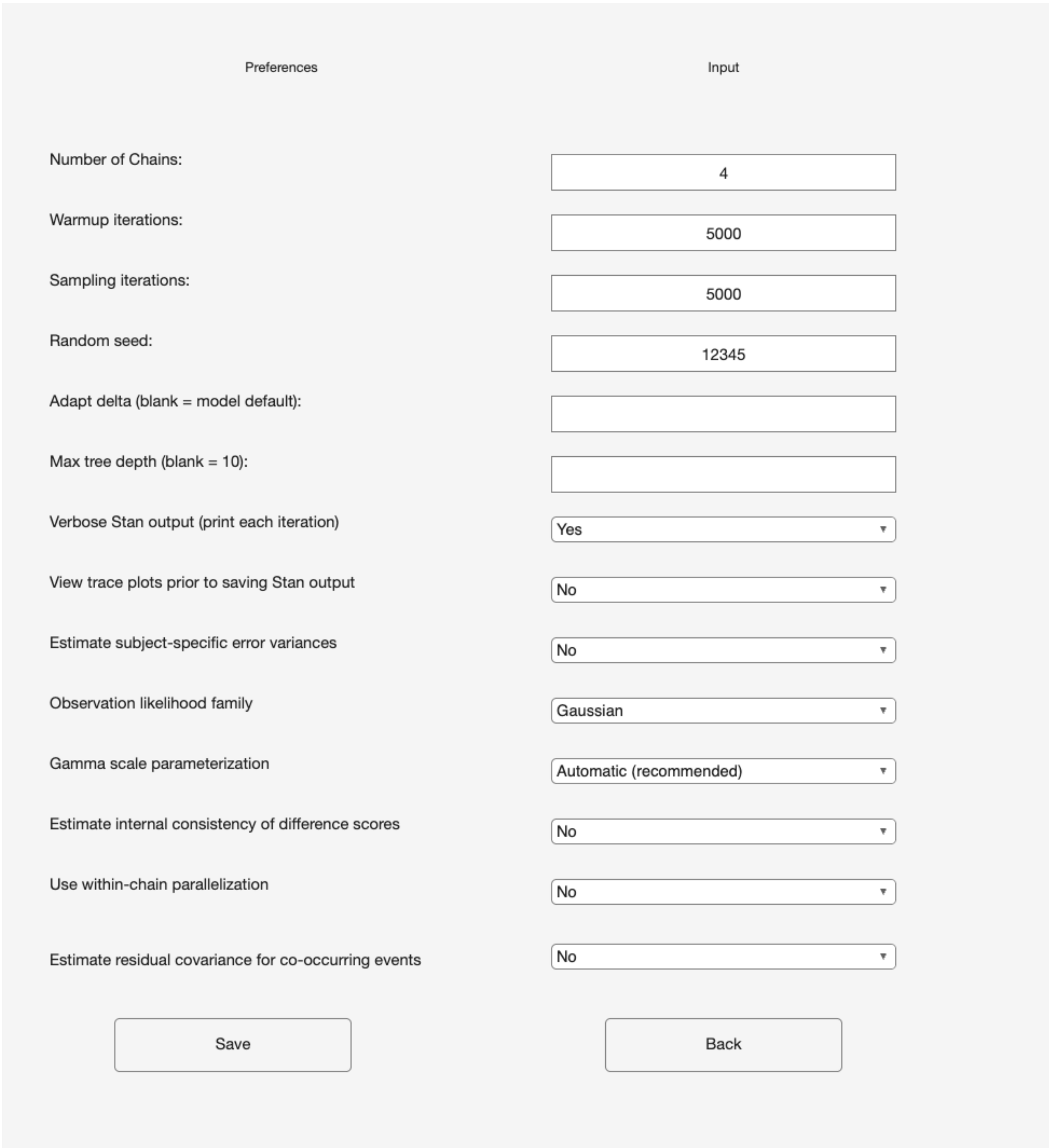


Figure 14: The Specify Processing Preferences screen, reached from the Preferences button. Score type, error level, residual coupling, and the likelihood family are all set here.

one fit, so leave it on Automatic unless you have a reason to do otherwise and are willing to say which one you used.

The Estimation engine popup on Specify Inputs picks how the model is fit. Automatic (recommended) uses CmdStan whenever it is installed, which is the configuration everything in this manual assumes. The popup is reachable only once CmdStan is found, because `psyrat_start` stops at its dependency check without it (Chapter 18); from a script, `psyrat_run` with the `engine` option at its automatic default falls back to the first native MATLAB engine that implements your design, preferring native HMC (Hamiltonian Monte Carlo, a genuine Bayesian fit, though not bit-identical to CmdStan) over native fitlme (restricted maximum likelihood, REML, with a parametric bootstrap, whose intervals are approximate and not credible intervals). Which engine is legal depends on the design, not on your preference: the plain one-facet designs and the nonparallel splits are fitlme-only, the location-scale and multivariate designs are HMC-only, plain test-retest is the one design both native engines can fit, and several designs have no native implementation at all. Every card in Chapter 16 names its engines, and Chapter 3 covers what each one needs installed. One dependency is worth knowing before you plan a gamma run: the concurrent gamma difference designs need the Statistics and Machine Learning Toolbox even on the CmdStan path, and the failure comes after sampling finishes rather than before it starts.

5.3 Working through it

Answer the questions in this order and you land on a card number. The first question is the one people skip, so it is first.

- **Are your rows split means rather than single trials?**
  - Yes, and every split holds the same number of items. Declare Parallel; the run reuses analyses 1 to 5 with trial relabeled as split (see the parallel splits card).
  - Yes, and the counts differ. Declare Nonparallel: analysis 23 with one occasion, analysis 24 with an occasion column.
  - No. Continue.
- **Do you want reliability reported as a function of a continuous participant characteristic?**
  - No. Take the static branch below.
  - Yes. Select Dimension 1 and take the dynamic branch.

Static branch

- **A single score.**
  - One occasion, one error variance for the sample: analysis 1; analyses 2, 3, and 4 once a group column, an event column, or both are selected.
  - One occasion, an error variance per participant: analysis 6.
  - Two occasions, one error variance for the sample: analysis 5.
  - Two occasions, an error variance per participant: analysis 25.
- **A two-event difference score.**
  - One occasion, group level: analysis 7 (the same number with residual covariance switched on).
  - One occasion, per participant: analysis 8.
  - With an occasion column: analysis 10, group level only.

- **A four-event difference of differences.** Analysis 9. One occasion, group level, no residual covariance.

Dynamic branch

- **A single score.** One facet: analysis 11 at the group level, 26 per participant. With an occasion column: analysis 14 at the group level, 27 per participant.
- **A two-event difference score, one facet.**
  - Group level: analysis 12, or 17 with residual covariance.
  - Per participant: analysis 13; with residual covariance, 19 for a shared correlation and 21 for a per-participant correlation.
- **A two-event difference score, with an occasion column.**
  - Group level: analysis 15, or 18 with residual covariance.
  - Per participant: analysis 16; with residual covariance, 20 for a shared correlation and 22 for a per-participant correlation.
- **A four-event difference of differences.** Analysis 28 with one occasion, 29 with an occasion column. Both are person-specific and non-concurrent, so subject-specific error variances must be on and residual covariance must be off.

Combinations that are not in those lists are rejected, with a dialog naming the setting to change. The rejection happens before sampling starts in almost every case. The exception worth planning around is the gamma extraction dependency named above, which fires after the fit.

Every analysis in those lists can be launched from the Analyze step. A few options have no screen control and are set only from a script: custom priors (Chapter 15) and the gamma copula residual-dispersion option (`dispersion`) are passed as `psyrat_run` arguments. Chapter 14 covers scripted runs in general.

You do not have to remember which analysis a set of settings produced. Every run writes a `*_runconfig.json` sidecar next to its `.psyrat` output, recording the seed, the sampler settings, the column mapping, and every design flag on this page, and every exported table carries the same information in its header. That sidecar is what makes a methods section reproducible, and it is what to read first when you come back to a result six months later.

5.4 What is supported, and what carries a caveat

Most of the grid is supported in the plain sense: covered by the unit and accuracy suites and, where a live recovery test exists, checked against known variance components in a live CmdStan run. The rest carries a label. The experimental arm says so in the header of every table it exports but nowhere on screen; the tractability-checked arm says in its export header that its one check established tractability, not robustness; the development-only cell says so nowhere at all. All of them are stated here and in the CHANGELOG's design maturity table, which uses the same five labels.

Designs	Configuration	Status
One-facet and test-retest reliability, subject-level error variances, non-concurrent two-event difference scores, the four-event difference of differences, reliability from splits, the dynamic family apart from analysis 20 under gamma, and the person-specific dynamic difference of differences	Gaussian throughout, and gamma wherever gamma is offered	Supported
Two-event difference scores with correlated residuals (analyses 7, 8, and 10)	Gamma, with a separate dispersion parameter for each participant (`dispersion` 1, the only gamma arm analysis 8 has)	Experimental
Two-event difference scores with correlated residuals (analyses 7 and 10)	Gamma, one global dispersion parameter (`dispersion` 2, the default for these two)	Tractability checked, robustness not
Two-facet subject-level concurrent dynamic difference (analysis 20)	Gamma, location-scale	Development-only
Analyses 15, 16, 17, 18, 21, and 22 under gamma; every splits design under gamma; residual covariance on the four-event difference of differences in any family	any	Not implemented: requesting them stops with a message

Supported does not mean that every cell has a live CmdStan recovery test of its own. The Gaussian four-event difference of differences (analysis 9) is recovered live only through the native HMC engine; the Gaussian subject-level dynamic designs (analyses 26 and 27) share the fits of analyses 11 and 14 and have no recovery test of their own; the gamma arm of analysis 19 is checked only by a replay of an external reference fit; and the splits designs (analyses 23 and 24) have no independent-oracle row. The CHANGELOG records the same list.

Experimental means one specific thing here. Convergence for those fits has not been established in this release: an in-repo subject-level fit reached an R-hat of about 2.5 with an effective sample size of about 2, and it did not improve when the simulated participant and trial counts were raised. Treat any estimate from that configuration as unverified until you have read its own convergence verdict, divergent-transition count, and per-parameter diagnostics. Analysis 8 has no alternative under gamma, because a per-participant coefficient needs a per-participant dispersion parameter. Analyses 7 and 10 default instead to a single dispersion parameter for the sample, which converged cleanly on one simulated one-facet dataset at one choice of sampler settings, while the two-facet check still produced divergent transitions. That is a statement about tractability, not about robustness, and its own disclosure text says so.

Development-only is stronger. The two-facet configuration of analysis 20 under gamma has no empirical anchor at all, so nothing outside the toolbox has ever been reproduced with it. Do not report from it.

Two more disclosures belong with these. Analyses 19 and 20 under the gamma family are fitted by a modular cut workflow: the margins are fitted first without the residual coupling, and the coupling is estimated afterward. The result is a cut posterior rather than a full Bayesian one, and the exported provenance says so. Analyses 28 and 29 are explicitly not cut posteriors, since they have no dependence module to cut.

All of the Gaussian arms named in those rows are unaffected. If you are running ERP amplitudes with the default family, none of this applies to your run.

5.5 Why the version carries -beta

The label is about the interface, not about the arithmetic. The public argument names, the shape of the exported tables, and the layout of the saved result are still moving between releases, so a script written against this version may need edits against the next one. It is not a statement that the estimates are provisional, and the design maturity table above is per-design for exactly that reason.

There is a second, narrower reason, and it deserves saying plainly. The concurrent gamma difference designs ship with their convergence unvalidated in this repository. Shipping them at all is a judgment call I made in favor of letting people run them with the caveat attached rather than withholding them until the sampler problem is solved. The suffix is part of that bargain: it says the toolbox is usable and honest about which corner of it is not yet settled.

6. Preparing your data

PsyRAT reads one kind of file, and everything downstream depends on getting it right. Keep this chapter open while you export from EEGLAB, ERPLAB, or your own scripts. It is the reference the tutorials in Part IV assume you have read.

6.1 Long format: one row per single-trial score

The input is a long-format table with one row per single-trial score. Two columns are required: a participant identifier and a numeric measurement. Three more are optional: group, which splits the sample into separately modeled strata, and event (condition) and occasion, which enter the model as facets.

These ten rows are lifted verbatim from `test_data/manual_ern_singlesession.csv`, the file Tutorial 1 uses:

id	group	event	meas
101	GroupA	inc_err	-12.041412
101	GroupA	inc_err	-6.130699
101	GroupA	inc_err	-1.701634
101	GroupA	inc_err	-1.552103
101	GroupA	inc_err	-3.576956
101	GroupA	inc_cor	-3.551382
101	GroupA	inc_cor	-2.451203
101	GroupA	inc_cor	-11.254837
101	GroupA	inc_cor	-4.363838
101	GroupA	inc_cor	-4.467972

Read the columns left to right. `id` is the participant, `group` is the between-person grouping, `event` is the condition label, and `meas` is the single-trial ERN amplitude in microvolts. That last column is the only one the model treats as a score. The window shown here straddles a condition boundary: the first five rows are error trials on incongruent stimuli and the next five are correct trials on the same stimuli. Nothing marks the boundary except the `event` value, and nothing needs to. Rows may be interleaved across participants and conditions, but within a participant and condition the order of the rows matters: PsyRAT numbers each participant's trials in the order the rows appear, and that number is the trial level shared across participants in the trial main effect (a participant's third row within a condition is that condition's trial 3 for everyone). The concurrent difference designs pair trials the same way, the k-th trial of one event with the k-th trial of the other. Keep every participant's rows in the order the trials occurred, and do not shuffle rows within a participant.

Trial counts do not have to match. Participants routinely differ in how many artifact-free error trials they contribute, and the trial-count projection in the D-study exists precisely because they do. What must match across participants is the number of distinct facet levels, which Section 6.5 covers.

Do not average before loading. A file of participant means carries no within-person replicates, so the residual variance the model needs is not identified and the reliability estimate has nothing to be built from. The one exception is the splits format below, where each row is a mean of a known number of items and you declare how many.

6.2 Your column names, not mine

Header names are free. The GUI maps columns to roles through the popup menus on the Specify Inputs screen (Chapter 7), so a file whose headers read `subjid`, `cond`, and `ern_uv` works exactly as well as one written to the canonical names. Scripted callers pass the same mapping as arguments (`idcol`, `meascol`, `groupcol`, `eventcol`, `timecol`); see Chapter 14.

The canonical names buy one thing: they are what auto-detection finds first. When the Specify Inputs screen opens, PsyRAT preselects each role by matching headers case-insensitively against a short candidate list. Participant ID matches Subject, ID, Participant, SubjID, or Subj. Measurement matches Measurement or Meas. Group ID matches Group. Event Type matches Event or Type. Occasion ID matches Time, Occasion, or Session. A header has to match one of those candidates exactly (apart from case), so `subj_id` and `ERN` match nothing.

If no header matches Measurement, PsyRAT falls back to the LAST numeric column that no other role has already claimed. In a long-format export the score is the melted value column and usually does come last, so the fallback is right more often than it is wrong. It is still a guess. Confirm the Measurement popup before you click Analyze, because a run on the wrong column completes normally and reports the reliability of whatever you pointed at.

Dimension columns and the items-per-split column are never auto-detected. They stay on `(none)` until you pick them, which is deliberate: selecting either one changes which analysis runs.

One header caveat, and it will surprise you the first time. MATLAB coerces table headers into valid variable names as it reads the file, so `ERN (uV)` arrives as `ERN_uV_` (the space is dropped and each parenthesis becomes an underscore), and that coerced name is what the popups show. Your data are untouched. If you want the popups to read cleanly, export headers made of letters, digits, and underscores.

6.3 The splits format: one row per split, with a weight

Reliability from data splits takes a different input, and the difference is not cosmetic. Each row is one SPLIT rather than one trial, the measurement column holds the split MEAN, and a weight column reports how many items were averaged into that split. The weight column is required. Without it the split-level model cannot be identified, because the sampling variance of a split mean of `n_i` items is the per-item residual variance divided by `n_i`.

These ten rows come verbatim from `test_data/manual_ern_splits.csv`:

id	meas	weight
301	-4.171339	24
301	-6.509574	40
301	-4.995126	28
301	-4.531105	28
301	-4.46002	6
301	-2.156673	8
301	-4.495167	26
301	-5.342279	20
301	-8.377559	30
301	-3.916375	35

Participant 301 contributes sixteen splits; the ten rows shown are its first ten, with weights of 24, 40, 28, 28, 6, 8, 26, 20, 30, and 35 items. The mean of the 24 items in the first split is -4.17 microvolts. Because those counts are unequal, this file is declared Nonparallel on the Split type popup; a file whose splits all held the same number of items would be declared Parallel instead. The whole shipped file runs 4 to 40 items per split across 36 participants.

Every weight must be a positive whole number. Fractions, zeros, negatives, and non-finite entries are rejected at load time. Group, event, and occasion columns load here exactly as they do for single-trial input, and Parallel splits use them as single-trial input does. One restriction applies to Nonparallel splits: their single-occasion model is fitted as one pooled model over all participants, so a run with a Group ID or Event Type column selected stops with a message saying so rather than fitting one model per stratum. The fix the message names is to remove the column or to add an Occasion ID column, which routes to the group- and event-aware two-facet split design (Chapter 13).

The two split types are not interchangeable declarations of taste. Nonparallel splits are identified ONLY by variation in the weights, so a nonparallel run on all-equal weights is refused outright, and a run whose weights barely vary is allowed but flagged. Parallel splits assume equal weights and reuse the standard one-facet machinery with the split mean as the score. Declare what your splitting procedure actually produced. Chapter 13 walks the workflow end to end.

6.4 Dimension columns for dynamic reliability

Dynamic (conditional) reliability reports reliability as a function of one or two continuous between-person covariates, and you enable it by selecting a Dimension 1 column. Dimension 2 is optional and adds a second predictor with its interaction. Selecting Dimension 2 without Dimension 1 is refused.

A dimension column must be numeric, and it must be constant within participant. One value per person, repeated down that person's rows, is what the format expects: a questionnaire total, an age, a symptom score. If a participant's rows disagree, the run stops and names the participant and the column. If a participant has no finite value at all, the run stops and names that participant too. A column with no between-participant variance cannot be standardized and is also refused.

Nothing else about the file changes. The dimension column sits alongside the usual `id`, `meas`, and facet columns, one more column in the same long-format table. Chapter 15 covers what the analysis reports.

6.5 What loading checks, and what to do about it

The data checks do not run when the file picker closes. Opening the file only reads it into a table. The checks below run when you click Analyze, once per selected measurement, and (this catches people out) AFTER you have chosen a name and folder for the output. The one exception is the setup-screen wording quoted in the non-numeric row below, which fires once for all selected measurements before the save dialog opens. Chapter 7, Section 7.5, follows that sequence in order.

The table gives the message fragment you will see and the fix. Messages are quoted from what the toolbox prints, trimmed to the identifying phrase.

What you will see	How to fix
”Input data include missing cells, which CmdStan cannot process.”	Remove or impute the affected rows so every row has an id, a measurement, and a value for every facet you selected.
”Participants differ in the number of distinct events” (with a printed table of ids and `Number_of_Events`)	Read the printed table, find the ids whose count differs from the rest, and either restore the missing condition or drop those participants.
”Participants differ in the number of distinct occasion cells” (with a printed table of ids and `Number_of_Occasions`)	Same remedy, applied to the event-by-occasion grid (or, when no event column is selected, to occasions alone; the check runs either way).
”The data in the measurement column is not numeric” (or, from the setup screen, ”The following selected measurement columns are not numeric:”)	Point Measurement at a numeric column. A text or date column selected by mistake is the usual cause.
”Column 'group' is a logical (true/false) array, which cannot be used as a label column.”	Convert the column to numeric codes, a categorical, or text before loading, for example with `double(t.group)` or `categorical(t.group)`.
”has numeric values that convert to the SAME text label, so two or more distinct levels would be merged into one”	Supply that label column as text yourself, rounding or naming the levels the way you want them reported.
”The 'id' column contains missing values”	Give every row a label. A missing label cannot be filled in safely, because every missing entry would collapse into one level.
”The dataset is a table with column headers but no data rows.”	Re-export the file with its single-trial rows.

Two of those rows deserve more than a line.

The two participant-count checks compare COUNTS, not membership, and the error text says so itself. A participant with four distinct events passes even if they are not the same four events everybody else has. That is a real limit, not a subtlety I am glossing: uniform counts are what the check can enforce cheaply on a long-format table. A genuinely incomplete design still runs. When both an event and an occasion column are selected, PsyRAT warns rather than blocking, reporting how many participant-by-event-by-occasion cells hold no observations and naming the first few. In a single-session design there is no such warning, so a mismatched-but-equal-count event set runs with no notice. Estimation is valid with empty cells, because variance-component estimation does not require a complete design. The estimates for an empty cell are model-based rather than observed, and you should check that each empty cell reflects an intended exclusion, such as artifact rejection, and not a data-preparation error.

Two more warnings belong to the same non-blocking class. ”The dataset has only N participant(s), fewer than the recommended minimum of 20” fires below 20 participants, per group when a group column is selected, and ”Median trials per cell is X, fewer than the recommended minimum of 10” fires when the median trial count per participant, or per participant-by-event cell, falls below 10. None of the three stops the run. In the GUI they appear together in one modal dialog titled Preflight warnings, raised after the save dialog and before the first fit, and the run waits until you click OK; the same lines print in the Command Window, which is the only channel on the `psyrat_run` route (Chapter 14).

The numeric-label collision is the check people meet unprepared. MATLAB renders numbers with about five significant digits, so two event codes that differ only beyond that point convert to the same text label and would silently merge into one level. PsyRAT refuses instead. Consider this example: two numeric event codes of `0.123456789` and `0.123456111` both convert to the label `0.12346`, so two conditions that look distinct in the spreadsheet would arrive as one. Writing label columns as text (`t1` and `t2`, `cor` and `err`) in the first place avoids the whole question.

Two more checks never stop a run and only advise. PsyRAT warns when a modeled group has fewer than 20 participants, because variance components are unstable and prior-dominated at small N, and it warns when the median number of trials per cell falls below 10. Both leave the decision with you.

6.6 File types PsyRAT reads

The file picker offers a combined All Readable Files filter covering `.xlsx`, `.xls`, `.csv`, `.dat`, `.txt`, and `.ods`, plus one filter per type. Underneath, the reader also accepts `.xlsb`, `.xlsm`, `.xltm`, and `.xltx`, so those open if you type the name or widen the filter to All Files. Anything else is refused with a message about unsupported file types.

Delimited text does not need a declaration. PsyRAT first asks MATLAB to read the file on the strength of its extension, and if that succeeds but yields only one column it retries with a comma, then a space, then a tab, then a semicolon, then a pipe, stopping at the first delimiter that produces more than one column. A file that survives none of those raises a loading error. Comma-separated files are the safest choice, and they are what the tutorial datasets ship as.

6.7 Before you load

Run down this list once against a new export. It takes a minute and saves the round trip through a save dialog and a failed run.

1. One row per single-trial score, unaveraged.
2. A participant column and a numeric measurement column, both present in every row.
3. No blank cells in any column you intend to use.
4. Facet labels as text where you can manage it (`t1`, `err`), not as bare numbers.
5. The same number of distinct events, and of event-by-occasion cells, for every participant.
6. Plain headers: letters, digits, underscores.
7. For splits, a positive whole-number weight on every row, and a decision about Parallel or Nonparallel.
8. For dynamic reliability, one numeric Dimension value per participant, constant down that participant's rows.
9. A save location whose full path contains no spaces (a CmdStan restriction, covered in Section 7.5).

The last item is not about the data at all, and it is one of the mistakes that most often stop a first run. With the nine satisfied you have a file PsyRAT will load, and Chapter 7 picks up at the file picker.

7. Processing data, screen by screen

Running `psyrat_start` prints a startup banner to the MATLAB Command Window and opens the home screen. This chapter follows the Process New Data route from that screen to a saved result, one window at a time. The tutorials in Part IV point back here rather than repeating any of it, so if a tutorial says "set the Occasion ID popup" and you want to know what that popup does, this is where the answer lives.

Read Chapter 6 first. Every screen below assumes a file that already meets the input contract.

7.1 The home screen

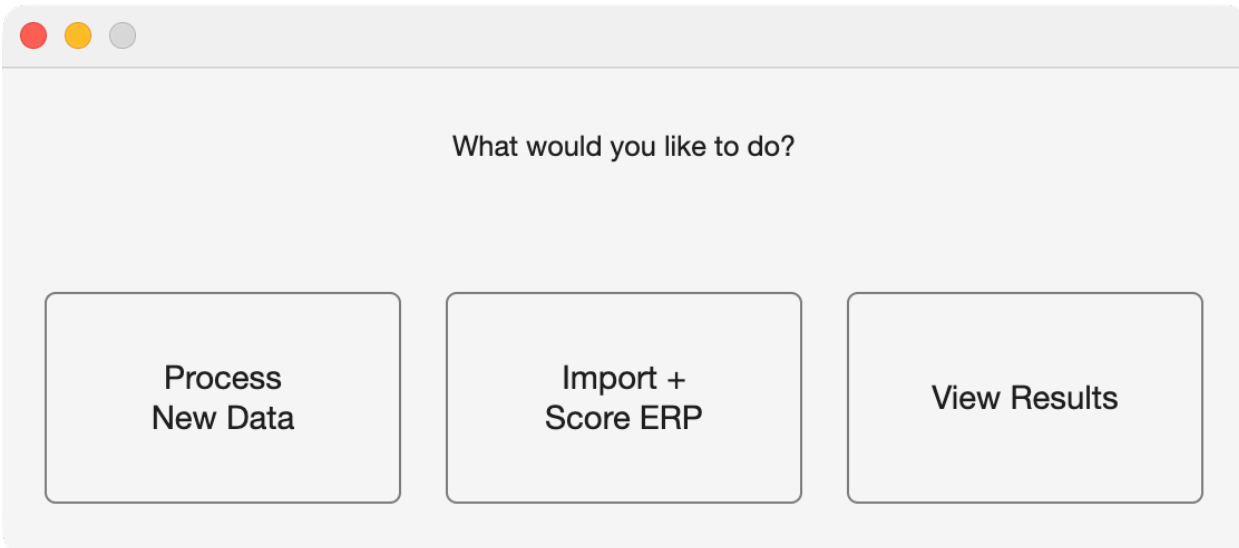


Figure 15: The PsyRAT home screen, with its three buttons.

This is the whole launcher. Three buttons, and each one starts a workflow that runs to completion on its own.

Process New Data: loads a table of ERP measurements and runs estimation, which is the route this chapter documents.

Import + Score ERP: imports ERP waveform files, defines component scoring, and exports a validated single-trial table, covered in Chapter 8.

View Results: opens a `.psyrat` file produced by an earlier run and displays its figures and tables, covered in Chapter 9.

7.2 Choosing a data file

Selecting the Process New Data button opens a file picker before it opens anything else. Its default filter, All Readable Files, matches `.xlsx`, `.xls`, `.csv`, `.dat`, `.txt`, and `.ods`, and the file-type popup offers a separate filter for each of those types. Section 6.6 lists the extensions the reader accepts beyond the filter.

The Command Window prints `Loading Data...` and a note that this may take a while. Large single-trial files take a moment, and no window reports progress while it happens.

If you cancel the picker, PsyRAT returns you to the home screen and raises a File Error dialog saying no data file was selected. Nothing is lost. Click Process New Data again.

Loading at this stage only reads the file into a table. None of the data checks in Section 6.5 have run yet.

7.3 Specify Inputs

The window is titled Specify Inputs, and it names your file at the top (`Dataset:` followed by the filename). Two columns run down the screen: Variable on the left, Input on the right. Each row assigns one of your source columns, or one processing mode, to a PsyRAT role. The buttons along the right edge open sub-screens for narrowing what gets processed.

Everything on this screen is a declaration about your data. Everything in Preferences (Section 7.4) is a declaration about the fit. The two interact, which is why some controls here are greyed out until Preferences is saved.

The eleven popups

Participant ID: the column holding the participant identifier, one level per person. Required, and preselected to the first column when no header matches. Every participant contributes one clustering unit to the model, so an identifier that repeats across genuinely different people silently pools them.

Measurement: the numeric column holding the single-trial score. Required. This popup lists your columns for a single selection; when you have chosen more than one measurement through the Select Measurements button it collapses to the words `multiple measurements` instead of a column name. Confirm this one every time. A run on the wrong numeric column completes normally and reports the reliability of whatever you pointed at.

Group ID: the column holding a between-person grouping, such as a diagnostic or task group. Optional. Estimation fits a separate model per group, so choose this when you want group-specific variance components rather than one pooled set. Leave it on `(none)` when the sample is one group.

Event Type: the column holding condition or event labels, such as correct and error trials. Optional for a plain reliability analysis, and required for every difference-score analysis. Choose it when reliability differs by condition, which for error-related measures it usually does.

Occasion ID: the column holding session or occasion labels. Optional. Selecting it turns the analysis into a test-retest design, which changes the model, the coefficients reported, and the iteration count you should expect to need. Leave it on `(none)` for single-session data. Once an occasion column is selected, opening Preferences with fewer than 10,000 total iterations raises a Recommended Iteration Increase advisory; Section 7.4 explains it.

Dimension 1 (dynamic): the first continuous between-person covariate for dynamic (conditional) reliability. Optional, and selecting it is what enables the dynamic-reliability analysis. The column must be numeric and constant within participant (Section 6.4). Leave it on `(none)` for a standard analysis.

Dimension 2 (dynamic): a second continuous between-person covariate, adding its main effect and its interaction with Dimension 1. Optional, and usable only alongside Dimension 1. Selecting Dimension 2 on its own is rejected at Analyze rather than silently ignored.

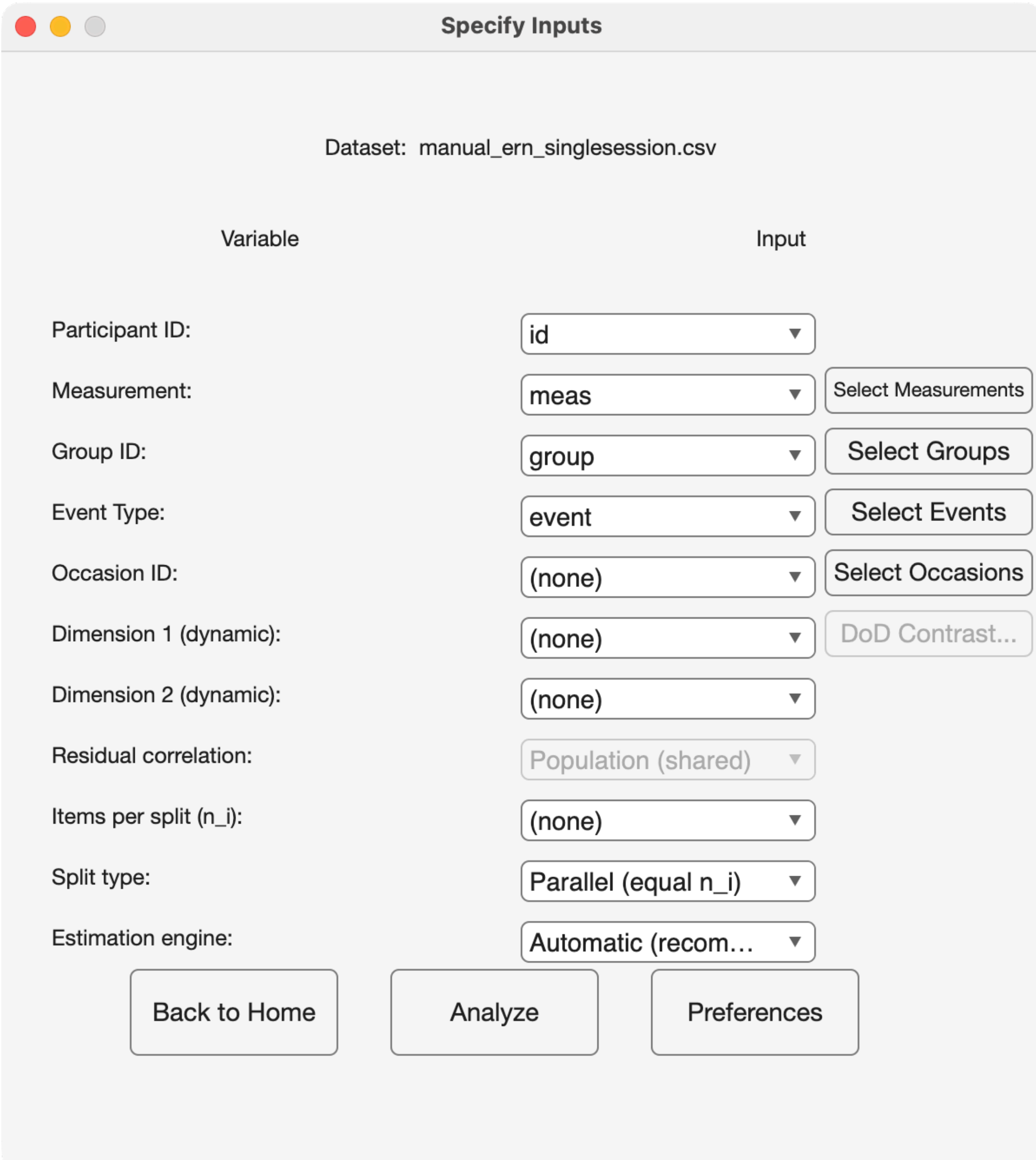


Figure 16: The Specify Inputs screen, showing the eleven role popups and the selection buttons.

Residual correlation: whether the residual correlation in the subject-level concurrent dynamic difference design is shared across participants (Population (shared)) or estimated separately for each participant (Per-subject). The default is Population (shared). This popup is greyed out unless three Preferences are all set: difference scores on Yes: 2-event difference, subject-specific error variances on Yes, and residual covariance for co-occurring events on Yes. It is inert for every other run, and a stale Per-subject selection on a run that does not support it is reset rather than carried through.

Items per split (n_i): the column holding the number of items averaged into each split mean. Optional, and selecting it declares that each row of your file is a split rather than a single trial, which switches the analysis to reliability from data splits. Leave it on `(none)` for single-trial input. Section 6.3 covers the format this expects.

Split type: whether the splits are Parallel (equal n_i) or Nonparallel (unequal n_i). The default is Parallel (equal n_i). This popup is consulted only when an items-per-split column is selected. Declare what your splitting procedure produced, because the two choices fit different models: Parallel reuses the standard one-facet or test-retest machinery with the split mean as the score, while Nonparallel fits the weighted observed-design model, which is identified only by variation in n_i.

Estimation engine: which engine estimates the variance components. Automatic (recommended) is the default and uses CmdStan whenever it is installed, falling back to a native MATLAB engine only when it is not. The other three choices, CmdStan only, Native HMC, and Native fitlme, force one engine and error clearly at run time if that engine is missing or cannot handle the design. A native option whose required toolbox is absent is annotated on screen with `(toolbox not installed)`. Leave this on Automatic (recommended). The native engines (HMC is Hamiltonian Monte Carlo; fitlme is MATLAB's linear mixed-effects fitter, which uses restricted maximum likelihood, REML) are not bit-identical to CmdStan, and the fitlme intervals are approximate frequentist bootstrap intervals rather than Bayesian credible intervals, so a forced native engine changes what your numbers mean.

The `(none)` sentinel

Group ID, Event Type, Occasion ID, both Dimension popups, and Items per split all carry an extra final entry reading `(none)`, which means the facet is not used. The parentheses are load-bearing. MATLAB coerces every header it reads into a valid variable name, and `(none)` is not one, so no column of yours can ever carry that label and the sentinel is unambiguous by construction. A column literally named `none` in your file stays selectable and behaves like any other column.

The sentinel is also the trap under the selection buttons. Clicking Select Groups, Select Events, or Select Occasions while the matching popup still reads `(none)` raises an error dialog naming the missing column, because there is no column to list levels from. Set the popup first, then click the button.

Select Measurements

Clicking Select Measurements opens Specify Which Measurements to Process, a multi-select list of every column in your file, prompted "Select one or more measurement columns to process:". Save records the selection and returns you to Specify Inputs; Back discards it.

Selecting more than one column runs a batch: PsyRAT fits each measurement in turn and writes one output file per measurement (Section 7.6). Use it when you have scored several components

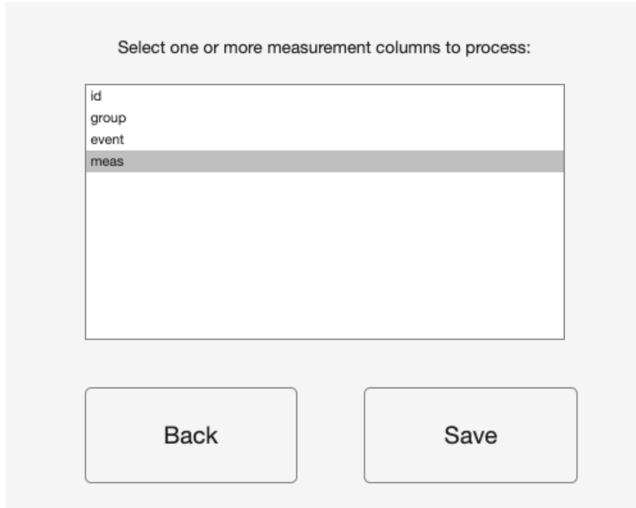


Figure 17: The Specify Which Measurements to Process screen.

or several electrodes into one file. Saving with nothing selected raises a Measurement Not Defined dialog.

Select Groups

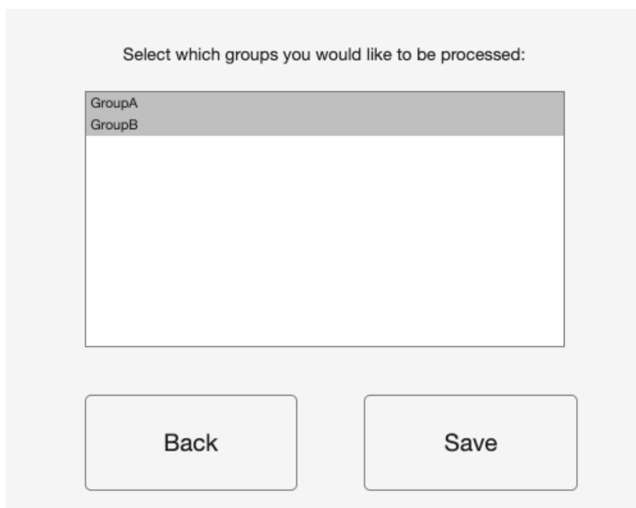


Figure 18: The Specify Which Groups to Process screen.

Clicking Select Groups opens Specify Which Groups to Process, a multi-select list of the distinct values found in your Group ID column, prompted ”Select which groups you would like to be processed:”. Everything is selected the first time. Deselect the groups you want left out of this run.

You do not have to open this screen at all. When you never touch it, PsyRAT processes every group it finds. Changing the Group ID popup to a different column resets the selection to all levels of the new column.

Figure 19: The Specify Which Events to Process screen.

Select Events

Clicking Select Events opens Specify Which Events to Process, the same list behavior applied to your Event Type column, prompted ”Select which events you would like to be processed:”. This is the screen that matters most for difference scores, which require exactly two selected events, and for the four-event difference-of-differences (DoD) contrast, which requires exactly four.

Saving here can discard work elsewhere. If the saved event set differs from the previous one, or if the event column itself changed, a configured difference-of-differences contrast is cleared and a modal DoD contrast cleared warning tells you so. Reconfigure the contrast before running. A save that leaves the same set of events selected, in any order, keeps the contrast intact.

Select Occasions

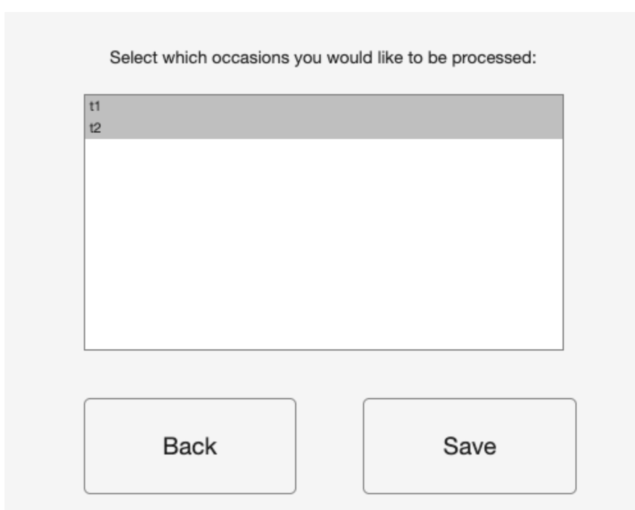


Figure 20: The Specify Which Occasions to Process screen.

Clicking Select Occasions opens Specify Which Occasions to Process, prompted ”Select which occasions you would like to be processed:”, and behaves exactly as the group and event screens do. Use it to drop a third session from a design you want analyzed as two occasions.

DoD Contrast...

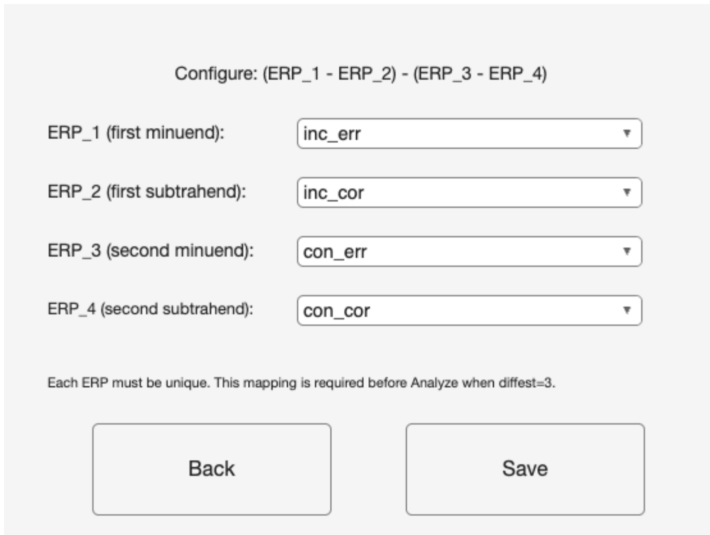


Figure 21: The Configure DoD Contrast screen, with its four ERP popups.

Clicking DoD Contrast... opens Configure DoD Contrast, which assigns your four event labels to the four terms of the contrast printed across the top of the window, `(ERP_1 - ERP_2) - (ERP_3 - ERP_4)`. Four popups do the assigning: ERP_1 (first minuend), ERP_2 (first subtrahend), ERP_3 (second minuend), and ERP_4 (second subtrahend). Each ERP must be a different event, and Save refuses a mapping that repeats one.

The button is greyed out until you set Estimate internal consistency of difference scores to Yes: 4-event DoD in Preferences and click Save there. It also refuses to open until an Event Type column is chosen and exactly four events are selected, naming whichever precondition is unmet. The button's tooltip reports the current state: `Needs event column`, `Needs exactly 4 selected events (current: N)`, `Not configured`, or `Configured:` followed by the assembled contrast.

The mapping is required. A four-event run reaching Analyze without one is rejected before the save dialog opens, so nothing is lost but the click.

The three action buttons

Back to Home: closes Specify Inputs and returns to the home screen, and discards everything you set on the way, including saved Preferences, the column assignments, the subset selections, and any configured DoD contrast. The next Process New Data run starts from the defaults. If you want to keep your settings, run the analysis or leave the window open.

Analyze: validates every selection on this screen, asks where to save, and starts estimation. Section 7.5 follows what happens next.

Preferences: opens Specify Processing Preferences, carrying your current column assignments with it.

7.4 Specify Processing Preferences

Clicking the Preferences button opens Specify Processing Preferences. Fourteen rows, then Save and Back. Save writes the settings and redraws Specify Inputs; Back returns without writing. Nothing on this screen touches your data file. The settings live for the current MATLAB session only: PsyRAT rebuilds its defaults every time it starts, so a changed chain count or family has to be entered again after a restart. The settings a run actually used are recorded in its run-config sidecar (Section 7.6), which `psyrat_run` can replay.

The defaults are the ones I run, and the entries below say where a default is a considered choice and where it is a convention I inherited.

Number of Chains: how many independent Markov chains CmdStan runs. The default is 4, and 3 is the enforced minimum. Anything below 3 is refused with an Invalid Number of Chains dialog, because the convergence check compares chains against each other and needs enough of them for the comparison to mean anything. Four is also the number most desktop machines can run in parallel without contention.

Warmup iterations: how many adaptation iterations each chain runs before draws are kept. The default is 5,000. Warmup draws are discarded, so this number buys adaptation quality rather than posterior precision.

Sampling iterations: how many post-warmup draws each chain keeps. The default is 5,000, giving 20,000 retained draws across four chains. That is generous for the one-facet models and, in my experience, about right for the test-retest ones. Use as many iterations as you need to get your model to converge.

If an occasion column is selected on Specify Inputs and the total (warmup plus sampling) is under 10,000, opening this screen raises a Recommended Iteration Increase advisory over the finished window. It is worded as advice, not as a refusal, though MATLAB draws it in the same style as an error dialog. At the defaults it never appears, because 5,000 plus 5,000 is exactly 10,000.

Random seed: the random number generator (RNG) seed handed to CmdStan. The default is 12345. Reusing the same seed on the same data at the same settings reproduces the same draws and therefore the same estimates to the last digit, which is what makes a PsyRAT analysis reproducible rather than merely repeatable. Record it. The sidecar in Section 7.6 records it for you. Changing it is a legitimate robustness check and should be reported as one.

Adapt delta (blank = model default): the target acceptance rate of the No-U-Turn Sampler (NUTS). Leave it blank, which is the default. Blank means the model's own default: most designs carry a tuned value (0.90 to 0.98, chosen per model rather than picked once), and the plain one-facet designs (analyses 1 through 4) run at CmdStan's default of 0.80. Enter a value between 0 and 1, typically 0.95 or 0.99, to reduce divergent transitions at the cost of slower sampling. A malformed entry is refused with an Invalid Adapt Delta dialog.

Max tree depth (blank = 10): the NUTS maximum tree depth. Leave it blank, which is the default and means CmdStan's own default of 10. Raise it to 12 or 15 when a hard posterior is terminating trajectories early. Both this and adapt delta change step-size adaptation during warmup, so a run that sets either one does not reproduce a run at the same seed that did not.

Verbose Stan output (print each iteration): whether CmdStan's raw console log is echoed into the MATLAB Command Window. The default is Yes. If yes, you see iteration counts and can tell a

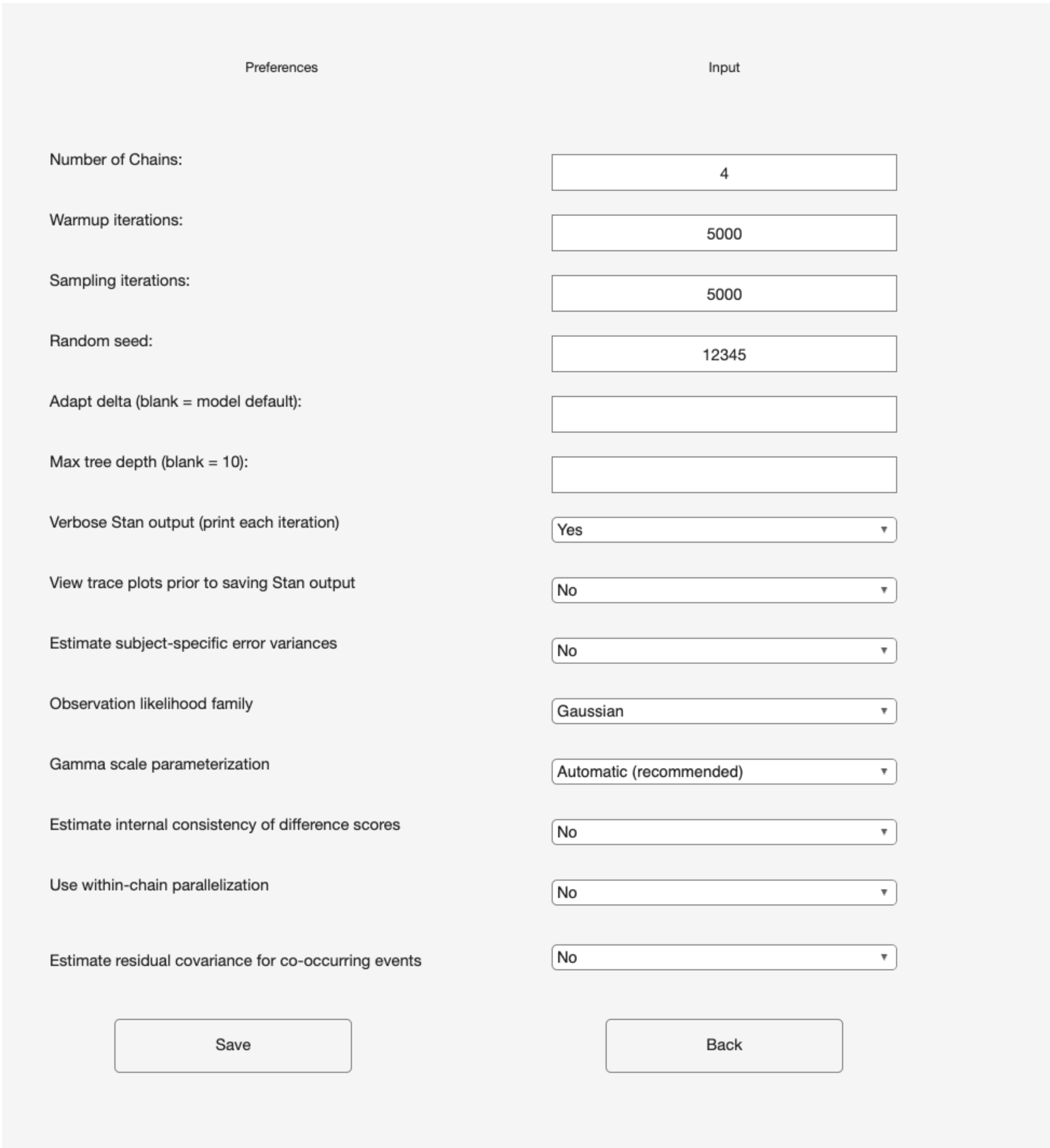


Figure 22: The Specify Processing Preferences screen, at its defaults.

slow run from a stalled one. If no, the Command Window goes quiet for the length of the fit, which on a long model is indistinguishable from a hang.

View trace plots prior to saving Stan output: whether trace plots are drawn before the result is saved, with a chance to rerun. The default is No, and the limitation is worth stating plainly: plots are rendered only for the plain one-facet internal-consistency analysis. For every other analysis, setting this to Yes prints a console line saying trace plots are not supported or not rendered for that analysis, and continues without plotting. Section 7.5 covers what you see when plots are rendered.

Estimate subject-specific error variances: whether the model estimates a separate residual variance per participant, which is what subject-level reliability estimates are computed from. The default is No. Turning it on costs sampling time and constrains which designs are available: with an occasion column present, it is supported for the non-difference test-retest design and for the dynamic-reliability designs, and it is not supported for non-dynamic difference scores in a test-retest workflow.

Observation likelihood family: the likelihood for the single-trial scores. The options are Gaussian and Gamma, and the default is Gaussian. Gaussian suits ERP amplitudes, which are routinely negative. Gamma is the scaled chi-square, log-linked family for strictly positive, right-skewed data such as single-trial time-frequency power, it is available for a specific subset of designs, and it always runs on CmdStan. Non-positive scores are rejected under Gamma before estimation starts. Chapter 15 sets out which designs accept it.

Gamma scale parameterization: which quantity the Gamma dispersion submodel is built on. The options are Automatic (recommended), Log-nu (relative dispersion / CV), and Location-scale (absolute residual SD), and the default is Automatic (recommended). Leave it there. Automatic lets the analysis choose location-scale on every design that supports it and log-nu on the rest, which is the choice you want when output is read as precision or compared against a Gaussian analysis. The two parameterizations are different estimands rather than two views of one fit, so results are not comparable across the setting. Under the Gaussian family this control is not gated: Automatic and Log-nu are ignored, but Location-scale is rejected and stops the run.

Estimate internal consistency of difference scores: whether reliability is estimated for a difference score, and of which kind. The options are No, Yes: 2-event difference, and Yes: 4-event DoD, and the default is No. Yes: 2-event difference estimates the reliability of X minus Y and needs exactly two selected events. Yes: 4-event DoD estimates the four-event contrast and needs exactly four selected events plus the mapping from the DoD Contrast... screen (Section 7.3). Difference-score modes are available for single-session data, with the dynamic-reliability designs as the documented exception.

Use within-chain parallelization: whether CmdStan splits each chain's likelihood across threads. The default is No. If yes, PsyRAT picks threads per chain automatically so at least two CPU cores stay free after chains times threads. If no, each chain runs single-threaded. Turning it on helps most on large datasets and least on small ones, where the threading overhead can outweigh the gain. Threaded runs are not bit-reproducible: the partition of the likelihood sum is chosen at run time, so the same seed does not guarantee the same draws (Chapter 18). The thread count a run resolved to is recorded in its sidecar and in every exported table header.

Estimate residual covariance for co-occurring events: whether the within-person residual covariance between two co-occurring events (ipsi and contra recordings of the same trial, say) is estimated.

The default is No, which fixes that covariance at zero. Turn it on when the two events are literally the same trial scored twice, because forcing their residuals to be independent then misstates the error variance of the difference. It is not available for the four-event contrast in any family, and requesting both is rejected at Analyze.

7.5 What happens when you click Analyze

Clicking Analyze runs a gauntlet of checks, asks where to save, and only then starts estimation. The order matters, and it is not the order most people expect.

The checks you can still fail

Every check below runs before the save dialog opens, so a failure here costs you a click and nothing else. Most test your selections; the numeric-column and event-count checks read the loaded table. The remaining data checks (section 6.5) run later, after you have named the output; those later checks can also raise the non-fatal Preflight warnings dialog described there, which waits for OK and then lets estimation start. Each dialog names its remedy on a "How to fix" line. The table gives the dialog title and the condition.

Dialog title	What tripped it	How to fix
Measurement Not Defined	No measurement column selected	Select one or more numeric data columns for Measurement.
Dynamic reliability	Dimension 2 selected without Dimension 1	Select a Dimension 1 column, or set both to `(none)`.
Dynamic reliability	Four-event DoD under dynamic reliability without subject-level error variances	Set subject-specific error variances to Yes, choose a different difference-score mode, or clear the Dimension columns.
Reliability with splits	Splits combined with dynamic reliability, difference scores, or subject-specific error variances	Turn off the conflicting option, or set Items per split (n_i) to `(none)`.
Unique variable names not provided	One source column assigned to two roles	Assign unique columns for ID, Group, Event, Occasion, Dimension, and Items-per-split.
Measurement column conflict	A selected measurement column is also serving as a facet	Remove the overlapping column from the Measurement selection.
Measurement data not numeric	A selected measurement column is text	Choose only numeric columns for Measurement.
Single-subject error variance not supported for retest	Subject-specific error variances plus difference scores plus an occasion column, outside dynamic reliability	Turn off subject-level estimation, remove the occasion facet, or turn off difference scores.

Dialog title	What tripped it	How to fix
Difference score reliability is not supported for retest	Difference-of-differences plus an occasion column, outside the dynamic person-specific design	Use a two-event difference score, set difference scores to No, or remove the occasion facet.
Event column not specified	Difference scores requested with no Event Type column	Choose the column that stores event labels.
Only 1 event / Only 2 events for difference score	Not exactly two events selected for a two-event difference	Select exactly two events.
Need 4 events for difference-of-differences / Need exactly 4 events for difference-of-differences	Fewer than four events selected for the DoD contrast (first title) or more than four (second)	Select exactly four events.
Residual covariance not supported for DoD	Residual covariance requested with the four-event contrast	Set Estimate residual covariance for co-occurring events to No.
Configure DoD contrast before Analyze	Four-event mode with no saved ERP_1 to ERP_4 mapping	Click DoD Contrast... and assign four unique events.

Read one row to see the pattern. Unique variable names not provided lists the offending column by name and puts the cursor back on the first popup that claims it, so the fix is one click away rather than a hunt.

Naming the output, and the whitespace rule

Once the checks pass, a save dialog titled Save output file as opens on the folder your input file came from, filtered to `*.psyrat`. Choose a name and a folder. Cancelling raises a File Error dialog and leaves you on Specify Inputs with nothing lost.

THE PATH MUST CONTAIN NO WHITESPACE, filename and folders alike. CmdStan does not accept whitespace in filenames or paths, so PsyRAT checks the pick and, if it finds a space, raises a Whitespace detected dialog reading "The selected filename/path contains whitespace." and reopens the save dialog. It will keep reopening until you give it a clean path or cancel, so it is worth setting up an analysis folder without spaces in its name before you start rather than discovering the rule at the save dialog. On macOS, `~/Documents/ERP Analyses/` fails and `~/Documents/erp_analyses/` works.

If you selected several measurements and set the four-event contrast, a modal notice tells you that this mode processes one measurement at a time and names the one it will use, which is the first in your selection. You must dismiss it before the run starts.

The run itself

There is no progress bar. PsyRAT reports progress to the MATLAB Command Window and nowhere else, so the only place to look is the console. You will see `Preparing data for analysis...`; then, for multi-event and multi-group designs, a `Working on event 1 of 4` line (or `Working on group ...` or `Working on event/group ...`, depending on the design) as each chunk starts;

then `Model is being run in cmdstan`, repeated per chunk (most designs precede it with a warning that this may take a while; some print the announcement alone). With Verbose Stan output on you also get CmdStan's own per-iteration log.

Every run includes a compilation step. PsyRAT generates a Stan model specific to your design and names it uniquely per run, and CmdStan compiles that model to C++ before it can sample, so the console sits at the compile step for several seconds per generated model (about 8 s was measured on the capture machine; a run over several events compiles one model per event) with nothing else to show. This happens on every run, not just the first (Chapter 18 lists it among the expected slow points).

MATLAB will report itself as busy for the whole fit, and it genuinely is. Do not click into the MATLAB window expecting a response.

Two families of CmdStan message appear with verbose output on, and both are expected. Runs of the location-scale, difference-score, dynamic-reliability, and residual-correlation designs print repeated `Informational Message: The current Metropolis proposal is about to be rejected` lines mentioning `lkj_corr_cholesky_lpdf`, early in warmup. Those designs estimate a correlation matrix through its Cholesky factor, an occasional early-warmup draw pushes a diagonal element to zero, and the sampler discards that proposal and continues. CmdStan may also print a warning that the data file is being read as an `RDump` file and that the format is deprecated. Neither message affects the reported variance components, coefficients, SEMs, or convergence diagnostics. Chapter 18 collects the console messages that DO mean something.

> CAUTION. Terminating MATLAB's processing with ctrl+c does not terminate the models. They run in C++, as separate CmdStan processes, one per chain. Interrupting MATLAB leaves those processes sampling in the background, consuming cores and writing output nobody will read. To stop a run for real, interrupt MATLAB and then kill each CmdStan process yourself in Activity Monitor on macOS or Task Manager on Windows. There will be one per chain, so at the default four chains, four processes.

Convergence, and the rerun offer

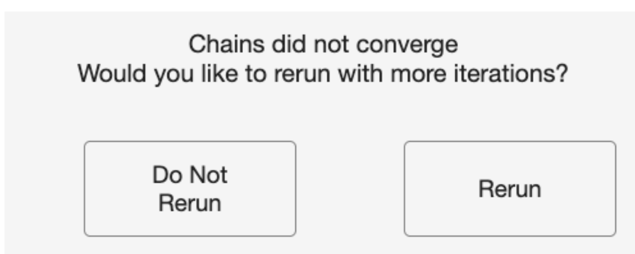


Figure 23: The rerun dialog raised when the chains did not converge.

When sampling finishes, PsyRAT checks convergence on every monitored parameter. Two conditions must hold: no potential scale reduction factor (R-hat) may reach 1.1, and no effective sample size may fall below ten times the number of chains. Failing either one marks the fit as not converged.

Divergent transitions are counted separately and reported in the console when there are any. They never trigger a rerun on their own, and they never block saving, but they do signal possible bias in the posterior even when R-hat looks healthy. The console message that reports them says as much and recommends more warmup iterations or stronger priors; raising adapt delta is the other standard lever. Treat a non-trivial divergence count as a reason to refit, not as noise.

If the fit did not converge, this dialog appears, on two lines: ”Chains did not converge” and ”Would you like to rerun with more iterations?”, with two buttons, Do Not Rerun and Rerun. Clicking Rerun DOUBLES both warmup and sampling and fits again from scratch (a doubled total below 1,000 is raised to 500 warmup and 500 sampling instead). The offer repeats after each attempt, so a run that starts at 5,000 and 5,000 and needs three reruns ends at 40,000 and 40,000, and takes correspondingly longer each time. Clicking Do Not Rerun saves the non-converged result and prints a console line saying so; closing the dialog with the window's close box counts as Do Not Rerun. That result is still written to disk, and it is still not trustworthy.

Trace plots

If View trace plots prior to saving Stan output is set to Yes and the analysis is the plain one-facet internal-consistency one, a window titled Trace Plots of Stan Parameters opens before saving. Each panel plots one parameter's draws against iteration number, with one line per chain. What you want is a dense, stationary band in which the chains overlap each other and neither drifts across the x-axis. Wandering, or one chain sitting apart from the rest, means the fit has not settled, whatever R-hat said.

A second prompt follows, ”Would you like to rerun with more iterations?”, with the same Do Not Rerun and Rerun buttons and the same doubling behavior. This is the one place the toolbox lets you refit on visual evidence rather than on the convergence rule alone.

For every other analysis, setting this to Yes prints a console line instead of plotting: family-specific text for the test-retest, subject-level, difference-score, and dynamic-reliability designs, and a generic line naming the analysis for anything else. The run then continues to saving.

After the run

The console prints `Saving Processed Data...` and the result is written. For a single measurement that succeeded, PsyRAT opens the viewer for you.

One thing on that path looks like a hang and is not. The viewer raises up to two notices before it draws anything: one titled Native estimation engine, when a native MATLAB engine rather than CmdStan produced the fit, and one titled Estimation diagnostics, when the fit failed to converge or produced divergent transitions. Each blocks until dismissed.

> CAUTION. The post-run notice, titled Estimation diagnostics, blocks MATLAB until it is dismissed, and it can open behind another window or on a second display. MATLAB then reads as busy at near-zero CPU with the output file already written, which looks exactly like a hang. It is not one. Check your other windows and displays for the dialog, click OK, and the viewer opens.

7.6 What gets saved

The `.psyrat` file

The result is a MAT-file carrying the `.psyrat` extension, holding one variable, `psyrat_data`. Everything the viewer needs is inside it: the loaded data, the column mapping, the posterior draws for every variance component, the convergence tables, and the reliability output. You can load it

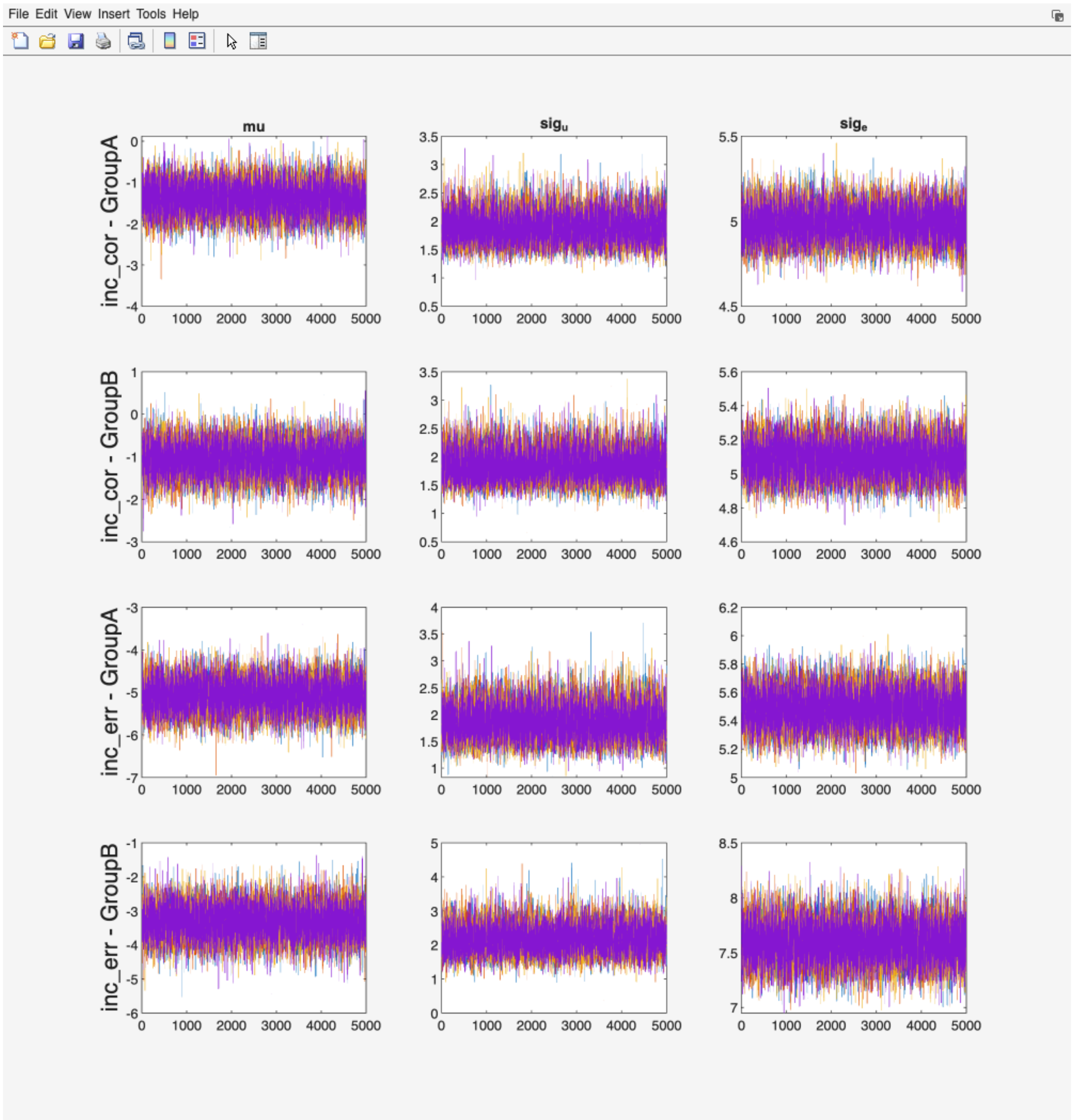


Figure 24: Trace plots of Stan parameters for a converged one-facet fit.

from the command line with `load('myrun.psyrat','-mat')` if you want to work with the draws directly.

Expect it to be large. A one-facet single-session fit runs from a few hundred kilobytes to a few megabytes. A test-retest fit runs to a few megabytes. The subject-level dynamic-reliability designs reach tens of megabytes, and the four-event difference-of-differences designs are the extreme case: recorded runs of the person-specific contrast have produced files of roughly 150 MB single-occasion and 300 MB with an occasion facet. Size scales with the number of retained draws times the number of monitored parameters, so doubling iterations through the rerun dialog roughly doubles the file. Plan disk accordingly before a DoD run.

A temporary directory named `psyrat_stan_` plus a random suffix appears next to your output while CmdStan works, and is removed when the run finishes. That is the other reason the save folder must be writable and free of whitespace.

The run-config sidecar

Alongside every result PsyRAT writes `<name>_runconfig.json`, a human-readable JSON file recording what produced the fit. This is the file to keep with a manuscript, and the file to attach to a bug report.

It records the schema version, a UTC timestamp, the PsyRAT, MATLAB, and detected CmdStan versions, and the source revision of the toolbox tree that ran. It records the analysis that ran and the output location. Under `columns` it stores your source column names for every role, including the dimension and items-per-split columns, plus the group, event, and occasion selections. Under `data` it stores the input file name, a SHA-256 hash of that file, a content hash of the loaded table, and the number of rows that entered estimation. Under `estimation` it stores the seed, chains, warmup, sampling, total iterations, verbose and within-chain settings, adapt delta, max tree depth, the bootstrap and gradient settings the native engines use, and both the requested engine and the engine that actually ran. Under `design` it stores every design flag: subject-level estimation, difference-score mode, residual covariance, dynamic reliability, splits, residual-correlation parameterization, dispersion, likelihood family, gamma scale parameterization, and the DoD mapping. Under `view` it stores the coefficient selectors in effect. Under `priors` it stores the complete prior set the run fit (a partial override is merged over the defaults before fitting, and the merged result is recorded).

That is enough to replay the run. `psyrat_run('config', '<name>_runconfig.json')` reads it back and re-executes the same settings; see Chapter 14. Two honest limits. The viewer block is record-only, because the scripted runner exposes no options for it, and it captures the viewer settings in effect AT ESTIMATION TIME, so changing a coefficient in the viewer afterwards is not re-recorded. Writing the sidecar is also best-effort: if it fails, the console says so and your results are kept regardless.

Batches, failures, and error reports

Selecting several measurements runs them in sequence, and each gets its own output file. The names are built from the one you typed by appending the measurement's column name, sanitized for filesystem use, so saving as `study.psyrat` with `ERN` and `Pe` selected produces `study_ERN.psyrat` and `study_Pe.psyrat`, each with its own sidecar.

A measurement that fails does not stop the batch. PsyRAT catches the failure, writes a `.txt` error report next to where the output would have gone, and moves to the next measurement. The console prints the measurement name, the reason, and the report path. The report itself records the measurement, the requested output filename, a timestamp, the error identifier, the full message, and the stack trace, which is exactly what makes a bug report actionable.

When the batch ends, a Processing Complete box reports how many runs completed, how many succeeded, how many failed, and the folder they went to. If every measurement failed, an All Measurements Failed dialog points you at the `.txt` reports instead. A single-measurement run that succeeds skips the summary box and opens the viewer directly, which is the ordinary single-measurement path Section 7.5 describes.

8. Importing and scoring ERP data

Everything in Chapter 7 starts from a long-format table of single-trial scores. Some readers arrive with epoched ERP files and no table at all. For them the toolbox will read those files, measure a component on every trial, and hand back a table the processing workflow accepts.

8.1 When to use this workflow

If your single-trial scores are already in a spreadsheet, you do not need this chapter at all. Chapter 6 describes the columns PsyRAT expects, and Chapter 7 walks the processing screens. Bring your own table whenever you have one. Most labs already score components in EEGLAB, ERPLAB, or a local script, and there is no advantage to re-measuring the same waveforms here.

Use the import workflow when the scores do not exist yet. You have epoched single-trial files, you know the component window and the channels, and you would rather not write the export loop yourself. The workflow reads the waveforms, applies one measurement rule to every trial, and builds the table for you.

The workflow does not replace preprocessing. Filtering, re-referencing, artifact rejection, epoching, and baseline correction all happen upstream in your usual pipeline. Single trials that arrive here are assumed to be already baseline-adjusted, and the scorer performs no baseline correction of its own. (There is deliberately no baseline field on the scoring form. If your epochs are not baseline-corrected, correct them before importing, or the amplitudes you get will carry whatever offset the raw epoch had.) Artifact rejection has to be carried out, not only marked: the importer reads every epoch in a file and scores all of them, and it does not read EEGLAB's rejection marks (`EEG.reject`) or ERPLAB's artifact flags, so an epoch that was flagged but not removed is scored and enters the reliability estimate. Remove rejected epochs from the dataset before importing.

Projects

Work in this screen lives in a project, saved as a `.psyratproj` file. A project holds the imported sources (recorded by path and content hash), the canonical single-trial data assembled from them, the trial metadata table, your scoring specifications, every scoring run's per-trial results, and an audit log of what happened and when. Saving and reopening a project resumes an import session where you left it.

Three buttons manage that file. New Project opens a one-field dialog (Project name:) and creates an empty project in memory, with nothing on disk yet. Open Project browses for an existing `.psyratproj` file, validates its structure, and refuses a file whose schema major version does not match this build. Save Project writes the current state, prompting for a location the first time and overwriting the same file after that.

Project files are large. The canonical single-trial array holds every imported trial at every channel and sample, and it is saved along with everything else, so a project covering a full study is measured in gigabytes rather than megabytes. Save it somewhere with room.

Opening the screen

From the PsyRAT home screen, click Import + Score ERP. The window that opens is titled PsyRAT: Import + Score ERP.

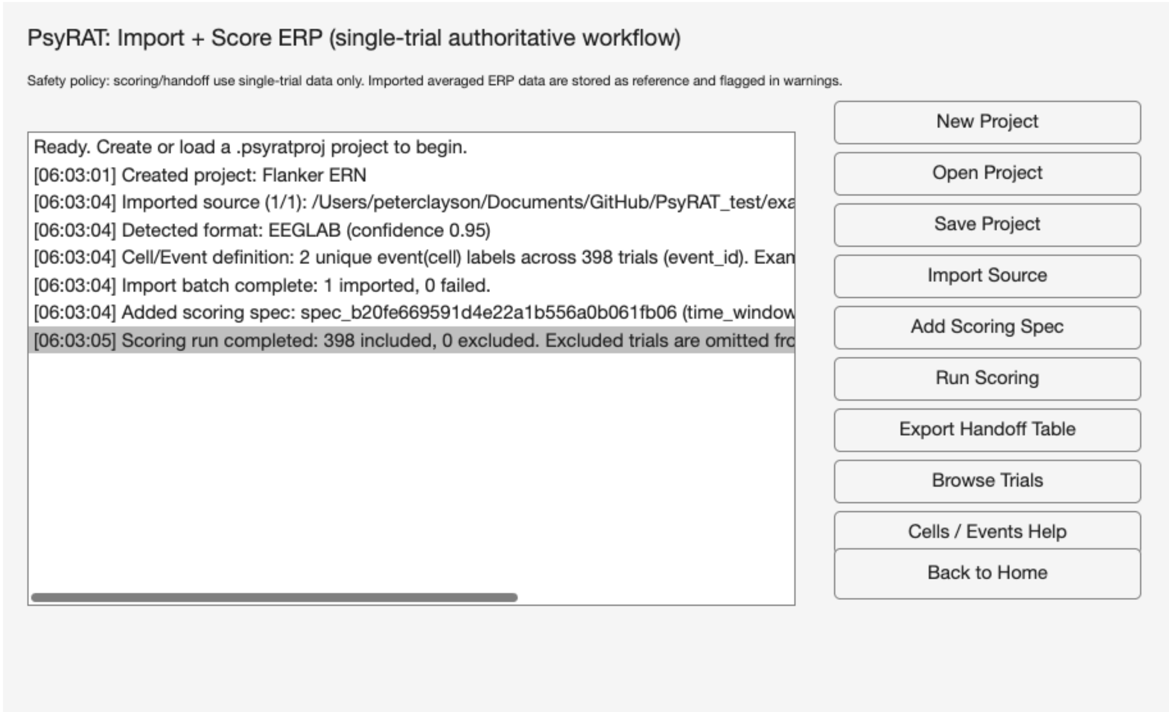


Figure 25: The Import + Score ERP screen after a single EEGLAB .set file has been imported and one scoring spec has been run. The status listbox on the left records each action in order; the buttons on the right run the workflow top to bottom.

The heading reads PsyRAT: Import + Score ERP (single-trial authoritative workflow), and one line under it states the safety policy that governs the whole screen (Section 8.2). The large listbox on the left is the status log. Every action appends a timestamped line, newest at the bottom, and the box scrolls to the newest line on its own. Read it after every click; it is where the toolbox reports what it detected, what it warned about, and what it counted.

The buttons on the right run the workflow from top to bottom:

- New Project: create an empty project in memory.
- Open Project: load an existing `.psyratproj` file.
- Save Project: write the current project to disk.
- Import Source: read one or more ERP source files into the project.
- Add Scoring Spec: define a measurement method and its parameters.
- Run Scoring: score every included trial with the selected spec.
- Export Handoff Table: build the process-data table and send it to the workspace.
- Browse Trials: open the Trial Browser on the imported single trials.
- Cells / Events Help: explain how event (cell) labels were assigned.
- Back to Home: close this screen and return to the home screen.

Save Project, Import Source, Add Scoring Spec, Run Scoring, Export Handoff Table, and Browse Trials all report the same error when no project is loaded, naming New Project or Open Project as the fix. Create the project first.

8.2 What the toolbox reads

The safety policy

The line under the heading states the rule the rest of the screen follows:

> Safety policy: scoring/handoff use single-trial data only. Imported averaged ERP data are stored as reference and flagged in warnings.

That is the whole policy, and it is enforced in code rather than by convention. Averaged waveforms are kept in the project as reference data and flagged, and the scoring routine refuses to run at all when a project holds no single-trial data. Generalizability theory partitions variance across trials. An average has already spent the trial variance, so there is nothing left to partition.

The three adapters

Three readers are registered, and the file extension decides which one runs.

EEGLAB `.set` is the format that supports the full workflow. A dataset whose data array is three-dimensional with more than one epoch is read as single-trial data: trials, channels, and samples, with the time axis taken from `EEG.times` (or reconstructed from `EEG.srate` and `EEG.xmin` when `times` is absent) and channel labels taken from `EEG.chanlocs`. A `.set` file that was never epoched, or that holds a single epoch, contributes averaged reference data instead and warns that it cannot be used for scoring or handoff. Reading a `.set` requires EEGLAB on the MATLAB path; without `pop_loadset` the import stops with a message saying so.

ERPLAB `.erp` is always reference-only. An ERPset stores bin-averaged waveforms (channel by sample by bin), not single trials, so the adapter imports it as averaged data with a warning and the scoring pipeline then refuses it. This is not a limitation the toolbox can work around. If you need single-trial scores from an ERPLAB pipeline, import the epoched `.set` the ERPset was averaged from.

EP Toolkit `.mat` and `.ept` files are read from the `EPdata` struct (or from the first struct in the file carrying a `data` field alongside `chanNames` or `timeNames`). The file is treated as single-trial when its `dataType` field says `single_trial`, and otherwise whenever it holds more than one wave-by-participant slice. That second condition is a heuristic, so check the trial count reported in the status log against what you expect. Extra factor, reliability, or frequency dimensions are not read; only the first slice of each is used, and a warning says so.

Detection, confidence, and alternates

Selecting the Import Source button will bring up a file dialog titled Import ERP Source(s), filtered to Supported ERP Sources (*.set, *.erp, *.mat, *.ept), with multi-select enabled. Once you choose files, every registered adapter is asked how well it recognizes each one. The results are sorted by confidence, the winner reads the file, and the rest are kept as alternates in the import report.

Confidence is driven by the extension and, for `.mat` and `.ept` files, by the variables inside. A `.set` scores .95 for EEGLAB and a `.erp` scores .95 for ERPLAB. A file containing a variable named `EPdata` scores .98 for the EP reader, other variable names suggesting EP content score .45, and a generic `.mat` file scores .15 (.05 when its variable list cannot be read at all). The status log prints the winner in the form `Detected format: EEGLAB (confidence 0.95)`. Import stops with an unsupported-format error only when no adapter scores above zero.

That floor of .15 has a consequence worth knowing. Since the EEGLAB and ERPLAB adapters both score zero on a `.mat` file, any `.mat` you select is routed to the EP reader, even one that has nothing to do with EP Toolkit. It will fail there with a message about not locating an EP Toolkit-like struct, which is correct but arrives one step later than you might expect.

Merging several files

Import Source accepts several files at once, and you can also click it repeatedly to add more sources to the same project. Each file is imported in turn and merged into one canonical single-trial store, which is what makes a multi-participant dataset possible. A merge is refused unless the incoming file matches the existing data on time axis, sampling rate, and channel labels.

A file that fails is reported and skipped rather than aborting the batch. The log ends with a line counting successes and failures, and adds a note naming a time-axis, sampling-rate, or channel mismatch as the usual cause. A skipped file leaves the project exactly as it was, so nothing partial is carried forward.

Where the trial metadata come from

Each imported trial carries metadata that later becomes the facet structure of your table. From an EEGLAB source, `subject_id` comes from `EEG.subject`, and falls back to the file name without its extension when that field is empty. `group_id` comes from `EEG.group` and `occasion_id` from `EEG.session`, both empty when absent.

The event (cell) label is the time-locking event of each epoch, resolved as the event whose latency is closest to zero rather than the first event listed. That distinction matters whenever an epoch captures more than one event, for example a preceding stimulus or a response falling inside the baseline. Taking the first listed event in that case assigns the wrong condition to the trial and feeds the wrong label into the event facet of every downstream estimate. When an epoch holds several events and no usable latency list, the first event is used and the affected epochs are counted in a warning telling you to verify the labels.

From an EP Toolkit source, `subject_id` comes from `subNames` and the event (cell) label from `cellNames`; no group or occasion is assigned. After each import batch the status log summarizes what it found, counting the unique event (cell) labels across all trials and previewing up to five of them.

8.3 The seven scoring methods

Add Scoring Spec asks for the method first, in a picker whose prompt reads Select scoring method:. The seven entries appear in this order.

absolute_peak_amplitude. The amplitude of the largest deflection anywhere in the window (the maximum for positive polarity, the minimum for negative, the largest absolute value for both). Use it when the component is a sharp, reliably present peak and you want peak amplitude rather than mean amplitude.

local_peak_amplitude. The most extreme sample in the window (largest for positive polarity, smallest for negative, largest in absolute value for both) that is also an extremum relative to a fixed number of neighboring samples on each side. Use it when a plain peak search would be captured by monotone drift toward the edge of the window; a trial with no qualifying local peak is excluded rather than scored.

time_window_mean_amplitude. The mean amplitude across the window. Use it as the default for single-trial amplitude, since a mean over a window is far less sensitive to single-trial noise than a peak is, and its statistical behavior is better understood.

peak_latency. The latency of the window extremum, located exactly as absolute_peak_amplitude locates it. The score itself is the latency in milliseconds, so a table exported from this method holds latencies in `meas`, not amplitudes.

centroid_latency. The amplitude-weighted mean latency of the window, with weights taken from the positive part, the negative part, or the absolute value. It is nonstandard for ERP component timing, and I would reach for fractional_area_latency instead for anything you plan to publish; it is kept because the weighted centroid is occasionally the quantity someone wants.

fractional_area_latency. The latency at which the cumulative area of the polarity-selected waveform reaches a specified fraction of the total window area, interpolated between samples. At the default fraction of .5 this is the conventional 50% area latency, and it is the timing measure to prefer. Set polarity to the component's own polarity; `both` rectifies the waveform and mixes lobes across a zero crossing.

integrated_area_amplitude. The time integral of the polarity-selected waveform over the window, in microvolt-milliseconds. Use it when you want area rather than mean amplitude, and remember that it is sensitive to the baseline your epochs already carry.

The parameter form

Choosing a method opens a form titled Scoring Spec. Five fields are always present, in this order:

- Spec name: a label for this measurement, shown wherever specs are listed.
- Window start (ms): the earliest sample included, in milliseconds.
- Window end (ms): the latest sample included, in milliseconds.
- Channel selector (index/label; comma-separated ROI allowed): one channel or several.
- Polarity (positive/negative/both): which direction of deflection counts.

Three methods add one field each. local_peak_amplitude adds Neighborhood samples (local peak), defaulting to 3, which sets how many samples on each side a candidate must beat; a window holding fewer than twice that many samples plus one can contain no local peak at all, and every trial is then excluded. centroid_latency adds Centroid weight mode (positive_only/negative_only/absolute), defaulting to absolute. fractional_area_latency adds Area fraction (0-1; fractional area latency), defaulting to 0.5.

The channel selector takes numeric indices or channel labels, separated by commas. A single entry such as `FCz` or `1` scores that channel. Several entries such as `FCz,Cz,Fz` define a region of interest, and each channel is scored separately before the per-channel scores are averaged into the trial's score (latencies are averaged the same way). A label that matches no channel in the data stops the scoring run with an error naming the label, so check your spelling against the labels the file actually carries.

One consequence of the region-of-interest rule deserves its own paragraph. Consider this example: you score a three-channel region of interest with local_peak_amplitude, and on one trial two channels find a local peak while the third does not. That trial is excluded entirely rather than scored from the two surviving channels. Averaging a different subset of channels on different trials would let the spatial sampling vary trial to trial, which adds nuisance variance to exactly the quantity being estimated. The exclusion reason names each failing channel.

Windows are checked strictly. A window extending past either end of the available time axis stops the run with a message printing both the requested window and the axis. Ties within a window resolve to the earliest sample. When absolute_peak_amplitude or peak_latency places the extremum on the first or last sample of the window, a warning counts the affected trials and suggests the window is too narrow, which is usually right (the other peak and latency methods do not emit this warning). That warning goes to the MATLAB Command Window, not to the status log, so look there after a peak-based run.

Polarity has no effect on time_window_mean_amplitude. The mean is taken over the signed samples in the window whatever you enter in that field, so a negative-going component scored this way returns a negative number, as it should. The field is still on the form because the form is shared.

8.4 A worked walkthrough

Follow this walkthrough with any epoched single-trial `.set` of your own. The capture behind it imports one epoched EEGLAB `.set` file of response-locked ERP data recorded during a flanker task, scores ERN as the mean amplitude from 0 to 100 ms at a frontocentral channel, and inspects the result trial by trial. That file is not distributed with the toolbox, so substitute your own window and channel. The clicks are the same either way.

1. Open the screen. From the home screen click Import + Score ERP.

2. Create the project. Click New Project. A dialog titled New PsyRAT Project asks for Project name:, prefilled with Untitled PsyRAT ERP Project. Type a name you will recognize later and click OK. The status log reports `Created project: <your name>`. Nothing is on disk yet.

3. Import the file. Click Import Source, then choose your `.set` file in the Import ERP Source(s) dialog. The log then records the file that was imported and its position in the batch, the detected format and its confidence (`Detected format: EEGLAB (confidence 0.95)`), any warnings the adapter raised, a summary of the event (cell) labels found, and a closing count of imports and failures. In the capture shown above the file yielded 398 trials carrying 2 unique event (cell) labels, with one file imported and none failed.

Check the event labels in that summary before going on. They become the event facet, and a label that looks wrong here will look exactly as wrong three screens later.

4. Define the spec. Click Add Scoring Spec. The picker titled Select scoring method: lists the seven methods; select time_window_mean_amplitude (third in the list) and click OK. The Scoring Spec form then opens with five fields. Fill them in as:

- Spec name: `ERN`
- Window start (ms): `0`
- Window end (ms): `100`
- Channel selector (index/label; comma-separated ROI allowed): `FCz`
- Polarity (positive/negative/both): `negative`

Click OK. The log reports `Added scoring spec:` followed by the generated spec id (a `spec_` prefix and a hexadecimal token) and the method name. If your montage labels the frontocentral site differently, enter that label instead; the selector is matched against the channel labels in the file, case insensitively.

5. Run the scoring. Click Run Scoring. With one spec defined there is no picker, and the spec runs directly. The log reports the run as

```
Scoring run completed: N included, M excluded. Excluded trials are omitted from
↪ final handoff dataset.
```

The capture shows 398 included and 0 excluded. Time-window mean amplitude excludes a trial only when the window holds no finite samples, so a nonzero exclusion count here points at bad epochs rather than at a bad window.

6. Inspect trials. Click Browse Trials.

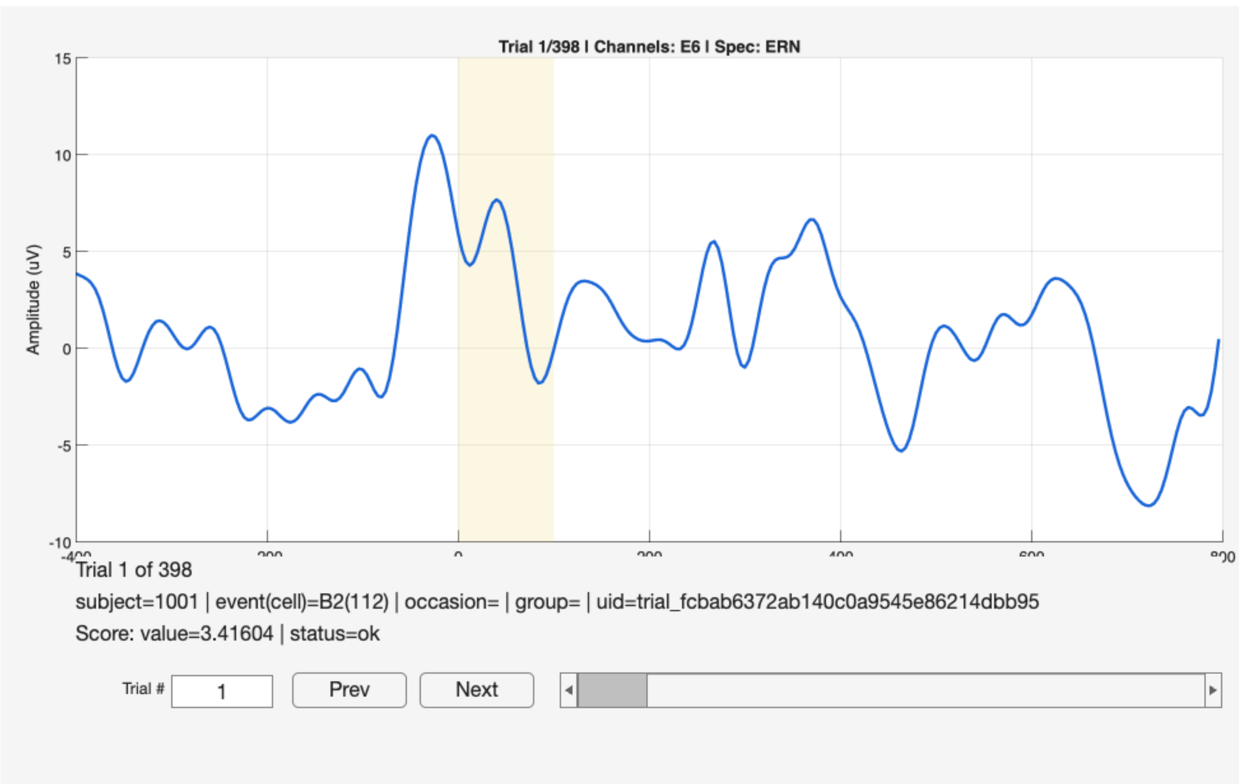


Figure 26: The Trial Browser on the first trial of the imported flanker dataset, with the ERN spec active. The shaded band marks the 0 to 100 ms scoring window; the lines under the axes give the trial number, its metadata, and its score. This capture used an EGI montage, where the frontocentral site is E6.

The axes plot the current trial in microvolts against time in milliseconds, and the title names the trial, the channels being drawn, and the active spec. When the spec selects one channel you see that channel alone; when it selects several you see each channel in its own color with their mean overplotted in black. The shaded band marks the scoring window, which is the fastest way to confirm that the window you typed sits where you meant it to.

Check the channels named in the title. A channel label that matches nothing in the data falls back to the first channel here without complaint, so an unexpected channel in the title means the label in your spec did not match. (Run Scoring is stricter and stops with an error naming the label, which is why you are unlikely to see this after a successful run.)

Four controls move between trials. Trial # is an edit box: type a number, press Enter, and the browser jumps there (out-of-range numbers clamp to the ends). Prev and Next step one trial. The slider under them drags through the whole set, one trial per arrow click and up to fifty per page click. The arrow keys step too: right and down move forward, left and up move back.

Three text lines sit under the axes. The first repeats the trial position as `Trial 1 of N`. The second gives the metadata in the form

```
subject=<subject_id> | event(cell)=<event_id> | occasion=<occasion_id> |
 ↪  group=<group_id> | uid=<trial_uid>
```

An empty field prints its label with nothing after it, which is what you will see for occasion and group in a single-session import. A third line reports the score: `Score: value=<v> | status=ok` for a method with no latency, `Score: value=<v> | latency=<t> ms | status=ok` when a latency was measured, and `Score: EXCLUDED | status=failed | reason=<why>` for a trial that failed. For the first trial of the capture that line reads `Score: value=3.41604 | status=ok`.

The browser takes a snapshot of the project when it opens. Scores computed after that do not appear in an already-open browser, so run the scoring first, and close and reopen the browser after any re-scoring. This failure is silent: an open browser keeps showing the scores from the run that was current when it opened, with nothing to mark them as stale. (Before any scoring run at all, every trial instead reads `Score: not available (run scoring to see trial-level results). [trial index N]`.)

7. Export and save. Click Export Handoff Table, then Save Project. Success! The project on disk now contains the imported waveforms, the ERN spec, the per-trial scores, and the audit trail, and the workspace contains a table ready for Chapter 7.

8.5 Export Handoff Table

Export Handoff Table converts the scored trials into the long-format table Chapter 6 specifies. If the project holds more than one spec, a picker whose prompt reads Select scoring spec to export: appears with the active spec preselected; with one spec it exports that spec without asking.

Only the most recent run of the chosen spec is exported, regardless of which spec you scored last. Re-running a spec marks its earlier results stale, and stale rows are dropped here. Within that run, a trial reaches the table only when its status is `ok` and it was not excluded, so every trial that failed to score is left out and counted.

What lands in the workspace

Two variables are assigned in the base workspace:

- `psyrat_import_proc_table`: the process-data table itself.
- `psyrat_import_handoff_meta`: the accompanying metadata.

The metadata record the project, spec, and run identifiers, the UTC timestamp, the number of input, included, and excluded trials, a tally of exclusion reasons with their counts, and the full preflight report. The status log prints the included and excluded counts and then the name of the table variable. It does not print the metadata variable name, and it does not print preflight warnings; those live in `psyrat_import_handoff_meta.preflight.warnings` and are worth reading, because that is where a warning about too few participants or thin trial counts will be sitting.

The column mapping

The table is built with these columns, in this order:

Column	Source	Present when
`id`	`subject_id` from the trial table	always
`meas`	the scored value for the trial	always
`group`	`group_id`	any trial has a nonempty `group_id`
`event`	`event_id`, the event (cell) label	any trial has a nonempty `event_id`
`time`	`occasion_id`	any trial has a nonempty `occasion_id`

`id`, `group`, `event`, and `time` are text; `meas` is numeric. A facet column whose values are empty on every trial is left out entirely rather than shipped as a column of blanks, which is why a single-session import with no group information produces a three-column table (`id`, `meas`, `event`). That is the correct shape for a one-facet design.

Three guards run before the table is returned. Scores computed from data that no longer match the project's canonical single-trial store are refused, with a message telling you to recompute. A table in which every trial was excluded is refused rather than returned empty. Any non-finite or missing value in `meas` is refused, since PsyRAT requires complete measurements. The table then goes through the same preflight validation the processing workflow uses, and a blocking error there stops the export with that error's own message.

Note what the table does not contain. There is no items-per-split (`n_i`) column, so the data-splits analyses in Chapter 13 need a table you build yourself. There is also nothing in the export that groups trials, so if your design has a group facet it has to come from the source files (`EEG.group`), not from this screen.

Continuing into Process New Data

Process New Data reads a data file rather than a workspace variable, so write the table out first:

```
writetable(psyrat_import_proc_table, '/path/to/ern_scores.csv')
```

Then return to the home screen (Back to Home), click Process New Data, and select that file. From there Chapter 7 takes over: you assign the columns, choose the analysis, and set the estimation preferences.

If you would rather stay in code, `psyrat_run` accepts the table directly, with no intermediate file:

```
psyrat_data = psyrat_run('data', psyrat_import_proc_table, ...
    'savepath', '/path/with_no_spaces', 'savename', 'ern_study.psyrat');
```

Chapter 14 covers the rest of the arguments.

Cells / Events Help

The Cells / Events Help button opens a reference panel explaining where event (cell) labels come from, since that is the field people most often need to check. It states that `event(cell)` is the trial-level condition label stored in the canonical trial table's `event_id`; that EEGLAB imports take it from the time-locking event of each epoch (falling back to the first listed event when no latency is available); that EP Toolkit imports take it from the EP cell names; and that ERPLAB `.erp` imports are averaged and reference-only, with no single-trial event cells to score. It closes by listing the handoff mapping (`subject_id` to `id`, `event_id` to `event`, `occasion_id` to `time`, `group_id` to `group`) and restating the safety policy.

Limits

Scoring operates on single-trial data, and only on single-trial data. Averaged ERP input, which is every `.erp` file plus any `.set` or EP file the adapter reads as averaged, is stored for reference and flagged, and Run Scoring refuses a project that holds nothing else. Browse Trials refuses it too, naming a single-trial import as the fix. One nuance about the flag: every single-trial import also stores a derived average for reference, so the averaged-data notice appears on ordinary imports as well; on its own it is not a sign that scoring will be refused.

Two smaller limits follow from how sources are merged. Files must agree on time axis, sampling rate, and channel labels to join one project, so datasets epoched with different windows have to be re-epoched before they can be analyzed together. And because `subject_id` falls back to the file name, a batch of `.set` files without an `EEG.subject` field will be identified by their file names, which is fine as long as the file names are unique and meaningful. Occasions have the same shape without the fallback: `occasion_id` comes from `EEG.session`, and when that field is empty every file lands in one occasion, so two sessions imported from files without `EEG.session` produce a single-occasion table and a single-session analysis rather than a test-retest one. Set `EEG.session` before importing when you intend a test-retest design. One file gives you one participant, and one participant is not a reliability study; import every participant's file into the same project before exporting. If an import or a scoring run fails, Chapter 18 lists the messages by symptom.

9. Viewing results

Estimation leaves you with a single `.psyrat` file. Everything in this chapter reads that file back and turns it into figures and tables. Nothing here re-estimates anything. You can reopen the same results file as often as you like, try a different reliability threshold, save a table, and close the windows again without disturbing the fit.

If you have not run an analysis yet, Chapter 7 covers processing, and Chapter 2 walks the shortest path from raw data to a first set of results.

9.1 What every viewer has in common

Opening a results file

Selecting View Results on the home screen brings up a file picker titled Data, filtered to PsyRAT Toolbox files (*.psyrat). Choose the file you want. If you cancel the picker instead, the toolbox reopens the home screen and shows a File Error dialog saying that no PsyRAT results file was selected. That is a notification, not a failure, and clicking View Results again picks up where you left off.

Loading prints a short `Loading Data...` line in the MATLAB Command Window, then opens one of five setup screens.

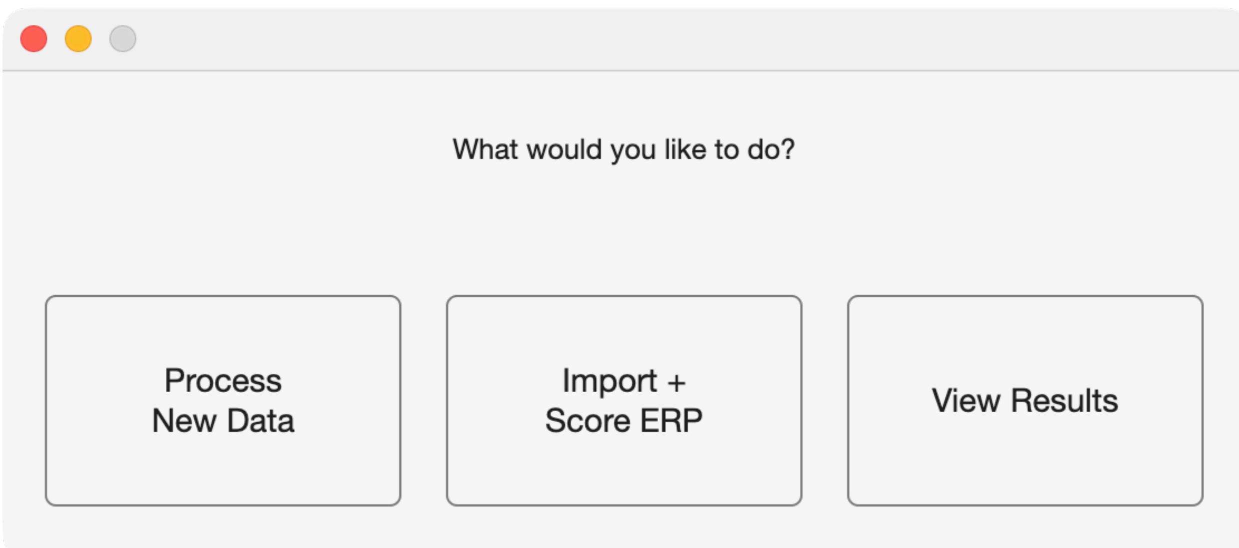


Figure 27: The home screen. View Results is the third button.

Which viewer opens, and why you do not choose it

The results file records the design that produced it, and the toolbox routes on that record. Five viewers exist, one per analysis family:

- Single-session (one-facet) results open View Results Setup.
- Test-retest (two-facet) results open View Results Setup, with extra controls.
- Difference-of-differences results open DoD Reliability Setup.
- Dynamic reliability results open Dynamic Reliability: View Results.

- Nonparallel data-splits results open Data-Splits Reliability: View Results.

The routing is not a preference and no control overrides it. A difference-score result opens the single-session viewer because a difference score in a single session is a one-facet design, not because anything guessed. If a results file carries a design label the toolbox does not recognize, no viewer opens at all and you are returned to a bare prompt, which happens only with a file written by a different version or edited by hand. One route does cross families on purpose: the DoD viewer can build a single-event or two-event view out of a DoD model and hand it to the single-session viewer (section 9.4).

Parallel data splits are a separate case. They reuse the single-session and test-retest viewers, with every trial-labeled control relabeled to split (Plot: Splits vs Reliability, Table: Split Cutoffs at Reliability Threshold, and so on). Only nonparallel splits get their own viewer. Chapter 13 explains the difference between the two.

Two notices that may appear before the viewer does

Some results files raise a warning dialog on open. Dismiss it and the viewer opens.

The first is titled Native estimation engine. It appears when the fit came from a native MATLAB engine rather than CmdStan, and its wording depends on which one. A native HMC fit is described as genuine Bayesian posterior draws that are not bit-identical to a CmdStan run. A native `fitlme` fit is described more sharply, because its intervals are approximate frequentist intervals rather than Bayesian credible intervals. Both notices tell you to re-run with CmdStan for a definitive analysis, and I would take that literally. Use the native engines to get a pipeline working. Report from CmdStan.

The second is titled Estimation diagnostics. It appears when the run recorded a convergence failure, one or more divergent transitions, or both, and it names which. Non-convergence means at least one measurement failed the R-hat and effective-sample-size criteria. Divergences are reported independently of that verdict, because sampling can diverge on a fit whose R-hat looks fine. In both cases the fix is another run with more warmup and sampling iterations, and sometimes stronger priors. Chapter 18 covers the diagnostics.

Both notices block. The toolbox waits for the click before it draws the viewer, which is deliberate: a non-blocking warning would be covered by the setup screen a fraction of a second later and never read. The side effect is that MATLAB can look hung while a notice waits, especially if the dialog opened behind another window. If the toolbox appears frozen just after you selected a `.psyrat` file, look for a warning dialog before you reach for ctrl+c.

Neither notice appears for a clean CmdStan run, and neither appears for results saved before these fields existed. Silence on open means there was nothing to flag, not that nothing was checked.

View Preferences

The Output Prefs button on the single-session and test-retest screens opens View Preferences. Four rows, then Save and Back:

- Line to plot for reliability: which of the three posterior summaries is drawn as the curve in the reliability-versus-trials figure. The choices are Lower Limit, Point Estimate, and Upper Limit.
- Number of trials to plot: how far the x-axis of that figure extends.

- Estimate to use for trial cutoffs: which posterior summary is compared against the Reliability Cutoff when the toolbox searches for the trial count that meets your threshold. Same three choices.
- Measure of central tendency for overall reliability calculations: Mean or Median.

The defaults are Point Estimate, 50, Point Estimate, and Mean. The cutoff estimate is the one to think about. Choosing Lower Limit there is the conservative reading, since it requires the whole credible interval to clear your threshold, and it will generally demand more trials than the point estimate does. That is a defensible choice and I use it when a decision hangs on the result. It is also a different estimand, so say in your methods which one you used. Fifty trials, meanwhile, is a plotting range and not a claim about your data. If your task delivers 200 usable trials, raise it, or the curve will stop before it reaches the part you care about.

To save the preferences click Save. To leave them as they were click Back. Either way you land back on the setup screen. Entering fewer than two trials to plot raises a dialog titled Invalid plot trial count and keeps you here, because a one-point curve is not a curve.

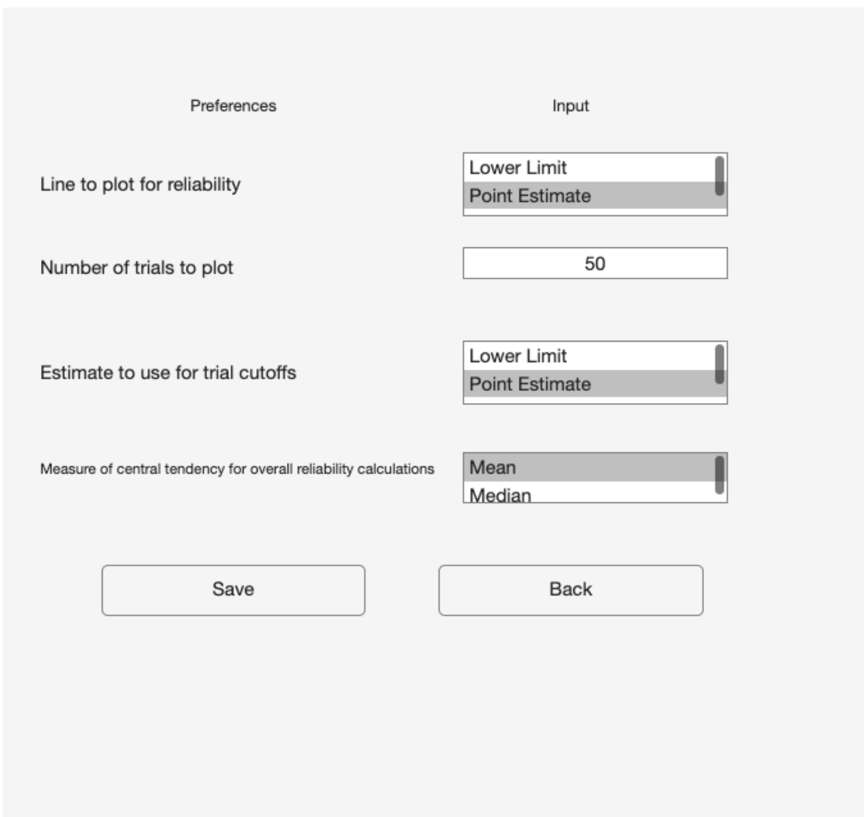


Figure 28: View Preferences, reached with the Output Prefs button.

Table windows and what a saved table carries

Every table the single-session, test-retest, criterion, and DoD viewers draw opens in its own window with a Save Table button at the bottom. Clicking it opens a save dialog offering Excel File (.xlsx) and Comma-Separated Value File (.csv); both write on every platform. (The dynamic-reliability and data-splits table windows carry no Save button of their own; you export those tables from the

Save buttons on their setup screens.)

Above the table itself, a saved file carries a header block. The table-specific part records when the table was generated, the toolbox version, the dataset file name, the reliability cutoff you set (to four decimal places, since the cutoff is an input you may need to reproduce exactly), and which posterior summary the cutoff was applied to.

Under that comes a standard provenance block, written by the same routine for every export. It records the estimation seed, the analysis design label, the estimation engine, whether the priors were the toolbox defaults or customized, the convergence verdict, and the divergent-transition count. Designs that need more get more: a test-retest export names which of the three coefficients produced the numbers, a non-Gaussian likelihood is labeled, and the designs with unusual estimands carry a sentence saying so. Anything the file does not record is written as ”not recorded” or ”not assessed” rather than guessed, so an old results file still exports and its header is never fabricated.

The block exists so a saved table can be interpreted a year later without the session that produced it. Keep the header when you circulate a table. Stripping it leaves a column of coefficients with no design attached.

Save IDs, and what it decides for you

The overall summary window (titled Dependability Analyses or Generalizability Analyses after the coefficient you selected, with Including All Trials and the occasion estimand appended on the test-retest viewer) carries a second button, Save IDs, next to Save Table. It writes two columns, Good IDs and Bad IDs, listing the participants whose data met your reliability threshold and those whose data did not. The same file formats are offered.

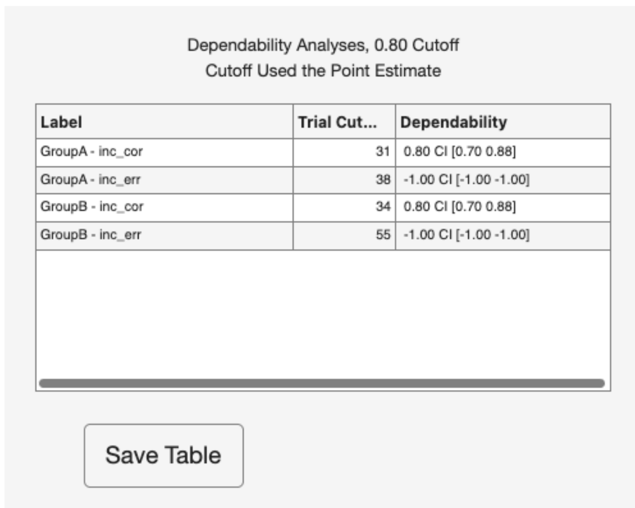
Two behaviors are worth knowing before you use that list.

The first is the all-events inclusion rule. When a group has more than one event, a participant reaches the Good IDs column only by meeting the threshold in every event. Falling short in one event is enough to move that participant into Bad IDs for the whole group, even where their data in the other events were fine. The rule has no exceptions: an event in which no participant clears the threshold empties the group's Good IDs entirely. Whenever an event empties a group this way, on any design, the console prints a line, the affected group's n Included reads 0, and the summary still opens, so the per-event rows stay available for diagnosis. Group-level difference coefficients keep rendering whenever a group ends up retaining nobody, whether through this rule or, on the difference routes, through the difference-score threshold itself; the coefficient is then evaluated at the full sample's mean (or median, per the central-tendency choice) trial counts. The per-event rows of a group that retained nobody are rendered the same way: their coefficient and trial-count columns describe the full sample, printed beside an n Included of 0, so read them as diagnostics rather than as the retained sample's reliability. The test-retest difference route uses the full sample's counts at every retention level.

Consider this example: a flanker analysis with error and correct trials, where a participant has 90 correct trials and 4 errors. The correct condition clears the threshold comfortably and the error condition does not, so the participant appears once, in Bad IDs. That is the right default for an analysis that uses both conditions, including any difference between them. It is the wrong default if you meant to screen each condition on its own, and the per-event lists are not what Save IDs writes.

The second is the -1 sentinel, which you will meet in the trial-cutoff table (the window titled Results of Increasing the Number of Trials on Dependability) rather than in the ID file. When no trial count anywhere in the projected range reaches your threshold, the toolbox stores -1 for that cell's trial cutoff and for the reliability point estimate and interval at the cutoff. A value of -1 indicates that an appropriate cutoff could not be realistically found given the specified reliability threshold. It is a flag, not a count, and averaging it into anything produces nonsense. When it appears, every participant in that group is placed in Bad IDs, since no attainable trial count qualified anyone. You will also see a dialog titled Cutoff not calculable saying the data are too variable or there are not enough trials. Lower the threshold, or accept that these scores do not support the decision you wanted to make with them.

A related dialog, Extrapolation beyond data, means the opposite problem: a cutoff was found, but only past the largest trial count anyone actually has. The number is an extrapolation from the model, and treating it as an observed result overstates what the data show.


Dependability Analyses, 0.80 Cutoff
Cutoff Used the Point Estimate

Label	Trial Cut...	Dependability
GroupA - inc_cor	31	0.80 CI [0.70 0.88]
GroupA - inc_err	38	-1.00 CI [-1.00 -1.00]
GroupB - inc_cor	34	0.80 CI [0.70 0.88]
GroupB - inc_err	55	-1.00 CI [-1.00 -1.00]

Save Table

Figure 29: The trial-cutoff table, where the -1 sentinel appears.

Note one asymmetry. Save Table writes the full provenance block. Save IDs writes the shorter header (generation date, version, dataset, cutoff, cutoff estimate, and the iteration line) without it. If you archive an ID list, archive the summary table alongside it.

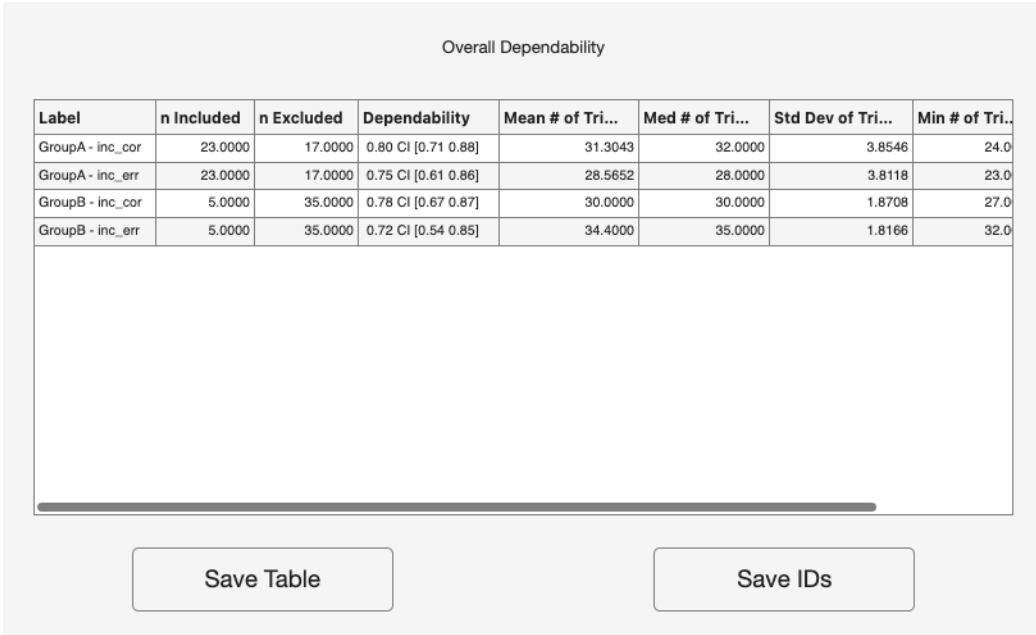
9.2 Single-session results

The one-facet viewer is titled View Results Setup. Its top rows are read-only: Dataset, Measurement, and, when the analysis is not a difference score and covers exactly one event, Cell/Event.

Two settings sit above the checkboxes.

- Reliability Cutoff: the reliability threshold used to retain data. Participants without enough trials to meet it are recommended for exclusion. The default is .8.
- G-Theory Coefficient: which coefficient the outputs report. Dependability uses absolute error variance, generalizability uses relative error variance.

The .8 default is conventional rather than derived, and I would not defend it as a universal standard. Pick the threshold your decision needs and state it.



Overall Dependability

Label	n Included	n Excluded	Dependability	Mean # of Tri...	Med # of Tri...	Std Dev of Tri...	Min # of Tri..
GroupA - inc_cor	23.0000	17.0000	0.80 CI [0.71 0.88]	31.3043	32.0000	3.8546	24.0
GroupA - inc_err	23.0000	17.0000	0.75 CI [0.61 0.86]	28.5652	28.0000	3.8118	23.0
GroupB - inc_cor	5.0000	35.0000	0.78 CI [0.67 0.87]	30.0000	30.0000	1.8708	27.0
GroupB - inc_err	5.0000	35.0000	0.72 CI [0.54 0.85]	34.4000	35.0000	1.8166	32.0

Figure 30: The overall summary window, with Save Table and Save IDs.

The coefficient control changes with the design. On a plain one-facet run and on a subject-level run the label reads G-Theory Coefficient; on a subject-level run its tooltip names the absolute coefficient phi_s and the relative coefficient G_s, which is the same distinction at the participant level. On either difference-score analysis the label instead reads Difference-Score Coefficient, and the choice governs the difference score. Only one coefficient control is shown at a time, and it is the one the summary actually consumes.

Below the Output Selection heading (marked "(checked = include)") sit six checkboxes, all on by default:

- Plot: Trials vs Reliability
- Plot: ICC Estimates
- Plot: Between-Person Standard Deviations
- Table: Trial Cutoffs at Reliability Threshold
- Table: Overall Reliability Summary
- Table: Sources of Variance

Table: Sources of Variance is the one that carries the within-person terms. It reports between- and within-person standard deviations, standard errors of measurement, and ICCs in one window. The between-person standard deviations also get their own figure; the within-person ones do not.

Three of those labels change on a subject-level run. Plot: Trials vs Reliability becomes Plot: Subject-Level Reliability, Plot: ICC Estimates becomes Plot: Subject-Level ICC Estimates, and Table: Trial Cutoffs at Reliability Threshold becomes Table: Subject-Level Reliability. The change is not cosmetic. A subject-level analysis estimates a per-participant residual, so a curve projecting the group's reliability over trial counts is not a quantity it produced, and the toolbox offers the per-participant outputs instead. On split data every "Trials" in these labels reads "Splits".

One route keeps the table as well. On the subject-level difference-score analysis the trial-cutoff

Group-Level Between- and Within-Person Standard Deviations, SEMs, and ICCs

Label	Between Std Dev	Trial Std Dev	Within Std Dev	SEM (dependabil...	ICC
GroupA - inc_cor	1.88 CI [1.41 2.47]	1.01 CI [0.62 1.46]	4.98 CI [4.79 5.19]	0.91 CI [0.87 0.95]	0.12
GroupA - inc_err	1.85 CI [1.30 2.54]	0.43 CI [0.02 1.04]	5.48 CI [5.23 5.75]	1.03 CI [0.98 1.08]	0.10
GroupB - inc_cor	1.84 CI [1.37 2.45]	1.16 CI [0.74 1.67]	5.09 CI [4.89 5.30]	0.95 CI [0.91 1.00]	0.11
GroupB - inc_err	2.17 CI [1.43 3.06]	1.14 CI [0.34 1.91]	7.59 CI [7.24 7.95]	1.31 CI [1.25 1.37]	0.08

Save Table

Figure 31: The Sources of Variance table.

table still opens alongside the subject-level outputs, and this screen offers no control over it: the relabeled checkbox governs the subject-level table instead, so the table appears whenever results are viewed here. Its contents there are observed quantities rather than projections: the per-event rows report the smallest trial count among the participants that event included at the threshold (or across everyone, when that event included nobody) next to the group-level coefficient, and the diff score row reports the observed-design difference coefficient with a --- placeholder in its count column, because this analysis runs no trial-cutoff search.

Six buttons run along the bottom:

- Back to Home
- Open Another .psyrat File
- Generate Figures/Tables
- Output Prefs
- Criterion Score Outputs
- Close PsyRAT Outputs

Generate Figures/Tables draws everything you checked, each in its own window. Close PsyRAT Outputs closes those windows and leaves your own MATLAB figures alone, which matters after the fourth or fifth pass through different settings.

Criterion Score Outputs is the one button that is not always there. It appears only for a plain one-facet analysis whose estimated universe-score mean is available. Subject-level analyses and one-facet difference scores do not show it, and neither does a view built from a DoD model that parameterizes a per-cell log mean instead of an observed-scale mean. When it is absent the row closes up and shows five buttons, which is the suppression working rather than a rendering fault. Section 9.7 covers the screen it opens.

Entering a reliability cutoff that is not numeric or not between 0 and 1 turns on a red line under the box reading "Reliability cutoff must be numeric and between 0 and 1 (inclusive)" and returns the cursor to the field. Nothing is generated until you fix it.

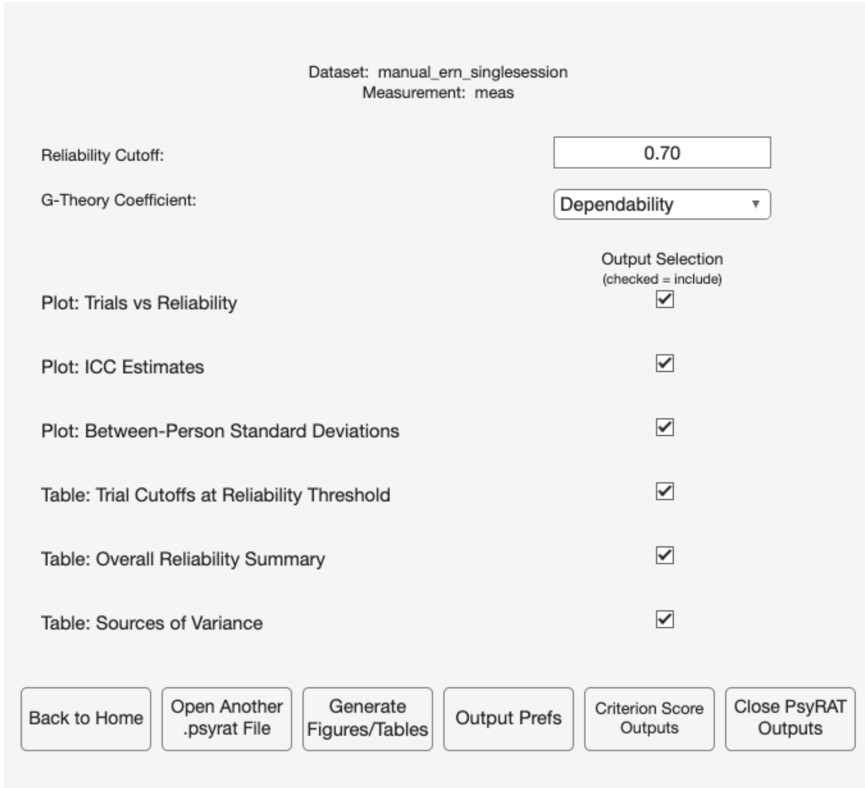


Figure 32: View Results Setup for a single-session analysis.

9.3 Test-retest results

The two-facet viewer carries the same title, View Results Setup, and the same Reliability Cutoff, G-Theory Coefficient, output checkboxes, and buttons. Read Section 9.2 first. What follows is what test-retest adds.

- Type of Reliability Coefficient: which coefficient to report. Equivalence is the internal-consistency coefficient, Stability is the test-retest coefficient, and Equivalence + Stability combines both. The default is Equivalence.
- Test-Retest Occasion Score: whether the reported coefficient describes a score from one occasion or a score averaged over several. The choices are Single-occasion (n'_o = 1) and Multi-occasion composite (n'_o = k).
- # Occasions in Composite (k): how many occasions the composite averages.

The three coefficients are not near-neighbors, and on real data they can differ by more than the gap between an acceptable result and an unusable one. Choose on the basis of the score you actually intend to use, then name the choice in your write-up. The exported provenance block records it, so a saved table is unambiguous even when the manuscript is not.

The occasion control needs the same care. Single-occasion is the toolbox's default estimand: the reliability of a score from one session, generalizing over the occasion facet. Multi-occasion composite reports the reliability of a score averaged across k occasions, which is a different quantity and almost always the larger of the two. Use it when your analysis really does average across sessions. Do not use it to make a single-session score look better than it is.

The k box is enabled only when Multi-occasion composite is selected, and greys out again the moment you switch back. It defaults to the number of occasions in your data. A blank, fractional, or negative entry falls back silently to that observed count rather than erroring, so check what you typed if a composite result looks unfamiliar.

Two-facet difference scores add a fourth control, Difference-Score Coefficient, choosing dependability or generalizability for the difference itself. It is independent of the per-condition G-Theory Coefficient above it.

The test-retest screen also adds two subject-level plot options and a subject-level table (Plot: Subject-Level Reliability, Plot: Subject-Level ICC, and Table: Subject-Level Reliability), inserted after Plot: ICC Estimates and before Plot: Between-Person Standard Deviations, for nine checkboxes in all.

Subject-level test-retest data gate five of those outputs. On such a run, Plot: Trials vs Reliability, Plot: ICC Estimates, Plot: Between-Person Standard Deviations, Table: Trial Cutoffs at Reliability Threshold, and Table: Overall Reliability Summary are unchecked and disabled, and hovering over any of them shows the reason: "Not available for subject-level test-retest data: this design estimates a per-participant residual, not the group-level reliability, ICC and variance terms this output needs." The subject-level outputs stay on. They are disabled rather than offered and then silently skipped, which is a distinction you notice the first time a checkbox produces no window.

Criterion Score Outputs appears on the test-retest screen only for the plain test-retest analysis. On subject-level test-retest data and on two-facet difference-score data the button is not offered, because the cut score computation needs group-level components those designs do not estimate; the row closes up to five buttons, the same behavior the single-session viewer shows for its subject-level and difference designs (Section 9.2).

9.4 Difference-of-differences results

DoD results open a screen titled DoD Reliability Setup, and it works differently from the two above. There are no output checkboxes. Each button produces one kind of output directly.

Under the dataset and measurement lines, the screen states the contrast form it will build: (ERP_1 - ERP_2) - (ERP_3 - ERP_4). Everything else on the screen fills in that expression.

- Group: which group the contrast is evaluated in. Shown as a popup when there is more than one group, and as the fixed text "Group: none" otherwise.
- Heatmap metric: Dependability or Generalizability, for the two heatmaps only.
- ERP_1 through ERP_4: four popups assigning one event to each slot of the contrast.
- Trials for ERP_1 through Trials for ERP_4: four boxes giving the trial count each slot is evaluated at.
- Heatmap trial grid min: fixed at 0 and shown for reference.
- Heatmap trial grid max: how far the two heatmap axes extend.

The four ERP popups default to the mapping you declared during processing when the file records one, and the four trial boxes default to the rounded mean recorded trial count for the selected group. Those trial defaults describe your data, so they are a reasonable starting point, but they are editable because the interesting question is usually what a different design would buy you.

The four slots must name four different events. Repeating one raises a dialog titled DoD mapping requires four unique events. Trial counts and the grid maximum must be whole numbers of at least

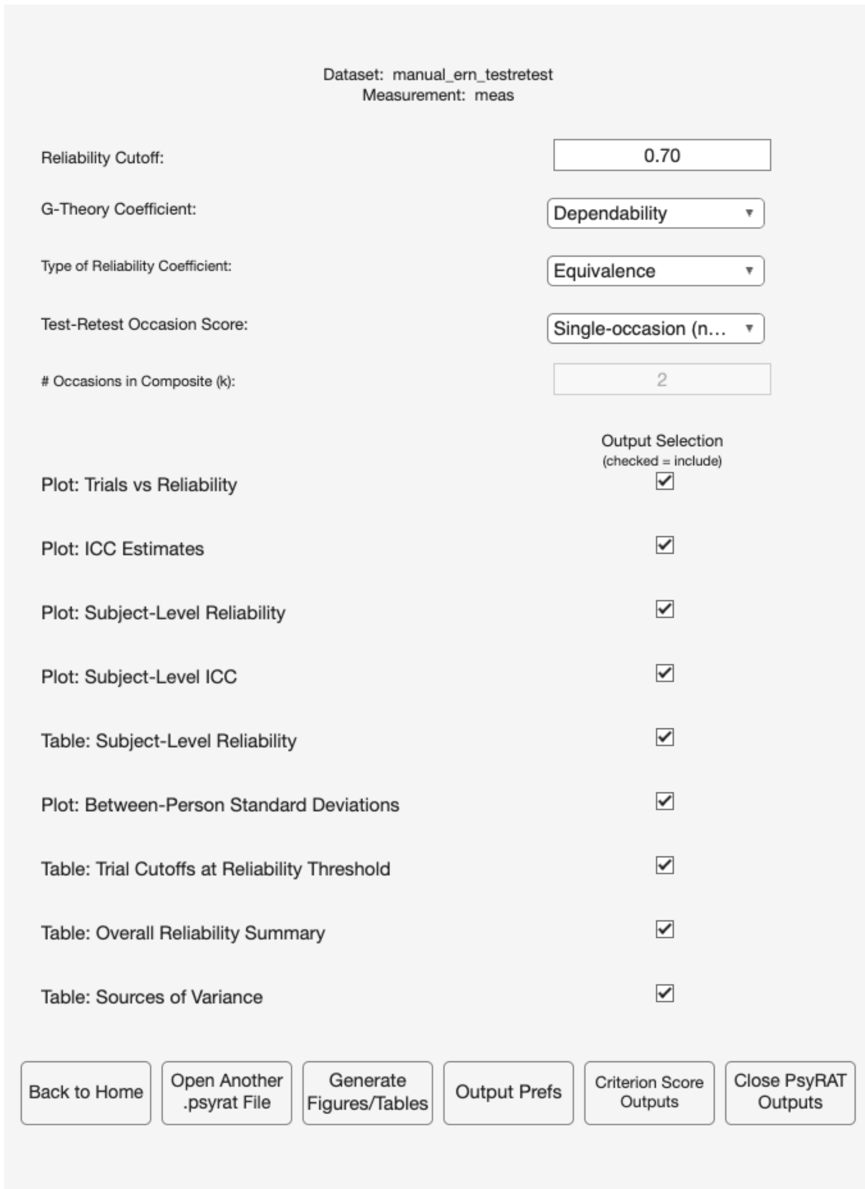


Figure 33: View Results Setup for a test-retest analysis.

0, and the grid maximum must not exceed 300, since the heatmaps evaluate the coefficient at every cell of a square grid and a larger one takes long enough to feel broken.

A line above the buttons states a difference you should carry into your reporting: heatmaps use posterior mean variance components for speed, and the summary table uses posterior draws. The table gives you a coefficient with a credible interval from the full posterior. The heatmaps give you a fast map of point estimates evaluated at a single plug-in value of each variance component, with no interval anywhere on them. Read a heatmap for shape and the table for numbers. Quoting a heatmap cell as a point estimate with the table's interval attached would be wrong.

Eight buttons run along the bottom:

- Back to Home
- Open Another .psyrat File
- Generate DoD Outputs
- Other Single-Session Metrics...
- Observed DoD Table
- Export DoD Var Components
- Refresh
- Close PsyRAT Outputs

Generate DoD Outputs opens one window titled DoD Reliability Outputs. It holds the group, the contrast written out with your event names, the trial counts you entered, the heatmap metric, a summary table of both coefficients with credible intervals and ICCs, and the two heatmaps. The ICC columns are computed at a single observation, which the window says on its face, so they do not move when you change the entered trial counts. If any entered count is 0, a red line notes that summary reliability is reported as 0 for that reason.

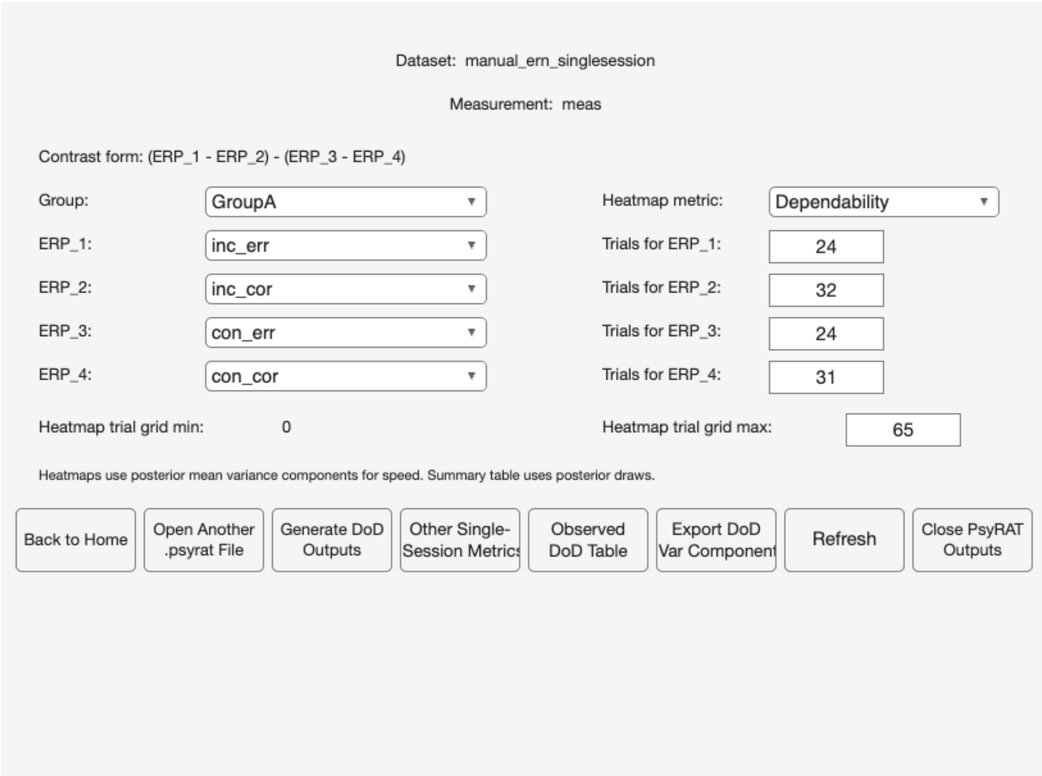
Observed DoD Table opens a window titled Observed Difference-of-Differences Reliability, holding one row evaluated at the mean recorded trial counts rather than at whatever you typed. It has its own Save Table button. Use it as the description of the design you ran, and the main outputs as the description of designs you might run.

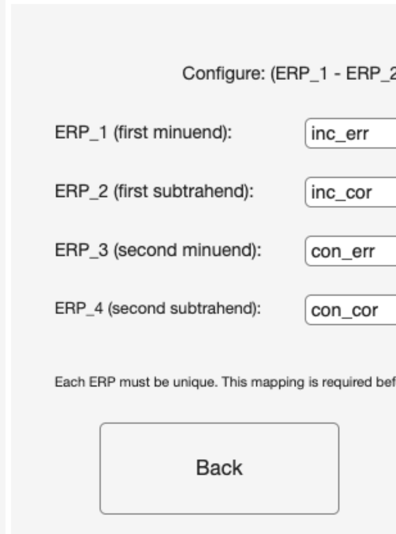
Export DoD Var Components goes straight to a save dialog. No window opens first. It writes the DoD contrast variance components with point estimates and credible intervals, then the two coefficients and their ICCs, at the observed mean trial counts.

Refresh rebuilds the screen from the current selections and resets the four trial counts to the selected group's defaults, which is how you get the trial defaults to follow a group change. The ERP mapping, the heatmap metric, and the grid maximum are kept as selected.

Other Single-Session Metrics... opens a small window titled DoD: Other Single-Session Metrics. It routes one group's DoD model into the standard single-session outputs. Mode offers Single-event reliability or 2-event difference reliability; Event 1 is always shown, and Event 2 appears only in difference mode. Open Standard Output GUI builds the requested view and hands it to the single-session viewer described in Section 9.2. Two different events are required in difference mode.

If any quantity that has to be a variance came out negative, a dialog titled Coefficient is not interpretable as a reliability appears after the output is drawn. Nothing was clipped or adjusted to produce the numbers on screen. The dialog explains them. A coefficient reported under that warning is not a reliability and should not be reported as one.





9.5 Dynamic reliability results

Dynamic reliability results open a screen titled Dynamic Reliability: View Results. Under the dataset line it lists the standardized dimensions the surface is drawn over. Like the DoD screen, it has no output checkboxes.

- Coefficient: Generalizability G(z) or Dependability D(z). The default is Generalizability G(z).
- Trials n' (blank = per-group median): the trial count the surface is evaluated at. Left blank, each stratum uses its own median.
- Credible interval (0-1): the interval width, defaulting to 0.95.
- Reference cutoff (blank = none): a reliability value to mark on the figure. Left blank, nothing is drawn.
- Estimand: Typical person (delta_p = 0) or Population average (lognormal).

I would keep the blank defaults on a first pass. A per-stratum median describes each stratum's own data instead of imposing one count on all of them, and the export header labels it as the default.

The Estimand control is enabled only for the non-difference variants. On every difference variant it is greyed out, because the reported surface is the typical-person reference and no population-average difference estimand is computed. That disabled control is the toolbox declining to imply a quantity it does not have.

Two-facet dynamic designs add two controls:

- Reliability coefficient: Equivalence (occasion fixed), Stability (trial fixed), or Equivalence + stability (both random). The default is the third.
- Occasions n': 1 occasion or Observed occasions. The default is 1 occasion.

Buttons come in pairs. Plot reliability surface draws the figure. Show variance components opens a table of per-stratum components with credible bounds. Save variance table (CSV) exports it. Done closes the viewer. Back to Home and Open Another .psyrat File sit at the bottom. Subject-level variants add Show per-participant table and Save per-participant table (CSV), giving each participant's own coefficient at their own dimension value.

The save buttons say CSV, and both offer Excel file (*.xlsx) first with CSV second. The label is narrower than the behavior. Either format writes the same header, naming the credible interval, the estimand, and the n' each stratum's surface was evaluated at, with the n' line saying whether that came from your override or from the per-stratum median default. That last line matters more than it looks: without it, an override is invisible in the file.

Some designs disclose an extra line on screen, under the variance-components table, and it belongs in any write-up of those results. A modular cut-posterior run states that the copula residual correlation is estimated conditionally on the retained margin draws with no copula feedback, so its intervals are not full joint-posterior credible intervals. A nonconcurrent difference-of-differences run states that the four constituents are treated as if measured on distinct trials, with all six conditional residual covariances fixed to zero by assumption, and that the result is neither a concurrent reliability nor a cut posterior. Both notes point at the exported header for the full provenance. Neither appears for designs it does not apply to.

The surface figure carries its own Save figure (PNG/PDF) button in the bottom left corner. The button hides itself while the image is written, so it does not appear in the saved file.

Figure 34: Dynamic Reliability: View Results.

9.6 Data-splits results

Nonparallel data-splits results open a screen titled Data-Splits Reliability: View Results. Under the dataset line sit two banners. The first names the analysis unit and states that each score is the mean of n_i items. The second is the caveat this design turns on:

> Dependability is exact; generalizability is a downward-biased lower bound.

That sentence is the toolbox's documented convention, and its reason is worth understanding rather than memorizing. When the input is split means rather than single trials, the only thing the model observes is the split-level score. The item-level partition inside each split is gone, so the item main effect cannot be separated from the residual and collapses into the split-level error term. Absolute error already includes that item variance, so dependability is unaffected and exact. Relative error is supposed to exclude it and here cannot, so the relative error variance is too large and the generalizability coefficient it produces is too small. The reported value is therefore a lower bound on the true generalizability, and the relative SEM correspondingly an upper bound. The generalizability-theory framework this rests on is developed in Rocha et al. (2026); the bound statement is the toolbox's convention for reporting what a split-mean design can and cannot identify.

The practical rule: quote dependability as an estimate and generalizability as a bound, in those words. The column headers help, since the coefficients table names its columns Dependability and Generalizability (lower bound), with SEM (abs) and SEM (rel, upper bound) beside them.

Three settings:

- Credible interval (0-1): the interval width, defaulting to 0.95.
- Splits n' (blank = observed median): the number of splits per person the coefficients are reported at. Blank uses each stratum's observed median.
- Occasions n' (blank = observed): the number of occasions, shown only for the multi-occasion design.

Five buttons: Show coefficients, Show variance components, Save coefficients (CSV), Save variance table (CSV), and Done. As on the dynamic screen, the save dialogs offer Excel file (*.xlsx) first and CSV second despite the labels.

There is no reliability-versus-trials plot here and no trial-cutoff table, because there is no D-study projection to draw. Unequal items per split make the splits nonparallel, and a projection over hypothetical item counts would require the item-level partition that split means discarded. The viewer reports the observed design only: the coefficients, their SEMs, and the variance components, at the split and occasion counts your data have. A projection over trial counts needs single-trial input, which is the analysis Chapter 10 runs.

The coefficients window repeats the bias note under the table, and every export writes it into the header, so the caveat travels with the numbers.

This screen has no Back to Home or Open Another .psyrat File button. Done closes it and returns you to the MATLAB prompt; run `psyrat_start` for the home screen.

9.7 Criterion Score Outputs

Everything above evaluates reliability for relative decisions or for absolute ones against the whole scale. Criterion Score Outputs evaluates it against a specific score. If your question is whether a participant falls above or below some value, the relevant quantity is reliability at that value, and it is not the same number as the overall coefficient.

The Criterion Score Outputs button on the single-session or test-retest screen opens Criterion Score Outputs Setup. Its top rows show the dataset and the measurement.

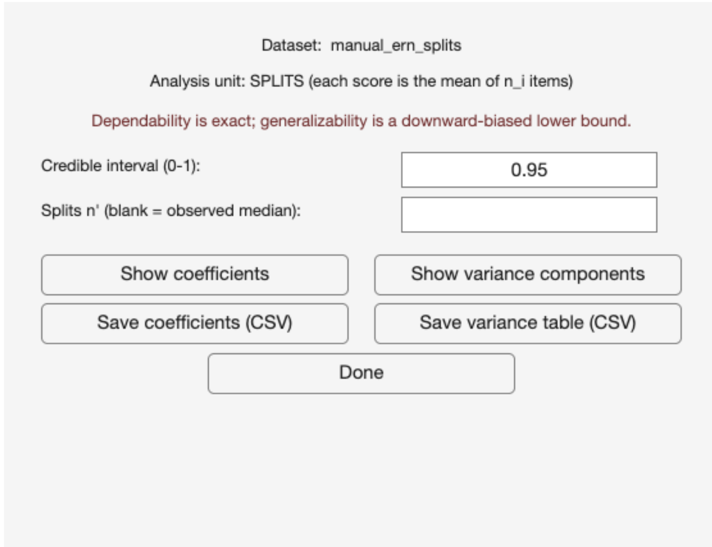


Figure 35: Data-Splits Reliability: View Results.

- Criterion Cutoff: the score threshold to evaluate. This is a value on the measurement scale, in microvolts or whatever your measurement column holds, and it is not a reliability threshold.
- Decision Coefficient: Dependability (Absolute). Fixed text, not a control.
- Average Score (mu): the mean score across the loaded data, shown to four decimal places. Also fixed text.
- Type of Reliability Coefficient: Equivalence, Stability, or Equivalence + Stability. Shown only when the results are test-retest.

Then two checkboxes under the Output Selection heading, both on by default:

- Plot: Trials vs Reliability at Criterion
- Table: Overall Reliability Summary at Criterion

Then three buttons: Back to Results Setup, Generate Criterion Outputs, and Close PsyRAT Outputs.

The Criterion Cutoff box opens at 0, not at the mean; the Average Score (mu) shown just above it is display text, and it marks the worst case. The criterion coefficient adds the squared distance between the mean and the cut score to both the numerator and the denominator, so a criterion sitting exactly at the mean adds nothing and reduces to the ordinary dependability, and every criterion further out returns a higher value. Enter the value your decision actually uses, and do not read either the 0 default or the displayed mean as a recommendation. A non-numeric entry turns on a red line reading "Criterion cutoff must be numeric" and nothing is generated.

Dependability is not a choice here, and the fixed wording is not an oversight. A cut score is an absolute decision by construction, so the relative coefficient is not a valid quantity for it, and asking for one raises an error further down. On test-retest data the criterion outputs are computed at one occasion, and the windows say so on their face. The route is offered only for the designs that estimate the components it needs: the plain one-facet analysis and the plain test-retest analysis. The subject-level and difference-score designs do not show the button (Sections 9.2 and 9.3).

Criterion Score Outputs is available only through the GUI. There is no scripted equivalent:

`psyrat_run` and the headless report writer described in Chapter 14 produce no criterion outputs, and no argument turns them on. If you need criterion results in a reproducible pipeline, the honest workflow is to generate them here, save the table, and record the criterion cutoff you used. The exported header writes that cutoff to four decimal places for exactly that reason.

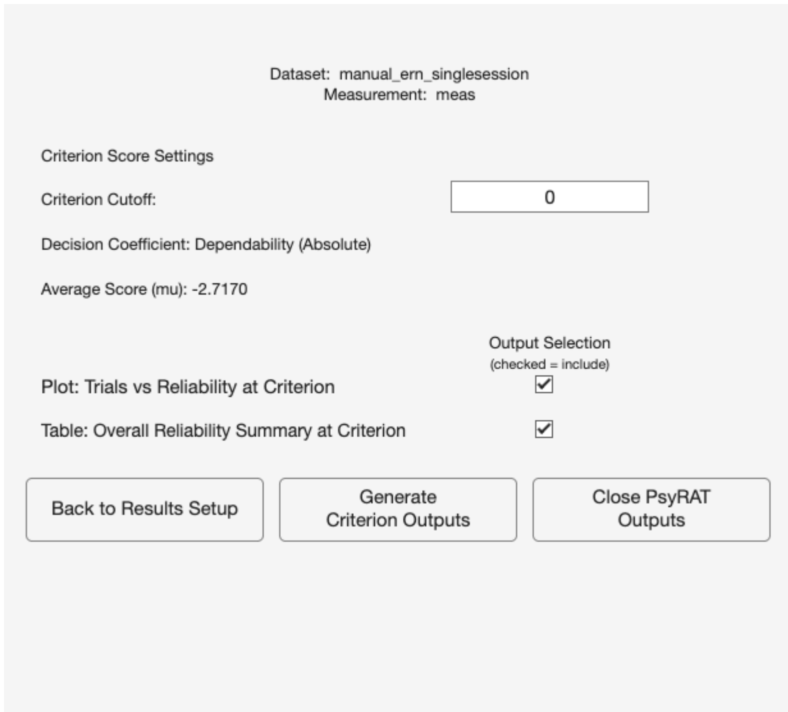


Figure 36: Criterion Score Outputs Setup.

9.8 A gallery of output figures

Each figure below is described the same way: what the axes are, then what to look for.

Reliability versus trials

The x-axis is the number of observations, running from 0 to the plotting range you set in View Preferences. The y-axis is the coefficient you selected, dependability or generalizability, on the full 0 to 1 scale. One curve is drawn per group, labeled by a legend when there is more than one, and one panel per event. The dotted blue horizontal line is your Reliability Cutoff. The curve itself is whichever posterior summary you chose, so a plot of the lower limit sits below a plot of the point estimate throughout.

Read where the curve crosses the dotted line, and how steeply. The crossing is the trial count the trial-cutoff table reports, but only when Line to plot for reliability and Estimate to use for trial cutoffs are set to the same summary. They are separate preferences, and with the line on the point estimate and the cutoff on the lower limit the table will report a larger trial count than the crossing you see. A curve that crosses early and then flattens says the design has headroom, and collecting more trials will not move reliability much. A curve still climbing steeply at the right edge says the opposite, and it also says your plotting range is too short to answer the question. A curve that

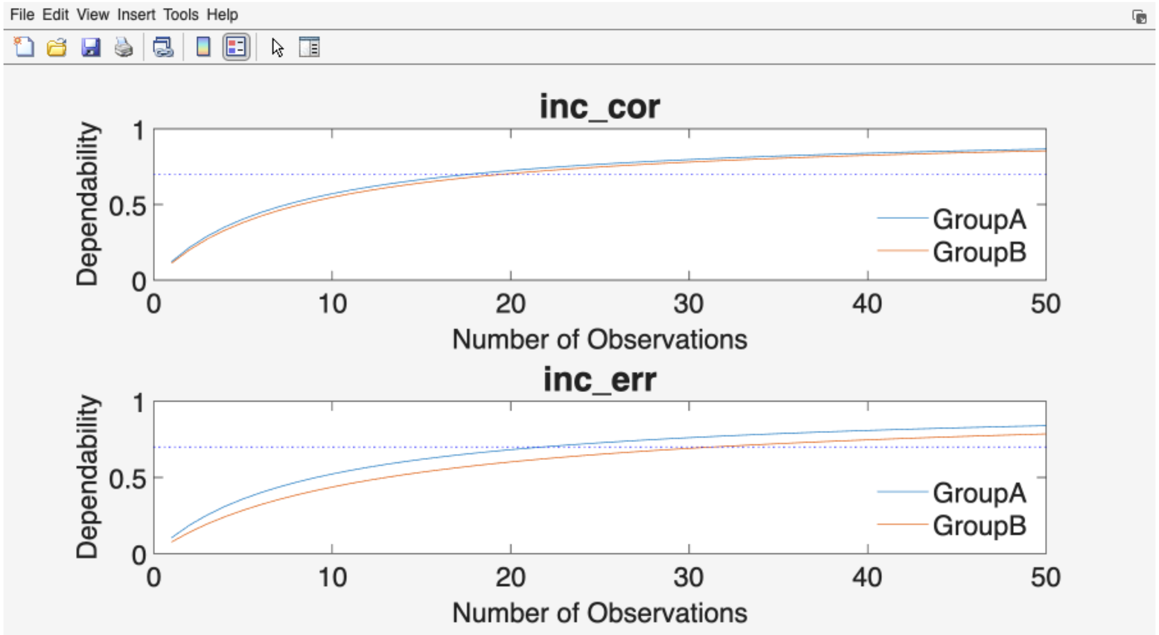


Figure 37: Dependability as a function of the number of trials retained, one panel per event and one curve per group, with the Reliability Cutoff drawn as a dotted horizontal line.

never reaches the line is the -1 case from Section 9.1, and it will be accompanied by the Cutoff not calculable dialog.

ICC estimates

Events are listed down the left, one row per event with one series per group, and the intraclass correlation coefficient runs along the bottom. Each point is a posterior point estimate and the bar through it is its credible interval.

Read the intervals before the points. A tight interval well away from zero says the between-person variance is estimated precisely and the measure separates people. A wide interval means the data did not pin the ICC down, and the point estimate on its own would mislead. Comparing groups, look at whether the intervals overlap rather than at whether the points differ. The figure is not a test, and a gap between two point estimates is not a group difference.

Between-person standard deviations

Same layout as the ICC figure. Events are listed down the left and the between-person standard deviation runs along the bottom, in the units of your measurement.

Reading this alongside the ICC figure usually explains a disappointing coefficient. Reliability can be low because within-person error is large or because between-person differences are small, and those call for different responses. A small between-person standard deviation with a tight interval says the sample is homogeneous on this measure, and no number of extra trials will fix that. The within-person standard deviations that complete the picture appear in the Sources of Variance table rather than as a plot.

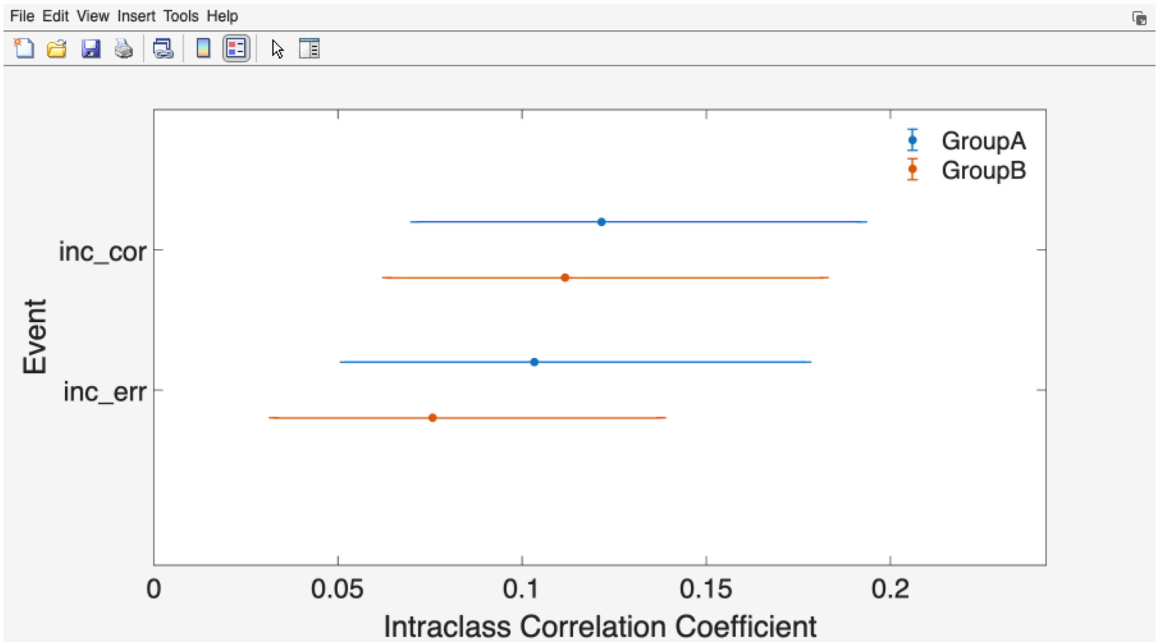


Figure 38: Intraclass correlation coefficients with credible intervals.

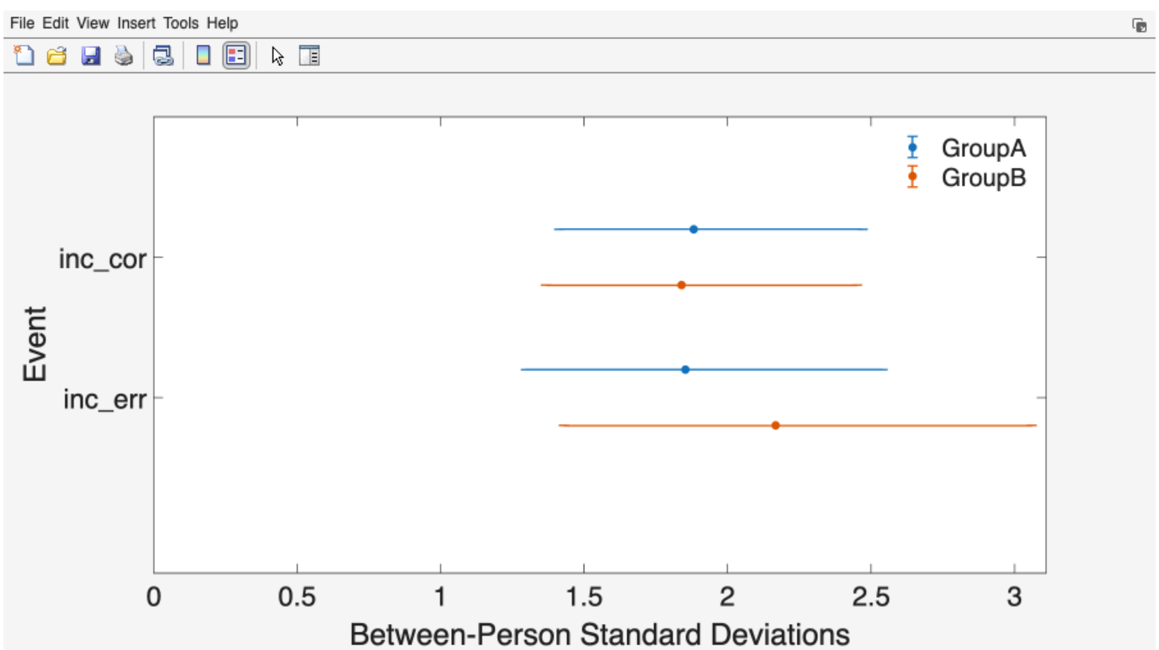


Figure 39: Between-person standard deviations with credible intervals.

Subject-level reliability

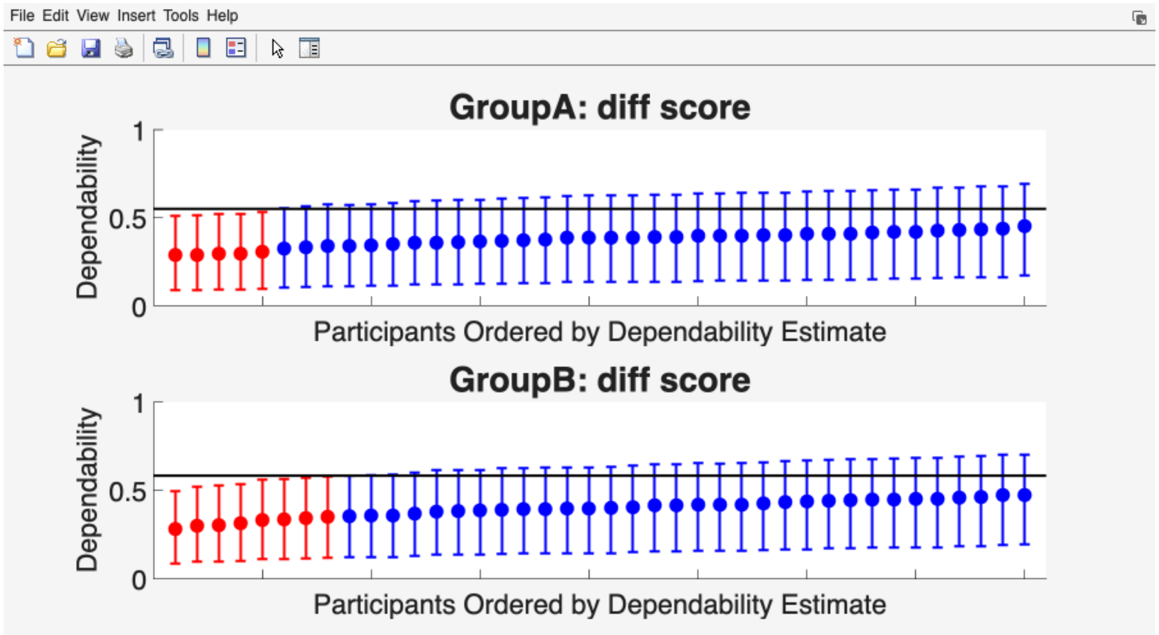


Figure 40: Subject-level reliability estimates with credible intervals.

One panel per stratum (per group, per event, or both). The x-axis is participants, ordered by their own estimate, with the tick labels suppressed because the ordering carries the information. The y-axis is the subject-level coefficient on the 0 to 1 scale. Each participant is one point with a credible interval. The solid black horizontal line is the group-level estimate, and participants whose interval excludes that line are drawn in red.

The color coding is the thing to read. When any participant's interval fails to cover the group estimate, the plot splits: those participants are drawn in red and the rest in blue. When every interval covers the group estimate, no split occurs and the points are drawn in the default single color. So red marks participants whose reliability is credibly different from the group's, and a figure with no red at all is a figure saying the group estimate describes everyone reasonably well.

Beyond the colors, look at the spread. A near-flat band of points across the line is the picture of a homogeneous sample. A long tail of red at the low end identifies specific participants whose scores carry more error than the group number implies, which is a screening result, not a diagnosis. The companion subject-level ICC plot has the same layout with ICC on the y-axis.

Dynamic reliability surface

With one dimension, the x-axis is that standardized dimension and the y-axis is the coefficient from 0 to 1. A solid line is the point estimate, the shaded band around it is the credible interval, and a dotted blue horizontal line marks the Reference cutoff if you set one. With two dimensions the figure becomes a filled contour over both standardized predictors, with the coefficient on the color bar and the reference cutoff drawn as a white contour. One panel per stratum, titled with the stratum label and the n' the surface was evaluated at.

Read the slope first. A surface that is flat across the dimension says reliability does not depend

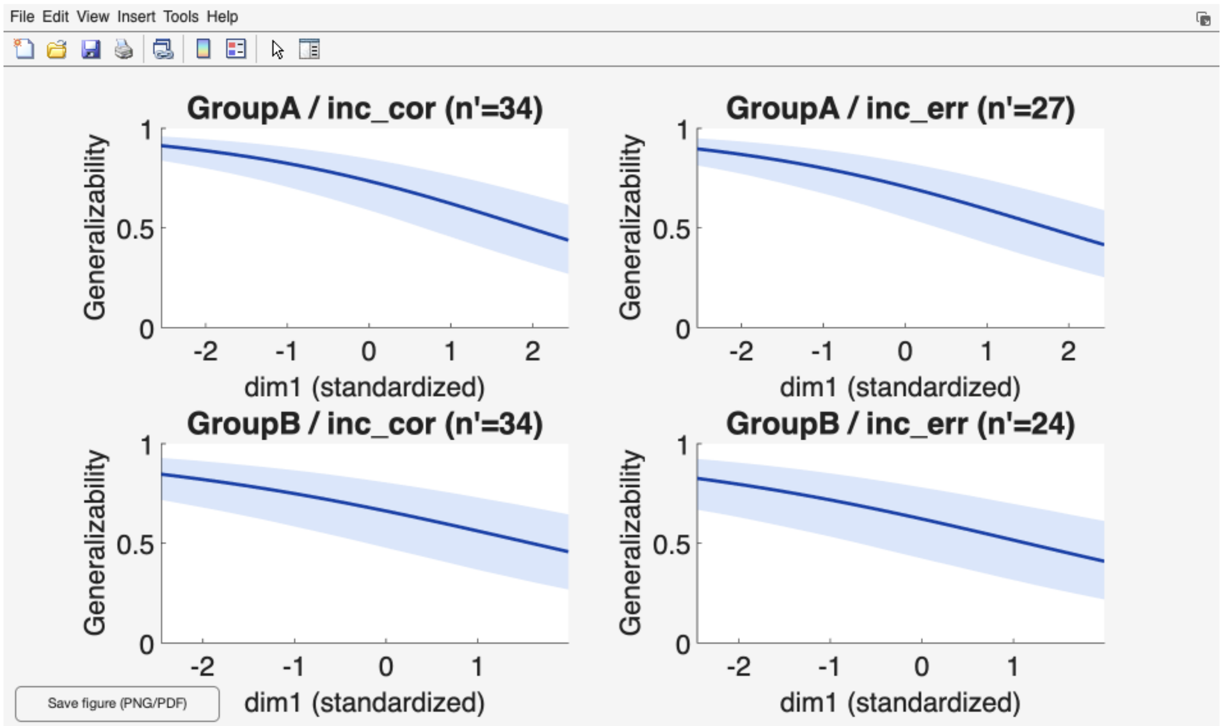


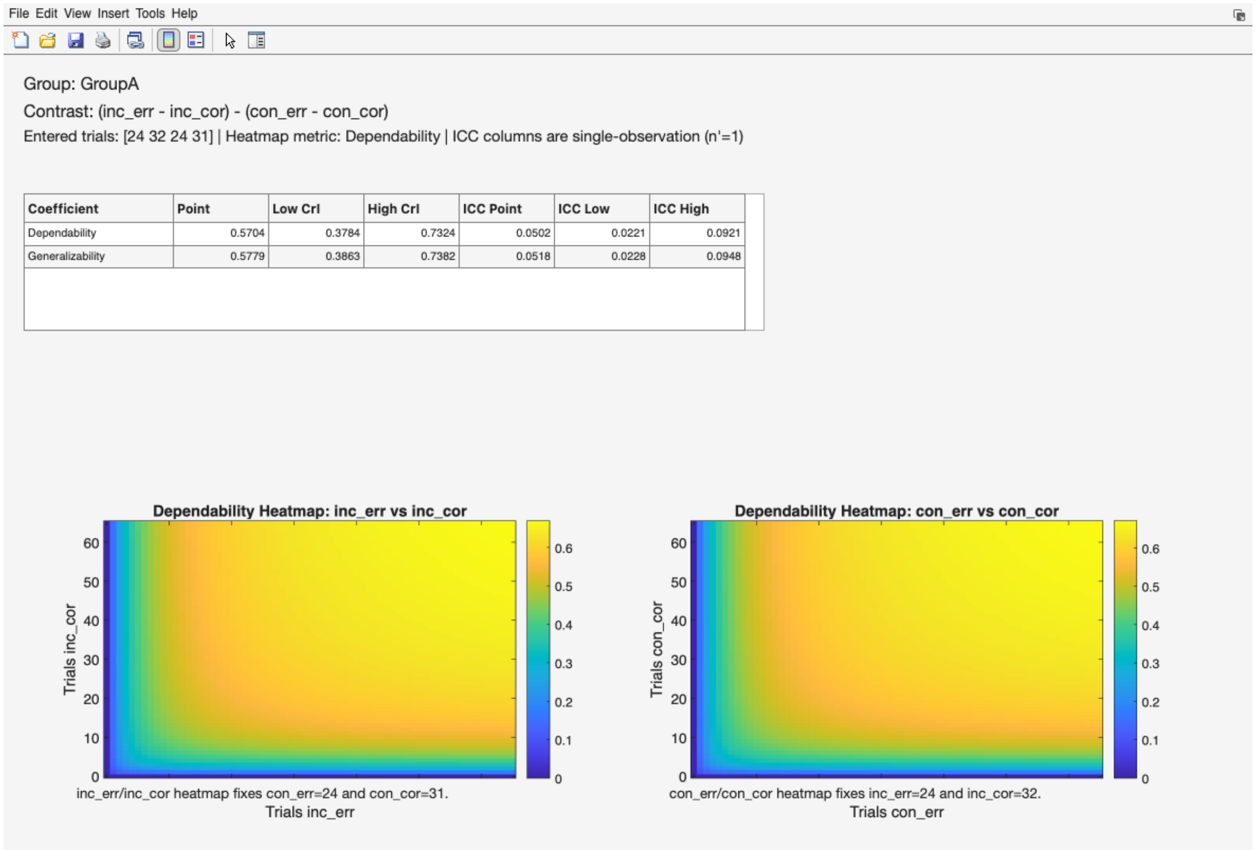
Figure 41: Dynamic reliability as a function of the standardized dimension.

on it, which is the assumption every ordinary coefficient makes. A surface that falls off at one end says the measure works better for some people, or in some conditions, than in others, and a single pooled coefficient is averaging over a real difference. Then read the band. A surface that slopes within a wide band has not established that dependence. Where a reference cutoff is drawn, the region of the dimension on the wrong side of it is the part of your sample the measure does not serve, and that is a sample-composition finding worth reporting.

Difference-of-differences heatmaps

Two heatmaps sit side by side under the summary table. On the left, the axes are the trial counts of ERP_1 and ERP_2, with the trial counts of ERP_3 and ERP_4 held fixed at the values you entered. On the right, the reverse. Both run from 0 to your Heatmap trial grid max. Color is the selected metric, dependability or generalizability, and each panel carries its own color bar, autoscaled to that panel's range, so read values off the bars rather than comparing shades across the two panels. A caption under each panel names the counts held fixed.

Read the gradient direction. Where the color changes fastest is where extra trials buy the most, and that is rarely symmetric: a rare condition usually dominates one axis. A plateau in the upper right corner tells you the design saturates, so the additional trials you were considering would be spent for almost nothing. Use these to choose a design, then take the coefficient and interval from the summary table above them, since the heatmaps carry point estimates from posterior means and no intervals at all.



Coefficient	Point	Low CrI	High CrI	ICC Point	ICC Low	ICC High
Dependability	0.5704	0.3784	0.7324	0.0502	0.0221	0.0921
Generalizability	0.5779	0.3863	0.7382	0.0518	0.0228	0.0948

Figure 42: Dependability heatmaps for the two halves of the DoD contrast.

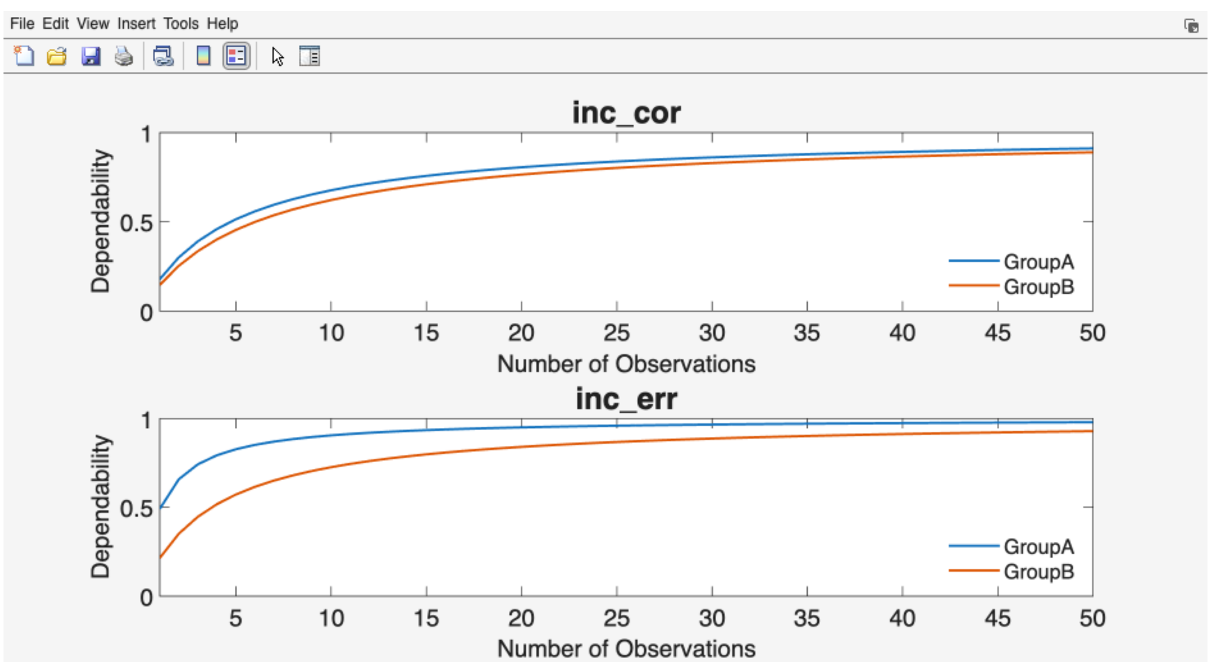


Figure 43: Dependability at the criterion cutoff as a function of trials.

Reliability at a criterion score

The x-axis is the number of observations and the y-axis is dependability at your criterion cutoff, from 0 to 1. One curve per group, one panel per event. The window title records the criterion cutoff to three decimal places, which is worth keeping in the screenshot when you circulate one.

Compare this curve against the corresponding reliability-versus-trials figure rather than reading it alone. The criterion curve can never fall below the ordinary dependability curve; it meets it when the cut score equals the mean score and rises above it as the cut score moves away. So a criterion curve sitting barely above the overall curve is telling you that your cut score falls near the middle of the distribution, which is the hardest place to classify people reliably. That is a finding about your decision rule rather than about your data quality, and moving the cut score is a legitimate response to it.

Trace plots

Trace plots belong to the processing stage rather than to any viewer. They appear at the end of a one-facet run when the trace-plot preference is on, and Chapter 7 covers the prompt that offers them. They are included here because they are the figure you will want when a viewer greets you with the Estimation diagnostics notice.

The x-axis is the number of samples and the y-axis is the sampled value of one parameter, with one panel per parameter (mu, the between-person variance, the within-person variance, and so on) and all chains overplotted.

What you want is dense, stationary, thoroughly mixed noise with the chains indistinguishable from one another. That is, these plots should look like a ”fat, hairy caterpillar” (Lunn et al., 2012). What you do not want is a chain drifting across the panel, a chain sitting in a band the others never visit, or a slow wandering that never settles, all of which mean the sampler has not converged and the reliability estimates built from those draws should not be reported. More warmup and sampling iterations are the first response. Chapter 18 covers the rest.

9.9 Where to go next

Reading these outputs scientifically, and reporting them in a manuscript, is Chapter 17. Producing the same tables without the GUI is Chapter 14. The four tutorials in Chapters 10 to 13 walk one complete pass through the viewers for each of the main designs, with the numbers you should expect to see.

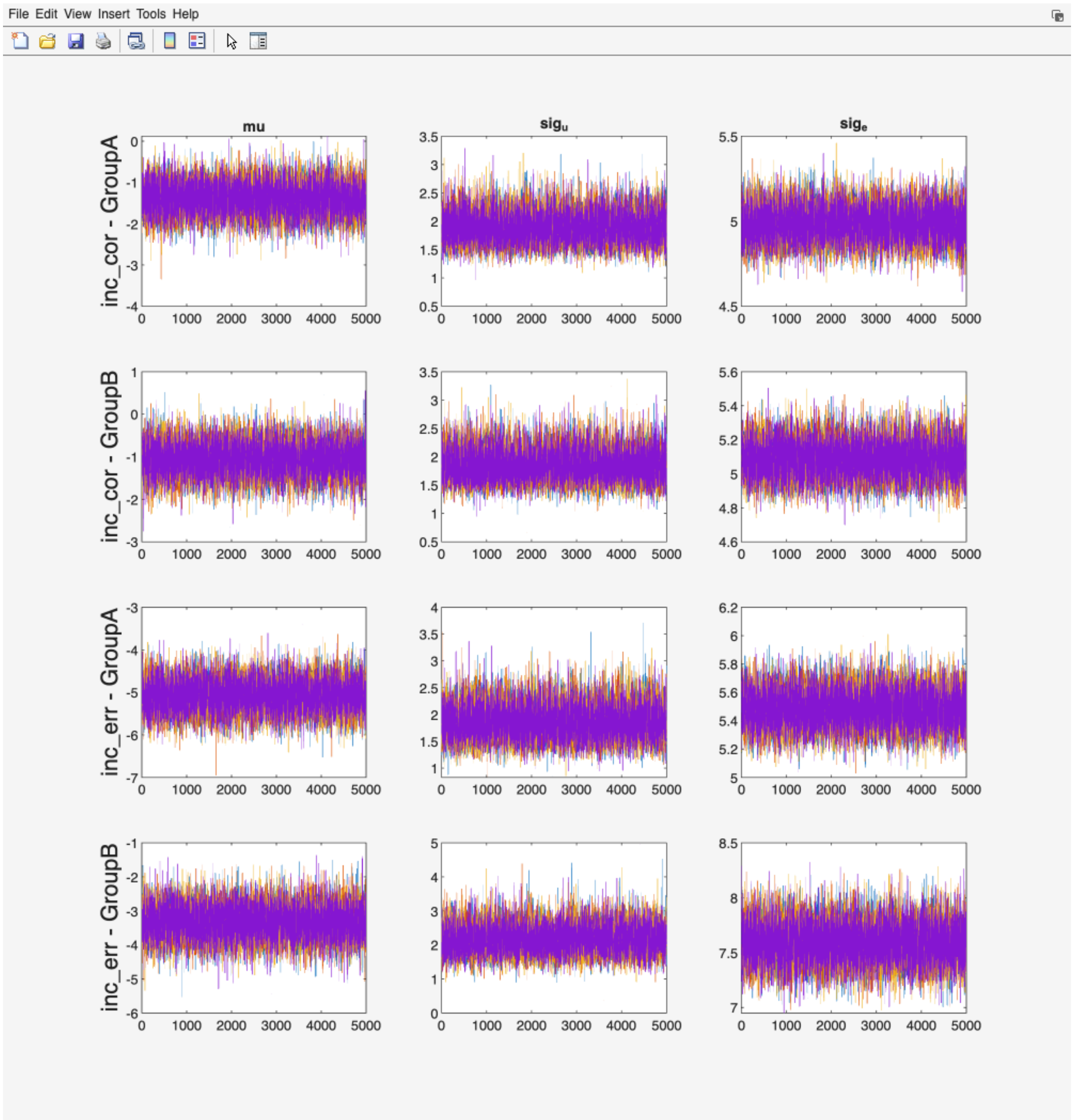


Figure 44: Trace plots of Stan parameters.

10. Tutorial 1: Single-session ERN reliability

A flanker task was run once, in two independent samples, and error-related negativity (ERN) was scored as a mean amplitude on every trial. Before anyone correlates ERN amplitude with a symptom measure, one question has to be answered: how much of the between-person variance in those averaged scores is person variance rather than trial-to-trial noise, and how many error trials does a participant need before the average is stable enough to use? This tutorial answers both for the file `test_data/manual_ern_singlesession.csv`. It reports the two groups separately and reads them against each other, because they differ in a way that shows up directly in the dependability estimates.

Work through this tutorial first. Tutorials 2, 3, and 4 assume you have done it and describe only what changes.

10.1 The dataset

`manual_ern_singlesession.csv` is simulated. No participants were recorded, and every generating parameter is documented in the generator that produced the file (`tests/helpers/psyrat_make_manual_singlesession.m`). That is the point: because the true variance components are known, the tutorial can print them next to what the toolbox recovers.

The file is long format, one row per single-trial score, with four columns: `id`, `group`, `event`, `meas`. There are 80 participants (ids 101 to 180) split evenly between two between-person groups, GroupA and GroupB. Four events cross flanker congruency with response accuracy: `inc_err`, `inc_cor`, `con_err`, `con_cor`. Trial counts differ across participants on purpose, as they do in real ERP files. Error trials are scarce (drawn between 12 and 36 per participant per cell) and correct trials are plentiful (24 to 40).

The tutorial processes two of the four events, `inc_err` and `inc_cor`. The reading of interest is incongruent-error dependability per group; incongruent correct trials come along because a second event makes the plots and tables worth looking at, and because Tutorial 3 builds a difference score from exactly this pair.

10.2 What the true dependability is

The generating model has one facet. Each single-trial score is a grand mean plus a person effect plus a trial main effect plus independent trial noise. Under that model the dependability of a mean of n trials is given by the one-facet D-coefficient in Rocha et al. (2026, Table 2), which for these data reduces to

```
D(n) = sigma_p^2 / (sigma_p^2 + (sigma_i^2 + sigma_e^2) / n)
```

In words: dependability is the share of the total variance in an n-trial average that is person variance, once the trial-level variance has been shrunk by averaging. Averaging divides the trial main effect and the noise by n but leaves the person variance untouched, so dependability rises with n and approaches 1 but never reaches it. The trial main effect (sigma_i) belongs to absolute error only: generalizability, the relative-error sibling, drops the sigma_i^2 term, which is why the two coefficients print different values for this file. Chapter 4 develops both in full.

The generator's values are these. Person SDs are 2.0 microvolts for GroupA error trials, 2.2 for GroupB error trials, and 1.7 for correct trials in both groups. Trial-noise SDs are 5.5 for GroupA error trials, 7.5 for GroupB error trials, and 5.0 for correct trials in both groups; the trial main effect is 1.0 in every cell. GroupB's error trials are the noisier ones, which is the whole comparison.

Take GroupA incongruent errors at 15 trials. The person variance is 2.0^2 = 4.00 and the trial-level variance is 1.0^2 + 5.5^2 = 31.25, so the absolute error variance of a 15-trial average is 31.25 / 15 = 2.083, and

```
D(15) = 4.00 / (4.00 + 2.083) = 4.00 / 6.083 = .658
```

Now GroupB, same event, same 15 trials. Person variance 2.2^2 = 4.84, trial-level variance 1.00 + 56.25 = 57.25, error variance 57.25 / 15 = 3.817, and

```
D(15) = 4.84 / (4.84 + 3.817) = 4.84 / 8.657 = .559
```

GroupB has slightly MORE person variance and still lands a full tenth lower, because its trial noise is so much larger. That is the comparative lesson of this tutorial, and it is worth stating plainly: a group can look less reliable without being less variable between people.

Repeating that arithmetic across trial counts gives the truth table.

Table 10.1. True dependability of the tutorial's cells, computed from the generator's parameters. The last two columns invert the formula: n = (c / (1 - c)) x ((sigma_i^2 + sigma_e^2) / sigma_p^2), rounded up to a whole trial. sigma_i is 1.0 in every cell.

Cell	sigma_p	sigma_e	D(1)	D(15)	D(20)	D(32)	D(50)	n for .70	n for .80
GroupA inc_err	2.0	5.5	.113	.658	.719	.804	.865	19	32
GroupB inc_err	2.2	7.5	.078	.559	.628	.730	.809	28	48
GroupA inc_cor	1.7	5.0	.100	.625	.690	.781	.848	21	36
GroupB inc_cor	1.7	5.0	.100	.625	.690	.781	.848	21	36

D(1) is the same quantity at a single trial, which is what the toolbox reports as the absolute ICC. It is low, and it should be: a single ERN trial is mostly noise (that is why anyone averages).

Two things to carry forward. GroupA needs 19 incongruent-error trials to reach .70 and 32 to reach .80. GroupB needs 28 and 48. The observed error-trial counts in this file reach 36 in both groups, so a .70 threshold is inside what these data supply; a .80 threshold on error trials is beyond the data in GroupB and at the very top of the range in GroupA, and the toolbox will say so.

10.3 About the numbers printed in this tutorial

Every value shown in a checkpoint or an output table below was produced from the committed CSV at the settings each tutorial states: seed 12345, 4 chains, 5,000 warmup iterations, and 5,000 sampling iterations, on a single machine (Tutorial 2 raises the iteration counts, and says so). Run the same file at the same settings on the same CmdStan version and you will reproduce them exactly, because the estimation is seeded and deterministic.

Across CmdStan versions, compilers, and platforms, expect agreement to about .02 on the reliability point estimates and a little more movement on the credible interval bounds. Larger discrepancies are worth investigating rather than shrugging at; Section 18.3 of Chapter 18 lists the usual causes.

One caveat that matters more than it looks. If the chains do not converge and you accept the rerun offer, the toolbox DOUBLES warmup and sampling and refits. Your run then no longer matches the settings above, and your numbers will not match the printed ones. That is the correct outcome (a converged fit at more iterations beats a matching number from a fit that failed), but note the change before you compare.

I state this policy once, here. Tutorials 2 through 4 inherit it.

10.4 Running the analysis

Step 1. Start the toolbox

At the MATLAB prompt, type `psyrat_start`. The Command Window prints a dependency check and the version banner, then the home screen opens with three buttons.

```
Ensuring dependents are found in the Matlab path
PsyRAT Toolbox files found
CmdStan, MatlabProcessManager, and MatlabStan files found
CmdStan version is current
MatlabProcessManager version is current
MatlabStan version is current



PsyRAT Version 0.1.0-beta
```

Figure 45: The startup banner. PsyRAT checks for CmdStan, MatlabProcessManager, and MatlabStan before the home screen appears.

Checkpoint: the Command Window should report that the PsyRAT Toolbox files were found and that the CmdStan, MatlabProcessManager, and MatlabStan files were found. If it instead prints a list of missing dependents, stop here and work through Chapter 3. Nothing in this tutorial will run without CmdStan.

Step 2. Load the data

Click the Process New Data button. A file dialog opens. Navigate to `test_data/` and select `manual_ern_singlesession.csv`.

The Command Window prints ”Loading Data...” and then the Specify Inputs screen opens with the file name across the top.

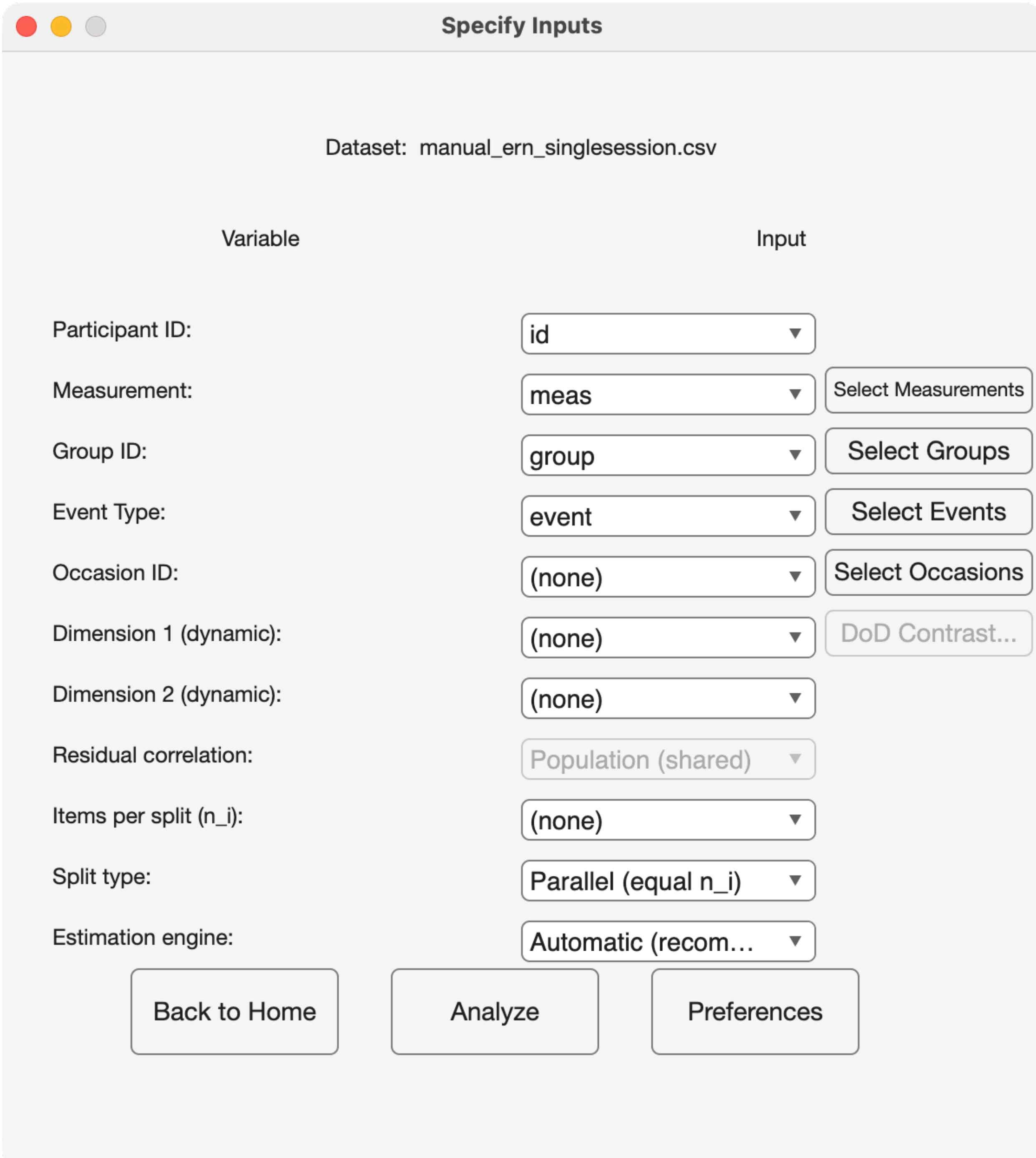


Figure 46: The Specify Inputs screen after loading manual_ern_singlesession.csv. The Variable column names each PsyRAT input; the Input column holds the column chosen for it.

Step 3. Check the auto-detected columns

Specify Inputs guesses the column mapping from the header names. This file uses the toolbox's own canonical headers, so the guesses are all correct and there is nothing to change. Confirm them anyway, because a wrong mapping here produces a plausible number rather than an error.

Checkpoint: the Specify Inputs screen should show these selections.

Control	Value
Participant ID:	`id`
Measurement:	`meas`
Group ID:	`group`
Event Type:	`event`
Occasion ID:	`(none)`
Items per split (n_i):	`(none)`
Estimation engine:	Automatic (recommended)

The `(none)` entry is a sentinel, not a column. Selecting it tells PsyRAT that the facet is absent from this design, and it is spelled with parentheses so that a real data column can never be confused with it. This dataset has one session, so Occasion ID stays on `(none)`; that is what makes this a single-session analysis rather than the test-retest analysis in Chapter 11.

Four more rows sit on this screen and are not in the table above: Dimension 1 (dynamic), Dimension 2 (dynamic), Residual correlation, and Split type. The first three belong to the dynamic-reliability analyses in Chapter 15 and are inert unless a Dimension column is selected; Split type matters only when Items per split (n_i) names a column (Chapter 13). Leave all four alone.

Step 4. Select the two events

Click the Select Events button. The Specify Which Events to Process screen opens. All four event labels are listed in alphabetical order (`con_cor`, `con_err`, `inc_cor`, `inc_err`), and the first time you open the screen all four are selected.

Select `inc_cor` and `inc_err`, and nothing else. The list takes multiple selections, so hold your platform's multi-select modifier key while clicking the second one, or the first will deselect. Click Save.

Specify Inputs returns. Hover over the Select Events button and its tooltip now ends with "(Selected: 2/4)".

Specify the Event Type column BEFORE clicking Select Events. Otherwise you will get an error if `(none)` is selected, or a long list of measurement values if you pointed Event Type at a numeric column. The same applies to Select Groups and Select Occasions.

Leave Select Groups alone. Both groups are wanted, and the default is all of them.

Step 5. Check the processing preferences

Click the Preferences button. The Specify Processing Preferences screen opens.

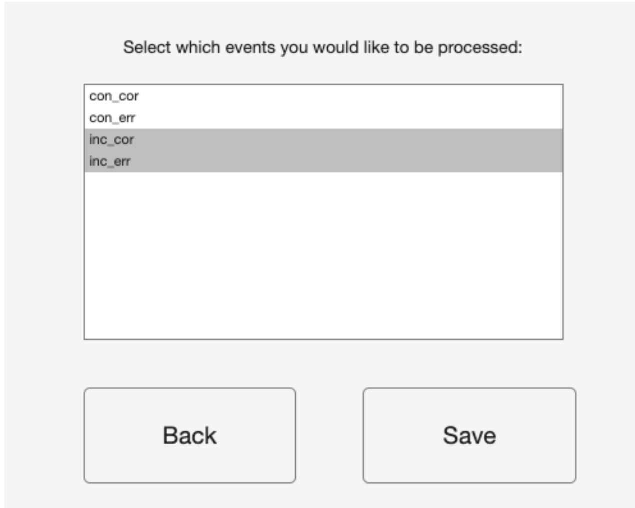


Figure 47: Specify Which Events to Process, with inc_err and inc_cor selected out of the four available events.

Nothing on this screen needs changing for this tutorial. Confirm that Number of Chains is 4, Warmup iterations is 5000, Sampling iterations is 5000, and Random seed is 12345. The seed is the reason two runs of the same file agree to the last digit, so leave it alone unless you are deliberately checking sensitivity to it.

Four chains at 5,000 warmup and 5,000 sampling is a defensible default rather than a law. It is enough for the one-facet models in this tutorial and it is not enough for every design in the toolbox; the test-retest chapter raises it. Use as many iterations as you need to get your model to converge.

Verbose Stan output is set to Yes by default, which means CmdStan's per-iteration progress scrolls through the Command Window during the fit. Leave it on the first time. Watching the sampler is the cheapest way to notice that something has gone wrong before the run finishes.

Click Save to return to Specify Inputs. Clicking Back instead discards anything you changed on that screen.

Step 6. Analyze

Click the Analyze button. A save dialog opens: choose a folder and a file name for the `.psyrat` output.

Choose a path with no spaces in it. CmdStan does not support whitespace in filenames or paths, so a path containing a space produces a modal Whitespace detected dialog and returns you to the save dialog. This is the single most common way a first run stalls, and it is worth checking before you click Save rather than after (see Chapter 7 for the full list of pre-run checks, and Chapter 2 for the short version).

While the run proceeds, PsyRAT raises modal dialogs for anything that requires a decision. A modal dialog blocks the rest of the interface until you dismiss it, so if the toolbox appears frozen, look for a dialog window behind the main figure.

Checkpoint: the Command Window should print, in order, "Preparing data for analysis..." once, then "Working on event 1 of 2", "Model is being run in cmdstan", and CmdStan's own progress

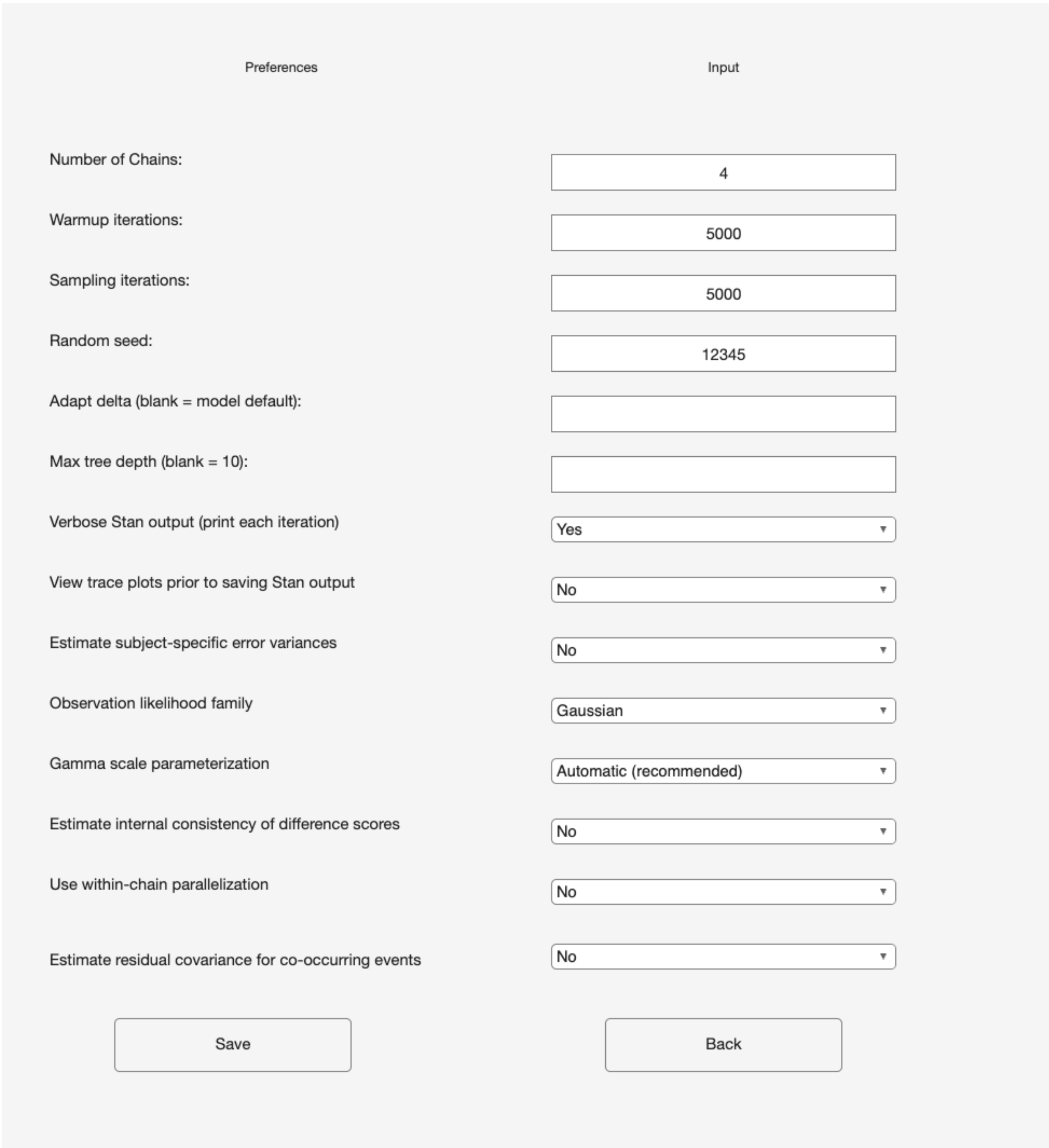


Figure 48: Specify Processing Preferences. Every value shown here is the shipped default; this tutorial changes none of them.

output, then ”Working on event 2 of 2” and ”Model is being run in cmdstan” again for the second fit. Each event is fitted in its own model, with both groups estimated jointly inside it, so a two-event run is two fits. When sampling finishes it should print ”Model converged”, then ”Saving Processed Data...”.

If it instead prints ”Chains did not converge” in a dialog offering Rerun and Do Not Rerun, the fit failed PsyRAT's convergence screen (any monitored parameter with an R-hat at or above 1.1, or an effective sample size below 40 at four chains). Click Rerun. See Section 10.3 for what that does to your numbers.

Checkpoint: the largest R-hat in the CmdStan summary printed to the Command Window should be 1.00, and the run should report 0 divergent transitions.

Success! There is now a `.psyrat` file on disk holding the posterior draws for every variance component in this design, together with a `_runconfig.json` sidecar recording the seed and settings that produced it. The viewer opens by itself.

10.5 Setting up the outputs

The View Results Setup screen opens automatically after a successful single measurement run. Its header names the dataset and the measurement column.

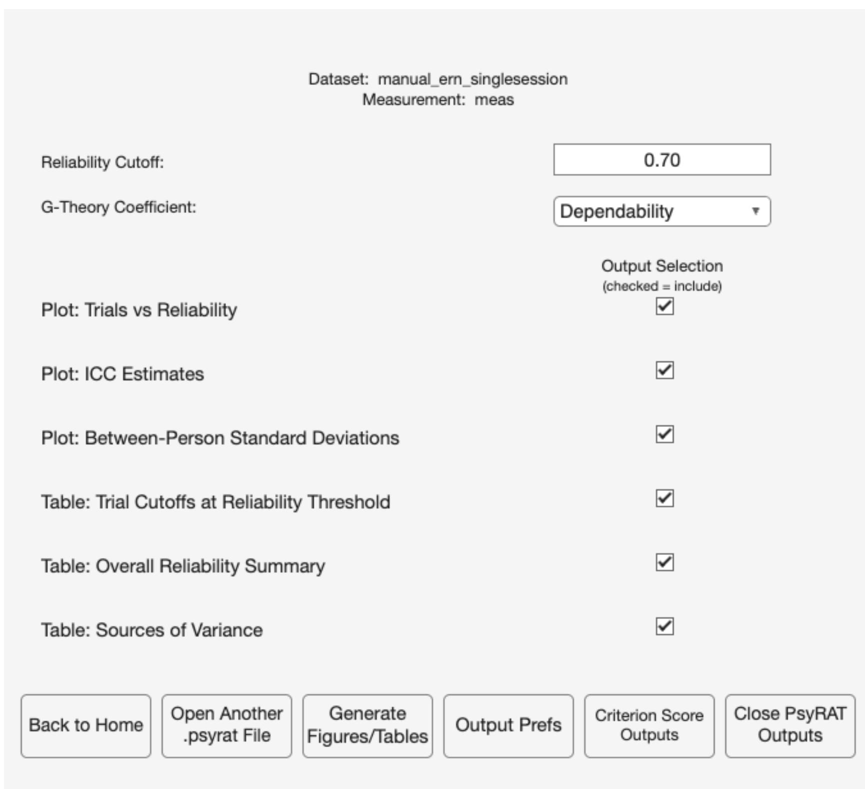


Figure 49: View Results Setup for a single-session run. The Reliability Cutoff box drives every trial-cutoff calculation below it.

Control	Value
Reliability Cutoff:	`0.70`

Everything else on this screen stays at its default: G-Theory Coefficient is Dependability, and all six output checkboxes are checked.

A word about that .70. Clayson and Miller (2017b) recommend .70 as a minimum for exploratory ERP work and .80 as more appropriate in most contexts, and they are explicit that a threshold is a decision about how much measurement error a study can tolerate rather than a property of the data. Table 10.1 shows why .70 is the value to walk through: every cell's cutoff lands inside the observed trial counts, so the table comes back with numbers in every cell, and the retention cost the threshold exacts (steep in GroupB) is visible on real rows. Section 10.8 then raises the threshold to .80 and shows what the sentinels look like when a cutoff lands beyond the data.

Click Generate Figures/Tables. Three figure windows and three table windows open.

10.6 Reading the output

The dependability-versus-trials plot

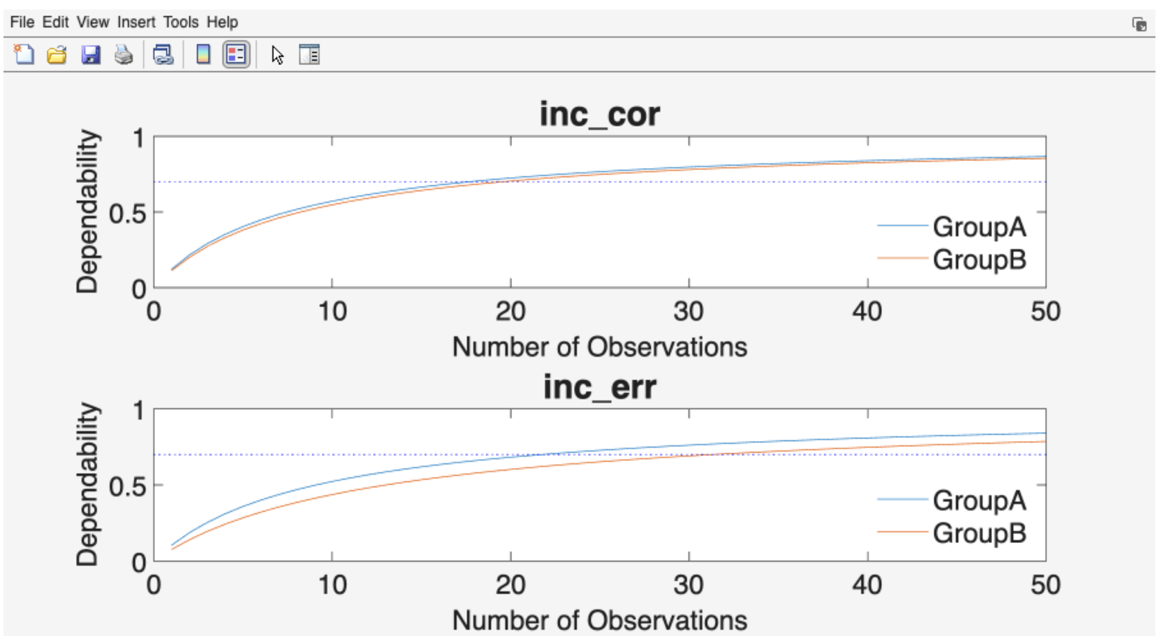


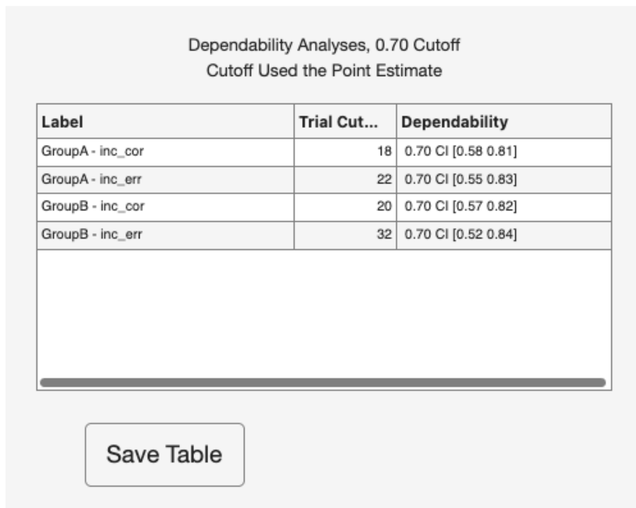
Figure 50: Dependability as a function of the number of trials retained for averaging, one panel per event, one line per group, with the reliability cutoff drawn as a dotted horizontal line.

The window is titled Dependability v Number of Trials: Point Estimate. The x axis is the number of observations averaged into the score, running from 1 to 50 (set by Number of trials to plot under Output Prefs). The y axis is dependability, fixed from 0 to 1. There is one panel per event, titled with the event label, and one line per group; the legend names them. The dotted horizontal line is the Reliability Cutoff you entered.

This is the D-study. Each curve answers "if every participant contributed exactly n trials, what would the dependability of their averages be?", evaluated at the posterior mean of the variance components. Read it left to right: the curves rise steeply over the first ten or fifteen trials, then flatten. That flattening is the practical message of generalizability theory for ERP work. Going from 5 to 15 trials buys a great deal; going from 40 to 50 buys almost nothing.

In the `inc_err` panel, the GroupA line reaches .84 at 50 trials and the GroupB line reaches .79. The gap between them is the whole comparison: the two groups' error-trial curves do not converge, because they differ in the trial noise that averaging is trying to beat down. In the `inc_cor` panel the two lines sit close together (.87 and .85 at 50 trials), which is what you expect from two cells the simulator gave identical parameters.

The trial-cutoff table



Dependability Analyses, 0.70 Cutoff
Cutoff Used the Point Estimate

Label	Trial Cut...	Dependability
GroupA - inc_cor	18	0.70 CI [0.58 0.81]
GroupA - inc_err	22	0.70 CI [0.55 0.83]
GroupB - inc_cor	20	0.70 CI [0.57 0.82]
GroupB - inc_err	32	0.70 CI [0.52 0.84]

Save Table

Figure 51: Results of Increasing the Number of Trials on Dependability. One row per group and event, with the trials needed to clear the threshold and the dependability achieved there.

The window is titled Results of Increasing the Number of Trials on Dependability, with a subtitle giving the cutoff and which estimate was used to find it (Point Estimate, the default). The columns are Label, Trial Cutoff, and Dependability.

Trial Cutoff is the smallest number of trials at which the dependability point estimate reaches the threshold. Dependability is the coefficient at that cutoff, printed as a point estimate with its 95% credible interval.

Label	Trial Cutoff	Dependability	True cutoff	True D there
GroupA - inc_cor	18	.70 [.58, .81]	21	.700
GroupA - inc_err	22	.70 [.55, .83]	19	.709
GroupB - inc_cor	20	.70 [.57, .82]	21	.700
GroupB - inc_err	32	.70 [.52, .84]	28	.703

The true cutoffs invert the formula at .70, with the trial main effect included in the numerator of the error ratio: (.7 / .3) x 7.813 = 18.2 for GroupA error trials, (.7 / .3) x 11.829 = 27.6 for

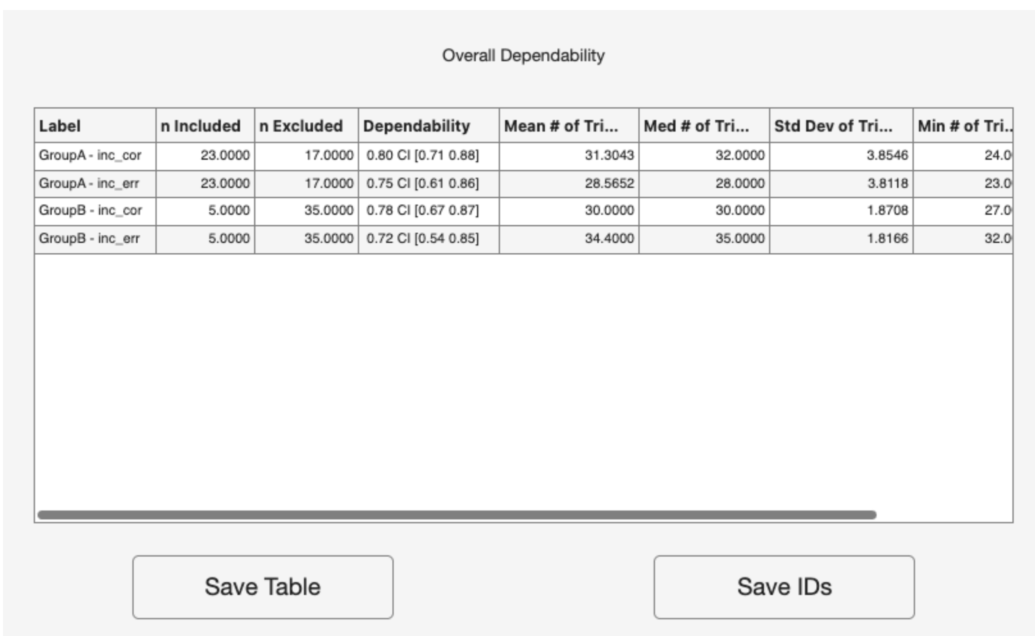
GroupB error trials, and (.7 / .3) x 8.997 = 21.0 for correct trials, each rounded up to the next whole trial. The cutoff is always a whole trial and the coefficient at it therefore sits at or a little above the threshold, which is why the True D column reads .709 rather than .700 for GroupA error trials.

There are two ways this table reports that a threshold is out of reach, and they are different conditions with different appearances. Sentinels get their own labeled paragraph below because both are easy to misread as data.

A value of -1 in the Trial Cutoff column indicates that an appropriate cutoff could not be realistically found given the specified reliability threshold. It is a sentinel, not a count, and it never means zero trials. PsyRAT searches out to the largest observed trial count plus a thousand, so this appears only when even that extended search finds nothing.

A value of -1.00 CI [-1.00 -1.00] in the Dependability column is the OTHER condition: a cutoff was found, and it is printed, but no participant in that cell supplied that many trials. Consider this example. At a .80 threshold GroupB incongruent errors need 48 trials at the true values (your run prints its own estimate of that count; Section 10.8 reads it). The search finds the count and prints it, but the most any GroupB participant supplied is 36, so the coefficient cell is blanked to -1 and PsyRAT raises a non-modal dialog titled Extrapolation beyond data reading "Not enough trials are present in the current data" and "Cutoffs represent an extrapolation beyond the data". Both conditions mean the same thing in practice: the score you have cannot be made that dependable by averaging the trials you collected. Section 10.8 walks into this case deliberately.

The overall summary



Overall Dependability

Label	n Included	n Excluded	Dependability	Mean # of Tri...	Med # of Tri...	Std Dev of Tri...	Min # of Tri..
GroupA - inc_cor	23.0000	17.0000	0.80 CI [0.71 0.88]	31.3043	32.0000	3.8546	24.0
GroupA - inc_err	23.0000	17.0000	0.75 CI [0.61 0.86]	28.5652	28.0000	3.8118	23.0
GroupB - inc_cor	5.0000	35.0000	0.78 CI [0.67 0.87]	30.0000	30.0000	1.8708	27.0
GroupB - inc_err	5.0000	35.0000	0.72 CI [0.54 0.85]	34.4000	35.0000	1.8166	32.0

Figure 52: Overall Dependability. Included and excluded counts per cell, the dependability of the retained data, and the trial-count distribution behind it.

The window is titled Dependability Analyses and the table is headed Overall Dependability. The columns are Label, n Included, n Excluded, Dependability, Mean # of Trials, Med # of Trials, Std

Dev of Trials, Min # of Trials, and Max # of Trials.

The n Included logic is worth pausing on, because it is a group-level quantity printed on a per-event row. A participant is retained for a group only if they meet the trial cutoff for EVERY event processed in that group. At this threshold both rows for GroupA therefore report the same n Included: participants who cleared the cutoff on `inc_err` but not on `inc_cor` are excluded from both. That is deliberate. If you intend to compare or combine the two conditions, a participant who is adequately measured in only one of them is not adequately measured for the comparison. Process one event at a time if you want per-event retention instead.

The rule has no exceptions, and Section 10.8 walks into its sharpest case on purpose: an event in which nobody clears the cutoff empties the group's retained set, so every row of that group prints n Included 0, n Excluded rises to the full group size, and for this analysis the two columns add up to the group size on every row. The console also prints a line saying the group retained nobody.

Dependability in this table is not the same quantity as in the cutoff table. Here it is the coefficient evaluated at the mean number of trials the RETAINED participants actually contributed, so it describes the scores you would carry forward, not a hypothetical count.

Label	n Included	n Excluded	Dependability	Mean # of Trials
GroupA - inc_cor	23	17	.80	31.3
GroupA - inc_err	23	17	.75	28.6
GroupB - inc_cor	5	35	.78	30.0
GroupB - inc_err	5	35	.72	34.4

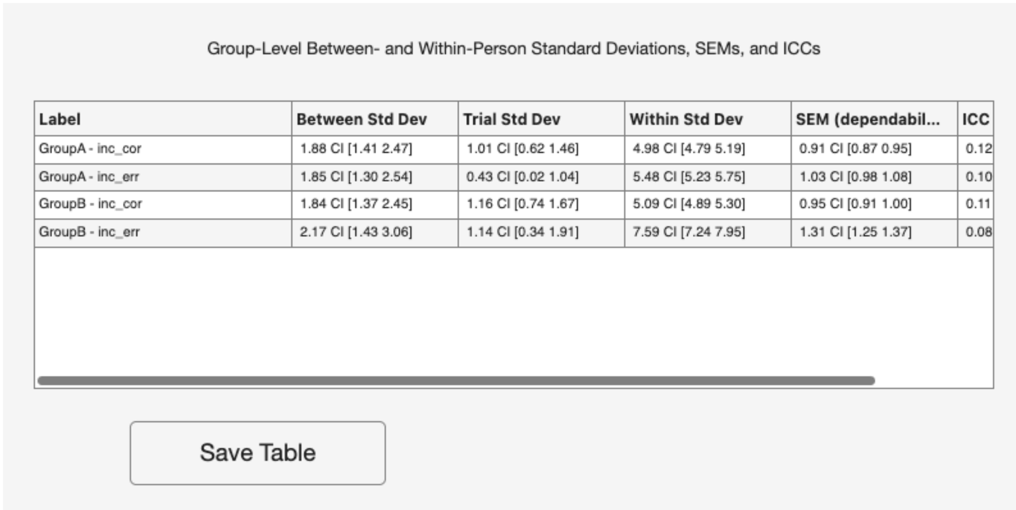
Applying the TRUE cutoffs from the table above to the observed trial counts retains 29 of 40 participants in GroupA and 13 of 40 in GroupB, at mean retained error-trial counts of 26.7 and 31.5. That retention gap is the comparison again, in the currency a study actually pays: GroupB's noisier error trials cost twenty-seven participants at a threshold GroupA clears with eleven exclusions. The counts your run reports follow the ESTIMATED cutoffs, and an estimated cutoff a few trials above or below the true one moves several participants across the line, so expect the printed counts to differ from these.

Two buttons sit under the table. Save Table writes the table with a provenance header (seed, analysis, engine, priors, convergence verdict, divergent transitions) to `.xlsx` or `.csv`. Save IDs writes the list of retained participant ids, which is what you feed back into your analysis pipeline as the inclusion list. Save both. The provenance header is what makes the exported table interpretable a year later.

Sources of variance

The window is titled Point and 95% Interval Estimates for the Between- and Within-Person Standard Deviations, SEMs, and ICCs. For a one-facet analysis the columns are Label, Between Std Dev, Trial Std Dev, Within Std Dev, SEM (dependability), and ICC (dependability). Every cell is a posterior mean with its 95% credible interval.

Between Std Dev is sigma_p, the between-person standard deviation. Trial Std Dev is sigma_i, the trial main effect: systematic drift across trial position, common to all participants. Within Std Dev is sigma_pi,e, the person-by-trial residual, which is the noise that averaging reduces. SEM



Group-Level Between- and Within-Person Standard Deviations, SEMs, and ICCs

Label	Between Std Dev	Trial Std Dev	Within Std Dev	SEM (dependabil...	ICC
GroupA - inc_cor	1.88 CI [1.41 2.47]	1.01 CI [0.62 1.46]	4.98 CI [4.79 5.19]	0.91 CI [0.87 0.95]	0.12
GroupA - inc_err	1.85 CI [1.30 2.54]	0.43 CI [0.02 1.04]	5.48 CI [5.23 5.75]	1.03 CI [0.98 1.08]	0.10
GroupB - inc_cor	1.84 CI [1.37 2.45]	1.16 CI [0.74 1.67]	5.09 CI [4.89 5.30]	0.95 CI [0.91 1.00]	0.11
GroupB - inc_err	2.17 CI [1.43 3.06]	1.14 CI [0.34 1.91]	7.59 CI [7.24 7.95]	1.31 CI [1.25 1.37]	0.08

Save Table

Figure 53: Point and 95% Interval Estimates for the Between- and Within-Person Standard Deviations, SEMs, and ICCs.

(dependability) is the absolute standard error of measurement of a score at the retained mean trial count, in microvolts, and it is the number to attach to an individual participant's score. ICC (dependability) is the absolute intraclass correlation, which is dependability at a single trial.

Label	Between Std Dev	Trial Std Dev	Within Std Dev	SEM (dep.)	ICC (dep.)
GroupA - inc_cor	1.88	1.01	4.98	0.91	.12
GroupA - inc_err	1.85	0.43	5.48	1.03	.10
GroupB - inc_cor	1.84	1.16	5.09	0.95	.11
GroupB - inc_err	2.17	1.14	7.59	1.31	.08

The simulator put a genuine trial main effect of 1.0 microvolt into every cell, shared by all participants at the same trial position, and the Trial Std Dev column is the model recovering it. Three of the four cells come back near that value (1.01, 1.14, and 1.16); GroupA inc_err comes back at 0.43, with a 95% credible interval of [0.02, 1.04] that still covers the generating value. The trial main effect is estimated from at most 36 shared trial positions per cell, so it is the least well determined component in this table, and Table 10.2 shows the same cell again. The trial main effect is also why the SEM and ICC in this table (absolute, dependability-based) differ from their relative-error counterparts, the SEM sitting above and the ICC below: sigma_i belongs to absolute error only, so absolute error is the larger error term. In your own data a substantial trial SD is drift worth investigating (fatigue, habituation, electrode impedance) rather than a nuisance to average away.

Between-person standard deviations and ICC estimates

Both figures put Event on the x axis, with one point per group at each event and a legend naming the groups. The first has Between-Person Standard Deviations on the y axis in microvolts; the second has Intraclass Correlation Coefficient. Each y axis starts at zero and runs to just above the largest upper bound plotted, so the ICC figure is not scaled to a fixed 0-to-1 range. Each point is a posterior mean and each whisker a 95% credible interval.

Read the two together, against Table 10.1. The generator gave GroupB the LARGER person standard deviation on error trials (2.2 against 2.0) and the LOWER true ICC (.078 against .113).

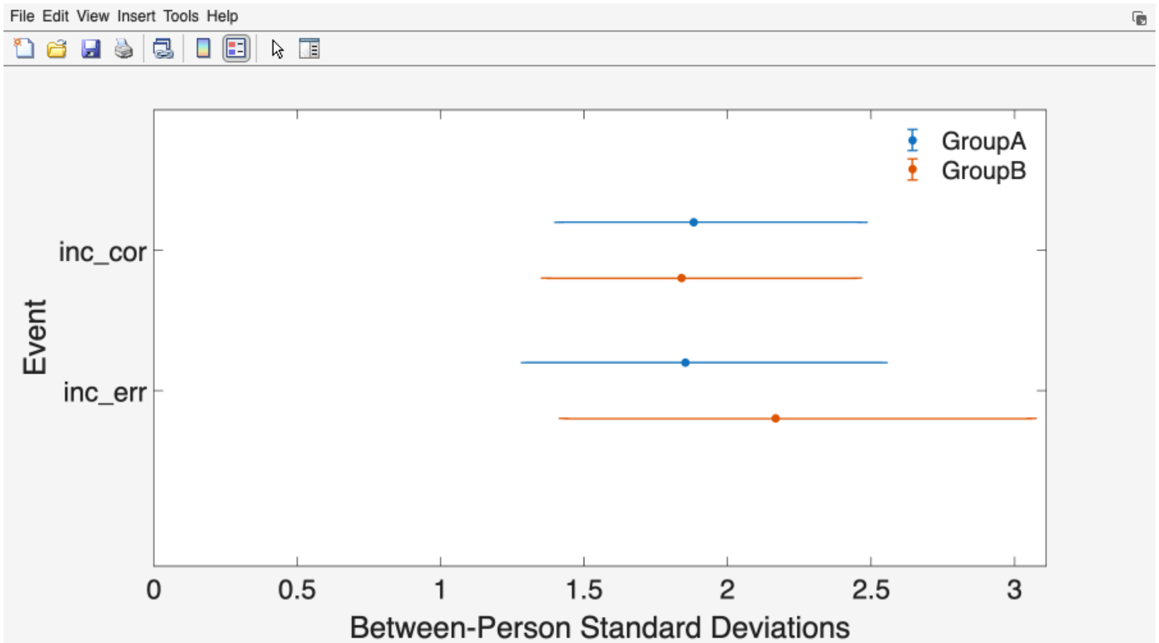


Figure 54: Between-Person Standard Deviations, one point estimate and 95% credible interval per group and event.

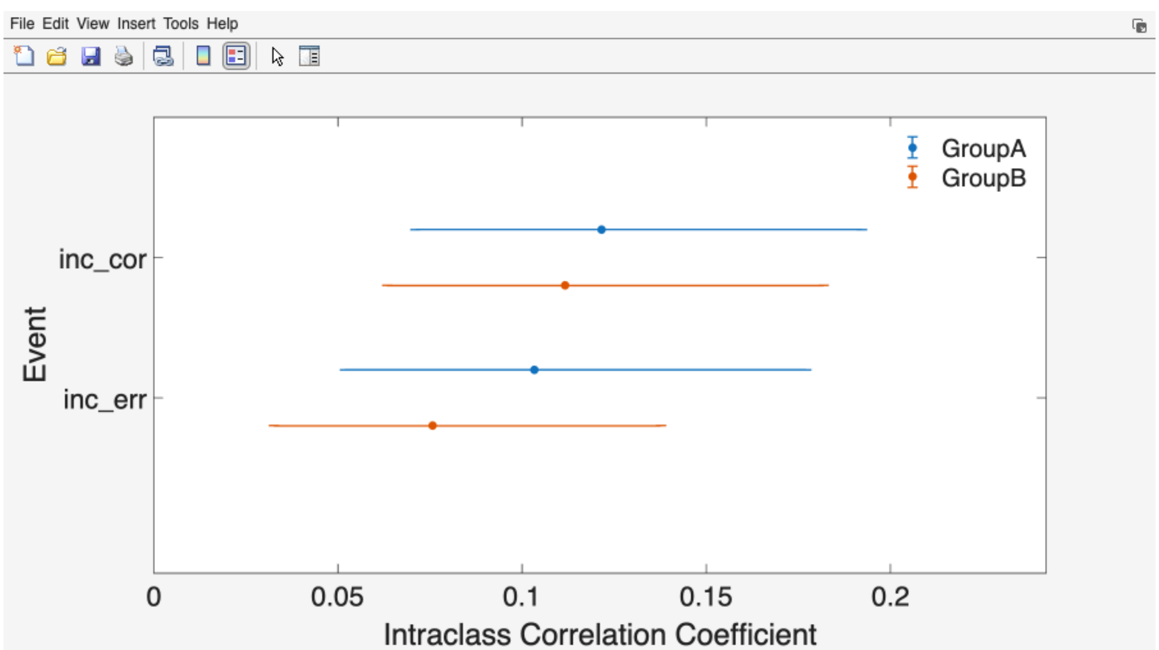


Figure 55: ICC Estimates, one point estimate and 95% credible interval per group and event.

situation for error-trial ERP scores, and it is the reason trial counts belong in a methods section. Baldwin, Larson, and Clayson (2015) reported the same pattern in real performance-monitoring data: within-person variance exceeded between-person variance for every component examined.

The practical consequence is that a study designed around 20 error trials per participant has built a measure whose scores carry substantial measurement error for a group like GroupB, and any correlation estimated from those scores is attenuated accordingly. The fix is either more error trials or a design that tolerates lower reliability with a correspondingly larger sample.

10.8 How many trials for .80?

Change Reliability Cutoff to `0.80` and click Generate Figures/Tables again. The posterior does not change; only the threshold applied to it does, so the pass costs nothing and re-answers the design question at the stricter standard.

Label	Trial Cutoff at .80	True
GroupA - inc_cor	31	36
GroupA - inc_err	38	32
GroupB - inc_cor	34	36
GroupB - inc_err	55	48

GroupB's true error-trial cutoff, 48, exceeds every observed count in its cell: the most any GroupB participant supplied is 36. GroupA's true cutoff, 32, sits inside the top of its range (the GroupA maximum is also 36), but the estimated cutoff in the table above lands past 36, so in this run both error rows come back blanked; a run whose GroupA estimate landed a few trials lower would keep a coefficient on that row, reached by a handful of participants. Correct trials are plentiful enough that their cutoffs stay inside the data. Where a cutoff lands outside, the Dependability cell for that row is blanked to -1.00 CI [-1.00 -1.00] and the Extrapolation beyond data dialog appears. The Trial Cutoff itself is still printed, because it is still the honest answer to ”how many would be needed”.

Now read the overall summary. In GroupB no participant supplies 48 error trials, so the error event retains nobody and the all-events rule empties the group: both GroupB rows print n Included 0, n Excluded rises to 40, and the Command Window prints a line saying no participants met the threshold for every event type. GroupA keeps only the participants who supplied at least 32 error trials AND at least 36 correct trials, which at the true cutoffs is one of 40; your run prints n Included 0 on both GroupA rows, following the estimated cutoffs. A row whose group retained nobody still shows a dependability figure, computed at the mean trial count over ALL participants; it describes the file, not a retained sample, so treat it accordingly.

That is the answer to the design question, and it is a blunt one. Reaching .70 on incongruent-error ERN takes 19 trials in GroupA, inside what most of its participants supplied, and 28 in GroupB, in the upper part of its range. Reaching .80 takes 32 and 48: within reach of only the best-measured GroupA participants, and a third again the maximum anyone in GroupB supplied. Plan the task accordingly, or report the reliability you actually achieved and interpret the effect sizes in light of it. The toolbox will not make that call for you.

10.9 Recovery: truth against estimate

Because the generating parameters are known, the estimates can be checked against them directly.

Table 10.2. Recovery of the generating parameters. Truth is from the generator; the estimate and interval are the posterior mean and 95% credible interval from the Sources of Variance table.

Quantity	Truth	Estimate	95% CrI
GroupA inc_err sigma_p	2.00	1.85	[1.30, 2.54]
GroupA inc_err sigma_pi,e	5.50	5.48	[5.23, 5.75]
GroupA inc_err sigma_i	1.00	0.43	[0.02, 1.04]
GroupB inc_err sigma_p	2.20	2.17	[1.43, 3.06]
GroupB inc_err sigma_pi,e	7.50	7.59	[7.24, 7.95]
GroupA inc_cor sigma_p	1.70	1.88	[1.41, 2.47]
GroupA inc_cor sigma_pi,e	5.00	4.98	[4.79, 5.19]
GroupB inc_cor sigma_p	1.70	1.84	[1.37, 2.45]
GroupB inc_cor sigma_pi,e	5.00	5.09	[4.89, 5.30]
GroupA inc_err dependability at .70 retention	.774	.75	(printed with the estimate)
GroupB inc_err dependability at .70 retention	.727	.72	(printed with the estimate)

The two dependability truths are the D(n) formula at the mean error-trial counts of the participants the TRUE .70 cutoffs retain, 26.724 for GroupA and 31.462 for GroupB: 4.00 / (4.00 + 31.25 / 26.724) = .774 and 4.84 / (4.84 + 57.25 / 31.462) = .727. Note what retention has done to the comparison. GroupB's surviving thirteen participants are its best-measured thirteen, so its reported dependability is much closer to GroupA's than the two groups' true curves are at any common trial count. Selecting on trial count selects on reliability, and a coefficient computed after that selection describes the retained sample, not the population you sampled from. Read 28.6 and 34.4 out of your own overall summary and recompute the truth at those counts if the estimated cutoffs retained a different set.

Read this table for what it teaches, not as a validation exercise. With 40 participants per group and 12 to 36 error trials each, the point estimates will NOT equal the true values, and the between-person standard deviations in particular are estimated from 40 people and carry wide intervals. A person SD estimated at 1.8 when the truth is 2.0 is not a bug in the toolbox and not a failure of the model. It is what 40 participants buys you, and the credible interval is the toolbox telling you so.

That is the pedagogical point, and it generalizes: when your own study reports a dependability of .68, the honest statement is not "reliability is .68" but "reliability is estimated at .68, with an interval that runs from here to here". Report the interval.

This recovery exercise is an illustration. It is not evidence that the toolbox's estimators are correct. PsyRAT's correctness claims rest on its validation suites, described in `README.md`, which check the formulas against fixed references and check estimation against simulated data with known components at sample sizes chosen for that purpose. A single 80-participant run is far too small a sample to license any conclusion about estimator accuracy, in either direction.

10.10 Reporting this analysis

Chapter 17 sets out what a reliability section should contain and why. Here is that chapter's template, pre-filled for this tutorial. Replace the placeholders with the values from your own run.

> Score reliability was evaluated with generalizability theory as implemented in PsyRAT, which estimates variance components in a Bayesian one-facet (persons x trials) design via CmdStan. Models were fitted with 4 chains, 5,000 warmup and 5,000 sampling iterations, and a random seed of 12345; convergence was assessed by the potential scale reduction factor (maximum R-hat = 1.00) and effective sample size, and 0 divergent transitions were observed. Dependability (the absolute-error coefficient) was estimated separately for each group and event. For incongruent-error trials, the between-person standard deviation was 1.85 microvolts in GroupA and 2.17 microvolts in GroupB, and the person-by-trial residual standard deviation was 5.48 and 7.59 microvolts, respectively. A decision study indicated that 22 incongruent-error trials were required to reach a dependability of .70 in GroupA and 32 in GroupB; 38 and 55 trials, respectively, were required to reach .80. A retention threshold of .70 was applied. Participants who did not supply enough trials to meet that threshold in every processed condition were excluded, retaining 23 of 40 participants in GroupA and 5 of 40 in GroupB. Among retained participants, incongruent-error scores had a dependability of .75 in GroupA and .72 in GroupB at mean trial counts of 28.6 and 34.4. The absolute standard error of measurement was 1.03 and 1.31 microvolts, respectively.

Report the trial counts, the threshold, the exclusion rule, and the interval, not just the coefficient. A dependability estimate without the trial count it was evaluated at is not interpretable, because the same variance components produce a different coefficient at every n.

Notice what the retention sentence had to do. The .70 threshold kept only a minority of GroupB, and the paragraph says so with the counts rather than burying them. A reliability coefficient reported after exclusions describes the retained sample; disclosing how many participants the threshold cost is part of reporting it, and quietly reporting the correct-trial reliability instead is not a defensible substitute.

10.11 Where to go next

- Two occasions instead of one: Chapter 11.
- The reliability of `inc_err` minus `inc_cor`: Chapter 12.
- Data that arrive as split means rather than single trials: Chapter 13.

11. Tutorial 2: Test-retest reliability

The same flanker task, run twice, two weeks apart. A score that is internally consistent within a session can still be unstable across sessions, and the two questions have different answers because they generalize over different things. Internal consistency asks whether a participant's average would be the same if you had recorded different trials on the same day. Test-retest reliability asks whether it would be the same if you had recorded the same participant on a different day. A measure intended to index a trait, or to serve as a baseline in a treatment study, needs the second answer, and a two-facet design is what supplies it.

This tutorial uses `test_data/manual_ern_testretest.csv`. It assumes you have worked through Chapter 10 and describes only what differs. The screen-by-screen reference lives in Chapter 7 and Chapter 9.

11.1 The dataset

Thirty participants (ids 201 to 230) completed the task at two occasions, `t1` and `t2`. Two events, `err` and `cor`. There is no group column. Trial counts are balanced by design: every participant contributed exactly 24 error and 40 correct trials at each occasion. The unequal-counts lesson belongs to Tutorial 1; this file needs a common trial index so that the occasion, trial, and interaction components are all identifiable.

The columns are `id`, `time`, `event`, `meas`. The file is simulated, and the generator (`tests/helpers/psyrat_make_manual_testretest.m`) documents every component.

11.2 What the true coefficients are

The generating model is the fully crossed persons x trials x occasions decomposition, which is the design Rocha et al. (2026, Table 3) tabulates. A score is a grand mean plus seven independent effects: person, occasion, trial, person x trial, person x occasion, occasion x trial, and residual.

Table 11.1. The seven generating components, in microvolts.

Component	Symbol	err	cor
Person	sigma_p	3.2	2.8
Occasion	sigma_o	0.6	0.6
Trial	sigma_i	1.0	1.0
Person x trial	sigma_pi	1.0	0.8
Person x occasion	sigma_po	1.7	1.4
Occasion x trial	sigma_oi	0.5	0.5
Residual (p x o x i)	sigma_poi,e	5.5	5.0

Three coefficients follow from those seven numbers, and they differ in which facet is held fixed. Squaring the SDs and taking the error event at its 24 trials, with one occasion:

Coefficient of equivalence (CE). Occasion is fixed, so person x occasion variance counts as universe-score variance rather than error. This is internal consistency at one session.

```
universe = sigma_p^2 + sigma_po^2 = 10.24 + 2.89 = 13.13
absolute error = (sigma_pi^2 + sigma_poi,e^2 + sigma_i^2 + sigma_oi^2) / 24
```

```
                = (1.00 + 30.25 + 1.00 + 0.25) / 24 = 32.50 / 24 = 1.354
D_CE = 13.13 / (13.13 + 1.354) = .907
```

Coefficient of stability (CS). Trials are fixed, so person x trial variance joins the universe score and occasion-related variance becomes error. This is test-retest reliability proper.

```
universe = sigma_p^2 + sigma_pi^2 / 24 = 10.24 + 0.042 = 10.282
absolute error = sigma_po^2 + sigma_o^2 + (sigma_poi,e^2 + sigma_oi^2) / 24
               = 2.89 + 0.36 + 30.50 / 24 = 4.521
D_CS = 10.282 / (10.282 + 4.521) = .695
```

Coefficient of equivalence and stability (CES). Both facets are random, so only person variance survives in the numerator. This is the reliability of a score meant to generalize across both trials and days.

```
universe = sigma_p^2 = 10.24
absolute error = 0.042 + 2.89 + 1.260 + 0.042 + 0.36 + 0.010 = 4.604
D_CES = 10.24 / (10.24 + 4.604) = .690
```

Table 11.2. True dependability and generalizability at the observed design (24 error trials, 40 correct trials, one occasion).

Coefficient	err D	err G	cor D	cor G
Equivalence (CE)	.907	.910	.936	.939
Stability (CS)	.695	.712	.727	.752
Equivalence + stability (CES)	.690	.710	.724	.751

Stability is far below equivalence for both events, and that ordering is the substantive finding. It is driven by two components: the occasion main effect (sigma_o = 0.6) and the person x occasion interaction (sigma_po = 1.7 for error trials and 1.4 for correct trials). Neither of them shrinks when you add trials, because they are not trial-level noise. A study that reports internal consistency and calls it reliability has reported the larger and less demanding of the two numbers.

That has a hard consequence for thresholds. As the trial count grows without bound, the stability and CES dependability coefficients approach 10.24 / (10.24 + 2.89 + 0.36) = .759 for error trials and 7.84 / (7.84 + 1.96 + 0.36) = .772 for correct trials. They never reach .80 at any trial count. Collecting more trials cannot fix instability across days; only more occasions can, and that is a different design.

11.3 What changes on the input screen

Load `manual_ern_testretest.csv` through Process New Data as before. Auto-detect finds the occasion column this time, because the header is `time`.

Checkpoint: the Specify Inputs screen should show these selections.

Control	Value
Participant ID:	`id`
Measurement:	`meas`
Group ID:	`(none)`

Control	Value
Event Type:	`event`
Occasion ID:	`time`
Items per split (n_i):	`(none)`

Occasion ID is the control that turns a one-facet analysis into a two-facet one. Selecting a column there is the entire difference at the input stage. Leave Group ID on `(none)`, since this file has no group column, and leave Select Events alone: both events are wanted.

11.4 The iteration advisory, and honoring it

The next time you open the Preferences screen with an Occasion ID column assigned, PsyRAT checks the iteration settings and may raise a non-modal dialog titled Recommended Iteration Increase:

> Test-retest workflows usually require at least 10,000 iterations. You currently selected fewer iterations, which may prevent convergence. How to fix: open Preferences and increase warmup and/or sampling iterations.

The dialog fires only when warmup plus sampling falls below 10,000. At the shipped defaults (5,000 plus 5,000) the total is exactly 10,000, so on a fresh installation you will not see it. You will see it if you have saved lower settings, or if a previous run left them lowered.

Honor the advice regardless. Seven variance components estimated from 30 participants is a harder posterior than the one-facet model in Tutorial 1, and the default that suffices there is the floor here, not a comfortable margin. Click Preferences and set both iteration fields higher.

Control	Value
Warmup iterations:	`10000`
Sampling iterations:	`10000`

Click Save. Number of Chains stays at 4 and Random seed stays at 12345. Those two settings plus these iteration counts are what the reference values in this chapter were produced at; record them, because a run at any other iteration count will not reproduce them exactly.

This run takes noticeably longer than Tutorial 1. Twice the iterations on a seven-component crossed model is not a small change, and the honest advice is to start it and go do something else.

Click Analyze, choose a whitespace-free output path, and wait.

Checkpoint: the Command Window should print ”Model converged” and then ”Saving Processed Data...”. The largest R-hat should be 1.00 and the run should report 909 divergent transitions.

That divergence count is not zero, and it will not be zero for you either. A persons x trials x occasions model with two occasions leaves the occasion standard deviation almost unidentified (two occasion effects cannot pin down a spread), and the sampler reports divergent transitions in that region even at the tighter target acceptance this model already runs at. R-hat is 1.00, the person and residual components are recovered (Section 11.8), and the equivalence coefficients this tutorial reads first do not carry the occasion term; the stability coefficients do, and Section 11.7

shows what that costs. The count is reported here rather than hidden, and the same message will appear on your own two-session data. The tutorial proceeds because the quantities it reads are insensitive to the offending term, but a run you intend to report should try the remedies in Section 18.3 of Chapter 18 first: more warmup, a higher Adapt delta, and, where the design allows it, more occasions.

Trace plots are not rendered for test-retest analyses. If you left View trace plots prior to saving Stan output on Yes, the Command Window prints "Viewing trace plots for trt analyses is not currently supported" and the run continues. That message is informational, not an error.

11.5 The test-retest viewer

Before the setup screen appears, a notice titled Estimation diagnostics repeats the divergence count from the checkpoint above. Click OK to continue.

The View Results Setup screen for a two-facet run carries three controls the single-session viewer does not.

Type of Reliability Coefficient: which of the three coefficients to compute. The options are Equivalence, Stability, and Equivalence + Stability, and they correspond to CE, CS, and CES in Rocha et al. (2026, Table 3). The choice is not cosmetic: it changes which variance components sit in the numerator and which count as error, so the three answer three different scientific questions.

Choose Equivalence when you want internal consistency at a single session, which is what a one-session study reports and what makes a test-retest run comparable to Tutorial 1. Choose Stability when the question is whether a participant's score holds across days, which is what a trait interpretation requires and what a pre-post treatment design depends on. Choose Equivalence + Stability when the score has to generalize across both, which is the right default for a measure proposed as an individual-difference index. Clayson, Carbine, et al. (2021) work through the three coefficients and their algorithms in detail, and their framing is the one this viewer implements.

Test-Retest Occasion Score: whether the reported coefficient describes a score from one occasion or a composite averaged over several. The options are Single-occasion (n'_o = 1) and Multi-occasion composite (n'_o = k). A box labeled "# Occasions in Composite (k):" sits below it and becomes relevant only for the second option.

The default is single-occasion, n'_o = 1, and that is the toolbox's convention rather than something a table forces. PsyRAT reports, by default, the reliability of a score taken at ONE session while generalizing over the occasion facet, because that is the score most studies actually analyze: participants are recorded twice so that occasion variance can be estimated, and then the analysis uses one session's data. Set n'_o = k only when your analysis really does average across k occasions. The reported SEM follows the same n'_o as the coefficient, so the two always describe the same score.

G-Theory Coefficient: Dependability or Generalizability, as in Tutorial 1.

Control	Value
Reliability Cutoff:	`0.70`

Everything else stays at its default: Type of Reliability Coefficient is Equivalence, G-Theory Coefficient is Dependability, and Test-Retest Occasion Score is Single-occasion (n'_o = 1). Section

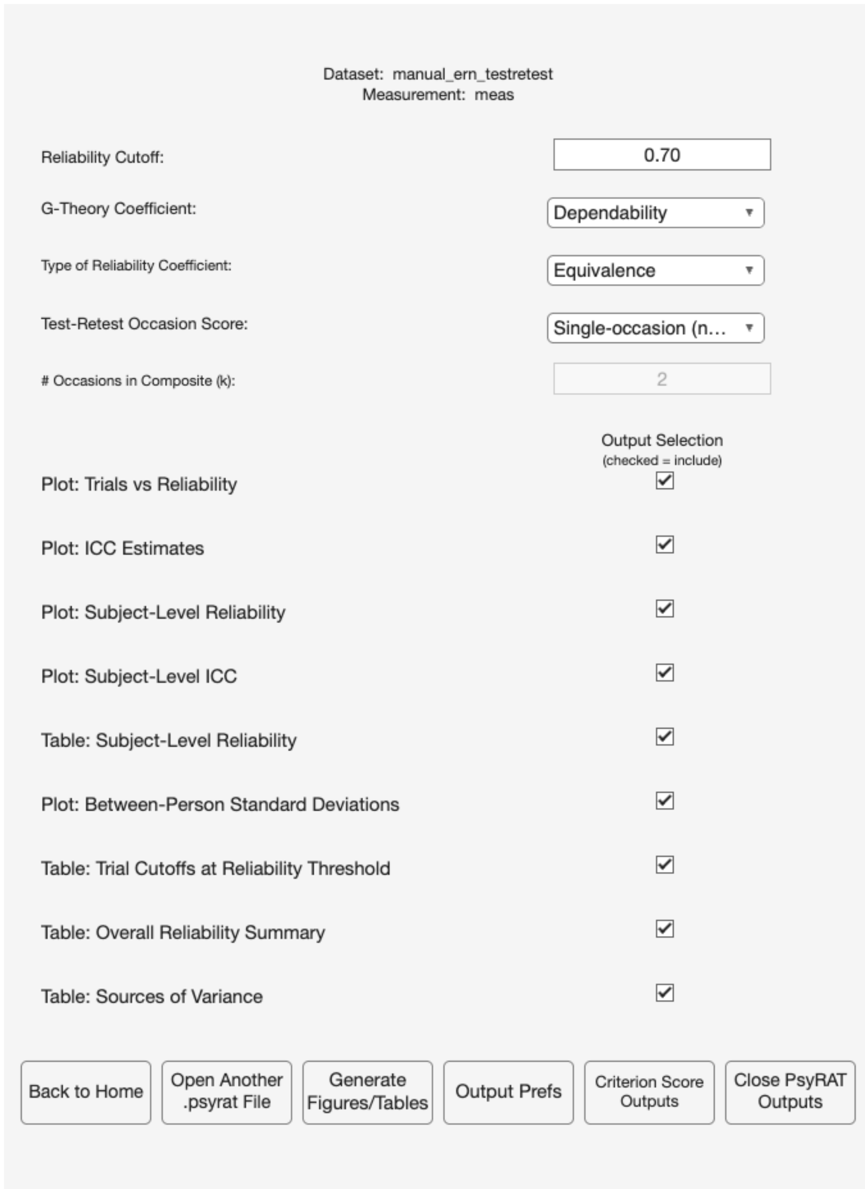


Figure 56: View Results Setup for a test-retest run. Type of Reliability Coefficient and Test-Retest Occasion Score are the two controls that define the estimand.

11.7 changes the first of those.

Three of the output checkboxes on this screen (Plot: Subject-Level Reliability, Plot: Subject-Level ICC, Table: Subject-Level Reliability) apply only to runs that estimated subject-specific error variances. This run did not, so leaving them checked produces nothing and costs nothing.

Click Generate Figures/Tables.

11.6 Reading the equivalence results

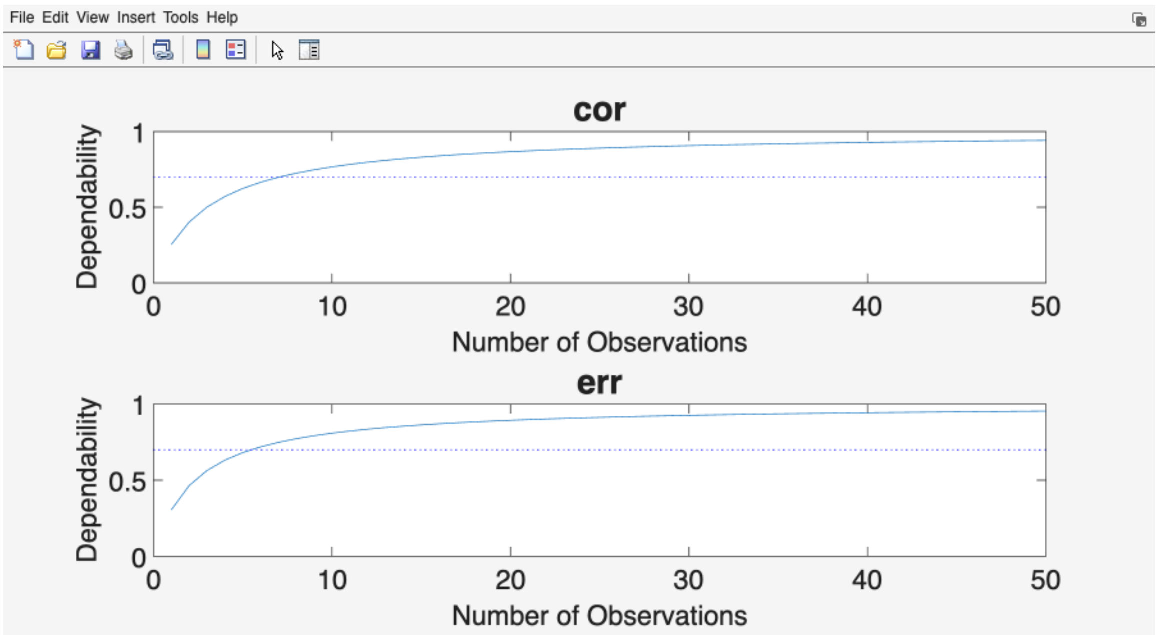


Figure 57: Dependability against the number of trials, one panel per event, for the test-retest design at the selected coefficient type.

The dependability-versus-trials plot reads exactly as in Tutorial 1: number of observations on the x axis, dependability from 0 to 1 on the y axis, one panel per event, and the cutoff drawn as a dotted line. With no group column there is no legend and a single line per panel.

The trial-cutoff table is titled Results of Increasing the Number of Trials on Dependability, exactly as in Tutorial 1. Nothing on that window names the coefficient type, so if you generate the table three times at three settings you will end up with three windows that look alike. Save each one as you go: the exported file's header block records the coefficient type, the occasion estimand, and the seed. Overwriting the wrong one is easily done and not easily noticed.

Label	Trial Cutoff	Dependability	True cutoff
err	6	.72 [.62, .82]	6
cor	8	.73 [.63, .82]	7

The true cutoffs come from inverting the CE dependability formula at .70: (.7 / .3) x 32.50 / 13.13 = 5.8 for error trials and (.7 / .3) x 26.89 / 9.80 = 6.4 for correct trials, each rounded up. Both are far inside what participants supplied (24 and 40), so every participant is retained. Equivalence is easy to reach in these data, and that is the point of the comparison that follows.

The overall summary reports 30 participants included and 0 excluded, with equivalence dependability of .91 for error trials at 24 trials and .93 for correct trials at 40.

The Sources of Variance table has one fewer column than in Tutorial 1: Label, Between Std Dev, Within Std Dev, SEM (dependability), and ICC (dependability), with the occasion estimand named in a line above the table. For a two-facet design Between Std Dev is sigma_p and Within Std Dev is the three-way residual sigma_poi,e. The other five components are estimated and used, but they are not printed here.

Label	Between Std Dev	Within Std Dev	SEM (dep.)	ICC (dep.)
err	3.47	5.42	1.20	.30
cor	2.61	5.05	0.83	.25

To see the generalizability coefficients instead, set G-Theory Coefficient to Generalizability and click Generate Figures/Tables again. Equivalence generalizability comes back as .91 for error trials and .93 for correct trials. The two coefficients are close here because the terms dependability adds (the trial and occasion x trial main effects) are small in these data, which is exactly what Table 11.1 predicts.

11.7 Stability, and a threshold that cannot be met

Set Type of Reliability Coefficient to Stability and click Generate Figures/Tables. Then do it again with Equivalence + Stability.

Lower Reliability Cutoff to `0.50` before you do, and here is why. Section 11.2 showed that stability and CES dependability top out at .759 and .772 as trials increase, so .80 is out of reach at any trial count, and .70 sits just under those ceilings: at the true values it takes 27 to 29 error trials (more than the 24 supplied) and 25 or 26 correct trials. The estimates sit lower still, and the reason is the occasion component. With two occasions the occasion main effect is estimated from two values, its posterior is pulled toward the prior and lands well above the true 0.6, and both stability coefficients carry sigma_o^2 in their error term (Section 11.8 shows the recovered value). Leave the cutoff at .70 and the Trial Cutoff column will report a count far larger than any participant supplied, PsyRAT will warn that the cutoff extrapolates beyond the data, and n Included will come back as zero. Nothing is broken when that happens. It is the software telling you that, as estimated from two sessions, no amount of averaging will make these day-to-day scores that stable.

At a .50 cutoff every cell clears the threshold well inside the observed data (5 trials for either coefficient and either event at the true values, and more than that in the estimate, for the same occasion-component reason), and the tables become readable.

Coefficient	Event	Trial Cutoff	Dependability	Truth (Table 11.2)
Stability	err	9	.56	.695
Stability	cor	29	.51	.727
Equivalence + stability	err	11	.55	.690
Equivalence + stability	cor	32	.51	.724

Read the three coefficients side by side. Equivalence sits near .91 and .93; stability sits near .56 and .51. Reporting only the first would overstate how reproducible these scores are across sessions by thirty-five points or more, and the overstatement is not a rounding matter. It is the difference between a measure you can use as a stable individual-difference index and one you cannot.

Two occasions is also the minimum for this design, not a comfortable number. With two sessions the occasion main effect is estimated from a single contrast, and its credible interval will be correspondingly wide. Treat a stability coefficient from two occasions as an estimate with real uncertainty attached, and report the interval.

11.8 Recovery of the seven components

The Sources of Variance table shows two of the seven components. To see all seven, click the View All Standard Deviations button at the bottom right of that window. A second window opens, titled Point Estimates of All Standard Deviation Components, with the columns Label, Between Person, Between Session, Between Trial, Person x Session, Person x Trial, Session x Trial, and Within Person (Error).

That window reports point estimates only, with no credible intervals, and its Save Table button exports them. Read them as posterior means: a component printed as 0.41 is the mean of its posterior, not a value the data pinned down to two decimals.

Table 11.3. Recovery of the seven generating components. Estimates are the posterior means shown in Point Estimates of All Standard Deviation Components.

Component	Column	Truth (err)	Estimate (err)	Truth (cor)	Estimate (cor)
Person	Between Person	3.2	3.47	2.8	2.61
Occasion	Between Session	0.6	2.58	0.6	2.16
Trial	Between Trial	1.0	1.25	1.0	1.12
Person x occasion	Person x Session	1.7	1.70	1.4	1.56
Person x trial	Person x Trial	1.0	1.73	0.8	0.75
Occasion x trial	Session x Trial	0.5	0.28	0.5	0.22
Residual	Within Person (Error)	5.5	5.42	5.0	5.05

The column order on screen puts Person x Session before Person x Trial, which is not the order Table 11.1 uses. Match on the column heading, not the position.

Expect the residual and the person component to land close to truth and the small components to land loosely. The occasion main effect is the extreme case: with two occasions there are two occasion effects, and estimating a standard deviation from two values is not something 30 participants can rescue. Its posterior will be pulled toward the prior. That is the correct behavior for a component the design barely identifies, and it is worth seeing once, because the same thing will happen in your own two-session data. The full posterior draws for all seven components are on the saved `.psyrat` file under `psyrat_data.rel.out`, if you want intervals rather than point estimates; see Chapter 14.

This is an illustration, not validation evidence. Thirty participants at two occasions cannot establish that an estimator is unbiased. PsyRAT's correctness claims rest on the validation suites described in `README.md`.

11.9 Reporting this analysis

The Chapter 17 template, pre-filled for this tutorial.

> Test-retest reliability was evaluated with generalizability theory as implemented in PsyRAT, using a fully crossed persons x trials x occasions design with Bayesian variance-component estimation in CmdStan. Models were fitted with 4 chains, 10,000 warmup and 10,000 sampling iterations, and a random seed of 12345 (maximum R-hat = 1.00; 909 divergent transitions). Coefficients are reported for a single-occasion score (n'_o = 1), generalizing over the occasion facet. For error trials at 24 trials, the coefficient of equivalence (dependability) was .91, the coefficient of stability was .56, and the coefficient of equivalence and stability was .55; the corresponding values for correct trials at 40 trials were .93, .51, and .51. The between-person standard deviation was 3.47 microvolts for error trials and 2.61 microvolts for correct trials, with a person x occasion standard deviation of 1.70 and 1.56, respectively. Applying a .70 threshold to the coefficient of equivalence retained 30 of 30 participants. The absolute standard error of measurement was 1.20 microvolts for error trials and 0.83 microvolts for correct trials.

Name the coefficient. "Test-retest reliability was .73" is ambiguous among three quantities, and the reader cannot recover which one you meant. In these data equivalence exceeds the other two by thirty-five points or more, so the ambiguity is not a small one. State whether it is equivalence, stability, or both, and state the n'_o the coefficient was evaluated at.

11.10 Where to go next

- The reliability of a difference between two conditions: Chapter 12.
- Data that arrive as split means: Chapter 13.
- Other two-facet and conditional designs: Chapter 15.

12. Tutorial 3: Difference-score reliability

ERN is very often analyzed as a difference: error-trial amplitude minus correct-trial amplitude, on the argument that subtracting the correct-trial response removes shared, non-specific activity. The subtraction has a psychometric cost, and it is a large one. This tutorial estimates it directly on the file from Chapter 10, so the difference score can be read next to the two scores it was built from.

The dataset, the loading steps, and the viewer are all as in Tutorial 1. Only the processing preference and the reading change.

12.1 Why a difference is less dependable than its parts

A difference score inherits the error of both constituents and keeps only the part of their person variance that they do NOT share. Both halves of that sentence work against it.

The universe-score variance of a difference is

```
sigma^2_diff(p) = sigma^2_X(p) + sigma^2_Y(p) - 2 sigma_XY(p)
```

and the covariance term is what does the damage. Write the covariance as r x sigma_X(p) x sigma_Y(p) and the more correlated the two conditions are across people, the less between-person variance survives the subtraction. Meanwhile the error variances ADD, because trial noise in the error condition is independent of trial noise in the correct condition. Rocha et al. (2026, Table 6) give the full one-facet coefficient, with each constituent's residual divided by its own trial count and the residual covariance divided by the harmonic mean of the two counts.

The simulator built these data with a person-effect correlation of r = .75 between every pair of events, which is a realistic value for correct-trial and error-trial ERN person effects. Take GroupA, `inc_err` minus `inc_cor`. The person SDs are 2.0 and 1.7, so

```
numerator = 2.0^2 + 1.7^2 - 2 x .75 x 2.0 x 1.7
          = 4.00 + 2.89 - 5.10
          = 1.79
```

Of the 6.89 microvolts squared of person variance that go in, 1.79 come out: 26% survives. For GroupB, with person SDs of 2.2 and 1.7,

```
numerator = 4.84 + 2.89 - 2 x .75 x 2.2 x 1.7 = 7.73 - 5.61 = 2.12
```

Now the error. At 15 error trials and 30 correct trials, the trial-level variance of each condition is its trial main effect plus its trial noise (1.0^2 + 5.5^2 = 31.25 for GroupA errors, 1.0^2 + 5.0^2 = 26.00 for corrects). The trial main effects were drawn independently per condition, so no covariance enters, and with no residual covariance between the two conditions,

```
absolute error = 31.25 / 15 + 26.00 / 30 = 2.083 + 0.867 = 2.950
D_diff = 1.79 / (1.79 + 2.950) = .378
```

The two constituents at those same trial counts are .658 and .769 (Table 10.1 and the same formula at n = 30). The difference lands at .378. Nothing has gone wrong. The subtraction threw away three quarters of the signal and kept all of the noise.

Table 12.1. True dependability of the `inc_err` minus `inc_cor` difference, by trial counts. n1 is error trials, n2 correct trials.

(n1, n2)	GroupA D	GroupB D
(10, 25)	.301	.239
(15, 30)	.378	.312
(20, 35)	.437	.370
(40, 40)	.556	.505
(100, 100)	.758	.718

Even at a hundred trials in each condition, neither group's difference score reaches .80. It would get there eventually, because the error term still shrinks with n, but slowly: solving for .80 gives 128 trials per condition in GroupA and 158 in GroupB. Compare those with the 32 and 48 error trials the constituent scores need for the same threshold (Table 10.1). Shrinking the numerator by three quarters roughly quadruples the trial count the difference requires.

Table 12.2. What the person-effect correlation costs, GroupA at 15 and 30 trials. Only r changes; everything else is held at the generator's values.

r	Numerator	D_diff
.00	6.89	.700
.50	3.49	.542
.75	1.79	.378
.90	0.77	.207

This is the table to keep in mind when someone reports that their difference score is unreliable. The usual response is to collect more trials. Table 12.2 says the binding constraint is often r, not n, and more trials cannot touch r.

12.2 Running the analysis

Load `test_data/manual_ern_singlesession.csv` and set up Specify Inputs exactly as in Tutorial 1 (auto-detect gets `id`, `meas`, `group`, and `event` right, and Occasion ID stays on `(none)`).

Click Select Events and choose `inc_cor` and `inc_err`, and nothing else. Exactly two events are required for a two-event difference score; PsyRAT rejects one or more than two before CmdStan launches.

The difference is formed as the FIRST selected event minus the second, where "first" means the order in which the two events first appear in the file after sorting by group and participant. For this file that is `inc_err` minus `inc_cor`, which is the conventional ERN difference. Direction does not change the coefficient (the universe-score and error terms are both symmetric), but it does determine the sign of the scores, so check it before you interpret a mean.

Click Preferences and change one setting.

Control	Value
Estimate internal consistency of difference scores	Yes: 2-event difference

Leave Estimate residual covariance for co-occurring events on No. That preference exists for events measured on the SAME trial (two components scored from one epoch, say), where the two residuals are correlated by construction. Incongruent error trials and incongruent correct trials are different trials, so their residuals are independent, and asking PsyRAT to estimate a covariance between them would fit a parameter the design cannot identify. Leave the iteration settings at their defaults (4 chains, 5,000 warmup, 5,000 sampling, seed 12345).

Click Save, then Analyze, and choose a whitespace-free output path.

Checkpoint: the Command Window should print "Model converged", with a maximum R-hat of 1.00 and 0 divergent transitions.

12.3 The viewer

The View Results Setup screen looks slightly different for a difference-score run. Where a plain single-session run offers G-Theory Coefficient, this one offers Difference-Score Coefficient, with the same two options.

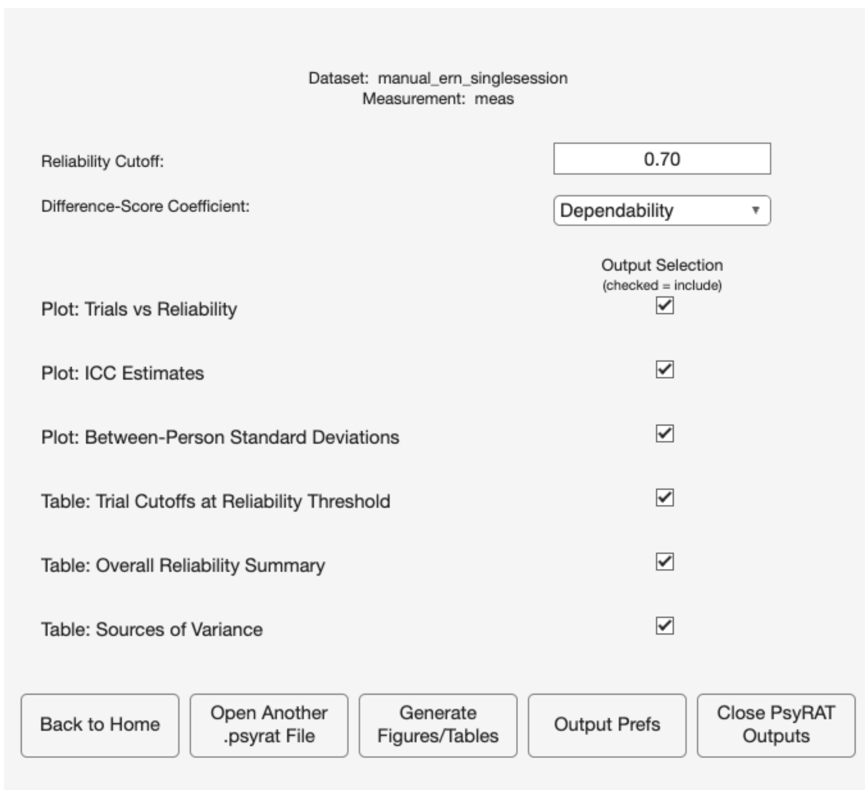


Figure 58: View Results Setup after a two-event difference-score run. The G-Theory Coefficient control is replaced by Difference-Score Coefficient.

Control	Value
Reliability Cutoff:	`0.70`

Use .70, as Tutorial 1 did: every per-event cutoff lands inside the observed trial counts, so those rows come back with numbers. The difference-score rows are another matter, and that is the lesson: the same threshold the constituents can meet sends the difference score's common cutoff far beyond the data. Those rows still print the cutoff and a coefficient at it, and the Extrapolation beyond data dialog is the honest report of it (Section 10.8 covers the -1 sentinels, which do not appear on these rows).

Difference-Score Coefficient stays on Dependability. On this analysis, that one visible control governs both the difference-score row and the per-event rows, so everything in the output follows the coefficient you pick here; the stored G-Theory Coefficient preference is not read.

Click Generate Figures/Tables.

12.4 Reading the output

The trial-cutoff table and the overall summary each carry an extra row per group, labeled `GroupA - diff score` and `GroupB - diff score`, alongside the per-event rows you already know. The Sources of Variance table does not: for a difference-score run its difference information lives in a separate window, described at the end of this section.

The trial-cutoff table

Label	Trial Cutoff	Dependability
GroupA - inc_err	22	.70 [.55, .82]
GroupA - inc_cor	19	.71 [.59, .81]
GroupA - diff score	114	.70 [.39, .87]
GroupB - inc_err	32	.71 [.53, .83]
GroupB - inc_cor	21	.71 [.58, .82]
GroupB - diff score	97	.70 [.40, .87]

The diff score row's Trial Cutoff is a single COMMON trial count, applied to both conditions at once, and not the pair of per-event cutoffs above it. That is a deliberate choice: a reader of this row wants to know how many trials the difference score needs, and two separately derived counts (each chosen so its own event clears the threshold) do not answer that question. Holding the two counts equal also removes the case where unequal counts make the error variance misbehave.

From the truth values, GroupA's difference needs a common count of 75 trials in each condition to reach .70, and GroupB's needs 92. Solve 1.79 x .3 = .7 x 57.25 / n for GroupA, and 2.12 x .3 = .7 x 83.25 / n for GroupB (each condition's error term is its trial main effect plus its trial noise). Both counts are far beyond what any participant supplied (the largest observed error-trial count in the file is 36), so PsyRAT raises a non-modal dialog titled Extrapolation beyond data:

> Not enough trials are present in the current data Cutoffs represent an extrapolation beyond the data

That dialog is the correct outcome and the substantive result. Read the number, note that it is an extrapolation, and do not report it as though participants supplied that many trials.

The overall summary

Label	Dependability	Mean # of Trials	True D
GroupA - inc_err	.75	28.6	.774
GroupA - inc_cor	.80	31.3	.781
GroupA - diff score	.40	(uses both counts)	.475
GroupB - inc_err	.72	34.4	.727
GroupB - inc_cor	.78	30.0	.783
GroupB - diff score	.46	(uses both counts)	.447

This is the table to read. The diff score row is the difference-score dependability evaluated at the mean trial counts the retained participants actually contributed in each condition, which is the score a study would carry forward. Put the three rows for a group side by side and the cost of the subtraction is visible in one line: two constituents between .72 and .80 produce a difference below .47.

The True D column is computed at the trial counts the TRUE .70 cutoffs retain: means of 26.724 and 32.000 in GroupA and 31.462 and 32.538 in GroupB, computed from the committed dataset (Tutorial 1 prints the error-trial pair; the correct-trial pair comes from the same retention rule). For GroupA's difference, 1.79 / (1.79 + 31.25 / 26.724 + 26 / 32.000) = 1.79 / 3.772 = .475. If your run's estimated cutoffs retained a different set, recompute at the Mean # of Trials it actually reports.

To see the relative-error coefficient instead, set Difference-Score Coefficient to Generalizability and regenerate. Generalizability comes back as .41 for GroupA and .47 for GroupB. The two coefficients differ by the two sigma_i^2 / n terms the absolute coefficient adds; with a trial main effect of 1.0 against trial noise five or more times larger, the gap is real but small (truth: .483 against .475 for GroupA). One caveat when reading the printed pair: switching the coefficient re-derives the trial cutoffs. In GroupB the generalizability cutoffs retain two more participants than the dependability cutoffs did, so GroupB's two values are evaluated at slightly different mean trial counts as well. GroupA's retained set is the same in both passes, and there the difference is the formula alone.

All standard deviation components

The Sources of Variance window carries a second button for a difference-score run: View All Standard Deviations. Click it and a window titled All Standard Deviation Components opens, with the columns Label, Between-Person Std Dev, BP Covariance Std Dev, Between-Trial Std Dev, BT Covariance Std Dev, Error Std Dev, Error Covariance Std Dev, and ICC (n'=1).

The covariance columns are populated on the `diff score` rows and dashed out on the per-event rows; the standard-deviation columns are the reverse. Every covariance is reported on the standard-deviation scale, as the square root of the covariance, so it can be read against the standard deviations beside it.

Quantity	Truth	Estimate
GroupA BP Covariance Std Dev	1.60	1.66
GroupB BP Covariance Std Dev	1.67	1.68
GroupA Error Covariance Std Dev	0.00	0.00
GroupA diff score ICC (n'=1)	.030	.02
GroupB diff score ICC (n'=1)	.025	.03

Quantity	Truth	Estimate

The truth values for the covariance columns are the square roots of the generating covariances: sqrt(.75 x 2.0 x 1.7) = sqrt(2.55) = 1.60 for GroupA and sqrt(.75 x 2.2 x 1.7) = sqrt(2.805) = 1.67 for GroupB. The error covariance is exactly zero because it was not estimated, and it prints as a bare number with no interval for the same reason.

The ICC column is labeled ICC (n'=1), and on the `diff score` rows it is a DERIVED single-replication quantity: the same variance ratio as the coefficient, evaluated at one trial in each condition. Rocha et al. (2026, Table 6) tabulate no ICC for difference scores, so this is the toolbox's own convention rather than a formula from that table, and the column heading says so. The truth values above were computed the same way: 1.79 / (1.79 + 31.25 + 26.00) = .030 for GroupA. Report it as a single-trial quantity if you report it at all.

12.5 When is a difference score worth it?

The psychometric answer is unhappily simple: a subtraction-based difference score is worth its cost only when the two conditions are weakly correlated across people, or when the difference is what the theory is actually about and the constituents are not.

Clayson, Baldwin, and Larson (2021) worked this through for ERN and reward positivity difference scores in the same sample and reached a conclusion worth repeating: generalizability theory gives more suitable internal-consistency estimates for subtraction-based difference scores than classical test theory does, and the estimates it gives are frequently low. Their paper also covers residualized difference scores, which are a different construction with a different psychometric profile, and is the reference to read before committing to one.

Three things follow for practice. Report the difference score's reliability separately, because the constituents' reliabilities do not imply it and are usually much higher. Report the person-effect correlation between conditions, or at least the covariance, because it is the quantity that determines whether the difference can be reliable at all. And when the difference score's reliability is low, say so and interpret the effect sizes accordingly rather than quietly switching to the constituent that reads better.

I have no general rule to offer about when to subtract. Whether a difference score is the right dependent variable is a theoretical question about what the subtraction is supposed to remove, and the psychometrics can only tell you what it will cost. The toolbox gives you the cost; the decision stays yours.

12.6 Reporting this analysis

The Chapter 17 template, pre-filled.

> The internal consistency of the incongruent error minus incongruent correct difference score was estimated with generalizability theory as implemented in PsyRAT, using the one-facet difference-score model (Rocha et al., 2026, Table 6) with Bayesian variance-component estimation in CmdStan (4 chains, 5,000 warmup and 5,000 sampling iterations, seed 12345; maximum R-hat = 1.00, 0 divergent transitions). Residual covariance between conditions was not estimated, because the two conditions comprise different

> trials. Dependability of the difference score was .40 in GroupA and .46 in GroupB at mean trial counts of 28.6 and 31.3 (GroupA) and 34.4 and 30.0 (GroupB). The constituent scores were considerably more dependable (.75 and .80 in GroupA; .72 and .78 in GroupB), reflecting the between-person covariance between conditions (covariance standard deviation 1.66 and 1.68 microvolts), which the subtraction removes. A decision study indicated that 114 and 97 trials per condition would be required for the difference score to reach a dependability of .70; both counts extrapolate beyond the observed trial counts and should be read as such.

Never report the constituents' reliabilities as though they described the difference. That substitution is common in the ERP literature, and Table 12.1 shows the size of the error it introduces.

12.7 Where to go next

- Data that arrive as split means rather than single trials: Chapter 13.
- Four-event differences of differences and other advanced contrasts: Chapter 15.

13. Tutorial 4: Reliability from data splits

Sometimes single-trial scores are gone. A collaborator sends averages from six blocks; a published dataset supplies odd-trial and even-trial means; a preprocessing pipeline exports per-run averages and nothing finer. The variance components a one-facet model needs are still recoverable from those averages, provided one extra piece of information travels with them: how many trials went into each average.

That number is what makes the analysis possible, and PsyRAT requires it. This tutorial runs `test_data/manual_ern_splits.csv` through the splits path. It assumes Chapter 10.

13.1 The splits input contract

A splits file is long format, one row per SPLIT rather than one row per trial, with three required columns.

- `id`: the participant identifier, as always.
- `meas`: the split MEAN, not a single-trial score.
- a weight column: n_i, the number of items (trials) averaged into that split.

Optional `group`, `event`, and `time` columns behave as they do elsewhere, subject to the design restriction in Section 13.7. Chapter 6 covers file preparation in general.

The weight column is not optional and it is not a precision hint. It is a structural parameter of the model: the residual variance of a split mean is the per-item residual variance divided by n_i, so the estimator cannot separate the item-level residual from the person-by-split interaction without variation in n_i. PsyRAT checks the column on load and requires positive whole numbers. A value of 0, a negative value, a fraction, or a non-numeric entry stops the run with an error naming the column.

`manual_ern_splits.csv` has 36 participants (ids 301 to 336), sixteen split means each, 576 rows, and weights drawn between 4 and 40 trials per split. The unequal weights are deliberate: they are what puts this file on the nonparallel path.

13.2 Parallel or nonparallel

The Split type control asks you to declare which kind of splits you have, and it is a declaration about the design rather than a request for a computation.

Parallel (equal n_i) means every split holds the same number of items. PsyRAT then reuses the standard one-facet or test-retest machinery unchanged: the split mean is simply the score, and n' becomes the number of splits instead of the number of trials. Every output you know from Tutorial 1 appears, with ”trial” relabeled ”split” throughout.

Nonparallel (unequal n_i) means the splits hold different numbers of items. PsyRAT then fits a weighted, observed-design model in which each split mean's residual is scaled by its own n_i. That model is identified ONLY by variation in n_i, which is why declaring nonparallel on a file with equal weights will not work, and why declaring parallel on this file would silently treat sixteen unequal averages as interchangeable.

Check the weights before you declare. This file has unequal weights, so it is nonparallel.

13.3 What the true values are

The generating model is persons crossed with splits on the item scale. Each split mean is a grand mean plus a person effect, a split main effect (shared by every participant at that split index), a person-by-split interaction, and residual noise whose variance is the per-item variance divided by that split's n_i:

```
meas(p, s) = mu + person(p) + split(s) + ps(p, s) + e(p, s),
             e ~ Normal(0, sigma_item^2 / n_i)
```

with mu = -3.5 microvolts, sigma_p = 2.0, sigma_s = 0.5, sigma_ps = 0.7, and sigma_item = 5.0. The two split-level components are small on purpose: large enough that the model has something real to find, small enough that the item-level noise dominates the residual, as it does in split data from an ERP task.

Dependability of an equal-weight mean of n_s split means uses the HARMONIC mean of the items per split, because that is the exact residual scale when split means of different sizes are averaged with equal weight. For this file the harmonic mean over all 576 splits is

```
n-bar_i = 576 / sum(1 / n_i) = 14.757
```

against an arithmetic mean of 21.87. The gap between those two numbers is the price of equal-weight averaging over unequal splits, and it is real: the design behaves like 16 x 14.757 = 236.1 items per participant rather than the 349.9 items that were actually averaged.

The true dependability at the observed design (16 splits) is

```
item part of one split mean's variance = 5.0^2 / 14.757 = 1.694
split-level part = sigma_ps^2 + sigma_s^2 = 0.49 + 0.25 = 0.74
absolute error variance of a 16-split mean = (1.694 + 0.74) / 16 = 0.152
D = 2.0^2 / (2.0^2 + 0.152) = 4.00 / 4.152 = .963
```

and the absolute standard error of measurement is sqrt(0.152) = 0.39 microvolts. The true generalizability coefficient drops the split main effect from the error, (1.694 + 0.49) / 16 = 0.137, and comes out at 4.00 / 4.137 = .967, with a relative standard error of measurement of 0.37. Section 13.6 explains why PsyRAT reports the second one as a bound.

For comparison, a straight mean of all 349.9 items, with no split structure at all, would have reached 4.00 / (4.00 + 25 / 349.9) = .982. The two hundredths between that and .963 split about evenly between the two split-level components, which averaging over sixteen splits shrinks by sixteen rather than by three hundred and fifty, and the harmonic-mean penalty for unequal weights. Both costs are small, and neither is zero.

13.4 Running the analysis

Load `test_data/manual_ern_splits.csv` through Process New Data.

Checkpoint: the Specify Inputs screen should show these selections.

Control	Value
Participant ID:	`id`
Measurement:	`meas`
Group ID:	`(none)`

Control	Value
Event Type:	`(none)`
Occasion ID:	`(none)`
Items per split (n_i):	`(none)`

Auto-detect finds `id` and `meas`. It does NOT find the weight column, and it never will: PsyRAT cannot tell a column of item counts from any other integer column, and guessing wrong would change the design silently. Set the last two controls yourself.

Control	Value
Items per split (n_i):	`weight`
Split type:	Nonparallel (unequal n_i)

Selecting a column under Items per split (n_i) is what declares that each row is a split mean rather than a single trial. Leave it on `(none)` and PsyRAT will run an ordinary one-facet analysis without complaint, because nothing in the file itself says otherwise.

What comes back is then mislabeled rather than obviously wrong, which is worse. The sixteen averages are treated as sixteen trials, so every "trial" in the output is really a split; the decision study projects over hypothetical numbers of SPLITS while calling them trials, and a trial cutoff of 9 would mean nine splits, not nine trials; and the per-item residual, which is the quantity the splits model exists to recover, is never estimated at all. The coefficient at the observed design happens to come out close either way, because a common residual fitted to split means lands near the same place. That coincidence is exactly why the mislabeling is easy to miss. Declaring the column costs one click.

Leave the processing preferences at their defaults: 4 chains, 5,000 warmup, 5,000 sampling, seed 12345.

Click Analyze and choose a whitespace-free output path.

Checkpoint: the Command Window should print "Model converged", with a maximum R-hat of 1.00 and 0 divergent transitions.

13.5 The splits viewer

A nonparallel splits run opens its own viewer, not the single-session one.

The window is titled Data-Splits Reliability: View Results. Under the dataset name sits a banner reading "Analysis unit: SPLITS (each score is the mean of n_i items)", and under that, in red, the caveat this chapter's Section 13.6 explains:

> Dependability is exact; generalizability is a downward-biased lower bound.

Two controls and five buttons follow.

- Credible interval (0-1): the interval width, default `0.95`.
- Splits n' (blank = observed median): the number of splits the coefficients are reported at. Blank uses each stratum's observed median, which is 16 here.

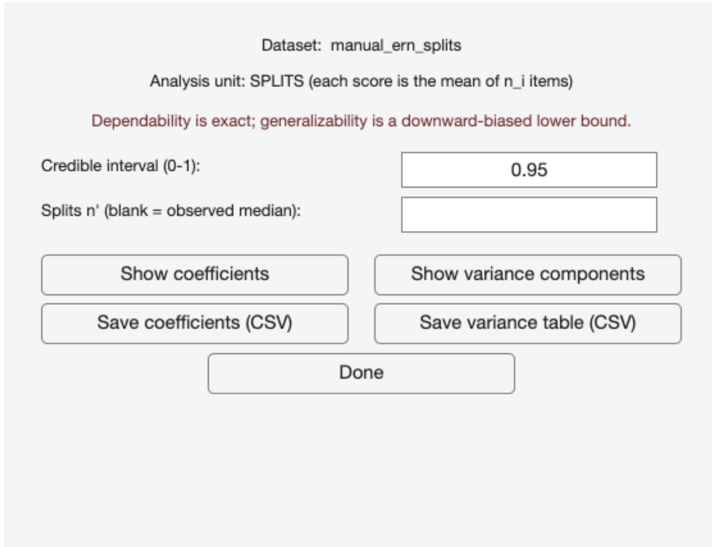


Figure 59: Data-Splits Reliability: View Results. The banner names the analysis unit and the red line states the bias caveat that applies to every relative coefficient below.

- Show coefficients, Show variance components, Save coefficients (CSV), Save variance table (CSV), and Done.

A third control, Occasions n' (blank = observed), appears only for the multi-occasion split design. This file has one occasion, so it is not there.

Leave both controls as they are and click Show coefficients.

Notice what this viewer does NOT have: no Reliability Cutoff, no dependability-versus-splits plot, no split-cutoff table, and no included and excluded counts. That absence is the design, not an omission, and Section 13.7 explains it.

The coefficients table

The window is titled Data-splits reliability: coefficients. Its columns are Stratum, Coefficient, n splits, n occasions, Dependability, D low, D high, Generalizability (lower bound), G low, G high, SEM (abs), and SEM (rel, upper bound). The bias note is repeated in full under the table.

There is one row, because this design has one stratum and one coefficient.

Column	Value	Truth
Stratum	measure	
Coefficient	Internal consistency (equivalence)	
n splits	16	16
n occasions	1	1
Dependability	.97	.963
D low, D high	.96, .98	
Generalizability (lower bound)	.97	.967
G low, G high	.96, .99	
SEM (abs)	0.39	0.39

Column	Value	Truth
SEM (rel, upper bound)	0.37	0.37

Dependability near .97 is high, and it should be. Sixteen splits of about fifteen effective items each is over two hundred effective items per participant, and the same components at 50 single trials would give only 4.00 / (4.00 + 25 / 50) = .889. The splits path is not doing anything more efficient than ordinary averaging. It is starting from data that have already been averaged a great deal.

The variance components table

Click Show variance components. The window is titled Data-splits reliability: variance components, with the columns Stratum, Component, Symbol, Estimate, CI lower, and CI upper.

These are VARIANCES, not standard deviations. The truth column below squares the generator's SDs accordingly.

Component	Symbol	Truth	Estimate	95% CrI
Between-person variance (universe score)	sigma^2_p	4.00	5.52	[3.37, 9.03]
Split main-effect variance	sigma^2_s	0.25	0.24	[0.08, 0.56]
Person x split variance	sigma^2_ps	0.49	0.71	[0.28, 1.13]
Per-item residual variance (item:split lumped)	sigma^2_(i:s),e	25.00	21.49	[13.78, 30.85]

The per-item residual is the interesting one. The data contain no item-level scores at all, and the model still recovers the item-level variance, because splits built from different numbers of items shrink at different rates and that pattern identifies the per-item scale. Recovering a per-item variance whose credible interval covers the true 25 from a file whose smallest unit is a split mean of anywhere from 4 to 40 trials is the whole trick of the nonparallel design, and it is worth a moment's appreciation before Section 13.6 explains what it cannot do.

The Save coefficients (CSV) and Save variance table (CSV) buttons export either table to `.xlsx` or `.csv` with a provenance header carrying the seed, the engine, the credible-interval width, the analysis-unit banner, the convergence verdict, and the bias note. Export both.

The two split-level components should come back small, with wide intervals: a split main effect is estimated from sixteen split indices and a person-by-split interaction from sixteen values per participant, so neither is pinned down tightly, and that is the correct amount of uncertainty for what sixteen splits can tell you. In your own data a substantial split main effect would mean the splits differ systematically (blocks recorded at different times, say), and a substantial person-by-split variance would mean participants differ in how they change across splits. Either is worth knowing about before you average the splits together.

This recovery is an illustration. Thirty-six participants cannot establish that an estimator is unbiased, and this run is not evidence that it is; PsyRAT's correctness claims rest on the validation suites described in `README.md`.

13.6 Why generalizability is only a lower bound

The caveat line in the viewer is not boilerplate, and it has a specific cause.

Dependability, the absolute-error coefficient, counts every source of within-person variation as error. Generalizability, the relative-error coefficient, excludes the item main effect, because a constant shift affecting all participants equally does not disturb their rank ordering. Excluding that term is what makes generalizability the larger of the two coefficients in an ordinary one-facet analysis (equal when the excluded term is zero).

From split means, the item main effect cannot be separated out. A split mean has already collapsed its items, so the item-level partition is gone, and the per-item residual PsyRAT estimates necessarily lumps the item main effect together with the person-involving item residual. The absolute coefficient does not care, because it was going to count both terms as error anyway, and it is computed exactly. The relative coefficient does care: it subtracts a term it cannot see, so it subtracts nothing, leaves too much in the error, and comes out too low. Its SEM is correspondingly too high.

Hence the labeling. Generalizability (lower bound) and SEM (rel, upper bound) are honest column headings, and the toolbox flags every relative coefficient it reports from split data. Use the dependability coefficient when you can. If your question genuinely requires the relative coefficient, report the value as a bound and say why.

In this file the bound happens to be tight, because the simulator included no item main effect for the estimator to miss. Do not read that as reassurance about your own data. The toolbox cannot tell a tight bound from a loose one, and neither can you, from split means alone.

13.7 No D-study for the nonparallel design

There is no dependability-versus-splits plot here and no split-cutoff table, and the reason is the same identifiability collapse.

A D-study projects reliability onto hypothetical designs: what would this score look like at 20 trials, at 40, at 100? That projection requires knowing how each variance component scales with the count being varied, and for the nonparallel split design the item-level partition needed to project over ITEM counts is not recoverable from split means. PsyRAT therefore reports coefficients at the OBSERVED design only, and offers exactly one projection it can defend: the Splits n' box, which reports the coefficients at a different number of SPLITS, holding the harmonic-mean items-per-split fixed at what the data supplied.

Try it. Enter `3` under Splits n' and click Show coefficients again, then try `12`. At the true components, the same arithmetic as Section 13.3 (2.434 of variance per split mean, item part plus split-level part) gives 4.00 / (4.00 + 2.434 / 3) = .831 at three splits and 4.00 / (4.00 + 2.434 / 12) = .952 at twelve. The viewer does not know the true components. It recomputes from the fitted ones, holding the harmonic-mean items-per-split fixed at what the data supplied, and the fitted between-person variance came out at 5.52 rather than 4.00, so the printed values run higher. Plugging the variance-components table into the same formula (0.24 + 0.71 + 21.49 / 14.757 = 2.41 of variance per split mean) gives 5.52 / (5.52 + 2.41 / 3) = .87 at three splits and 5.52 / (5.52 + 2.41 / 12) = .96 at twelve. The viewer prints .87 at three splits and .96 at twelve. It averages the coefficient over the posterior draws rather than plugging in point estimates, so it need not match the plug-in arithmetic to the second decimal. Those are legitimate projections because the number of splits is a count the design does identify.

What you cannot ask is "what if each split had 40 items instead of about 15", and the viewer does not offer it. If you need that answer, you need the single-trial data, and Chapter 10 is the analysis to run on them.

One further restriction, so it does not surprise you mid-run: the single-occasion nonparallel splits design does not support group or event facets. It is fitted as one pooled model over all participants. Adding an occasion column moves you to the two-facet split design, which IS group and event aware.

Point Group ID or Event Type at a real column on a single-occasion splits file and the run stops rather than pooling your groups silently. The Command Window prints "Measurement failed:" with the measurement column name, then a "Reason:" line, PsyRAT writes a `.txt` error report next to the output path, and a modal dialog reports that the measurement failed and points you at that report. Nothing is estimated, and nothing wrong is saved. Other combinations the splits path does not support (difference scores, subject-level error variances, dynamic reliability) are caught earlier, by a modal Reliability with splits dialog raised before CmdStan launches.

13.8 Reporting this analysis

The Chapter 17 template, pre-filled.

> Scores were available as block means rather than single trials, so reliability was estimated with the observed-design data-splits model implemented in PsyRAT, in which each score is the mean of n_i items and n_i is supplied per split (Bayesian estimation in CmdStan; 4 chains, 5,000 warmup and 5,000 sampling iterations, seed 12345; maximum R-hat = 1.00, 0 divergent transitions). Splits were nonparallel (items per split ranged from 4 to 40; harmonic mean 14.76). Coefficients are reported at the observed design of 16 splits per participant; no decision study over item counts is identified from split means. Dependability was .97, 95% CrI [.96, .98], with an absolute standard error of measurement of 0.39 microvolts. The between-person variance was 5.52 and the per-item residual variance 21.49. The generalizability coefficient (.97) is reported as a downward-biased lower bound, because the item main effect cannot be separated from the person-involving item residual when only split means are available.

State that the data were splits, state the items-per-split distribution, and state the design the coefficients were reported at. A dependability of .97 from sixteen averages of 4 to 40 trials is a different claim from a dependability of .97 from sixteen trials, and a reader who is given only the coefficient cannot tell them apart.

13.9 Where to go next

- The complete set of analyses and when each applies: Chapter 16.
- Running these analyses from a script instead of the GUI: Chapter 14.

14. Scripting with psyrat_run

Almost everything the Process New Data screens do can also be done from a script; Sections 14.3 and 14.6 name the exceptions. `psyrat_run` is the command-line entry point. It accepts the same settings the GUI collects, in the same units, with the same defaults, and it validates them against the same rules.

14.1 Why script an analysis

If a study has eight measurement columns and two datasets, the GUI asks for sixteen trips through the same screens. A loop asks once. Batch work is the most common reason to move to the command line, and the saving compounds every time a preprocessing change sends you back through all sixteen.

A script is also the record. It states the seed, the iteration counts, the column mapping, and the design flags in a form a reviewer can read and a collaborator can execute. Nothing has to be reconstructed from memory when the revise-and-resubmit arrives eleven months later.

The third reason is hardware. A cluster node or a remote session with no display cannot open a GUI at all. `psyrat_run` never opens one unless you ask it to.

None of that would be worth much if the scripted path were a second implementation of the analysis. It is not. `psyrat_run` builds the same preferences and data scaffold the GUI builds, then hands them to the same loader and the same estimation wrapper the GUI calls. Given the same inputs and the same seed, a scripted run produces the same variance components and the same convergence diagnostics as the GUI, and writes the same `.psyrat` file.

What a script needs installed is what the GUI needs installed, and nothing more: a MATLAB release the toolbox runs on, a working CmdStan for the full Bayesian path, and the patched MatlabStan that ships inside `bundled_dependents/` (Chapter 3). The scripted path adds no requirement of its own, and it removes none either.

Before scripting, get the toolbox on the MATLAB path by running `psyrat_start` once in the session. That adds the toolbox root, `subroutines/`, and `bundled_dependents/`. Do not use `addpath(genpath(...))` on the whole toolbox folder. Otherwise, stale copies sitting in tooling or artifact folders can shadow the real files, and you will be running code you cannot locate.

One more thing to know before the first call: `psyrat_run` analyzes one measurement column per call. The GUI can queue several measurements behind a single press. A script loops instead, which is what Section 14.3 shows.

14.2 Tutorial 1 as a script

Tutorial 1 (Chapter 10) walks the single-session ERN dataset through the GUI, and the script below runs that same analysis. Start with this preamble, because every block after it reuses these variables.

```
% Locate the toolbox root from a file that lives in it, rather than hardcoding
% a path. which() reports where MATLAB actually found psyrat_run, so this also
% fails loudly if the toolbox is not on the path yet.
psyratroot = fileparts(which('psyrat_run'));
datafile   = fullfile(psyratroot,'test_data','manual_ern_singlesession.csv');
```

```
% Output directory. It must already exist when psyrat_run is called: the
% toolbox writes into it, but it does not create it for you.
outdir = fullfile(tempdir,'psyrat_manual');
if ~exist(outdir,'dir')
    mkdir(outdir);
end

% CmdStan rejects whitespace anywhere in the output path or the output
% filename. Check here rather than after a failed model compile.
if any(isspace(outdir))
    error('Choose an output directory with no spaces in it: %s', outdir);
end
```

The estimation call comes next. The tutorial dataset uses the canonical column names (`id`, `meas`, plus `group` and `event`), so `idcol` and `meascol` could both be omitted. They are written out once here so the mechanism is visible: any file whose id or score column is named something else needs these two arguments, and most real files do.

```
psyrat_data = psyrat_run( ...
    'file',      datafile, ...     % long format: one row per single-trial score
    'idcol',     'id', ...         % participant id column (this IS the default)
    'meascol',   'meas', ...       % numeric score column to analyze (also default)
    'groupcol',  'group', ...      % between-person grouping column
    'eventcol',  'event', ...      % condition column
    'whichevents', {'inc_err', 'inc_cor'}, ... % the two events Tutorial 1
     ↪  selects
    'savepath',  outdir, ...       % must exist, and must contain no whitespace
    'savename',  'manual_ern.psyrat', ... % '.psyrat' is added if you omit it
    'chains',    4, ...            % default 4; 3 is the enforced floor for R-hat
    'warmup',    5000, ...         % default
    'sampling',  5000, ...         % default
    'seed',      12345);           % default; fix it, and report it in the paper
```

Success! Two files now exist in `outdir` and one structure exists in the workspace. `manual_ern.psyrat` holds the estimated result, `manual_ern_runconfig.json` records the settings that produced it (Section 14.4), and the returned `psyrat_data.rel` holds the variance components and the convergence diagnostics. Reopen the saved file later with the GUI View Results button, or read it back in a script. It is a MAT-file under a different extension, and it contains one variable:

```
S = load(fullfile(outdir,'manual_ern.psyrat'), '-mat');
psyrat_data = S.psyrat_data;
```

Budget for the first call to cost more than the ones after it. Every generated model is compiled once before it can be sampled, and a run that fits four events compiles four models (Chapter 3 records the measured compile time). Sampling time after that scales with the data and the iteration counts. The `psyrat_relsummary` and `psyrat_report` steps below add seconds rather than minutes, because neither one samples.

Two variations on the input are worth having before the first real dataset. Passing `data` with a table already in the workspace replaces `file`; supply exactly one of the two, never both. This is the route to take when the scores are built or filtered in MATLAB rather than read from disk, and it is how the import and scoring screens hand their result to an analysis (Chapter 8).

```
tbl = readtable(datafile);      % or any long-format table you built yourself
psyrat_data = psyrat_run('data', tbl, ...
    'savepath', outdir, ...
    'savename', 'manual_ern_inmemory.psyrat');
```

The second is `timecol`, which is the argument that selects a test-retest design. Naming an occasion column declares the occasion facet, and the analysis changes accordingly; the tutorial file here is single-occasion, so the examples in this chapter stay one-facet. On a repeated-measures file the change is `'timecol','session'` alongside the other column arguments (Chapter 11).

Estimation stops at the variance components. Two more calls turn those into the tables the results viewer shows, and both run headlessly. `psyrat_relsummary` is the compute layer: it runs the D-study and applies the trial-count cutoff, which determines the participants and conditions that clear the threshold. The estimand choices are made there, not later.

`psyrat_report` then assembles the result into MATLAB tables and, when given an `outdir`, writes them to disk. It never re-estimates and never re-runs the D-study; nothing it does opens a window. Every exported table carries a header recording the seed, the analysis, the engine, the priors preference, the convergence verdict, and the divergent transition count, so a saved table can be read on its own. Tables built from a summary also record the credible-interval width the summary was built at; the families that never build one (data splits, dynamic reliability, difference-of-differences) carry no width line. `psyrat_report` computes its own tables at that same stored width unless you pass `'CI'` explicitly; an explicit width that differs from the stored one is disclosed by an extra header note naming both.

```
% Step 1: the D-study. 'analysis' is a dispatch key (see the table below), not
% the name of the analysis that was estimated.
[psyrat_data, relerr] = psyrat_relsummary('psyrat_data', psyrat_data, ...
    'analysis',    'sing', ... % one-facet route
    'depcutoff',   .80, ...    % dependability threshold for the trial cutoff
    'meascutoff',  2, ...      % 1 lower CI limit, 2 point estimate, 3 upper
    'depcentmeas', 1, ...      % 1 mean, 2 median, for the overall estimate
    'gcoeff',      1);         % 1 dependability, 2 generalizability

% Stop before reporting if the summary aborted. The GUI raises a dialog at
% that point; a script has to test the flag itself. Note the flag's scope:
% on most designs an empty threshold result no longer aborts (the summary
% returns in full with a console line and n Included 0 rows), so also check
% that someone was retained before treating the tables as informative.
if relerr.nogooddata == 1
    error('The summary aborted, so there is nothing to report.');
end
if isempty(psyrat_data.relsummary.group(1).goodids)
    warning('No participant cleared the threshold; the tables report a zeroed
     ↪  group.');
end
```

```
% Step 2: assemble and export. Omit 'outdir' to get the tables back without
% writing any files.
report = psyrat_report(psyrat_data, ...
    'outdir', fullfile(outdir,'tables'), ...
    'format', '.csv');          % '.xlsx' is the default

disp(report.tables.overall);    % also .cutoff and .curve for this family
```

A summary can also finish without aborting and still have no cutoff to report. When no trial count in the projected range reaches the threshold, or when the count it finds lies beyond every participant's observed trials, the affected row of the cutoff table prints `-1` in place of the cutoff and the coefficient. The GUI raises a dialog for each case (Cutoff not calculable, Extrapolation beyond data); a script sees no dialog, so `psyrat_report` returns the explanation in `report.cutoff_note` and writes the same sentence into the header of every table it exports. Read that note before quoting a cutoff; it is empty for a clean run.

A `.psyrat` file saved by the GUI carries no record of how it will be summarized later, so `psyrat_relsummary` treats it as interactive and can raise the threshold-failure dialog when no participant clears the cutoff. Inside a batch loop, pass `'interactive', false` to `psyrat_relsummary`; the condition is still reported through `relerr.nogooddata`, and files written by `psyrat_run` already carry the headless flag.

Which `analysis` value, and which arguments

`psyrat_relsummary`'s `analysis` argument is a dispatch key, and for most designs it is not the string stored in `psyrat_data.rel.analysis`. Passing the stored string instead of the key is the single most common scripted mistake. The mapping:

`psyrat_data.rel.analysis`	Design	`analysis` to pass	Required arguments	Optional
`ic` (or field absent)	one-facet	`sing`	`depcutoff`, `meascutoff`, `depcentmeas`	`gcoeff` (1)
`ic_sserrvar`	one-facet, subject-level	`sing_sserr`	`depcutoff`, `meascutoff`, `depcentmeas`	`gcoeff` (1)
`ic_diff`	one-facet difference	`sing_diff`	`depcutoff`, `meascutoff`, `depcentmeas`	`gcoeff` (1), `diffgcoeff` (1)
`ic_diff_sserrvar`	one-facet difference, subject-level	`sing_diff_sserr`	`depcutoff`, `meascutoff`, `depcentmeas`	`gcoeff` (1), `diffgcoeff` (1)
`trt`	test-retest	`trt`	`gcoeff`, `reltype`, `relcutoff`, `meascutoff`, `relcentmeas`	`noccmode` (1), `nocc`

psyrat_ data.rel. analysis	Design	analysis to pass	Required arguments	Optional
trt_ sserrvar	test-retest, subject-level	trt_sserr	reltype, depcutoff, meascutoff	gcoeff (1), noccmode (1), nocc
trt_diff	test-retest difference	trt_diff	gcoeff, reltype, relcutoff, meascutoff, relcentmeas	diffgcoeff (1), noccmode (1), nocc
ic_dodiff	difference-of-differences	do not call		
ic_splits, trt_splits	nonparallel data splits	do not call		
the *dynrel* variants	dynamic/conditional reliability	do not call		

Read that table before writing the call. Its last two columns are where scripted calls go wrong, and four of the entries there are worth spelling out.

depcutoff and relcutoff carry the same GUI input under two names, and each branch accepts only its own spelling. Pass the wrong one and the branch errors saying the argument was not specified, which reads like a missing argument rather than a misspelled one. The subject-level test-retest route is the odd case: it is a two-facet design that still takes depcutoff and meascutoff, because it inherits the subject-level participant-inclusion contract instead of the test-retest one.

gcoeff is required on the trt and trt_diff routes and defaults to 1 (dependability) everywhere else. depcentmeas is required by all four sing* routes, including sing_sserr, where the function's own help notes that its value is never used. Omitting it there still errors. CI (default .95) is accepted by every route.

The last three families never reach psyrat_relsummary at all. Difference-of-differences, nonparallel data splits, and dynamic reliability read psyrat_data.rel directly, so for those you call psyrat_report immediately after psyrat_run. There is no branch in psyrat_relsummary for those strings. Passing one is not merely unnecessary: it falls through the dispatch. Parallel splits are the exception inside the exception, because they reuse the one-facet and test-retest models and therefore carry rel.analysis of ic or trt and follow the ordinary two-step path above (Chapter 13).

Which tables come back depends on the family. The one-facet and test-retest families return .cutoff, .overall, and .curve, with .diffcurve added for trt_diff. The subject-level family returns .variance and .sscoeffs.

Data splits return .coefficients and .varcomp. Dynamic reliability returns .surface and .varcomp, plus .ssrel for the subject-level variants, and difference-of-differences returns .dod_observed and .dod_varcomp.

With the same two events selected, the numbers this script produces are the numbers Tutorial 1 prints. Nothing differs between the two runs except which surface issued the call (leave whichevents out and the script fits all four events, and the retention counts change with the events processed). A

GUI run and a `psyrat_run` call at identical settings produce bit-identical posterior draws, because both surfaces call the same estimation function.

14.3 Running headless

`psyrat_run` defaults to `'interactive', false`, which means no dialogs, ever. Nothing waits for a click. If a model does not converge, the run automatically doubles both the warmup and the sampling iterations and fits again, up to four automatic reruns, then stops with an error naming the measurement that failed. Starting from the default 5,000 and 5,000, that means the last automatic attempt runs at 80,000 and 80,000, so budget the wall-clock time accordingly before launching an overnight batch.

If that error fires, no `.psyrat` file is written. The error is raised before the save, so a non-converged headless run leaves nothing behind that could later be mistaken for a finished analysis. Pass `'interactive', true` to opt back into the GUI rerun prompt, which offers the choice to accept and save the non-converged fit instead. Trace-plot review stays GUI-only on both settings.

A batch loop therefore needs one thing the GUI gives you for free: somewhere for the failures to go. Wrap each call in `try/catch`, record which ones failed, and report at the end rather than watching the console.

```
% One entry per measurement column to analyze. The tutorial file has a single
% score column; a real study file usually has several, and the loop is the same.
meascols = {'meas'};
failed   = {};        % names that errored, reported once when the loop ends

for k = 1:numel(meascols)
    thiscol = meascols{k};
    try
        % Each measurement gets its own output file, so one failure cannot
        % overwrite or invalidate a run that already finished.
        psyrat_run('file', datafile, ...
            'meascol',  thiscol, ...
            'groupcol', 'group', ...
            'eventcol', 'event', ...
            'savepath', outdir, ...
            'savename', sprintf('batch_%s.psyrat', thiscol), ...
            'seed',     12345);
    catch ME
        % Non-convergence after the automatic reruns arrives here, as does any
        % data-validation rejection. Keep going; report the whole set below.
        failed{end+1} = thiscol; %#ok<AGROW>
        fprintf('Skipped %s: %s\n', thiscol, ME.message);
    end
end

if isempty(failed)
    fprintf('All %d measurements finished.\n', numel(meascols));
else
    fprintf('%d of %d measurements failed: %s\n', ...
```

```
        numel(failed), numel(meascols), strjoin(failed, ', '));
end
```

Looping over files instead of columns takes the same shape: build a cellstr of paths, and give each one its own `savename`. Data-level validation (numeric measurement column, event counts, supported design combinations) is enforced by the same validator the GUI uses, `psyrat_preflight_validate`, so a script and the GUI reject the same inputs on the same rules. The wording can differ: the Process New Data screens carry their own earlier copies of several checks so they can warn before a run starts, and those speak in the GUI's voice (Chapter 18).

14.4 Reproducing a run from its sidecar

Every run writes a `*_runconfig.json` sidecar next to its `.psyrat` output, named after it, whether the run came from the GUI or from `psyrat_run`. This is what makes a GUI analysis reproducible from a script.

The sidecar records the estimation settings (seed, chains, warmup, sampling, total iterations, within-chain parallelization, verbosity, and the optional `adapt_delta` and `max_treedepth` overrides), the design flags, the column mapping, the priors structure, and the output location. The priors entry records the full prior set the run actually fit: a partial override passed to `psyrat_run` is merged over the defaults before fitting, and the merged result is what the sidecar stores, so a replay reproduces the same model (see Chapter 15 for setting priors).

The two sampler overrides are recorded for a specific reason, though not quite the same reason. `adapt_delta` sets the dual-averaging target during warmup, so raising it changes the step size the sampler settles on and therefore changes the draws at a fixed seed. `max_treedepth` is a cap on the NUTS trajectory rather than an adaptation setting, so it changes the draws only when the cap actually binds; when trajectories end on the U-turn criterion below the cap, raising it changes nothing. Either way, neither is a cosmetic setting that can be left out of a replay and reconstructed later, which is why the sidecar restores both rather than reverting to the defaults. It also records the provenance a reader needs to judge whether a replay is really a replication: the PsyRAT and MATLAB versions, the CmdStan version detected on the machine that ran it, the source filename with content hashes of both the file and the loaded table, and the row count that entered estimation. The engine appears twice, as the preference that was requested and as the engine that actually ran.

A separate viewer block records the settings that select which coefficient the output was reported under: `gcoeff`, `reltype`, `diffgcoeff`, `noccmode` and `nocc`, and the two viewer cutoffs. That block is record-only. `psyrat_run` exposes no options for it, so a replay reads past it. It is there for the methods section and for anyone reconstructing what a saved table's numbers actually were.

Replaying adds one argument, the sidecar path; the data source and output location are still required:

```
% Settings come from the sidecar. The data source and the output location are
% still supplied normally, so a recorded configuration can be pointed at new
% data without editing the JSON.
psyrat_run('config', fullfile(outdir,'manual_ern_runconfig.json'), ...
    'file',     datafile, ...
    'savepath', outdir, ...
    'savename', 'manual_ern_replay.psyrat');
```

Any option passed explicitly beats the recorded value, so one configuration can be reused across datasets, or its iteration counts raised while everything else stays fixed. The precedence is worth stating the other way round too: an option you do not pass is filled in from the sidecar, including options you may have forgotten it recorded.

Two differences make a replay something less than a replication, and both raise a warning rather than stopping the run. If the CmdStan version detected here differs from the recorded one, or if no CmdStan can be detected at all, the warning names both versions. NUTS draws are not guaranteed identical across CmdStan versions at a fixed seed, so treat a version mismatch as a reason to re-estimate rather than to compare numbers across the two runs. The second warning fires when the recorded `engine_used` is not what the restored preference will select here. A replay restores the engine preference, never the engine that ran, and under the default automatic preference the cascade is re-derived from the engines installed on this machine (Section 14.5).

CAUTION. Gamma-family sidecars written before August 3, 2026 need one argument on replay. The `gammascale` setting picks one of two parameterizations of the Gamma dispersion submodel. The two are different estimands, and their variance components are not comparable. On the designs that support it, the default changed on August 7, 2026, from log-nu dispersion (1) to log residual SD (2).

Runs from August 3, 2026 onward pin the resolved value into the sidecar, so replaying one of those reproduces its own estimand. A sidecar written before that date carries no `gammascale` field, and the replay falls through to whichever default the running toolbox has. Pass `'gammascale', 1` explicitly when replaying one. Otherwise you will silently fit a different model from the one the sidecar describes, and two sets of numbers that are not comparable will look as though they are.

Five designs reject an explicit 1 because they are location-scale only: the subject-level one-facet dynamic difference (analysis 13), the subject-level dynamic concurrent differences at one and two facets (19 and 20), and the person-specific dynamic difference of differences at one and two facets (28 and 29). None of it touches Gaussian runs. Chapter 15 covers the gamma family and what each parameterization means.

14.5 Choosing an engine when scripting

The `engine` option accepts `auto`, `cmdstan`, `fitlme`, or `hmc` (equivalently 0, 1, 2, or 3), and defaults to `auto`. Automatic tries CmdStan first, then the native HMC engine, then the native fitlme engine, considering only engines installed on this machine and warning loudly whenever it falls back off CmdStan. When CmdStan is present and new enough, the cascade takes it silently and the run is exactly the CmdStan run. A CmdStan below the supported minimum is the one case that neither runs nor falls back: `auto` stops with an error telling you to upgrade, on the reasoning that quietly substituting an approximate engine for one the user has installed is the wrong default. Forcing a native engine remains available.

Which engines are legal depends on the design, and that is the thing to settle before reaching for the option at all:

Design	Native engines that implement it
One-facet internal consistency, with or without group and event facets	fitlme only
Nonparallel data splits, single-occasion and test-retest	fitlme only

Design	Native engines that implement it
Plain test-retest	HMC and fitlme, cascade prefers HMC
Subject-level error variance	HMC only
Dynamic/conditional reliability	HMC only
Difference scores, difference-of-differences, and their two-facet and dynamic variants	HMC only

Plain test-retest is the one design both native engines can fit, and the cascade prefers HMC there because it is the more faithful fit and because fitlme is weak when the between-occasion variance is poorly identified at small occasion counts. Either stays reachable by forcing it. Any design absent from the table has no native implementation and is CmdStan-only, and the gamma family is pinned to CmdStan by the engine itself: forcing `fitlme` or `hmc` alongside `'family', 'gamma'` is rejected outright rather than quietly fitting a Gaussian model instead.

Forcing an engine is an explicit opt-in, and it errors naming the design if that engine cannot fit it: `The native HMC engine does not yet implement this analysis (design 'sing')`. It also errors if the engine's toolbox is not installed. The native engines are not bit-identical to CmdStan, and the fitlme engine's intervals are approximate frequentist bootstrap intervals rather than Bayesian credible intervals, which is why every fallback and every forced fitlme run prints a banner saying so.

```
% Force the native fitlme engine. The tutorial dataset is a plain one-facet
% design, so fitlme is its ONLY native option; forcing 'hmc' here would error.
% The parametric bootstrap dominates the runtime at 1000 resamples.
psyrat_data = psyrat_run('file', datafile, ...
    'groupcol', 'group', 'eventcol', 'event', ...
    'savepath', outdir, 'savename', 'manual_ern_fitlme.psyrat', ...
    'engine',         'fitlme', ...
    'bootstrap_reps', 1000);      % default; ignored by CmdStan and HMC
```

Forcing native HMC takes the same shape, `'engine','hmc'`, but it has to be paired with a design HMC implements: on this tutorial dataset that means turning on a design from the HMC-only rows of the table above, `'ssubjrel', 2` for subject-level reliability being the shortest route. One companion option belongs with it. `hmc_gradient` chooses how the sampler obtains gradients, and its default of `auto` uses automatic differentiation when the Deep Learning Toolbox is installed and drops to numerical gradients on its own when it is not, so `auto` needs no help on a machine without that toolbox. Pass `'hmc_gradient','numer'` to force the numerical path deliberately, or `'ad'` to insist on automatic differentiation and get an error rather than a silent fallback when the toolbox is missing. The CmdStan and fitlme engines ignore the option entirely.

No worked HMC example is printed here, and the omission is deliberate rather than an oversight: the native sampler is far slower per draw than CmdStan, so at the default four chains and 5,000 warmup and 5,000 sampling iterations a forced HMC run is not a demonstration anyone would sit through. Budget for it on real data, and prefer CmdStan wherever it is installed.

A native-engine result feeds `psyrat_relsummary` and `psyrat_report` through the same two-step chain and returns the same tables. The exported provenance header names the engine that actually ran, so a saved table says on its face whether it came from CmdStan. I would still use CmdStan whenever it is installed. The native engines exist so that an analysis is possible without it, not because they are as good.

14.6 The Formula Guide

Typing `psyrat_formula_guide` at the MATLAB prompt opens a window listing the reliability formulas the toolbox implements, one row per formula, with the paper table it comes from, the design, the coefficient, the decision type (absolute or relative), the formula in plain text, the function that computes it, and the place in the toolbox where it is reachable.

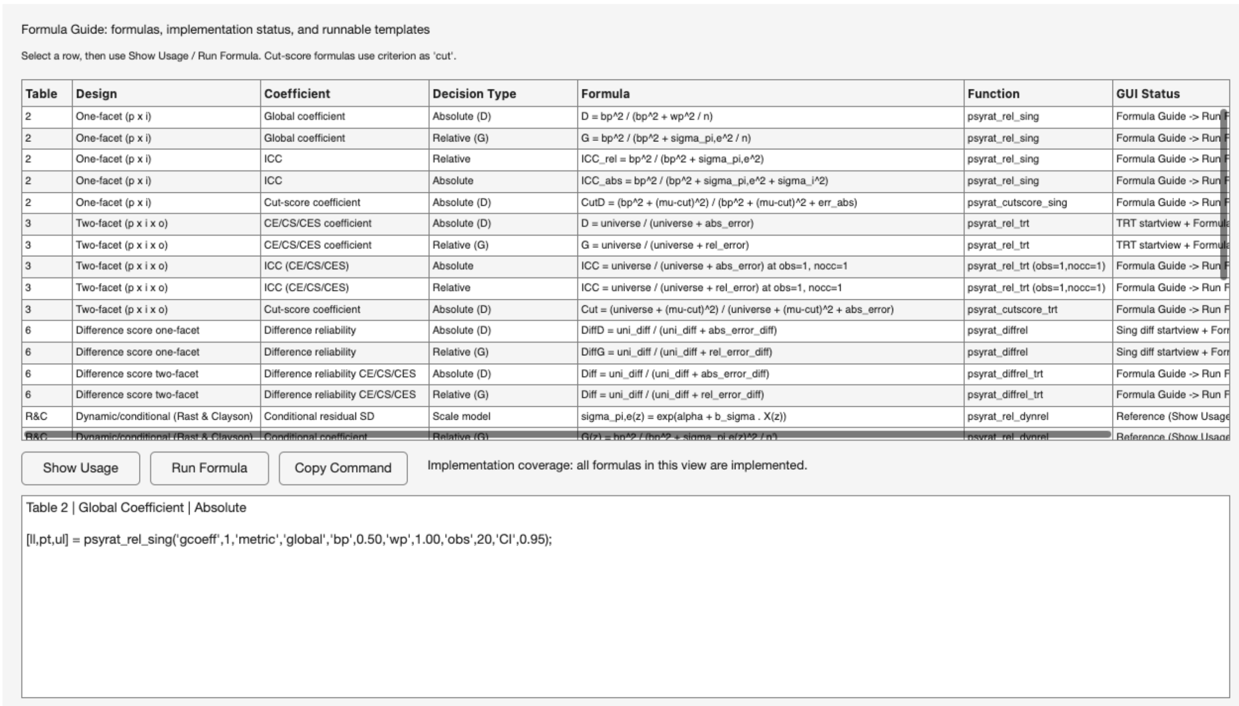


Figure 60: The Formula Guide window, opened by typing psyrat_formula_guide. Selecting a row and clicking Show Usage fills the panel at the bottom with a runnable call.

No button anywhere in the GUI opens this window. It is installed on the path and runnable, and typing its name is the only way in. Select a row first; the three buttons all act on the selection.

- Show Usage: fills the panel at the bottom of the window with a short label for the selected formula (its paper table, coefficient, and decision type) and a runnable command template beneath it.
- Run Formula: opens a small input form, runs the selected formula on the values entered, and replaces the panel text with the result.
- Copy Command: copies the current usage command to the system clipboard.

An optional argument narrows the list to one family, which is faster than scrolling when you know what you are after:

```
psyrat_formula_guide                       % every formula in the catalog
psyrat_formula_guide('analysis','trt')     % test-retest rows only
```

The accepted values are `all` (the default), `sing`, `sing_diff`, `sing_sserr`, `trt`, `dynrel`, and `splits`. Cut score formulas take the criterion as `cut`.

That leaves one thing a script cannot do. Criterion-score (cut score) reliability has no headless entry point: its outputs are built by a function that draws a figure and a table unconditionally, and `psyrat_report` has no criterion branch. Cut score results currently require the GUI (Chapter 9). Trace plots stay in the GUI for the same reason, and `psyrat_run` never renders them on either interactive setting. Everything else in this manual can be produced from a script.

14.7 Option reference

`help psyrat_run` prints the authoritative list with the full reasoning behind each entry, and the design flags in particular carry caveats no table can hold. What follows is the working summary: every option `psyrat_run` accepts, what it selects, and what it does when left out.

Data source and output. Supply exactly one data source, never both.

Option	Meaning	Default
`file`	path to a long-format data file	(one source required)
`data`	long-format table in the workspace	(one source required)
`savepath`	existing output dir; no whitespace	(required)
`savename`	output filename; no whitespace	(required)
`config`	sidecar to replay; see Section 14.4	`''`

Column mapping. The empty default means the facet is absent, not that the toolbox will guess at it.

Option	Meaning	Default
`idcol`	participant id column	`'id'`
`meascol`	numeric score column to analyze	`'meas'`
`eventcol`	event/condition column	`''` (single event)
`groupcol`	between-person group column	`''` (single group)
`timecol`	occasion column; selects a test-retest design	`''` (one occasion)
`weightcol`	items-per-split (n_i) column	`''`
`dim1col`, `dim2col`	dynamic-reliability covariate columns	`''`
`whichevents`, `whichgroups`, `whichtimes`	restrict which labels are analyzed	`{}` (all)

Estimation settings.

Option	Meaning	Default
`chains`	CmdStan chains; integer, 3 or higher	`4`
`warmup`	warmup iterations per chain	`5000`

Option	Meaning	Default
`sampling`	post-warmup sampling iterations per chain	`5000`
`seed`	RNG seed; fix it and report it	`12345`
`adapt_delta`	NUTS target acceptance rate (`delta=`)	`[]` (model's tuned value)
`max_treedepth`	NUTS tree-depth cap (`max_depth=`)	`[]` (CmdStan's 10)
`verbose`	1 quiet, 2 print CmdStan iterations	`2`
`withinchain`	1 off, 2 within-chain parallelization	`1`
`priors`	prior structure (see below)	`psyrat_default_priors`
`engine`	`auto/cmdstan/fitlme/hmc` (Section 14.5)	`auto`
`bootstrap_reps`	bootstrap resamples; fitlme only	`1000`
`hmc_gradient`	`auto/ad/numer`; native HMC only	`auto`
`family`	`gaussian` or `gamma`	`gaussian`

The priors defaults deserve a paragraph of their own, because they are the one setting whose default is wrong for some perfectly ordinary data, and because they are not all on one scale. The Gaussian-family defaults are weakly informative on the ERP microvolt scale, and they are NOT scale-free, so data on another measurement scale needs values matched to that scale. The gamma-family defaults are not microvolt priors at all: they live on the log expected-score and log degrees-of-freedom scales and are calibrated to a single-trial time-frequency-power reference, so they are already aimed at a different modality and should be re-derived from that starting point rather than from the Gaussian one. The dynamic-reliability dimension slopes are a third case, set on the standardized predictor scale. Chapter 15 shows how to set priors, and the priors and sensitivity note documents each block and what moves when it changes.

Design and analysis flags. These select the model, so a wrong value here is not a tuning mistake but a different analysis.

Option	Meaning	Default
`ssubjrel`	subject-level reliability: 1 off, 2 on	`1`
`diffest`	difference scores: 1 off, 2 two-event, 3 difference-of-differences	`1`
`dodmap`	1x4 cell of event labels, ERP_1 to ERP_4 order	`{}`
`diffrescor` / `diffwpcov`	one setting, two names: residual covariance for co-occurring events, 1 off, 2 on	`1`
`dynrel`	dynamic/conditional reliability: 1 off, 2 on	`1`, or 2 via `dim1col`
`splits`	splits: 1 single-trial, 2 parallel, 3 nonparallel	`1`
`ssrescor`	residual correlation: 1 population, 2 per-subject	`1`
`dispersion`	copula dispersion: 1 per-person log-nu, 2 global	`[]` (per design)
`gammascale`	Gamma dispersion: 1 log-nu, 2 log residual SD	`[]` (per design)

Six of those rows carry a condition the table cannot hold, and each one is a real trap rather than a refinement.

`dynrel` cannot simply be switched on. Naming `dim1col` sets it to 2 by itself, so the documented default of 1 is not the effective value whenever a dimension column is named, and `'dynrel', 2`

WITHOUT `dim1col` is an error rather than a plain dynamic run. `dim2col` without `dim1col` is an error too: `dim1col` is the one that switches the analysis on.

`dodmap` is required when `diffest = 3`, and `splits` values 2 and 3 both require `weightcol`. `ssubjrel` also answers to `sserrvar`, a legacy alias kept for older scripts.

`diffrescor` and `diffwpcov` are one setting under two names, so pass either and the other follows. Passing both with contradictory values is an error, because the two values select structurally different models and there is no safe way to guess which was meant. The setting is meaningful only when `diffest = 2`; a `diffest = 3` run rejects a concurrent request outright rather than ignoring it.

`gammascale` selects between two parameterizations that are DIFFERENT ESTIMANDS, and their variance components are not comparable. Its default is per design rather than global, and Section 14.4 covers what that means when replaying an older sidecar.

Execution control.

Option	Meaning	Default
`interactive`	allow the GUI rerun dialog on non-convergence	`false`

`interactive` enables the rerun dialog and nothing else. It does not turn on trace-plot review: `psyrat_run` pins that preference off and exposes no option to change it, so trace plots stay GUI-only on both settings (Section 14.3). `psyrat_relsummary` takes an option of the same name that governs its own threshold-failure dialog (Section 14.2); it defaults to the flag the `.psyrat` carries, and to interactive when the file was saved by the GUI.

`psyrat_report` options. The reporting call takes eight of its own. Note the export format: `.xlsx` is the default, so a script that wants CSV has to say so.

Option	Meaning	Default
`outdir`	where to write tables; omit to return only	`''`
`format`	`.xlsx` or `.csv`	`.xlsx`
`CI`	credible-interval width	the summary's width, else `.95`
`marginal`	dynrel: 0 typical-person, 1 population-average	`0`
`reltype`	dynrel two-facet: 1 equiv., 2 stability, 3 both	`3`
`nocc`	dynrel two-facet: occasion n', or `'observed'`	`[]` (1)
`ngrid`	dynrel: grid points per dimension axis	`25`
`ntrials`	one-facet/test-retest: max trial count for the curve	`50`

15. Advanced designs

Six settings change what PsyRAT estimates rather than how it estimates it. Five of them live on the Specify Inputs and Specify Processing Preferences screens described in Chapter 7, and one of them is not a screen control at all. Each section below is a delta from the standard workflow: the design in plain terms, the controls to set, the output it produces, and how to read that output.

Read the section you need. None of these designs is a prerequisite for another, and none is required for a conventional internal-consistency or test-retest analysis. Chapter 16 records which combinations are implemented, and the toolbox refuses an unimplemented one before it starts sampling rather than after.

Two things apply throughout. Every design here estimates more parameters than its standard counterpart, so it samples more slowly and needs more iterations to converge; budget for that before you start a run you cannot interrupt. And several carry a status caveat or a disclosure that is printed in the exported table headers. Where a design carries one, this chapter says so on the page rather than leaving you to find it in a file header.

15.1 Subject-level error variances

Group-level internal consistency yields a single coefficient for a sample: the ratio of between-person variance to the precision with which those persons' scores were measured. That coefficient is a property of the group, and it can conceal wide differences among the individuals who compose it. A participant who contributes few artifact-free trials, or unusually variable ones, may hold a score far less precise than the sample estimate implies (Clayson, Brush, & Hajcak, 2021).

Subject-level estimation models the residual variance separately for each participant. Every participant receives their own residual standard deviation, and their coefficient combines that residual variance, divided by their own retained trial count, with the between-person variance shared across the sample. The result is one estimate per participant rather than one estimate per group. A researcher can then ask which participants were measured precisely enough for a given purpose, instead of assuming that the sample-level answer describes each of them.

The estimand follows the same absolute-versus-relative distinction as the group-level analysis. Selecting Dependability gives each participant an absolute coefficient, whose error term keeps the trial main effect. Selecting Generalizability gives a relative coefficient, whose error term excludes it. The choice is made in the viewer, not at processing time, so one saved run answers both questions.

Control (Chapter 7)	Value
Estimate subject-specific error variances	Yes
Estimate internal consistency of difference scores	No, or Yes: 2-event difference
Occasion ID	(none), or an occasion column for the test-retest design
Dimension 1 (dynamic)	(none) for the static subject-level design
Observation likelihood family	Gaussian, or Gamma for strictly positive scores (Section 15.4)

An occasion column narrows what is available. With one selected, subject-specific error variances are supported for the non-difference test-retest design and for the dynamic designs of Section 15.2. They are not supported for non-dynamic difference scores in a test-retest workflow, and PsyRAT refuses that combination when you click Analyze, with a dialog naming difference scores and the occasion/time facet. Set Estimate subject-specific error variances to No, or drop the occasion column, if you meet that message.

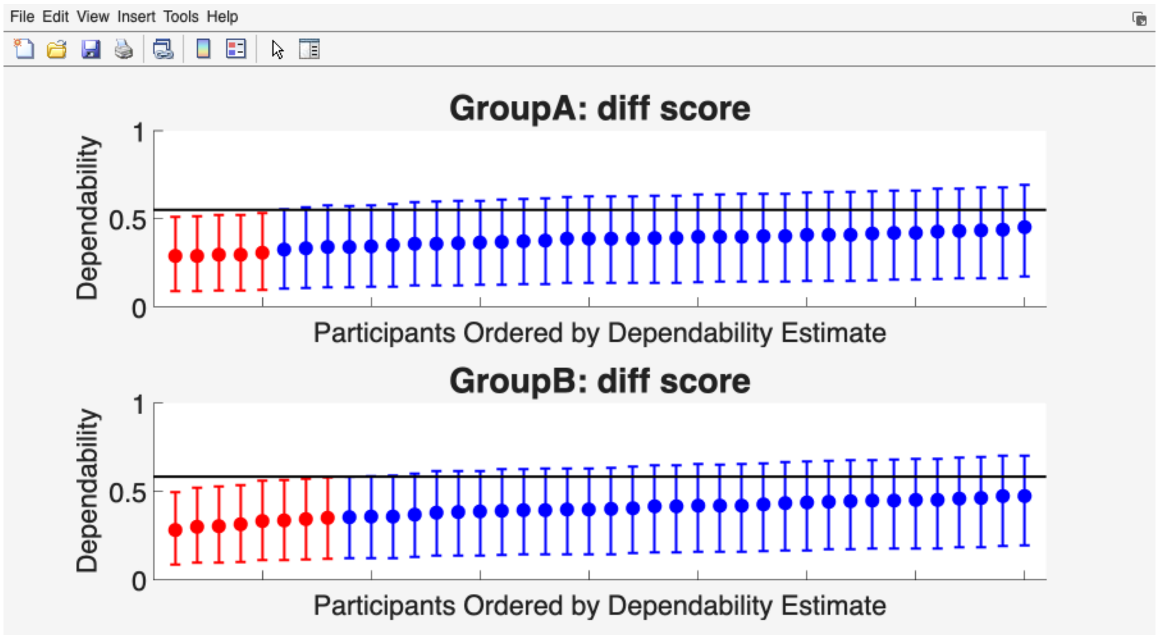


Figure 61: Subject-Level Dependability Estimates. One panel per stratum (here one per group, from a difference-score run); each marker is one participant's point estimate with its credible interval, ordered left to right by point estimate. The horizontal line is the group-level estimate at the mean retained trial count. Participants whose interval excludes that line are drawn in red.

The x axis is participants ordered by their own coefficient, lowest first, with the tick labels suppressed because the ordering rather than the identity is the point. The y axis runs from 0 to 1 and carries whichever coefficient you selected. The black horizontal line is the group-level estimate computed at the mean retained trial count. Markers in red are the participants whose credible interval lies entirely above or entirely below that line, which is the plot's way of saying that the group number does not describe them (the remaining markers turn blue when at least one participant is flagged; a fully clean sample draws in a single default color). A well-behaved sample gives a gently rising band of markers straddling the line; a long red tail on the left is the result the analysis exists to find.

Choosing Yes here also relabels the viewer, which is worth knowing before you go looking for a control that is no longer there. On the View Results Setup screen, Plot: Trials vs Reliability becomes Plot: Subject-Level Reliability, Plot: ICC Estimates becomes Plot: Subject-Level ICC Estimates, and Table: Trial Cutoffs at Reliability Threshold becomes Table: Subject-Level Reliability. The Criterion Score Outputs button is not offered, because the cut score coefficient is built from the group-level universe-score mean and has no per-participant form. Table: Overall Reliability Summary and Table: Sources of Variance are unchanged. On the difference-score variant the trial-cutoff table itself still opens, and this screen offers no control over it (the relabeled checkbox governs the subject-level table instead); Chapter 9 describes what its rows report on that route.

The subject-level table is where participant inclusion is decided. The window itself shows the group-level summary; the per-participant rows are written by its Save Subject Coefficients button, one row per participant with the coefficient, the intraclass correlation, the standard error of measurement, each with a credible interval, the retained trial count, the error variance, and a flag marking whether the participant meets the cutoff. The flag compares the participant's coefficient against the Reliability Cutoff you entered, using whichever bound the Estimate to use for trial cutoffs preference names: the lower limit of the credible interval, the point estimate, or the upper limit. Selecting the lower limit is the conservative choice and the one I use, because it excludes a participant unless the data support inclusion; the point estimate is the more common convention in published work. Whichever you pick, report it, since the three bounds do not select the same people.

Excluding on a reliability criterion changes who remains in the sample, and in a clinical sample that change can be systematic: a cutoff of .80 applied to error-related ERP scores excluded patients who made fewer task errors and reported fewer symptoms, while cutoffs below .80 shrank both between-group and within-group effect sizes (Heindorf et al., 2025). Those authors recommend examining the characteristics of the participants a threshold excludes and justifying the threshold rather than inheriting it. Decide the rule before you see the coefficients. And remember that a participant's coefficient depends on their retained trial count, so a low estimate describes that participant's data in hand rather than the participant.

Sources: Clayson, Brush, and Hajcak (2021) for the subject-level internal-consistency metric and its use; Rocha et al. (2026), Table 2, for the one-facet coefficient whose per-participant form this is.

15.2 Dynamic (conditional) reliability

Standard generalizability coefficients assume that residual variance is homoscedastic: one error variance describes every observation, so one coefficient describes every person and condition. That assumption is testable and often false. When residual variance depends on some measured quantity, reliability depends on it too, and a single coefficient averages over a range within which measurement precision genuinely differs.

Rast and Clayson (in press, p. 7, "we define a condition-specific generalizability coefficient") address this by giving the residual variance its own log-linear submodel, so that residual variance becomes a function of covariates. The coefficient defined on that model is the usual ratio of universe-score variance to universe-score variance plus residual variance divided by the number of replications, evaluated at each value of the covariate rather than once for the sample. Because the covariate enters the residual variance smoothly, the coefficient becomes a continuous function rather than a set of isolated numbers. That coefficient appears as an unnumbered display in the source; the equation numbered (9) beside it is the scale submodel, not the coefficient.

PsyRAT implements this with between-person dimensions. A dimension is a continuous covariate with one value per participant, such as an age, a questionnaire total, or a symptom score. Selecting a Dimension 1 column turns on the analysis, and the output is reliability as a function of that dimension: where on the dimension scores are measured precisely, and where they are not.

Control (Chapter 7)	Value
Dimension 1 (dynamic)	the covariate column (this selection enables the analysis)

Control (Chapter 7)	Value
Dimension 2 (dynamic)	(none), or a second covariate column
Occasion ID	(none) for the one-facet design; an occasion column for the trial-by-occasion design
Estimate subject-specific error variances	No for the population surface; Yes to add per-participant estimates
Estimate internal consistency of difference scores	No, or Yes: 2-event difference
Items per split (n_i)	(none); splits and dynamic reliability cannot be combined

PsyRAT standardizes each dimension before fitting, and that step determines what the axis on the resulting plot means. One value per participant is collapsed out of the column, those values are z-scored across participants using the sample standard deviation, and the standardized value is written back to every row belonging to that participant. Standardization happens ONCE, across the whole sample, before any splitting by group or event. So z = 0 is the overall sample mean of the dimension, not the mean of whichever group a panel shows, and the panels of a multi-group figure share one predictor scale and are directly comparable.

Section 6.4 lists the three conditions that stop the run: a dimension that is not constant within a participant, a participant with no finite value, and a column with no between-participant variance. A fourth only warns. In a multi-group run, a group with no between-participant variation in the dimension, or with fewer than two participants, has no identified scale slope; PsyRAT says so before the run starts rather than after the sampler finishes, and draws that group's surface across a nominal range. Treat such a panel as undefined rather than flat.

The figure below comes from `test_data/manual_ern_dynrel.csv`, which ships with the toolbox: two groups, the two incongruent-trial events, and one questionnaire-style score per participant. Like the other tutorial files it is simulated, and it was generated so that trial-to-trial noise grows with the score, which is the pattern the figure is there to show.

The x axis is the standardized dimension, so 0 is the sample mean of the covariate and 1 is one standard deviation above it. The y axis is the selected coefficient on a fixed 0 to 1 scale, the solid line is the posterior point estimate across the grid, and the shaded band is the credible interval at the width you set. A curve that falls as the dimension rises says that participants higher on the dimension were measured less precisely, which is a measurement finding in its own right and a confound for any correlation between that dimension and the ERP score. If you entered a Reference cutoff, it is drawn as a horizontal dotted line, and the point where the curve crosses it is the value of the dimension beyond which scores stop meeting your threshold. The axis, the viewer's Dimensions line, and the variance-component tables label the dimension `dim1` (and `dim2`): those are the toolbox's names for the Dimension 1 and 2 columns you selected, not your column headers.

With two dimensions the panel becomes a filled contour instead of a curve. The x axis is the standardized Dimension 1, the y axis is the standardized Dimension 2, and color is the coefficient, with a reference cutoff drawn as a dashed white contour. A second dimension adds its own slope AND the interaction of the two, so the model gains three scale terms rather than two, and the surface can bend rather than tilt. Add a second dimension when you have a reason to expect the two to interact. Otherwise two separate one-dimension runs are easier to read and easier to sample.

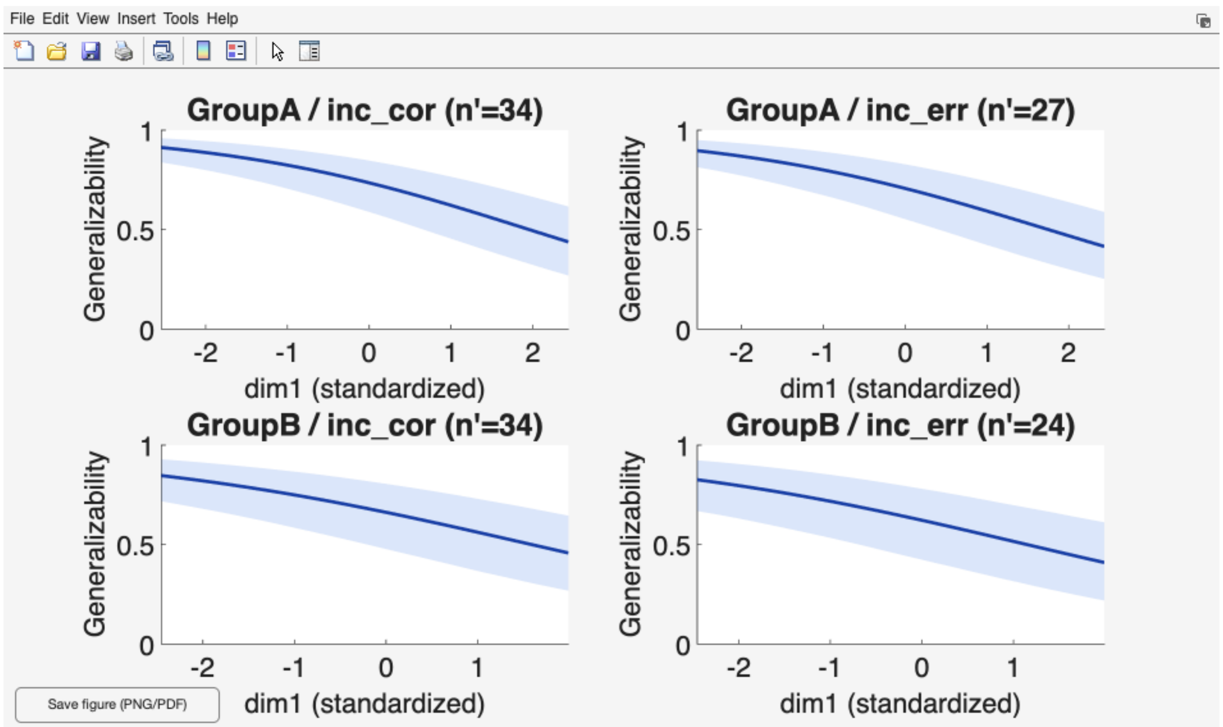


Figure 62: Dynamic Generalizability as a function of the dimension(s). One panel per group-by-event stratum. In the one-dimension case each panel is a curve of the coefficient against the standardized dimension with a shaded credible band; the panel title reports the trial count at which the curve was evaluated.

Four controls on the Dynamic Reliability: View Results screen shape the figure. Coefficient switches between Generalizability G(z) and Dependability D(z). Credible interval (0-1) sets the band width, and Reference cutoff (blank = none) draws the threshold line. Trials n' (blank = per-group median) sets the replication count the coefficient is evaluated at; blank uses each stratum's own median retained trial count, which describes the data in hand, and a number projects the surface to a design you are contemplating. The grid itself is fixed at 25 points per axis, spanning the observed range of the dimension within each stratum.

The Estimand popup chooses between Typical person (delta_p = 0), a participant with no residual deviation of their own, and Population average (lognormal), which averages over the person-level residual scale. The two differ whenever participants vary in residual variance, so say which one you report. The popup is disabled for the difference-score variants, which compute the typical-person surface only. A trial-by-occasion run adds two further controls. Reliability coefficient offers Equivalence (occasion fixed), Stability (trial fixed), or Equivalence + stability (both random) and defaults to the third. Occasions n' offers 1 occasion or Observed occasions and defaults to one, which is the coefficient for a score measured on a single occasion rather than a composite.

Turning on Estimate subject-specific error variances adds a per-participant layer. Each participant's own conditional coefficient is computed at their own position on the dimension, scattered over the population surface, and tabulated alongside it. The surface then serves as a reference against which individuals are read.

Availability, as implemented: dynamic reliability runs on the one-facet design and, with an Occasion ID column, on the trial-by-occasion two-facet design; for single scores and for two-event difference scores; at the group level and the subject level; and for the four-event contrast of Section 15.3 in its person-specific form only. It cannot be combined with data splits. The gamma family of Section 15.4 covers a subset of these, and Chapter 16 gives the cell-by-cell picture.

Sources: Rast and Clayson (in press), pp. 7 to 8, for the condition-specific generalizability and dependability coefficients and the log-linear scale submodel they are evaluated on.

15.3 Difference-of-differences

An interaction contrast is a difference between two difference scores. A researcher comparing the congruency effect between two blocks, or an error-minus-correct difference between two tasks, is asking about the quantity (ERP_1 - ERP_2) - (ERP_3 - ERP_4), and the psychometric properties of that quantity are not those of any of its four constituents. Difference scores inherit the error of both terms while cancelling the shared variance that made each term reliable, and a difference of differences applies that arithmetic twice.

PsyRAT estimates the four constituent scores jointly and forms the contrast from their posterior draws, so the reliability of the contrast is computed from a model of the four events rather than from four separate models. Dependability of the contrast keeps the between-trial variance of the contrast in the error term; generalizability excludes that whole block. Because the contrast weights sum to zero, a trial effect common to all four events cancels out of it, so what dependability retains is trial variance that differs across the four events. The two intraclass correlations reported alongside them are single-observation quantities, computed at one trial rather than at the trial counts you entered.

Control (Chapter 7)	Value
Estimate internal consistency of difference scores	Yes: 4-event DoD
Event Type	the column holding the four condition labels
Select Events	exactly four events
DoD Contrast...	assign ERP_1 through ERP_4 (required)
Occasion ID	(none), unless Dimension 1 and subject-specific error variances are both set
Estimate residual covariance for co-occurring events	No (a concurrent request is rejected, not ignored)
Estimate subject-specific error variances	No, unless Dimension 1 is also set

The contrast mapping is not optional and has no default that PsyRAT will guess for you. Click DoD Contrast... on the Specify Inputs screen and assign each of the four selected events to a slot: ERP_1 (first minuend), ERP_2 (first subtrahend), ERP_3 (second minuend), and ERP_4 (second subtrahend). Each event must appear once. Clicking Analyze without a valid mapping stops with a dialog telling you to configure the contrast first, and changing the event selection afterward clears the mapping so that a stale assignment cannot survive into a run. Set the events FIRST, then the mapping.

Coverage requirements differ between the two forms of this design, and the difference matters when you are deciding whom to drop. The group-level contrast requires exactly four event types in the data but does not require any individual participant to have all four, because its D-study divisors are group-level trial counts. The person-specific dynamic form of Section 15.2 does require every participant to have trials in all four events, because each participant's own per-cell counts are that design's divisors and a missing cell leaves their coefficient undefined. PsyRAT checks that before fitting and names the participants who fail, which spares you a completed multi-hour run whose results cannot be viewed.

Viewing a saved DoD run opens the DoD Reliability Setup screen rather than the usual View Results Setup. Choose the group, set Heatmap metric to Dependability or Generalizability, confirm the four event assignments, enter a trial count for each of ERP_1 through ERP_4, set Heatmap trial grid max, and click Generate DoD Outputs.

The contrast is printed at the top, then the coefficient table for the trial counts you entered, then two heatmaps. In the left heatmap the x axis is the trial count for ERP_1 and the y axis the trial count for ERP_2, with ERP_3 and ERP_4 held at their entered values; the right heatmap does the same for ERP_3 and ERP_4 with the first pair held fixed. Color is the selected coefficient, clipped to the 0 to 1 range. Read the captions beneath each panel, which name the counts held fixed, because the same contrast produces different surfaces at different fixed counts. What the pair of panels answers is where additional trials would buy the most reliability: a steep gradient along one axis is an argument for collecting more of that condition.

Two expectations about output size. The heatmaps evaluate the coefficient at every point of a square grid running from 0 to Heatmap trial grid max, which defaults to the largest observed trial count in the selected group plus 25. They are computed at the posterior MEAN of the variance components rather than over the draws, which is why they render quickly while the summary table above them, which does use the draws, does not. Raising the grid maximum raises the cost

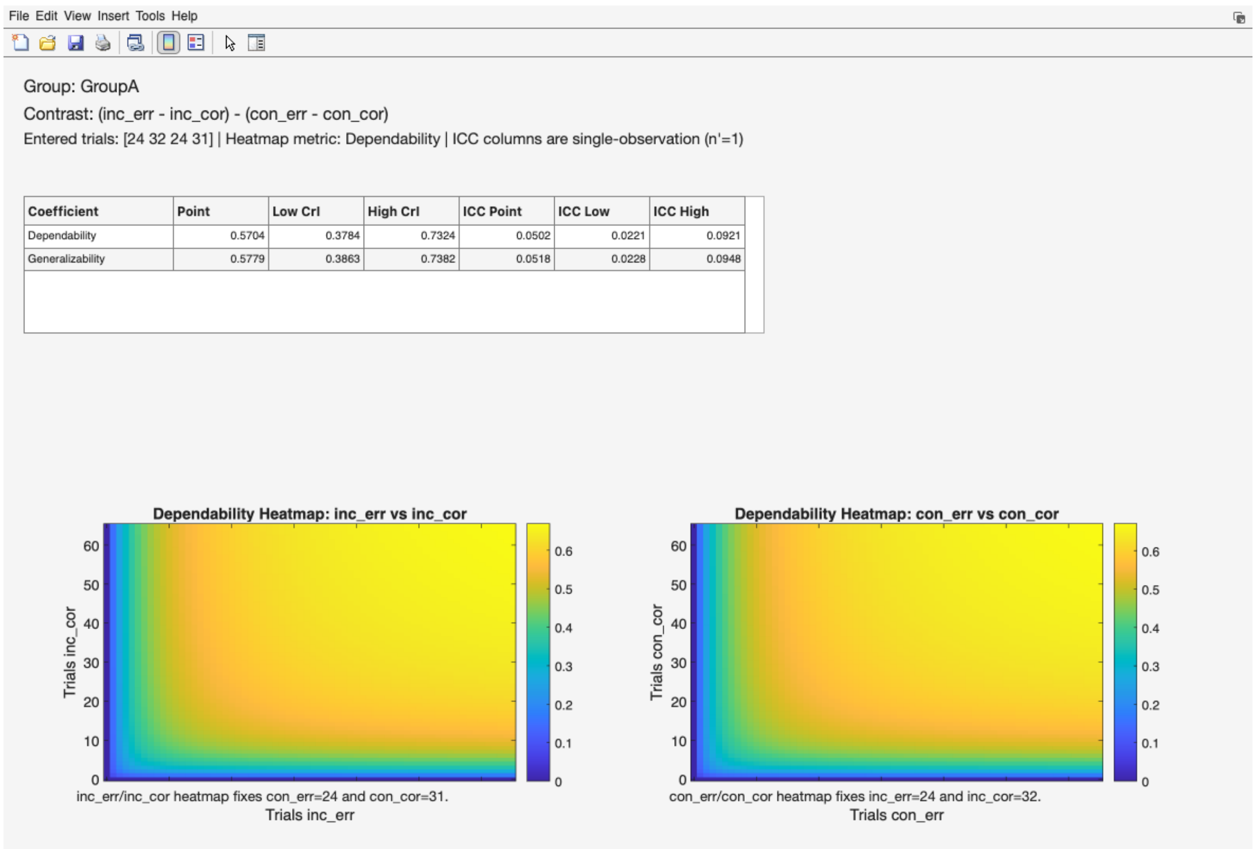


Figure 63: DoD Reliability Outputs. The summary table of dependability and generalizability with credible intervals and single-observation ICCs, above two trial-count heatmaps.

quadratically, so a maximum of 400 is four times the work of one at 200. The exported variance-component table is small by comparison, nine rows per group covering the contrast variances and the four coefficients, and outputs are produced one group at a time, so a multi-group dataset means one pass through Generate DoD Outputs per group.

Three estimand choices in this design are the toolbox's conventions rather than entries in any published table, and they should be reported as such. First, the four constituents are treated as measured on distinct trials, so all six pairwise residual covariances are fixed to exactly zero; the contrast's residual variance is the weighted sum of the four event residual variances. A concurrent variant does not exist and a request for one is refused rather than silently ignored. Second, the off-diagonal entries of the between-trial covariance block are scaled by the harmonic mean of the two trial counts involved, since the four events rarely share a trial count. Third, the intraclass correlations are single-observation quantities and are labeled as such in the table and in the window; they are not the coefficients at your entered trial counts. Where the person-specific dynamic form is used, the viewer prints the nonconcurrent assumption beneath its variance-component table, and the exported header repeats it, so the assumption travels with the numbers.

Sources: Rocha et al. (2026), Table 6, for the two-event difference-score coefficients this contrast generalizes; the four-event contrast and its scaling conventions are the toolbox's, and are derived rather than tabulated there.

15.4 The gamma (scaled chi-square) likelihood family

The Gaussian likelihood is the right default for ERP amplitudes, which are signed, roughly symmetric, and routinely negative. It is the wrong model for quantities that are strictly positive and right-skewed, where the variance grows with the mean. Single-trial time-frequency power is the clearest case: a normal model placed on it puts probability mass below zero and misstates the variance at both ends of the distribution.

The gamma family, in its scaled chi-square form, models such scores on their own support with a log link, so that the conditional variance scales with the square of the mean. Its parameters are estimated on that scale and converted back to the observed scale for reporting, so the reliability coefficients remain comparable in meaning to their Gaussian counterparts even though the model underneath is different.

Control (Chapter 7)	Value
Observation likelihood family	Gamma
Gamma scale parameterization	Automatic (recommended)
Estimation engine:	Automatic (recommended); gamma runs on CmdStan whatever this says

Every score in the measurement column must be strictly positive. PsyRAT checks this before estimation and stops on any value at or below zero, naming the column. Shift or transform the scores, or switch back to Gaussian, if your data include zeros or negatives. Do not shift a distribution to clear the check without a reason to defend, since the amount you add changes the mean-variance relationship the family is there to represent.

Gamma scale parameterization decides which quantity the dispersion submodel is built on, and its two named options are different estimands rather than two views of one fit. Log-nu models

RELATIVE dispersion, a constant coefficient of variation, so a contrast on it blends amplitude with error. Location-scale models the absolute residual standard deviation directly, which is what you want when the output will be read as precision, as individual error variance, or against a Gaussian analysis of the same data. Leave the control on Automatic unless you have a specific reason not to. Automatic selects location-scale on every design that supports it and log-nu on the rest, and it is the only setting that does not require you to know which designs are which. Under the Gaussian family Automatic and Log-nu are ignored, but Location-scale is rejected and stops the run, so leave it on Automatic there too.

> CAUTION. The default changed on August 7, 2026. Before that date, gamma runs used the log-nu parameterization everywhere. They now default to location-scale on every design that supports it. A script or a saved configuration that did not name a gamma scale parameterization therefore fits a DIFFERENT estimand today than it did before that date, and the two results are not comparable. Runs from August 3, 2026 onward stamp the resolved setting into the saved result and into the `*_runconfig.json` sidecar, so replaying one of those reproduces its own run. A sidecar written before that date carries no gamma scale parameterization at all, and replaying it takes the CURRENT default, silently. To reproduce an older gamma run, pass `'gammascale', 1` explicitly to `psyrat_run` (see Chapter 14) or select Log-nu (relative dispersion / CV) in the GUI. The exceptions are the location-scale-only designs, which reject an explicit log-nu request: the subject-level dynamic difference score, the subject-level dynamic concurrent difference scores of Section 15.5, and the person-specific dynamic difference of differences. Only the static concurrent difference designs still default to log-nu and are unaffected.

The provenance block below is the one written into the variance-table export of a gamma run of the dynamic one-facet design (Section 15.2) on `test_data/manual_power_concurrent.csv`, a simulated single-trial power dataset that ships with the toolbox. The lines are as the export wrote them; the CSV form wraps a line that contains a comma in quotation marks, and those are omitted here.

```
Seed: 12345
Analysis: ic_dynrel
Estimation engine: cmdstan
Likelihood family: Gamma / scaled chi-square (observed-score-scale variance
 ↪  components)
Gamma scale parameterization: LOCATION-SCALE (log residual SD). The residual is
 ↪  modeled directly as sigma(z) rather than through the dispersion nu, so the
 ↪  dimension slope b_sigma is a pure residual-SD effect. This is a DIFFERENT
 ↪  ESTIMAND from the log-nu parameterization (gammascale=1); results from the
 ↪  two are not comparable.
Gamma dispersion (person log residual-SD SD, s_sd): median 0.216
Gamma mean-scale correlation (rho_p,sigma): median 0.376 - estimated and
 ↪  reported, but it does NOT enter the observed-scale conversion, because the
 ↪  residual is independent of the mean under this parameterization
Marginal residual correction factor F = exp(2*s_sd^2): median 1.098 (F = 1 is the
 ↪  homogeneous-residual plug-in)
```

```
Gamma dynamic reliability (location-scale): every signal component carries
 ↪  exp(2*alpha(z)), but the residual also carries an amplitude-FREE term
 ↪  exp(2*log sigma(z) + 2*s_sd^2) that the signal side has no counterpart for,
 ↪  so BOTH the mean slope (b) and the residual-SD slope (b_sigma) move the
 ↪  conditional coefficient. The mean-slope cancellation that holds under the
 ↪  log-nu parameterization does not hold here; report both slope posteriors.
Priors: PsyRAT defaults
Converged: Yes
Divergent transitions: 0
```

Read the header of any exported table to confirm what actually ran. The seed, engine, prior line, and convergence verdict appear on every export; the family line appears for non-Gaussian runs, and a scale line appears when location-scale was resolved (a run resolved to log-nu carries no scale line, so consult the sidecar's design block for that case). The header is the record to quote in a methods section, and what it prints is what the model used, not what you requested, which is the distinction that matters when Automatic did the choosing.

One convention of the gamma subject-level designs belongs here, because it decides what a participant's own coefficient means. A person-specific coefficient needs a person-specific residual, and under the log link that residual has two parts: the participant's own conditional variance, which the model evaluates at their own mean and dispersion effects, and the person-by-trial term of the mean surface, which the model carries once for the population. PsyRAT pools the population form of that second term into every participant's residual rather than rescaling it by the participant's own mean effect (audit item S20, owner ruling, September 2026). The choice keeps the per-participant residual on the same footing as the group residual, which carries the same term, at a known price: the pooled term understates the residual, and so overstates the coefficient, for a participant whose mean effect lies above the population value, and does the reverse below it. The export header of a run that pools records the convention in a line marked POOLED, so it travels with the numbers (the two-facet subject-level design carries the person-by-trial term as its own component and pools nothing); report it when you report per-participant gamma coefficients.

Status: gamma is shipped and supported, on CmdStan only, for a subset of the designs in this chapter. The native MATLAB engines are family-blind, so the toolbox pins every gamma run to CmdStan rather than risk fitting a gamma request as a Gaussian model; a gamma run fails on a machine without CmdStan instead of falling back. Gamma is permanently unavailable for the split designs, because a split score is the mean of several items and the scaled chi-square mean-variance relationship does not hold on a mean. It is also unavailable for some of the two-facet dynamic difference designs, where the work is simply unfinished. Chapter 16 lists the combinations, and a request for one that is not implemented is refused before sampling starts, with a message naming the design.

Sources: Rast and Clayson (in press) for the location-scale formulation the gamma scale submodel follows; the observed-scale conversions and the parameterization defaults are the toolbox's conventions.

15.5 Residual coupling for co-occurring events

Two events recorded on the same trials are not independent at the residual level. Ipsilateral and contralateral measurements from one epoch, or two components scored from one waveform, share whatever trial-level noise the epoch carried, so the residual for one is informative about the residual for the other. A difference score built from such a pair has less error variance than the sum of its parts implies, because part of the noise cancels in the subtraction.

Fixing that covariance at zero when the events genuinely co-occur therefore biases the difference-score coefficient downward, while estimating it lets the model use the cancellation the design provides. The opposite error is available too: estimating a residual covariance for events measured on separate trials fits a parameter that should be zero and spends sampling effort on it.

Control (Chapter 7)	Value
Estimate internal consistency of difference scores	Yes: 2-event difference
Estimate residual covariance for co-occurring events	Yes
Residual correlation:	Population (shared), or Per-subject
Estimate subject-specific error variances	No, or Yes
Dimension 1 (dynamic)	(none), or a covariate column

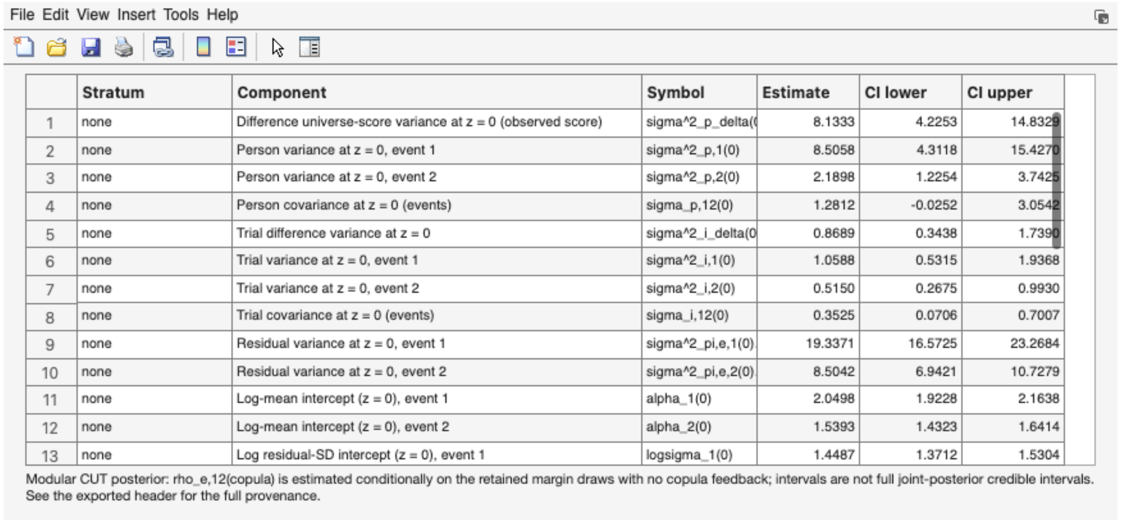
Set Estimate residual covariance for co-occurring events to Yes only when the two events were scored from the same trials. If you set it to No the within-person residual covariance is fixed to zero, which is the correct model for events measured on separate trials and the default. A four-event contrast (Section 15.3) rejects a request for residual covariance outright rather than ignoring it.

Residual correlation: on the Specify Inputs screen chooses between one residual correlation shared across participants and a separate residual correlation per participant. It has an effect on two designs, the one-facet and two-facet subject-level concurrent dynamic differences, which require four things at once: difference scores on, subject-specific error variances on, residual covariance on, and a Dimension 1 column selected (the two-facet form also requires an Occasion ID column). The popup stays disabled until the first three of those are set in Specify Processing Preferences, and a Per-subject selection left over from an earlier configuration is reset to Population for any run that does not qualify, so the setting cannot leak into a design with no per-participant correlation to estimate. Population (shared) is the default and the one to use unless each participant contributes enough paired trials to identify a correlation on their own.

The window below is from a run of that two-stage design under the gamma family on `test_data/manual_power_concurrent.csv`, a simulated dataset that ships with the toolbox: theta-band and alpha-band power scored from the same trials, with one questionnaire-style score per participant as the dimension.

The variance-component window lists one row per component with its name, its symbol, a point estimate, and a credible interval. The coupling appears as Residual correlation (events), symbol rho_e,12, alongside the residual covariance it implies at the mean of the dimension. The per-subject variant reports the population-average correlation in that row and adds SD(rho_e,s) for the spread across participants. A run fitted by the two-stage procedure described below renames the row Copula residual correlation (events), symbol rho_e,12(copula), omits the implied-covariance row, and a note sits beneath the table saying what that means; the exported header carries the full provenance the note points at. Read the correlation against the residual standard deviations in the same table rather than on its own, since a large correlation between two small residuals moves the coefficient very little. The per-subject variant reports SD(rho_e,s), the between-person spread of the correlation, rather than a single value.

Two disclosures belong on this page.



File Edit View Insert Tools Help

	Stratum	Component	Symbol	Estimate	CI lower	CI upper
1	none	Difference universe-score variance at z = 0 (observed score)	sigma^2_p_delta(	8.1333	4.2253	14.8329
2	none	Person variance at z = 0, event 1	sigma^2_p,1(0)	8.5058	4.3118	15.4270
3	none	Person variance at z = 0, event 2	sigma^2_p,2(0)	2.1898	1.2254	3.7425
4	none	Person covariance at z = 0 (events)	sigma_p,12(0)	1.2812	-0.0252	3.0542
5	none	Trial difference variance at z = 0	sigma^2_i_delta(0	0.8689	0.3438	1.7390
6	none	Trial variance at z = 0, event 1	sigma^2_i,1(0)	1.0588	0.5315	1.9368
7	none	Trial variance at z = 0, event 2	sigma^2_i,2(0)	0.5150	0.2675	0.9930
8	none	Trial covariance at z = 0 (events)	sigma_i,12(0)	0.3525	0.0706	0.7007
9	none	Residual variance at z = 0, event 1	sigma^2_pi,e,1(0)	19.3371	16.5725	23.2684
10	none	Residual variance at z = 0, event 2	sigma^2_pi,e,2(0)	8.5042	6.9421	10.7279
11	none	Log-mean intercept (z = 0), event 1	alpha_1(0)	2.0498	1.9228	2.1638
12	none	Log-mean intercept (z = 0), event 2	alpha_2(0)	1.5393	1.4323	1.6414
13	none	Log residual-SD intercept (z = 0), event 1	logsigma_1(0)	1.4487	1.3712	1.5304

Modular CUT posterior: rho_e,12(copula) is estimated conditionally on the retained margin draws with no copula feedback; intervals are not full joint-posterior credible intervals. See the exported header for the full provenance.

Figure 64: The Dynamic reliability variance components window for a modular cut posterior run, with the on-screen disclosure beneath the table.

> EXPERIMENTAL. Convergence for this design has not been validated in this release; exported tables carry an EXPERIMENTAL flag in their headers.

That admonition applies to the concurrent difference designs under the gamma family when the residual dispersion is modeled per participant. The per-person dispersion term gives the posterior a geometry the sampler handles poorly. The one check run against this design did not reach acceptable convergence diagnostics, and it did not improve when the simulated sample and trial counts were raised. That is a single fit rather than a proven property of the model, so judge your own run on its own convergence verdict, its divergent-transition count, and the per-parameter diagnostics before interpreting anything. Long warmup helps. The subject-level form has no non-experimental alternative, because a per-participant coefficient requires a per-participant dispersion term. The group-level forms default instead to a single shared dispersion parameter, which samples well enough to be usable but has itself been checked on one simulated dataset at one choice of sampler settings, with divergent transitions still appearing in the two-facet check. Neither status carries an on-screen label, which is a reason to export the tables rather than read numbers off the screen.

The second disclosure concerns how the coupling is estimated for the subject-level dynamic concurrent designs under the gamma family. Those runs are fitted in two stages: the first fits the per-event models with no coupling term at all, and the second estimates the residual correlation conditionally on the draws the first retained. Nothing from the second stage feeds back into the first. The consequence is specific. The residual correlation is estimated conditionally on the retained first-stage draws, so it inherits their uncertainty, but the copula stage cannot inform the margins in return: the margins' posteriors are those of an uncoupled fit, and no interval in the table is a full joint-posterior credible interval. The recorded provenance also notes that the correlation is known to attenuate at moderate designs, so read its point estimate as conservative. The viewer prints a short note beneath the variance-component table and points at the exported header for the full statement. The two-facet member of the pair has no completed external reference fit behind it, unlike its one-facet sibling, so treat its results as provisional and check them against a one-facet run on the same data where that is possible.

Sources: Rocha et al. (2026), Table 6, for the two-event difference-score coefficients; the residual-coupling parameterizations, the two-stage procedure, and the status flags are the toolbox's conventions.

15.6 Priors, and what they assume about your scale

Bayesian variance-component estimation needs priors, and PsyRAT supplies a set of weakly informative ones so that a first run needs no decisions about them. Those defaults for the Gaussian family are weakly informative ON THE MICROVOLT SCALE. They are not scale-free. The shipped between-person prior is a half-Cauchy with scale 40, generous for standard deviations of a few microvolts but informative for a quantity measured in thousands of units, and informative in the wrong direction for one measured in thousandths. (The gamma family's priors live on log scales of their own, and the dynamic-reliability slope priors on the standardized-predictor scale; the microvolt caution belongs to the Gaussian block.)

The practical rule follows from that. If your scores are ERP amplitudes in microvolts, the defaults are appropriate and you can leave them alone. If they are on any other scale (standardized values, time-frequency power, pupil diameter, or an ERP measure expressed in other units), compare the scale of your data against the scale the defaults assume before you trust a posterior. The check is quick: if the between-person standard deviation you expect is more than roughly an order of magnitude away from a few units, the defaults are doing work you did not intend.

No screen control sets priors, so the settings for this section are arguments rather than menu selections.

Setting	Value
GUI control	none; priors are set programmatically
`psyrat_default_priors`	returns the documented defaults as a struct
`'priors'` name-value	the edited struct, passed to `psyrat_run` or `psyrat_computevarcomp`

The recipe is three steps. Call `psyrat_default_priors` to get the defaults as a struct, edit the fields you want to change, and pass the result as the `'priors'` name-value to `psyrat_run` (or to a direct `psyrat_computevarcomp` call). Supplied values are merged OVER the defaults, so a partial override is safe and any family you leave out keeps its default:

```
pr = psyrat_default_priors;       % the documented defaults, as a struct
pr.single.sig_u = 200;            % wider half-Cauchy scale for a larger measure
psyrat_run('file', 'mydata.csv', 'priors', pr, ...
    'savepath', pwd, 'savename', 'wide_priors.psyrat');
```

A run started this way records the merged prior set in its run-config sidecar, so replaying the sidecar reproduces the same model.

With the defaults untouched, the generated Stan model is byte-identical to the hard-coded models of earlier releases, so leaving them alone changes nothing about your results. What is exposed is the population-level scales: the intercepts and means and the variance-component standard deviations. The structural priors of the non-centered reparameterization are not exposed, and the Gaussian-family LKJ correlation priors are fixed (the Gaussian difference model's residual-correlation location and scale are exposed, and the four-event difference of differences exposes its correlation-prior shape as `priors.dod.lkj` alongside its cell-mean, cell-residual, person, and trial scales, which until this release were fixed inside the model). The gamma family exposes its two correlation-prior shapes.

```
Stan is sampling with 4 chains...
output-1.csv: Warning: file '/Users/peterclayson/Desktop/PsyRAT_CmdStan/temp.data.R' is being read as an 'RDump'
output-1.csv:   This format is deprecated and will not receive new features.
output-1.csv:   Consider saving your data in JSON format instead.
output-1.csv: Informational Message: The current Metropolis proposal is about to be rejected because of the foll
output-1.csv: Exception: lkj_corr_cholesky_lpdf: Random variable[2] is 0, but must be positive! (in '/Users/pete
output-1.csv: If this warning occurs sporadically, such as for highly constrained variable types like covariance
output-1.csv: but if this warning occurs often then your model may be either severely ill-conditioned or misspec
output-1.csv: Informational Message: The current Metropolis proposal is about to be rejected because of the foll
output-1.csv: Exception: lkj_corr_cholesky_lpdf: Random variable[2] is 0, but must be positive! (in '/Users/pete
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output-1.csv: If this warning occurs sporadically, such as for highly constrained variable types like covariance
output-1.csv: but if this warning occurs often then your model may be either severely ill-conditioned or misspec
```

Figure 65: The MATLAB Command Window during a run with Verbose Stan output enabled, showing the LKJ warmup rejection lines and the RDump deprecation notice.

Two families of console message appear during estimation, and both are expected. They print only when Verbose Stan output is set to Yes, which is the default, so most users will see them. The first is a run of Informational Message lines mentioning `lkj_corr_cholesky_lpdf` and a random variable that is zero but must be positive. Designs that estimate a correlation matrix do so through its Cholesky factor, and an occasional early-warmup draw pushes a diagonal element of that factor to underflow; the sampler discards that proposal and continues. Sporadic rejections confined to early warmup are normal. Rejections that persist throughout sampling are not, and would point at an ill-conditioned model. The second is a warning that a `temp.data.R` file is being read as an RDump file and that the format is deprecated. The bundled interface writes data in that legacy format, CmdStan still reads it across the supported version range, and the results are identical. Neither message affects the reported estimates or the convergence diagnostics, which are computed from the retained post-warmup draws.

One prior change is worth knowing about because it moved published-looking numbers. The test-retest occasion priors were widened after their earlier values were found to force the occasion main effect toward zero. Since that effect enters the ABSOLUTE error term, pinning it near zero removed a real source of measurement error and inflated the dependability coefficients that include it, while the generalizability coefficients, which use relative error, changed very little. The sensitivity analysis, the component-by-component comparison, and the full list of configurable prior fields are in priors_and_sensitivity.md.

Report your priors. The exported table header records whether a run used the PsyRAT defaults or a custom set, and the run-config sidecar carries the full specification, so the information needed for a methods section is already written next to your results.

Sources: Rast and Clayson (in press), p. 10, ”The priors adopted here are weakly informative”, for weakly informative prior specification in the location-scale model; the microvolt-scale defaults, the sensitivity analysis, and the merge behavior are the toolbox's own.

16. Analysis reference

One card per routed analysis, plus a card for parallel splits, which reuses other analyses rather than receiving its own number. Use Chapter 5 to find the number; use this chapter to check what it needs, what fits it, and where the results open.

Every card follows the same shape. Status is the maturity label from Chapter 5. Data lists the columns the design requires and the constraints it enforces. GUI names the exact on-screen controls that reach it, on the Specify Inputs screen and the Specify Processing Preferences screen behind the Preferences button. Scripted lists the `psyrat_run` name-value pairs that differ from the defaults; every call also needs a data source, `savepath`, `savename`, and the column-mapping arguments (`idcol` and `meascol` when your headers differ from their defaults of `id` and `meas`, and whichever of `eventcol`, `groupcol`, `timecol`, `dim1col`, `dim2col`, `weightcol` the design uses). Engines names which fitters implement the design. Opens in names the results window. Notes and See carry the caveats and the source of the formulas.

Three conventions run through every card. `ssubjrel` is the current name for subject-level reliability, and `sserrvar` is accepted as a legacy alias. `diffrescor` and `diffwpcov` are one setting under two names, so pass either. The gamma family runs only on CmdStan: the Automatic engine setting is pinned to it, and a forced Native HMC or Native fitlme selection is rejected with an error rather than redirected.

Subject-level analyses are not collected in a section of their own. Each one sits with the design it modifies, and its Use when line says that it reports a coefficient per participant.

16.1 One-facet reliability, single occasion

These are the internal-consistency designs: persons crossed with trials, all data from one session. They are the oldest and best-covered part of the toolbox, and the only difference among the first four is which facet columns you selected.

Analysis 1: One-facet reliability

Status: Supported.

Use when: All the data come from one session, there are no group or condition splits, and you want one dependability coefficient for the sample.

Data: `id` and `meas`, one row per single-trial score. No group, event, or occasion column.

GUI: Specify Inputs, Participant ID and Measurement set; Group ID, Event Type, Occasion ID, Dimension 1 (dynamic), and Items per split (n_i) all left on none. Preferences: Estimate internal consistency of difference scores set to No, Estimate subject-specific error variances set to No.

Scripted: No design flags at all. The defaults (`diffest` 1, `ssubjrel` 1, `dynrel` 1, `splits` 1, `family 'gaussian'`) select this analysis.

Engines: CmdStan, or native fitlme. There is no native HMC implementation.

Opens in: View Results Setup, the single-session viewer.

Notes: This is the design the D-study machinery was built for. You get the dependability-against-trials curve, the trial-count cutoff, and the participant-inclusion table. Gamma is available; the location-scale parameterization is not, because a single-mean design gives the same coefficients under either one.

See: Rocha et al. (2026), Table 2.

Analysis 2: One-facet reliability by group

Status: Supported.

Use when: You have the analysis 1 design plus a grouping variable, and you want a separate coefficient for each group.

Data: `id`, `meas`, and `group`. No event or occasion column.

GUI: As analysis 1, with Group ID set to the grouping column. Select Groups restricts which groups are processed.

Scripted: Pass `groupcol`. Add `whichgroups` to restrict the levels.

Engines: CmdStan, or native fitlme.

Opens in: View Results Setup, the single-session viewer.

Notes: Each group is fitted as a separate stratum rather than as a fixed effect, so the groups do not borrow strength from one another and a small group gets wide intervals of its own. Everything else matches analysis 1.

See: Rocha et al. (2026), Table 2.

Analysis 3: One-facet reliability by event

Status: Supported.

Use when: You have two or more conditions and want each one's reliability separately, not the reliability of their difference.

Data: `id`, `meas`, and `event`. No group or occasion column.

GUI: As analysis 1, with Event Type set to the condition column. Select Events restricts which events are processed. Leave Estimate internal consistency of difference scores on No.

Scripted: Pass `eventcol`. Add `whichevents` to restrict the levels.

Engines: CmdStan, or native fitlme.

Opens in: View Results Setup, the single-session viewer.

Notes: Any number of events is allowed here, unlike the difference designs, which require exactly two or exactly four. If you want the difference rather than the constituents, you want analysis 7.

See: Rocha et al. (2026), Table 2.

Analysis 4: One-facet reliability by group and event

Status: Supported.

Use when: Both a grouping variable and a condition variable are present and you want a coefficient for each combination.

Data: `id`, `meas`, `group`, and `event`. No occasion column.

GUI: As analysis 1, with both Group ID and Event Type set.

Scripted: Pass both `groupcol` and `eventcol`.

Engines: CmdStan, or native fitlme.

Opens in: View Results Setup, the single-session viewer.

Notes: The number of strata is the product of the two level counts, so a design with three groups and four events fits twelve separate models and takes roughly twelve times as long. Trim with Select Groups and Select Events before committing to a long run.

See: Rocha et al. (2026), Table 2.

Analysis 6: Subject-level one-facet reliability

Status: Supported.

Use when: You want a dependability coefficient for each participant rather than one for the sample, from a single session.

Data: `id` and `meas`; optional `group` and `event`. No occasion column.

GUI: As analysis 1, with Preferences, Estimate subject-specific error variances set to Yes.

Scripted: `'ssubjrel', 2`.

Engines: CmdStan, or native HMC. There is no fitlme implementation, because the model gives each participant their own residual standard deviation.

Opens in: View Results Setup, the single-session viewer.

Notes: The fit is slower than analysis 1 by more than the parameter count suggests, since per-participant scale is harder to identify than a pooled one. Participants with few trials get wide per-person intervals, and that is the honest answer rather than a defect.

See: Rocha et al. (2026), Table 2, applied per participant (the toolbox's convention for the per-person read-out).

16.2 Test-retest reliability

Adding an occasion column crosses persons with trials and occasions. Both cards here are static and non-difference; the difference and dynamic two-facet designs live in later sections.

Analysis 5: Test-retest reliability

Status: Supported.

Use when: Participants were measured on two or more occasions and you want coefficients that generalize over occasions as well as trials.

Data: `id`, `meas`, and `time`; optional `group` and `event`.

GUI: Specify Inputs, Occasion ID set to the occasion column. Select Occasions restricts which occasions are processed. Preferences: difference scores No, subject-specific error variances No.

Scripted: Pass `timecol`. Add `whichtimes` to restrict the levels.

Engines: CmdStan, native HMC, or native fitlme. This is the only design both native engines implement, and the automatic cascade prefers HMC.

Opens in: View Results Setup, the test-retest viewer.

Notes: The viewer reports coefficients of equivalence, of stability, and of equivalence and stability, and you pick which one the tables use. Occasion-level variance components are estimated from as many levels as you have occasions, so at two occasions they stay wide and are prior-influenced. Group and event columns are fitted as separate strata, as in analyses 2 to 4.

See: Rocha et al. (2026), Table 3.

Analysis 25: Subject-level test-retest reliability

Status: Supported.

Use when: You want a per-participant coefficient from a multi-occasion design, without difference scores and without a dimension.

Data: `id`, `meas`, and `time`; optional `group` and `event`.

GUI: As analysis 5, with Preferences, Estimate subject-specific error variances set to Yes.

Scripted: Pass `timecol` and `'ssubjrel', 2`.

Engines: CmdStan only. No native engine implements it yet, and the toolbox refuses a native fallback rather than fitting a different model quietly.

Opens in: View Results Setup, the test-retest viewer.

Notes: This is the direct two-facet analog of analysis 6. It inherits the subject-level participant-inclusion contract rather than the test-retest one, which matters if you drive the summary layer from a script. Criterion Score Outputs is not offered for this design (Section 9.7).

See: Rocha et al. (2026), Table 3, applied per participant (the toolbox's convention for the per-person read-out).

16.3 Difference scores

Two conditions are modeled together, and the reported coefficient belongs to their difference. The covariance between the constituents enters the universe-score variance, which is why a difference is often much less reliable than either score behind it.

Analysis 7: One-facet difference-score reliability

Status: Supported under the Gaussian family. Under the gamma family with correlated residuals, the per-person dispersion arm (`'dispersion', 1`) is Experimental; the default single-dispersion arm is tractability-checked but not robustness-checked.

> EXPERIMENTAL under gamma with per-participant dispersion (`dispersion` 1): convergence has not been validated in this release, and exported tables from that arm carry an EXPERIMENTAL line in their headers. The default global-dispersion arm

> (`dispersion` 2) carries a header line saying its one check established tractability, not robustness. Gaussian runs of this design are Supported and carry no such line.

Use when: One session, exactly two conditions, and the score of interest is the difference between them.

Data: `id`, `meas`, and `event` with exactly two unique event types; optional `group`. No occasion column.

GUI: Specify Inputs, Event Type set to the condition column, with exactly two events selected under Select Events. Preferences: Estimate internal consistency of difference scores set to Yes: 2-event difference. Set Estimate residual covariance for co-occurring events to Yes when both conditions come from the same trials.

Scripted: `'diffest', 2` (and `'diffrescor', 2` for the concurrent variant). Under gamma with residual covariance on, `'dispersion', 1` selects the experimental per-person parameterization; the default for that variant is `2`. The `dispersion` setting does not apply to the non-concurrent variant.

Engines: CmdStan, or native HMC.

Opens in: View Results Setup, the single-session viewer.

Notes: The concurrent variant keeps the same analysis number and changes the model behind it, so a run with residual covariance on is not comparable to one with it off. Under gamma, the concurrent variant additionally requires the Statistics and Machine Learning Toolbox even on the CmdStan path, and the missing-function failure lands after sampling has finished.

See: Rocha et al. (2026), Table 6.

Analysis 8: Subject-level one-facet difference-score reliability

Status: Supported under the Gaussian family. Experimental under the gamma family with correlated residuals; the non-concurrent gamma variant is Supported.

> EXPERIMENTAL under gamma with correlated residuals: that variant's only gamma arm uses per-participant dispersion, convergence has not been validated in this release, and its exported tables carry an EXPERIMENTAL line in their headers. The non-concurrent gamma variant and Gaussian runs of this design are Supported and carry no such line.

Use when: You want each participant's own difference-score reliability from a single session.

Data: `id`, `meas`, and `event` with exactly two unique event types; optional `group`. No occasion column.

GUI: As analysis 7, with Preferences, Estimate subject-specific error variances set to Yes.

Scripted: `'diffest', 2, 'ssubjrel', 2` (and `'diffrescor', 2` for the concurrent variant).

Engines: CmdStan, or native HMC.

Opens in: View Results Setup, the single-session viewer.

Notes: Under the Gaussian family the concurrent variant estimates a residual correlation for each participant, unlike analysis 7's single population correlation; under gamma it delegates to analysis

7's population copula and folds the per-person cross-covariance in downstream. The gamma concurrent variant also requires the Statistics and Machine Learning Toolbox even on the CmdStan path, and a missing toolbox surfaces only after sampling finishes. This design has no non-experimental gamma parameterization: a per-participant coefficient needs a per-participant dispersion parameter, and a single sample-wide dispersion is a group-level estimand. Run it under the Gaussian family unless you have a specific reason not to.

See: Rocha et al. (2026), Table 6.

Analysis 10: Test-retest difference-score reliability

Status: Supported under the Gaussian family. Under the gamma family with correlated residuals, the per-person dispersion arm (`'dispersion', 1`) is Experimental; the default single-dispersion arm is tractability-checked but not robustness-checked.

> EXPERIMENTAL under gamma with per-participant dispersion (`dispersion` 1): convergence has not been validated in this release, and exported tables from that arm carry an EXPERIMENTAL line in their headers. The default global-dispersion arm (`dispersion` 2) carries a header line saying its one check established tractability, not robustness. Gaussian runs of this design are Supported and carry no such line.

Use when: Two conditions measured on two or more occasions, and the score of interest is the difference.

Data: `id`, `meas`, `time`, and `event` with exactly two unique event types; optional `group`.

GUI: As analysis 7, with an Occasion ID column assigned on the input screen (subject-level reliability off).

Scripted: Pass `timecol` with `'diffest', 2` (and `'diffrescor', 2` for the concurrent variant; under gamma the same `'dispersion'` arms and the same Statistics and Machine Learning Toolbox requirement as analysis 7 apply).

Engines: CmdStan, or native HMC.

Opens in: View Results Setup, the test-retest viewer.

Notes: Six crossed variance components are estimated for the difference, and the facet main effects (occasions especially) are the weakly identified ones at realistic occasion counts. Read the person and person-by-facet components first. Criterion Score Outputs is not offered for this design (Section 9.7).

See: Rocha et al. (2026), Table 6, two-facet form.

16.4 Difference of differences

One four-event contrast, (ERP_1 - ERP_2) - (ERP_3 - ERP_4), reported as a single score. The static form is here; the person-specific dynamic forms have their own section.

Analysis 9: Difference-of-differences reliability

Status: Supported.

Use when: The quantity of interest is a contrast between two difference scores, such as a condition effect compared across two tasks or two electrode sites.

Data: `id`, `meas`, and `event` with exactly four unique event types; optional `group`. No occasion column. The four events must be assigned to ERP_1 through ERP_4 before the run.

GUI: Preferences, Estimate internal consistency of difference scores set to Yes: 4-event DoD, and Estimate residual covariance for co-occurring events set to No. On Specify Inputs, Event Type set to the condition column with exactly four events selected, then click DoD Contrast... and assign four unique events on the Configure DoD Contrast screen.

Scripted: `'diffest', 3` plus `'dodmap', {...}`, a 1-by-4 cell of event labels in ERP_1 to ERP_4 order.

Engines: CmdStan, or native HMC.

Opens in: DoD Reliability Setup.

Notes: Residual covariance is unavailable in every family, and a concurrent request is rejected rather than ignored: the contrast is defined with all four constituents treated as measured on distinct trials. The default contrast weights are 1, -1, -1, 1. The intraclass correlations this design reports are derived quantities rather than entries in the published difference-score table, and the exported headers label them that way.

See: Toolbox convention, built on the Rocha et al. (2026) Table 6 difference-score partitioning.

16.5 Dynamic reliability

Selecting a Dimension 1 column makes the mean and the residual scale functions of a standardized between-person covariate, so the result is a reliability surface over that dimension rather than a single number. The dimension is standardized once across the whole sample before any group or event stratum is split off, so zero is the overall mean everywhere. All fourteen cards in this section open in the same viewer.

Analysis 11: Dynamic one-facet reliability

Status: Supported.

Use when: One session, one score, and you want reliability reported as a function of a participant characteristic.

Data: `id`, `meas`, and a continuous between-person `dim1` (one value per participant); optional `dim2`, `group`, and `event`. No occasion column.

GUI: Specify Inputs, Dimension 1 (dynamic) set to the covariate column, and optionally Dimension 2 (dynamic). Preferences: difference scores No, subject-specific error variances No.

Scripted: Pass `dim1col` (and `dim2col`) with `'dynrel', 2`.

Engines: CmdStan, or native HMC.

Opens in: Dynamic Reliability: View Results.

Notes: Dimension 2 adds a second predictor and its interaction with the first, which turns the reported surface two-dimensional and roughly doubles what the viewer has to render. Group and event columns are fitted as separate strata.

See: Rast and Clayson (in press).

Analysis 26: Subject-level dynamic one-facet reliability

Status: Supported.

Use when: You want each participant's own conditional reliability from the one-facet dynamic model rather than one population surface.

Data: As analysis 11.

GUI: As analysis 11, with Preferences, Estimate subject-specific error variances set to Yes.

Scripted: `'dynrel', 2, 'ssubjrel', 2` plus `dim1col`.

Engines: CmdStan only.

Opens in: Dynamic Reliability: View Results.

Notes: The model and the posterior draws are identical to analysis 11. What changes is the read-out: each participant's own residual heterogeneity becomes their own conditional coefficient instead of being pooled into one surface. If you want both views, one fit gives you both.

See: Rast and Clayson (in press).

Analysis 14: Dynamic test-retest reliability

Status: Supported.

Use when: Multi-occasion data, one score, and reliability reported as a function of a participant characteristic.

Data: `id`, `meas`, `time`, and `dim1`; optional `dim2`, `group`, and `event`.

GUI: As analysis 11, with Occasion ID also set.

Scripted: Pass `timecol` and `dim1col` with `'dynrel', 2`.

Engines: CmdStan, or native HMC.

Opens in: Dynamic Reliability: View Results.

Notes: Five crossed mean components sit on top of the one-facet dynamic model. Occasion-level components stay weakly identified at two occasions, and the viewer prints them as diagnostics rather than as findings.

See: Rast and Clayson (in press).

Analysis 27: Subject-level dynamic test-retest reliability

Status: Supported.

Use when: You want per-participant conditional reliability from the two-facet dynamic model.

Data: As analysis 14.

GUI: As analysis 14, with Preferences, Estimate subject-specific error variances set to Yes.

Scripted: Pass `timecol` and `dim1col` with `'dynrel', 2, 'ssubjrel', 2`.

Engines: CmdStan only.

Opens in: Dynamic Reliability: View Results.

Notes: Same relationship to analysis 14 that analysis 26 has to analysis 11: one model, one set of draws, a per-participant read-out instead of a pooled surface.

See: Rast and Clayson (in press).

Analysis 12: Dynamic difference-score reliability

Status: Supported.

Use when: One session, a two-event difference score, and reliability reported as a function of a participant characteristic.

Data: `id`, `meas`, `dim1`, and `event` with exactly two unique event types; optional `dim2` and `group`. No occasion column.

GUI: Specify Inputs, Dimension 1 (dynamic) set; Preferences, Estimate internal consistency of difference scores set to Yes: 2-event difference, with Estimate residual covariance for co-occurring events set to No.

Scripted: `'dynrel', 2, 'diffest', 2` plus `dim1col`.

Engines: CmdStan, or native HMC.

Opens in: Dynamic Reliability: View Results.

Notes: The dimension carries event-specific slopes on both the mean and the per-event residual scale, so a dimension can move one condition's error more than the other's. Both gamma parameterizations are available here, which is not true of the subject-level sibling.

See: Rast and Clayson (in press), with the difference-score partitioning of Rocha et al. (2026), Table 6.

Analysis 13: Subject-level dynamic difference-score reliability

Status: Supported.

Use when: You want each participant's conditional reliability for a two-event difference score, from one session.

Data: As analysis 12.

GUI: As analysis 12, with Preferences, Estimate subject-specific error variances set to Yes.

Scripted: `'dynrel', 2, 'diffest', 2, 'ssubjrel', 2` plus `dim1col`.

Engines: CmdStan, or native HMC.

Opens in: Dynamic Reliability: View Results.

Notes: Under gamma this design accepts only the location-scale parameterization, and an explicit request for the other one is rejected rather than silently substituted. The Gaussian family has no such restriction.

See: Rast and Clayson (in press), with the difference-score partitioning of Rocha et al. (2026), Table 6.

Analysis 17: Dynamic difference-score reliability, correlated residuals

Status: Supported.

Use when: As analysis 12, and the two conditions are recorded on the same trials so their residuals are coupled.

Data: As analysis 12.

GUI: As analysis 12, with Estimate residual covariance for co-occurring events set to Yes.

Scripted: `'dynrel', 2, 'diffest', 2, 'diffrescor', 2` plus `dim1col`.

Engines: CmdStan, or native HMC.

Opens in: Dynamic Reliability: View Results.

Notes: Gaussian only. The gamma family has no implementation of this design, and a gamma request is rejected before the fit starts. Unlike the static difference designs, switching residual covariance on here changes the analysis number, so the two runs are labeled differently in the output.

See: Rast and Clayson (in press), with the difference-score partitioning of Rocha et al. (2026), Table 6.

Analysis 19: Subject-level dynamic difference-score reliability, correlated residuals

Status: Supported.

Use when: As analysis 13, with co-occurring conditions, and one residual correlation shared across the sample.

Data: As analysis 12.

GUI: As analysis 13, with Estimate residual covariance for co-occurring events set to Yes, and Specify Inputs, Residual correlation left on Population (shared).

Scripted: `'dynrel', 2, 'diffest', 2, 'ssubjrel', 2, 'diffrescor', 2` plus `dim1col`.

Engines: CmdStan for either family; native HMC for the Gaussian arm only.

Opens in: Dynamic Reliability: View Results.

Notes: Under gamma this design is fitted by a modular cut workflow: the margins are fitted first without the coupling, and the coupling is estimated afterward, so the result is a cut posterior rather than a full Bayesian one. The exported provenance says so on every table. The Gaussian arm is an ordinary joint fit.

See: Rast and Clayson (in press), with the difference-score partitioning of Rocha et al. (2026), Table 6.

Analysis 21: Subject-level dynamic difference-score reliability, per-participant residual correlation

Status: Supported.

Use when: As analysis 19, but the residual coupling itself is expected to differ across participants.

Data: As analysis 12.

GUI: As analysis 19, with Specify Inputs, Residual correlation set to Per-subject. That popup stays greyed out until difference scores, subject-specific error variances, and residual covariance are all on.

Scripted: `'dynrel', 2, 'diffest', 2, 'ssubjrel', 2, 'diffrescor', 2, 'ssrescor', 2` plus `dim1col`.

Engines: CmdStan, or native HMC.

Opens in: Dynamic Reliability: View Results.

Notes: Gaussian only; the gamma family has no implementation. A separate correlation per participant is the most demanding thing this family asks of the sampler, so budget warmup accordingly and read the diagnostics before the estimates.

See: Rast and Clayson (in press), with the difference-score partitioning of Rocha et al. (2026), Table 6.

Analysis 15: Dynamic test-retest difference-score reliability

Status: Supported.

Use when: As analysis 12, with an occasion facet.

Data: `id`, `meas`, `time`, `dim1`, and `event` with exactly two unique event types; optional `dim2` and `group`.

GUI: As analysis 12, with Occasion ID also set.

Scripted: Pass `timecol` and `dim1col` with `'dynrel', 2, 'diffest', 2`.

Engines: CmdStan, or native HMC.

Opens in: Dynamic Reliability: View Results.

Notes: Gaussian only; the gamma family has no implementation of the two-facet dynamic difference designs. Six crossed cross-condition blocks are estimated on top of the dimension slopes, which makes this one of the slower fits in the toolbox.

See: Rast and Clayson (in press), with the difference-score partitioning of Rocha et al. (2026), Table 6.

Analysis 16: Subject-level dynamic test-retest difference-score reliability

Status: Supported.

Use when: As analysis 13, with an occasion facet.

Data: As analysis 15.

GUI: As analysis 15, with Preferences, Estimate subject-specific error variances set to Yes.

Scripted: Pass `timecol` and `dim1col` with `'dynrel', 2, 'diffest', 2, 'ssubjrel', 2`.

Engines: CmdStan, or native HMC.

Opens in: Dynamic Reliability: View Results.

Notes: Gaussian only. The dynamic branch is the only route by which a subject-level difference score reaches an occasion facet (this analysis and its concurrent siblings, analyses 20 and 22); the static equivalent is rejected before the fit starts.

See: Rast and Clayson (in press), with the difference-score partitioning of Rocha et al. (2026), Table 6.

Analysis 18: Dynamic test-retest difference-score reliability, correlated residuals

Status: Supported.

Use when: As analysis 15, with co-occurring conditions.

Data: As analysis 15.

GUI: As analysis 15, with Estimate residual covariance for co-occurring events set to Yes.

Scripted: Pass `timecol` and `dim1col` with `'dynrel', 2, 'diffest', 2, 'diffrescor', 2`.

Engines: CmdStan, or native HMC.

Opens in: Dynamic Reliability: View Results.

Notes: Gaussian only. The bivariate residual is evaluated row by row, since its scale moves with the dimension, which is the main reason this design costs more per draw than analysis 15.

See: Rast and Clayson (in press), with the difference-score partitioning of Rocha et al. (2026), Table 6.

Analysis 20: Subject-level dynamic test-retest difference-score reliability, correlated residuals

Status: Supported under the Gaussian family. Development-only under the gamma family.

Use when: As analysis 19, with an occasion facet.

Data: As analysis 15.

GUI: As analysis 16, with Estimate residual covariance for co-occurring events set to Yes and Residual correlation left on Population (shared).

Scripted: Pass `timecol` and `dim1col` with `'dynrel', 2, 'diffest', 2, 'ssubjrel', 2, 'diffrescor', 2`.

Engines: CmdStan for either family; native HMC for the Gaussian arm only.

Opens in: Dynamic Reliability: View Results.

Notes: The gamma arm carries no user-facing status label anywhere in the software, so this card is the disclosure: its two-facet configuration has no empirical anchor, nothing outside the toolbox has been reproduced with it, and it should not be reported from. Like analysis 19 under gamma, it is a cut posterior. The Gaussian arm is ordinary and supported.

See: Rast and Clayson (in press), with the difference-score partitioning of Rocha et al. (2026), Table 6.

Analysis 22: Subject-level dynamic test-retest difference-score reliability, per-participant residual correlation

Status: Supported.

Use when: As analysis 21, with an occasion facet.

Data: As analysis 15.

GUI: As analysis 20, with Residual correlation set to Per-subject.

Scripted: Pass `timecol` and `dim1col` with `'dynrel', 2, 'diffest', 2, 'ssubjrel', 2, 'diffrescor', 2, 'ssrescor', 2`.

Engines: CmdStan, or native HMC.

Opens in: Dynamic Reliability: View Results.

Notes: Gaussian only. This is the most demanding model in the two-event dynamic family: a per-participant location and scale block, five crossed cross-condition blocks, dimension slopes, and a per-participant residual correlation. Treat a first run as a pilot and check the diagnostics before planning around the numbers.

See: Rast and Clayson (in press), with the difference-score partitioning of Rocha et al. (2026), Table 6.

16.6 Dynamic difference of differences

These two designs report each participant's own conditional reliability for a four-event contrast. They are person-specific by definition, so subject-level error variances must be on, and they are non-concurrent by definition, so residual covariance cannot be turned on.

Analysis 28: Dynamic difference-of-differences reliability

Status: Supported.

Use when: One session, a four-event contrast, and you want each participant's own conditional reliability for it.

Data: `id`, `meas`, `dim1`, and `event` with exactly four unique event types, assigned to ERP_1 through ERP_4; optional `dim2` and `group`. No occasion column. Every participant needs at least one trial in all four cells.

GUI: Specify Inputs, Dimension 1 (dynamic) set, Event Type set with exactly four events selected, and the DoD Contrast... assignment saved. Preferences: difference scores Yes: 4-event DoD, subject-specific error variances Yes, residual covariance No.

Scripted: `'dynrel', 2, 'diffest', 3, 'ssubjrel', 2` plus `dim1col` and `dodmap`.

Engines: CmdStan only, in both likelihood families.

Opens in: Dynamic Reliability: View Results, not the static DoD viewer.

Notes: All six pairwise residual covariances are fixed at zero by the estimand, so this is a full Bayesian posterior rather than a cut one. The per-participant coverage check runs before the fit: a participant missing a cell entirely would make their coefficient undefined, and finding that out after a multi-hour run is worse than being stopped at the start. Expect a large saved result, since the posterior carries four margins and a per-participant block.

See: Toolbox convention, built on Rast and Clayson (in press) and the Rocha et al. (2026) Table 6 difference-score partitioning.

Analysis 29: Dynamic test-retest difference-of-differences reliability

Status: Supported.

Use when: As analysis 28, with an occasion facet.

Data: As analysis 28, plus `time`.

GUI: As analysis 28, with Occasion ID also set.

Scripted: Pass `timecol` and `dim1col` with `'dynrel', 2, 'diffest', 3, 'ssubjrel', 2` and `dodmap`.

Engines: CmdStan only, in both likelihood families.

Opens in: Dynamic Reliability: View Results.

Notes: The largest and slowest routed design. Occasion standard deviations are prior-dominated at two occasions and stay printed as diagnostics. One observed run of this design took about half an hour and wrote a result file of roughly 300 MB, and loading a file that size back into the viewer takes noticeably longer than the fit's last iteration might lead you to expect.

See: Toolbox convention, built on Rast and Clayson (in press) and the Rocha et al. (2026) Table 6 difference-score partitioning.

16.7 Reliability from splits

Each row is the mean of several items rather than one trial, and a weight column says how many items went into it. Declaring the splits parallel reuses the ordinary machinery; declaring them nonparallel fits a weighted observed-design model with no projection.

Parallel splits (no analysis number)

Status: Supported.

Use when: Your rows are split means and every split holds the same number of items, as with an even split of the trials into halves or quarters.

Data: `id`, `meas` (the split mean), and a required items-per-split `weight` column; optional `group`, `event`, and `time`.

GUI: Specify Inputs, Items per split (n_i) set to the weight column, and Split type set to Parallel (equal n_i). Everything else as for the analysis it reuses. Preferences must keep difference scores No and subject-specific error variances No.

Scripted: Pass `weightcol` with `'splits', 2`.

Engines: Whatever the reused analysis offers, since the model is unchanged.

Opens in: View Results Setup, the single-session or test-retest viewer, with trial relabeled as split throughout the controls and tables.

Notes: No analysis number of its own. A single-occasion run reuses analyses 1 to 4 and a multi-occasion run reuses analysis 5, with the split mean as the score and the split count in place of the trial count. Unequal counts are allowed but warned about, on the grounds that the nonparallel model probably fits your data better. Gamma is permanently unavailable for any split design: the family models single-trial dispersion, and a split mean is not a single trial.

See: Rocha et al. (2026), Tables 4 and 5 for the split framing; the coefficients computed are the Table 2 and Table 3 forms with the split count in place of the trial count.

Analysis 23: Nonparallel splits, one occasion

Status: Supported.

Use when: Your rows are split means from one session and the splits hold different numbers of items.

Data: `id`, `meas`, and `weight`, with genuine variation in `weight` across splits. No group, event, or occasion column.

GUI: Specify Inputs, Items per split (n_i) set and Split type set to Nonparallel (unequal n_i), with Group ID, Event Type, Occasion ID, and Dimension 1 (dynamic) all left on none.

Scripted: Pass `weightcol` with `'splits', 3`.

Engines: CmdStan, or native fitlme.

Opens in: Data-Splits Reliability: View Results.

Notes: Equal item counts are rejected outright, because the per-item residual and the person-by-split interaction are identified only through the variation in the counts. Low variation is allowed with a warning, and the warning is worth heeding. Dependability is exact for this design; generalizability is a downward-biased lower bound (and its standard error a corresponding upper bound), because a split mean has already lost the item-level partition the relative coefficient needs. Every relative coefficient in the output carries a flag saying so. There is no D-study projection and no trial-count cutoff. A group or event column is rejected rather than pooled.

See: Rocha et al. (2026), Table 4.

Analysis 24: Nonparallel splits, test-retest

Status: Supported.

Use when: As analysis 23, with two or more occasions.

Data: `id`, `meas`, `weight`, and `time`; optional `group` and `event`.

GUI: As analysis 23, with Occasion ID also set. Group ID and Event Type are allowed here.

Scripted: Pass `weightcol` and `timecol` with `'splits', 3`.

Engines: CmdStan, or native fitlme.

Opens in: Data-Splits Reliability: View Results.

Notes: Unlike analysis 23, this design is group and event aware and fits each stratum separately. It reports the three test-retest coefficients at the observed design only, with no projection and no cutoff. The same exactness asymmetry applies: dependability is exact, generalizability is a lower bound, and the flag travels with every relative coefficient.

See: Rocha et al. (2026), Table 5.

17. Interpreting and reporting results

A dependability estimate earns its place in a manuscript at the point where it changes something: how many trials a task presents, which participants enter an analysis, or whether a score is treated as adequate for the question being asked. Psychometric reporting is part of the analysis rather than an appendix to it, and the case for treating it that way has been made at length for psychophysiology (Clayson, 2024). The material below covers the decision rules that a reliability estimate can legitimately support, paragraph templates for a methods section, and the record a reviewer needs in order to check the work.

17.1 Decision rules

What counts as adequate dependability

For internal consistency, a coefficient of .80 is the working standard once a measure has proven promising, with group-difference research held to it, and .70 is the floor for early-stage or exploratory work (Clayson & Miller, 2017b, p. 62). The reason is not convention. Reliability bounds the correlation an observed score can show with an external variable, and it lowers statistical power as it falls. Against a criterion whose own reliability is .90, a score with a reliability of .70 cannot correlate above about .79, whereas a score with a reliability of .80 can reach about .85 (Clayson & Miller, 2017b, p. 61, Table 1; the attenuation bound is defined on relative-error reliability, so substituting a dependability estimate gives a conservative bound). A study whose scores sit near the floor is therefore spending sample size on measurement error, and a null result from such a study is difficult to interpret.

The exception runs upward, not downward. When a decision is made about an individual score rather than about a group mean, the standard summarized by Clayson and Miller (2017b, p. 62) sets .90 as a bare minimum and prefers .95. Screening and treatment selection both fall under that heading, and the gap between a coefficient adequate for a group contrast and one adequate for an individual decision is the reason psychometric readiness is a precondition for biomarker claims rather than a consequence of them (Clayson, 2025).

The illegitimate reason to move a threshold is that too few participants survive it. Clayson and Miller (2017b, p. 62) note that the .70 standard is frequently cited to justify a single cutoff for every stage of research, which the source of that standard does not support. The threshold belongs to the question, and it should be fixed and recorded before any effect of interest is inspected. A threshold chosen after inspecting how much data it costs is a researcher degree of freedom, and reporting it as though it had been set in advance misdescribes the analysis.

Planning trial counts from the D-study projection

The dependability-versus-trials plot and the trial-cutoff table project the coefficient forward under the variance components estimated from the data in hand, and they answer a design question: how many trials per participant would be needed to reach the stated threshold. The projection holds the estimated components fixed, so it describes what dependability would be if scores at the larger trial count behaved like the scores observed. It does not certify that a future sample will reach the projected value, because reliability is a property of scores in a context rather than of the measure (Clayson & Miller, 2017b), and a dependability estimate obtained in one sample can fail to transfer to another (Baldwin et al., 2015).

A trial cutoff reported as -1 means that no trial count within the projected range reached the specified threshold. That result is informative rather than broken. It says the between-person variance in these scores is small relative to the within-person variance, and that the remedy lies in the task or the scoring pipeline rather than in collecting more trials of the same kind. The projection is also what lets you satisfy the third of the three reporting guidelines Clayson and Miller (2017b, p. 62) set out, reporting how the trial cutoff for data exclusion was justified, from your own data rather than from a number inherited from a prior study; applied test-retest work has followed the same practice (Carbine et al., 2021).

Participant inclusion and the reporting of exclusions

Subject-level estimates give a dependability coefficient for each participant rather than one coefficient for the group, so they support an inclusion rule that a group-level estimate cannot: a participant whose own scores are too imprecise for the intended use can be identified as such, even when the group coefficient is adequate (Clayson, Brush, & Hajcak, 2021). The toolbox reports the retained and excluded identifiers together with the coefficient that produced the split, and both belong in the manuscript.

Three things should be reported, and reported in this order relative to the analysis. The threshold and the rule come first, fixed before the effects of interest are examined. The counts come second: how many participants met the rule, how many did not, and what the trial distribution looked like on each side of the line. The estimates come third. Reporting the exclusions without the threshold, or the threshold without the count, leaves a reader unable to tell whether the inclusion rule was a measurement decision or an outcome-dependent one. Where the analysis was preregistered, the threshold is the element most worth registering, because it is the one whose post hoc adjustment is hardest to detect from the published record.

17.2 Methods paragraphs

Each block below is a complete paragraph with bracketed slots for the reader's own numbers. Replace every bracketed span, delete the alternatives that do not apply, and check the resulting coefficient name against the label printed in the exported table header, which states the estimand the numbers belong to. Every version number, chain count, and seed value should be copied from the run rather than from this page: the toolbox version appears in the startup banner and in the header of every exported table, and the CmdStan version and the sampler settings appear in the run-configuration sidecar described in Section 17.3.

Single-session (internal consistency, one facet).

```
Internal consistency of [COMPONENT] scores was evaluated within a
generalizability-theory framework treating trials as the sole facet of
measurement (Rocha et al., 2026). Variance components were estimated in a
Bayesian framework with the PsyRAT Toolbox (version [X.Y.Z];
https://github.com/peclayson/PsyRAT), which fit the models in CmdStan
[CMDSTAN VERSION] using [4] chains with [5,000] warmup and [5,000] sampling
iterations per chain and a random seed of [12345]. Dependability (the
absolute-error coefficient) of [COMPONENT] scores averaged over [X] trials was
[.XX], 95% CrI [[.XX], [.XX]]. A decision study indicated that [X] trials were
required to reach the dependability threshold of [.XX], which was specified
before the data were examined. Applying that trial minimum retained [N] of [N]
```

participants; [N] participants were excluded for contributing too few usable trials.

Test-retest (two facets: trials and occasions).

Test-retest reliability of [COMPONENT] scores across [N] sessions separated by [INTERVAL] was evaluated within a generalizability-theory framework treating trials and occasions as facets (Clayson, Carbine, et al., 2021; Rocha et al., 2026). Variance components were estimated in a Bayesian framework with the PsyRAT Toolbox (version [X.Y.Z]; https://github.com/peclayson/PsyRAT), which fit the models in CmdStan [CMDSTAN VERSION] using [4] chains with [5,000] warmup and [5,000] sampling iterations per chain and a random seed of [12345]. The coefficient reported is the coefficient of [equivalence and stability (CES) / equivalence (CE) / stability (CS)] under the [absolute-error (dependability) / relative-error (generalizability)] definition, evaluated for a single occasion (n'_o = 1). For scores averaged over [X] trials per occasion, the coefficient was [.XX], 95% CrI [[.XX], [.XX]].

Difference scores.

Internal consistency of the [CONDITION A] minus [CONDITION B] difference score was estimated with the generalizability-theory difference-score expressions of Rocha et al. (2026), which model the two constituent scores jointly and carry their covariance rather than treating them as independent (Clayson, Baldwin, & Larson, 2021). Variance components were estimated in a Bayesian framework with the PsyRAT Toolbox (version [X.Y.Z]; https://github.com/peclayson/PsyRAT), which fit the models in CmdStan [CMDSTAN VERSION] using [4] chains with [5,000] warmup and [5,000] sampling iterations per chain and a random seed of [12345]. Dependability of the difference score at [X] trials per condition was [.XX], 95% CrI [[.XX], [.XX]]; dependability of the constituent [CONDITION A] and [CONDITION B] scores was [.XX], 95% CrI [[.XX], [.XX]], and [.XX], 95% CrI [[.XX], [.XX]], respectively. The intraclass correlation reported alongside the difference score is a single-observation (n' = 1) quantity derived from the same variance components rather than a published difference-score expression.

Reliability from data splits.

Reliability was estimated from [N] splits of [X] items each rather than from single trials, with the number of items per split supplied as a weight, using the observed-design generalizability-theory expressions of Rocha et al. (2026); see Vispoel, Xu, and Schneider (2022) for splits rather than items as the unit of analysis in generalizability theory, and Vispoel et al. (2018) for the broader framework. The splits were [parallel (equal items per split) /
↪ nonparallel
(unequal items per split)]. Variance components were estimated in a Bayesian framework with the PsyRAT Toolbox (version [X.Y.Z];
https://github.com/peclayson/PsyRAT), which fit the models in CmdStan [CMDSTAN VERSION] using [4] chains with [5,000] warmup and [5,000] sampling iterations per chain and a random seed of [12345]. Dependability (absolute error) was [.XX], 95% CrI [[.XX], [.XX]]. Because only split means were observed, the item main effect within splits could not be separated from the

```
person-by-split residual; the generalizability (relative-error) coefficient is
therefore a lower bound and the relative standard error of measurement an upper
bound, whereas dependability is exact.
```

Dynamic (conditional) reliability.

```
Reliability was modeled as a function of [DIMENSION] using the mixed-effects
location-scale extension of generalizability theory described by Rast and
Clayson (in press), in which the residual standard deviation varies
systematically across measurement conditions and persons rather than being held
constant. [DIMENSION] was standardized across the full sample, so a value of
zero corresponds to the sample mean. Variance components were estimated in a
Bayesian framework with the PsyRAT Toolbox (version [X.Y.Z];
https://github.com/peclayson/PsyRAT), which fit the models in CmdStan
[CMDSTAN VERSION] using [4] chains with [5,000] warmup and [5,000] sampling
iterations per chain and a random seed of [12345]. At [X] trials, dependability
was [.XX], 95% CrI [[.XX], [.XX]], at one standard deviation below the mean of
[DIMENSION]; [.XX], 95% CrI [[.XX], [.XX]], at the mean; and [.XX], 95% CrI
[[.XX], [.XX]], at one standard deviation above. The posterior mean of the
residual-scale slope on [DIMENSION] was [X.XX], 95% CrI [[X.XX], [X.XX]].
```

Two habits make these paragraphs harder to misread. Name the coefficient rather than calling it reliability, because dependability and generalizability differ by whether the trial main effect enters absolute error, and a reader cannot recover which was reported from the number alone. Report the interval with the point estimate in every case, since a dependability estimate from a small sample can sit above a threshold with an interval that spans it.

17.3 Reproducibility for reviewers

Four items constitute the archive for a reliability analysis. The first is the set of exported tables with their headers intact. Each exported table carries the toolbox version, the generation date, the estimation seed, the analysis label, the estimation engine, whether the priors were the toolbox defaults or customized, the convergence verdict, the divergent-transition count, and any design-specific disclosure that applies to the estimand. Tables built from a summary also record the credible-interval width the summary was built at; the splits, dynamic-reliability, and difference-of-differences families build none and carry no width line. Stripping the header rows before archiving removes the only record that travels with the numbers. The second is the `*_runconfig.json` sidecar written next to the saved result, which records the seed, chains, warmup and sampling counts, any sampler overrides, the column mapping, the design flags, the coefficient selectors in effect, the full prior specification, the detected CmdStan version, the MATLAB version, and content hashes of the input data. The third is the saved `.psyrat` result file, which allows the posterior summaries to be regenerated without refitting. The fourth is the analysis dataset itself, in the long format the toolbox reads, where participant-level data can be shared. Where it cannot, the sidecar's data hashes and row count still document what entered estimation.

The seed guarantees less than it appears to. Holding the data, the settings, and the CmdStan build fixed, the same seed reproduces the same posterior draws exactly, which is what makes a rerun on the same machine a genuine check. Outside that combination, on a different CmdStan build or a different machine, the draws are not guaranteed to be bit-identical, and the toolbox warns rather than stops when a replay detects a CmdStan version different from the one recorded. Two runs under different CmdStan builds should agree closely, but a version mismatch is a reason to re-estimate rather than a license to compare numbers across the two runs. A run produced by a native MATLAB engine instead of CmdStan is Bayesian but is not bit-identical to a CmdStan run, and the engine that produced a result is recorded in both the table header and the sidecar. Reporting the seed without the CmdStan version, the engine, the sampler settings, and the within-chain parallelization setting therefore claims a reproducibility the seed alone does not deliver.

Three toolbox conventions account for most of the questions a careful reader raises, and each is a choice the software makes rather than a result it found. Test-retest coefficients default to a single occasion, n'_o = 1, so the reported coefficient describes a score from one occasion rather than a composite averaged over occasions; the multi-occasion composite is available and is labeled as such in the output, and Chapter 9 covers the selector. Intraclass correlations reported alongside a difference score are derived from the fitted variance components at n' = 1 rather than taken from a published difference-score expression; the table column is labeled ICC (n'=1), and the derivation is documented here and in the tutorial rather than in the header; Chapter 12 and Chapter 15 give the context. Splits analyses report an exact dependability coefficient and a generalizability coefficient that is a downward-biased lower bound, because split means do not identify the item main effect within splits separately from the person-involving residual; the relative standard error of measurement is correspondingly an upper bound, and Chapter 13 works through the design. Disclosing these three in a supplement costs a paragraph and prevents a reviewer from reading a convention as an error.

17.4 Reporting the toolbox itself

Cite two things: Rocha et al. (2026), which is the source of the estimators the toolbox implements, and the tagged GitHub release of the version that produced the estimates. The paper describes the method, and the version identifies the code. The version string is returned by `psyrat_defineversion`, printed in the startup banner, written into the header of every exported table, and recorded in the run-configuration sidecar, so the correct value is available from any saved output rather than from memory. Machine-readable metadata for the software entry is in `CITATION.cff` in the repository, and the release page supplies the year a reference style may require. Analyses that use the dynamic or conditional designs should additionally cite Rast and Clayson (in press), which is the basis for the location-scale estimation those designs rest on.

Clayson and Miller (2017a) describes the ERA Toolbox, PsyRAT's predecessor. It is the right citation for ERA and the wrong one for PsyRAT, and the distinction matters because the two toolboxes do not implement the same set of designs.

Bug reports and questions go to the issue tracker at `https://github.com/peclayson/PsyRAT/issues`. A report that includes the version string and, where the analysis can be shared, the `*_runconfig.json` sidecar is reproducible on another machine; a report without them usually is not, because the sidecar carries the seed, the priors, the design flags, and the data hashes that determine the result. Chapter 18 covers the errors that have a known cause and a documented fix, and is worth checking first.

18. Troubleshooting and FAQ

Most problems announce themselves in a dialog box or in the MATLAB console, and most of those messages carry a ”How to fix:” line. Section 18.1 lists the messages by the point in the workflow where they appear. Section 18.2 covers the situations that produce no message at all, which are the ones that waste the most time. Section 18.3 deals with convergence, and 18.4 answers the questions that come up most often.

If a run failed and you want to reproduce the problem for someone else, keep the `*_runconfig.json` sidecar written next to the output file. It records the seed and every estimation setting.

18.1 Error-message lookup

Find the message by when it appeared. Each entry quotes the message, or the fragment of it that identifies the case, then says what the toolbox found and what to change. Where a message contains a name in quotes, the toolbox substitutes your column, participant, or file name there.

In viewer output, figures, and exported tables, messages that mention trials say ”splits” instead when the run analyzes data splits; the loading and validation messages below keep the word ”trials” regardless.

While the data file is loading and prepared

The first three rows appear at file selection. The rest fire when the table is prepared, which happens after you click Analyze and name the output file; in a batch, a failure there writes a `.txt` error report and moves to the next measurement.

Message	What it means	What to do
WARNING: File type not supported	The file extension is not one MATLAB's `readtable` handles. The accepted extensions are `.txt`, `.dat`, `.csv`, `.xls`, `.xlsx`, `.xlsb`, `.xlsm`, `.xltm`, `.xltx`, and `.ods`.	Re-export the data with one of those extensions.
WARNING: File type not successfully loaded	The file has a supported extension, but no delimiter the toolbox tried (comma, space, tab, semicolon, vertical bar) split it into more than one column.	Re-export as comma-separated or tab-separated text, or as a spreadsheet.
No data file was selected.	The file picker was cancelled.	Click Process New Data again and choose an input file.

Message	What it means	What to do
Error: Column headers not properly specified / Please specify the headers for	One or more of the roles you assigned points at a header the file does not contain. The message names the roles that failed (Subject ID, Measurement, Group, Event, Occasion, Dimension 1, Dimension 2, Split Weight (n_i)).	Correct the header spelling in the file, or reassign that role on the input screen.
Error: The data in the measurement column is not numeric	The column assigned to Measurement holds text. This usually means a measurement column was assigned to the wrong header, or the exported file carries a units string or a placeholder such as `NA` in that column.	Assign a numeric column (ERP amplitudes or latencies) and reload.
Error: The Dimension 1 header matches more than one column	Two columns in the file share the header you selected for Dimension 1. The same message exists for Dimension 2 and for Split Weight (n_i).	Give every column in the input file a unique header.
Error: The data in the Dimension 1 column is not numeric	A dimension predictor is a continuous between-person covariate and must be numeric. The same check runs on Dimension 2.	Convert the covariate to numeric, or select a different column.
Error: The Split Weight (n_i) column must contain positive whole numbers	The items-per-split column holds a zero, a negative value, a fraction, or a non-finite value.	Supply the number of items averaged into each split mean as a positive integer.
Error: Specified groups to process do not exist in data	The group selection carried over from an earlier file, or the group column changed. The same message exists for events and for occasions.	Reopen Select Groups (or Select Events, or Select Occasions) and choose from the labels the current file contains.
Error: Participants differ in the number of distinct events	The count of distinct event labels is not the same for every participant. The table of ids and counts printed above the error names the participants to check. This compares counts; it does not check that every participant has every event.	Drop the affected participants, or restrict the run to events every participant has.

Message	What it means	What to do
Error: Participants differ in the number of distinct occasion cells	The same check applied to occasions, or to event-by-occasion cells when both facets are present.	Drop the affected participants, or restrict the run to the occasions every participant completed.
The '...' column contains missing values	A label column (id, group, event, or time) has a blank, `NaN`, `<missing>`, or `<undefined>` entry. Filling one in is not safe: every missing entry would collapse into a single level and pool observations from different participants, groups, events, or occasions.	Supply a label for every row, or drop the affected rows before loading.
has numeric values that convert to the SAME text label	Two distinct numeric label values render as the same text, which would merge two levels into one. MATLAB renders numbers with about five significant digits, so values that differ only past that point become indistinguishable.	Supply the label column as text, rounding or naming the levels the way you want them reported.

After clicking Analyze

Most of the checks below run when you click Analyze, before the save dialog appears (the whitespace check and the DoD single-measurement notice come just after it), and a failure returns you to the setup screen. The rows drawn from the Select buttons, the DoD Contrast screen, and the Preferences screen fire on those screens instead.

Message (dialog title)	What it means	What to do
No measurement columns were selected. (Measurement Not Defined)	Nothing is selected for Measurement.	Select one or more numeric data columns for Measurement.
The same source column was assigned to multiple PsyRAT inputs: (Unique variable names not provided)	One column is doing two jobs, for example serving as both Event and Occasion. The dialog names the columns involved.	Assign a unique column to each of ID, Group, Event, Occasion, Dimension, and Items per split.
Measurement columns must be unique from ID, Group, Event, Occasion, Dimension, and Items-per-split columns. (Measurement column conflict)	A column selected as Measurement is also assigned to a facet or dimension role.	Remove the overlapping column from the Measurement selection.

Message (dialog title)	What it means	What to do
The following selected measurement columns are not numeric: (Measurement data not numeric)	At least one selected measurement column holds text. The dialog names each one.	Select only numeric columns for Measurement.
No group column is selected. (Group Column Not Defined)	Select Groups was clicked while Group is set to none. The event twin reads ”No event column is selected.” (Event Column Not Defined) and the occasion twin ”No occasion/time column is selected.” (Occasion Column Not Defined).	Assign the column first, then open the selection screen.
Dimension 2 was selected without Dimension 1. (Dynamic reliability)	The dynamic-reliability model requires Dimension 1; Dimension 2 is optional and cannot stand alone.	Select a Dimension 1 column, or set both dimension pickers to none.
Dynamic reliability supports difference-of-differences (DoD) scores only as the person-specific nonconcurrent design: subject-level reliability must be Yes. (Dynamic reliability)	A four-event difference-of-differences run was requested with dimension predictors but without subject-level error variances.	Set Estimate subject-specific error variances to Yes, switch to a two-event difference score, or clear the Dimension columns.
Reliability with data splits cannot be combined with dynamic reliability. (Reliability with splits)	An items-per-split column and a Dimension 1 column were both selected.	Clear the Dimension columns, or set Items per split (n_i) to none.
Reliability with data splits does not support difference scores. (Reliability with splits)	Difference-score estimation is on for a splits run.	Set difference scores to No, or set Items per split (n_i) to none.
Reliability with data splits does not support subject-level error variances. (Reliability with splits)	Subject-level error variances are on for a splits run.	Set Estimate subject-specific error variances to No, or set Items per split (n_i) to none.
Single-subject error variance estimation is not supported for DIFFERENCE scores when an occasion/time facet is present (Single-subject error variance not supported for retest)	Subject-level error variances were requested for a difference score in a test-retest design, without dimension predictors.	Turn off subject-level error variances, remove the Occasion column, or turn off difference scores. Non-difference subject-level test-retest runs are supported and are not blocked here.

Message (dialog title)	What it means	What to do
This difference-score configuration is not supported when an occasion/time facet is present (test-retest workflow). (Difference score reliability is not supported for retest)	A difference-of-differences contrast was requested alongside an Occasion column in a non-dynamic run. Two-event difference scores with an Occasion column are supported.	Use a two-event difference score, turn off difference-score estimation, or remove the Occasion column and run the single-session DoD mode.
Difference-score reliability requires an event/condition column. (Event column not specified)	Difference scores need two conditions to subtract, and no Event column is assigned. The DoD Contrast... button raises the same title with the body "DoD contrast setup requires an event column."	Assign the column that stores event labels.
DoD contrast setup requires exactly 4 selected events; current selection has N. (Need exactly 4 selected events)	The DoD Contrast... button was clicked with fewer or more than four events selected.	Click Select Events and choose exactly four, then configure the contrast.
Difference-score reliability requires exactly two event types. (Only 1 event / Only 2 events for difference score)	Fewer or more than two event types are selected.	Click Select Events and choose exactly two.
Difference-of-differences reliability requires exactly four event types. (Need 4 events for difference-of-differences / Need exactly 4 events for difference-of-differences)	Fewer or more than four event types are selected for a four-event contrast; the first title covers too few, the second too many.	Click Select Events and choose exactly four.
Residual covariance is not available for difference-of-differences. (Residual covariance not supported for DoD)	The four-event contrast is estimated with the within-person residual covariance defined to be zero, in every likelihood family.	In Preferences, set Estimate residual covariance for co-occurring events to No. Co-occurring events are supported for the two-event difference score, so you can instead switch to the two-event mode.
Difference-of-differences reliability requires an explicit contrast mapping. (Configure DoD contrast before Analyze)	The four selected events have not been assigned to the ERP_1 through ERP_4 slots, or the event column changed after they were.	Click DoD Contrast... and assign each slot before clicking Analyze.
ERP_1 to ERP_4 must map to four unique events. (DoD mapping requires unique events)	Two slots point at the same event label.	Set each ERP slot to a different event.

Message (dialog title)	What it means	What to do
The event selection or event column changed, so the configured difference-of-differences contrast was cleared. (DoD contrast cleared)	This is a warning, not a failure. Changing the events invalidates a saved slot mapping.	Reconfigure the contrast with the DoD Contrast button before running.
Number of chains is invalid. (Invalid Number of Chains)	The Number of Chains field on the Preferences screen holds a value below 3 or text that does not parse as a real number; this and the five rows below fire when you click Save on that screen.	Set it to an integer of 3 or higher. The default is 4.
Warmup iterations are invalid. (Invalid Warmup Iterations)	Warmup iterations must be a whole number of 0 or higher.	Correct the field.
Sampling iterations are invalid. (Invalid Sampling Iterations)	Sampling iterations must be a whole number greater than 0.	Correct the field.
Random seed is invalid. (Invalid Random Seed)	The seed must be a whole number of 0 or higher.	Correct the field. The default is 12345.
Adapt delta is invalid. (Invalid Adapt Delta)	Adapt delta was supplied but is not strictly between 0 and 1.	Leave the field blank to use the model's default (a per-model tuned value, or CmdStan's 0.80 for the plain one-facet designs), or enter something like 0.95.
Max tree depth is invalid. (Invalid Max Tree Depth)	Max tree depth was supplied but is not a positive whole number.	Leave the field blank to use CmdStan's default of 10, or enter a positive integer such as 12.
Test-retest workflows usually require at least 10,000 iterations. (Recommended Iteration Increase)	This is advice, not a rejection. It appears when the Preferences screen opens with an Occasion column assigned and the total iteration count below 10,000.	Raise warmup and sampling in Preferences, or proceed and watch the convergence report.
Output file location was not selected. (File Error)	The save dialog was cancelled.	Choose a filename and folder.
The selected filename/path contains whitespace. (Whitespace detected)	CmdStan does not accept spaces in filenames or paths.	Choose a filename and folder with no spaces. The dialog reopens so you can pick again.

Message (dialog title)	What it means	What to do
Difference-of-differences mode currently processes one measurement at a time. (DoD mode: single measurement only)	This is a warning that must be acknowledged before the run begins. More than one measurement column was selected, and only the first will be analyzed. The dialog names it.	Dismiss the dialog to continue with the first column, or cancel and select a single measurement.

During estimation

These checks run after the data file is loaded and before CmdStan starts. In a multi-measurement run, a failure here stops that measurement, writes a `.txt` error report next to the output, and the run continues with the next measurement.

Message	What it means	What to do
The dataset is a table with column headers but no data rows.	The file has headers and nothing else. A headers-only export loads without complaint, so the emptiness surfaces here.	Check that the file was exported with its single-trial rows, then reload.
Dataset is missing required column header(s):	The prepared table lacks `id` or `meas`.	Confirm the ID and Measurement assignments on the input screen.
Measurement column '...' is not numeric.	The prepared measurement column is not numeric.	Assign a numeric column.
Input data include missing cells, which CmdStan cannot process.	At least one cell anywhere in the prepared table is missing.	Remove or impute the affected rows so every row has id, measurement, and facet values.
Column '...' is a logical (true/false) array, which cannot be used as a label column.	A label column (id, group, event, or time) was supplied as a MATLAB logical array. Left alone, its downstream behavior depends on the analysis, so the type is rejected outright.	Convert the column to numeric codes, a categorical, or text labels, then rerun.

Message	What it means	What to do
Measurement column '...' contains non-positive values, which the Gamma / scaled-chi-square family does not support.	The Gamma family has strictly positive support. ERP amplitudes are routinely negative, which is why the default Gaussian family is unaffected.	Use the Gamma family only with strictly positive scores such as single-trial time-frequency power, or switch back to Gaussian.
Output filename contains whitespace / Output save path contains whitespace / Output save path does not exist	The save location fails the CmdStan filename rules, or the folder was removed between selection and the run.	Choose a space-free filename and an existing folder.
Reliability with data splits requires an items-per-split (n_i) column, but none was found.	A splits analysis was requested without the weight column. Both the parallel and the nonparallel model need it.	Select the split-weight column on the input screen.
Nonparallel splits were requested, but every split has the same items-per-split (n_i) count.	The nonparallel model identifies the per-item residual and the person-by-split interaction only through differences in the 1/n_i precision weights. Equal counts leave them confounded.	Supply splits with unequal n_i, or declare the splits parallel.
Items-per-split (n_i) variation is low (coefficient of variation ...)	A warning. The run proceeds. Below a coefficient of variation of 0.25 the per-item residual and the person-by-split interaction may be weakly identified.	Interpret the nonparallel estimates with caution, or build splits with more variable n_i.
Parallel splits were declared, but the items-per-split (n_i) counts are unequal across splits.	A warning. Your declaration is honored. Parallel splits assume equal n_i.	Consider declaring the splits nonparallel so the unequal weights are modeled.

Message	What it means	What to do
Person-specific dynamic difference-of-differences requires every participant to have trials in all four events	Each participant's own per-cell counts are the divisors for this design, so a participant with an empty cell has an undefined coefficient. The message names up to five participants.	Exclude those participants, or choose events every participant completed.
Difference-score reliability found N event type(s); exactly 2 are required.	The prepared table, after subsetting, does not carry exactly two event labels.	Restrict the event selection to two.
Difference-of-differences reliability found N event type(s); exactly 4 are required.	The prepared table does not carry exactly four event labels.	Restrict the event selection to four.
No pairable observations were found for residual covariance estimation.	Residual-covariance estimation needs at least one within-participant paired observation for the two events in each modeled group.	Turn off residual covariance, or supply paired trials.
Correlated residual estimation for difference scores requires paired observations. Event counts differ for at least one	The stricter check applied once estimation reshapes events into paired rows: the two events must have equal trial counts within each participant, not merely one pairable trial.	Equalize the per-participant trial counts across the two events, or turn off residual covariance.
The dataset has only N participant(s), fewer than the recommended minimum of 20.	A warning. Variance-component estimates, especially interaction terms, are unstable and prior-dominated at small N. With a group column, the count is evaluated per group.	Proceed with added caution about the intervals, or collect more participants.
Median trials per cell is X, fewer than the recommended minimum of 10	A warning. Within-person measurement error is poorly estimated when cells are thin.	Judge whether the trial counts are adequate for the scored component.

Message	What it means	What to do
The observed event x occasion grid is incomplete:	A warning. Some participant-by-event-by-occasion cells contain no observations. Estimation is valid with an incomplete design, but quantities for the empty cells are model-based rather than observed.	Verify the empty cells reflect intended exclusions, such as artifact rejection, rather than a data-preparation error.
The dimension column '...' was not found in the data table.	A dynamic-reliability run reached the standardization step without its covariate.	Reassign Dimension 1 on the input screen.
Participant '...' has no finite value for dimension '...'.	A dimension predictor is blank or non-finite for that participant.	Supply the covariate for every participant, or drop the participant.
Dimension '...' is not constant within participant '...'.	Dimension predictors are between-person covariates and must carry one value per participant, repeated on every row.	Correct the column so each participant has a single value.
Dimension '...' has a non-positive between-participant standard deviation	Every participant has the same covariate value, so there is nothing to standardize and no slope to estimate.	Use a covariate that varies across participants.
Detected CmdStan X, but PsyRAT requires CmdStan Y or newer.	The generated models use array declaration syntax introduced in Stan 2.26.	Install a newer CmdStan (`psyrat_installdependents` installs the tested version) and restart MATLAB.

Message	What it means	What to do
CmdStan is unavailable (...). Falling back to the native HMC engine	A warning, not a failure. CmdStan could not be found, and a native MATLAB engine implements this design. Native HMC results are Bayesian but are not bit-identical to CmdStan; native fitlme intervals are a frequentist parametric bootstrap rather than credible intervals.	Install CmdStan for the reference path, or accept the approximation and say which engine you used when reporting.
No estimation engine can run this analysis (design '...').	CmdStan is unavailable and no installed native engine implements this design.	Install CmdStan, or install the Statistics and Machine Learning Toolbox for the designs the native engines cover.
Warning: N divergent transition(s) detected.	Sampling produced divergences. This never triggers a rerun on its own and never changes the convergence flag, but it signals possible bias even when R-hat looks acceptable.	See Section 18.3.
Preflight warning (preflight:fewParticipants): The dataset has only N participant(s), fewer than the recommended minimum of 20.	Non-fatal. Fewer than 20 participants in the dataset, or in one group. The fit proceeds; in the GUI the line also appears in the modal Preflight warnings dialog before the fit.	Expect wide intervals on every between-person component; report the interval, and treat a cutoff from so few participants with caution.
Preflight warning (preflight:lowMedianTrials): Median trials per cell is X, fewer than the recommended minimum of 10.	Non-fatal. The median trial count per participant, or per participant-by-event cell, is below 10.	Read the reliability-versus-trials curve and the cutoff before reporting; the design may need more trials than most participants supplied.

Message	What it means	What to do
Preflight warning (preflight:incompleteEventTimeCells): The observed event x occasion grid is incomplete	Non-fatal. Some participant-by-event-by-occasion cells hold no trials (Chapter 6, section 6.5); estimates for those cells are model-based.	Verify each empty cell is an intended exclusion and not a data-preparation error.
Model did not converge for measurement '...'.	In a multi-measurement run, a non-converged fit stops that measurement rather than prompting.	Raise warmup and sampling and rerun that measurement.
No reliability output was generated for measurement '...'.	Estimation returned without a reliability structure.	Review the convergence report and the CmdStan logs for that run.
All N selected measurements failed to process. (All Measurements Failed)	Every measurement in a batch failed. The setup screen reopens behind the dialog.	Read the generated `.txt` error reports in the save folder; each names the reason for one measurement.
Temporary directory could not be removed.	The run finished and its results were saved. Only the scratch folder CmdStan wrote into could not be deleted, usually because a file in it is still open or locked.	Delete the `psyrat_stan_*` folder inside the save path the message prints. Nothing about the results is affected.
Run-config sidecar could not be written:	The results file was saved but the `*_runconfig.json` sidecar was not, usually a permissions problem in the save folder.	Rerun into a writable folder if you want the sidecar. The estimates already saved are unaffected.

An undefined-function error naming `normcdf` or `gaminv` means the Statistics and Machine Learning Toolbox is missing. Those functions are used only by the gamma concurrent-difference designs, and only after sampling has finished, so the failure arrives at the end of a long run. Install the toolbox before starting a gamma run with residual covariance turned on.

While viewing results

Message (dialog title)	What it means	What to do
No PsyRAT results file was selected. (File Error)	The file picker was cancelled.	Click View Results again and choose a `.psyrat` file.

Message (dialog title)	What it means	What to do
The estimation model did not converge for at least one measurement. (Estimation diagnostics)	The saved fit failed the convergence criteria and was kept anyway, either because a rerun was declined or because a headless run exhausted its automatic retries.	Treat the coefficients as provisional and rerun with more warmup and sampling. See Section 18.3.
Sampling produced N divergent transition(s). (Estimation diagnostics)	The same divergence count reported in the console, repeated here so a viewer-only user sees it.	Rerun with more warmup iterations or stronger priors before reporting the estimates.
These results were estimated with the native HMC engine (Native estimation engine)	The fit did not run through CmdStan. The equivalent notice exists for the native fitlme engine.	Record the engine alongside the estimates. Native results are not bit-identical to CmdStan.
ERP_1 to ERP_4 must map to four unique events. (DoD mapping requires four unique events)	Two of the four ERP popups on DoD Reliability Setup name the same event.	Set each slot to a different event.
Data do not reach reliability threshold after extrapolation / Set a lower reliability threshold (Data do not meet reliability threshold)	No participant retained data meeting the threshold you entered, so there is nothing to summarize and no output is drawn.	Lower the Reliability Cutoff, or accept that this measure does not reach that threshold in this dataset.
Trial cutoffs for adequate dependability could not be calculated / Data are too variable or there are not enough trials (Cutoff not calculable)	The projected dependability never reaches the threshold at any trial count the toolbox evaluated. The corresponding cell in the cutoff table shows -1.	Lower the threshold, or report that no cutoff exists at this threshold for these scores.
Not enough trials are present in the current data / Cutoffs represent an extrapolation beyond the data (Extrapolation beyond data)	A cutoff was found, but it lies above the largest trial count actually observed. The projection is a model-based extrapolation.	Report the cutoff as an extrapolation, or collect more trials before treating it as an inclusion rule.

Message (dialog title)	What it means	What to do
NOT INTERPRETABLE AS A RELIABILITY: one or more quantities that must be variances came out negative. (Coefficient is not interpretable as a reliability)	A difference-score quantity that must be a variance is negative in part of the posterior. Nothing was clipped; the coefficient shown is the formula evaluated exactly. The dialog lays out the two possible causes; which one applies depends on whether the two conditions were evaluated at equal trial counts, which you know and it does not.	Follow the instruction in the dialog. If the two conditions were evaluated at different trial counts, re-evaluate at equal counts. If they were already equal, treat the estimated components themselves as suspect.

A trial cutoff printed as -1 is not a count. It records that no cutoff could be found at the reliability threshold you specified.

18.2 Operational traps

MATLAB looks hung after a fit finishes

SYMPTOM: sampling completes, the console goes quiet, and no results window appears.

The viewer raises a blocking dialog before it builds anything; it is not modal (it does not stay in front of other windows), which is exactly why other windows can cover it. When the fit failed its convergence criteria or produced divergent transitions, that dialog is titled Estimation diagnostics; when the fit used a native engine, it is titled Native estimation engine. Execution waits until the dialog is dismissed. The dialog can end up behind the MATLAB desktop or on another display, which leaves nothing visible to click.

Bring every MATLAB window forward and look for the dialog, then dismiss it. The viewer opens immediately afterward.

MATLAB reports Busy while the processor is idle

SYMPTOM: the status bar says Busy, but the machine is doing no work.

This is the same cause as above, seen from a different angle. Several points in the workflow block until a dialog is dismissed: the two viewer notices, the Rerun prompt raised when chains fail to converge, and the acknowledgment that a difference-of-differences run will process only the first selected measurement. A blocked dialog uses no processor time, which is what distinguishes this from a genuinely slow fit.

Check for a hidden dialog before assuming a run is stuck. A real CmdStan fit keeps one processor core busy per chain (more with within-chain parallelization on).

Ctrl+C does not stop a run

SYMPTOM: interrupting MATLAB returns the prompt, but the machine stays loaded and the sampler keeps running.

The models are compiled to C++ and run as separate operating-system processes, one per chain. Interrupting MATLAB stops MATLAB's own execution; it does not signal those processes.

To stop a run, end the CmdStan processes yourself. On macOS, open Activity Monitor and quit each process whose name matches the model being fit. On Windows, do the same in Task Manager. You will find one process per chain, so expect four of them at the default setting. Delete the leftover `psyrat_stan_*` temporary folder in your save directory afterward; a run that is killed never reaches the cleanup step.

A small dataset sits a long time before sampling starts

SYMPTOM: the console announces the run, then nothing happens for a while, and only afterward do iteration counts appear.

Every fit writes generated Stan source and compiles it to a native executable before sampling. Each run writes that model under a new, timestamped name into the CmdStan artifact folder, so the compile is repeated rather than reused from a previous run. Compilation cost does not depend on how much data you have (about 8 s per generated model was measured on the capture machine), so on a small dataset it is a noticeable share of the run. With Verbose Stan output enabled the console announces the compile step; with it disabled the wait is silent.

The artifact folder is named `PsyRAT_CmdStan` and sits on your Desktop, or in MATLAB's current folder if you have no Desktop folder.

Wait it out, and count the compile into your estimate of how long a run takes. That folder accumulates one model and one executable per run. It holds build artifacts, not results, and nothing in the toolbox reads it back after a run finishes, so you can clear it when it grows.

Console messages that look like errors and are not

SYMPTOM: with Verbose Stan output enabled, the console fills with exception text early in sampling.

Two message families are expected and neither reflects a problem with the model or the reported estimates. The first is a series of informational messages saying a Metropolis proposal is about to be rejected, naming `lkj_corr_cholesky_lpdf` and a random variable that is 0 but must be positive. Designs that estimate a correlation matrix through its Cholesky factor produce these when an early warmup draw pushes a diagonal element to underflow. The sampler discards that proposal and continues, and adaptation moves initialization away from the boundary. Rejections confined to early warmup mean the sampler is working; rejections that persist throughout sampling would point to an ill-conditioned model instead.

The second is a warning that a `temp.data.R` file is being read as an RDump file, and that the format is deprecated. The bundled MatlabStan writes each model's data in that legacy format. CmdStan still reads it across the supported version range and returns identical results.

Neither message appears unless Verbose Stan output is selected. The full discussion is in priors_and_sensitivity.md.

A .psyrat file is enormous

SYMPTOM: a subject-level, dynamic-reliability, or difference-of-differences run writes an output file of hundreds of megabytes or more.

The output file is a MATLAB data file holding the whole analysis state: the raw input table as read from disk, the prepared analysis table, and the retained posterior draws for every stored quantity. Designs that report a quantity for each participant store one draw array per participant, so the file grows with the number of participants multiplied by the number of retained draws. Doubling sampling iterations doubles that part of the file.

Nothing inside the file can be removed selectively; it is one saved structure. For long-term storage, keep the `*_runconfig.json` sidecar and the tables you exported from the viewer. The sidecar records the seed and every setting, and `psyrat_run` can replay it, so the fit is reproducible without the file itself (see Chapter 14). Keep the `.psyrat` file only while you still want to reopen the viewer.

Replaying a gamma run configuration saved before August 3, 2026

SYMPTOM: replaying an old gamma-family configuration produces different variance components from the run it came from.

Two dates matter. On August 7, 2026 the default scale parameterization for the Gamma family changed from log-nu to location-scale on every design that supports location-scale. Sidecars written from August 3, 2026 onward record the resolved value, so replaying one of those reproduces its own run. A sidecar written before that date carries no value at all and replays under the current default, which for a movable design is now the other estimand. The two are not comparable.

Pass `'gammascale', 1` explicitly when replaying a configuration older than that date. The exceptions are the five location-scale-only designs (analyses 13, 19, 20, 28, and 29), which reject an explicit 1. Gaussian runs, which is to say almost all ERP amplitude work, are unaffected. The option reference in `help psyrat_run` lists which designs accept which parameterization.

The startup release check reports a problem

SYMPTOM: `psyrat_start` prints a warning about GitHub releases before the home screen opens.

The startup banner checks the installed version against the published releases. Several outcomes produce a warning rather than a failure. A message saying the installed version is newer than the latest stable release means you are running a beta or development build. A message saying no official stable releases were found and only prerelease (beta) versions have been published is the expected state while the project is in beta, and means the check worked. A message that the releases interface returned 404 means the check could not reach release metadata. A message that the check could not be completed at all names a network or firewall problem.

None of these prevents an analysis from running or changes any result. The toolbox loads and the home screen opens as usual. Installation problems announce themselves differently, through the dependency messages printed above the version banner (see Chapter 3).

18.3 Convergence clinic

Every fit is checked before its results are kept, and the check is automatic. It reads two diagnostics for every monitored parameter. The first is R-hat, the potential scale reduction factor, which compares variation between chains with variation within them; a fit fails when any parameter's R-hat is 1.1 or higher. The second is the effective sample size, which counts how many independent draws the correlated chain is worth; a fit fails when any parameter's effective sample size falls below ten times the number of chains, which is 40 at the default of four chains. A fit passes only when every parameter clears both.

Divergent transitions are counted and reported separately. They never change the pass or fail decision and never trigger a rerun. A divergence means the sampler could not follow the posterior geometry accurately at that point, so a nonzero count signals that variance components and coefficients may be biased even though R-hat looks acceptable. Treat a run with divergences as one to repeat with more warmup or stronger priors before its numbers are reported.

When the check fails in an interactive run, a dialog appears reading ”Chains did not converge” and asking whether to rerun with more iterations.

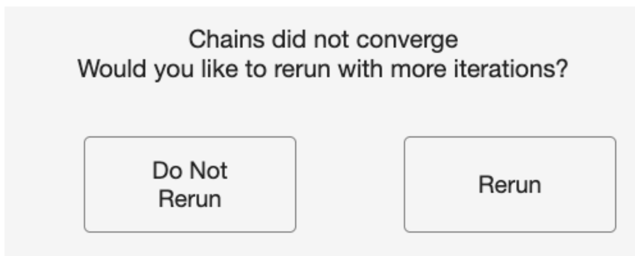

Figure 66: The dialog raised when a fit fails the convergence check. Rerun doubles both warmup and sampling; Do Not Rerun keeps the non-converged fit.

Clicking Rerun doubles the warmup count and doubles the sampling count, then fits the model again from the start with the same seed and data. If the doubled total still falls below 1,000 iterations, the toolbox uses 500 warmup and 500 sampling instead. The loop repeats for as long as you keep choosing Rerun. Clicking Do Not Rerun keeps the fit as it stands; the file is saved, and the viewer raises the Estimation diagnostics notice every time it is opened. Closing the dialog with the window's close box counts as Do Not Rerun.

Doubling is convenient rather than efficient. If a design is failing repeatedly, step out of the loop and work through the following in order.

Raise the iteration counts outright first, in Preferences, rather than doubling from a low starting point. The defaults are 5,000 warmup and 5,000 sampling. For test-retest designs I would not start below 10,000 total, and the toolbox says so when Preferences opens with an Occasion column assigned and fewer than 10,000 total iterations set. Use as many iterations as the model needs to converge.

If more iterations alone do not settle it, and especially if divergences are present, raise Adapt delta next. Leaving the field blank keeps each model's tuned value; a value such as 0.95 or 0.99 makes the sampler take smaller steps and accept fewer divergences, at the cost of a slower run. Raise Max tree depth only after that, and only if the console reports that the maximum tree depth was exceeded. The default is 10; 12 is a reasonable next value. Raising it when the problem is not tree depth buys nothing but run time.

If the sampler still struggles, reconsider the priors. The defaults are weakly informative on the ERP microvolt scale and are not scale-free, so scores on a different scale can make them either informative or far too loose, and either can produce a hard posterior geometry. Chapter 15 covers how to set them.

If none of that helps, reconsider the design. A model that will not converge is often estimating components the data cannot identify: an occasion variance from two occasions, an interaction from thin cells, or a nonparallel splits model from near-equal split sizes. The preflight warnings about participant counts, median trials per cell, and low items-per-split variation are pointing at exactly this. Chapter 5 covers which designs a given dataset can support.

My numbers differ slightly from the manual's

The reference values printed in the tutorials came from real runs on one machine, at the settings each tutorial states, using the datasets shipped in `test_data/`. Estimation is bit-reproducible for a fixed seed on a fixed platform and CmdStan version when within-chain parallelization is off (the default). With it on, the threaded models use Stan's `reduce_sum`, whose work partitioning the scheduler chooses at run time, so two runs at the same seed can differ in the trailing digits even on one machine; the export header and the sidecar record the thread count so that setting is visible. Across platforms or CmdStan versions the draws are not reproducible either: floating-point arithmetic and compiler behavior differ enough between builds to move the trailing decimals of a posterior summary.

Agreement within about .02 of a printed point estimate is expected and is not a sign that anything is wrong. A larger gap is worth checking before you conclude there is a problem. Start with the settings: the seed should be 12345, and the chain, warmup, and sampling counts should match what the tutorial specifies. Then confirm the run went through CmdStan rather than a native engine, because a native fit is a different estimator whose intervals are not credible intervals. Last, confirm the shipped dataset was loaded unmodified, with the same measurement column and the same event subset. If the settings and the data check out and the gap persists, report it (see Section 18.4).

18.4 Frequently asked questions

Why are dependability and generalizability different for the same data?

They answer different questions and use different error terms. Dependability is the absolute-error coefficient: it asks how well a participant's observed score estimates that participant's universe score on the measurement scale, so the error term includes the trial main effect divided by the number of trials. Generalizability is the relative-error coefficient: it asks how well the ordering of participants is reproduced, so the trial main effect drops out because it shifts everyone equally. Dependability is therefore the smaller of the two whenever the trial main effect is nonzero. Report the one that matches the decision you are making, and name it.

Why can a variance component not come out negative?

Every variance component in the estimation model is parameterized as a standard deviation constrained to be non-negative, and the toolbox squares it. A negative component of the kind familiar from method-of-moments ANOVA cannot occur. What can go negative is a derived quantity in a difference-score design, where a covariance term is subtracted from the variances that form it. The toolbox reports that value exactly as the formula produces it and raises a notice explaining what happened, rather than clipping it to zero and handing back a coefficient that looks ordinary and is not.

Why is there no progress bar?

Sampling runs in separate CmdStan processes rather than inside MATLAB, and the toolbox does not translate their output into a MATLAB progress indicator. Turn on Verbose Stan output in Preferences and CmdStan prints its own iteration counter to the console, which is the closest thing available. The trade is that verbose output also prints the warmup messages described in Section 18.2.

Why do trace plots appear only for some analyses?

Trace plots are rendered for the plain one-facet internal-consistency analysis only. For every other analysis family the toolbox prints a console line saying plots are not supported or not rendered, and draws nothing. Convergence is still checked for every analysis, by the criteria in Section 18.3; the plots are a visual supplement to that check, not a substitute for it.

Can I analyze scores that are not ERP amplitudes?

Yes. The estimators are general and take a numeric score column, so reaction times, single-trial power values, and questionnaire item scores all fit the input format. The caution is the priors: the defaults are weakly informative on the ERP microvolt scale and are not scale-free, so a measure on a much larger or much smaller scale can end up with priors that are informative in a way you did not intend. Set priors appropriate to your scale before trusting the output (Chapter 15).

Which viewer will open for my file?

The toolbox reads the analysis recorded in the results file and routes to the viewer for that analysis family, so the choice is not yours to make and does not depend on what you were doing when you opened the file. One-facet analyses open the single-session viewer, test-retest analyses the test-retest viewer, four-event difference-of-differences the DoD viewer, nonparallel splits the splits viewer, and every dynamic-reliability variant the dynamic-reliability viewer. Chapter 9 describes each one.

Where is the guidance on citing the toolbox?

In Chapter 17, together with what to report about the estimation settings and the design. Full references are in Appendix B.

How do I report a bug?

Open an issue on the project's GitHub issue tracker at `https://github.com/peclayson/PsyRAT/issues`. Include the version string printed in the startup banner, which is also written into the header of every exported table, and attach the `*_runconfig.json` sidecar saved next to the analysis output. The sidecar records the seed, the estimation settings, and the design, which together are usually enough to reproduce the problem without your data. If the failure produced a `.txt` error report in the save folder, include that too.

Why does the version say beta?

The label is deliberate. It marks a release whose interfaces and defaults may still change, and it flags that some designs are documented as experimental. `CHANGELOG.md` records what the current label covers, including which designs carry that status. Nothing about the label limits what the toolbox will run.

Can estimation run without CmdStan?

From a script, for a subset of designs, yes. The GUI, no. `psyrat_start` stops at its dependency check when CmdStan cannot be located, prints that the dependencies were not found, and offers the guided installer of Chapter 3, so without CmdStan there is no Specify Inputs screen, no Estimation engine popup, and no viewer. Two native MATLAB engines are available through `psyrat_run` (Chapter 14): Native fitlme, a frequentist restricted maximum likelihood (REML) fitter for the plain one-facet, test-retest, and nonparallel splits designs, and Native HMC, a Hamiltonian Monte Carlo sampler for many of the location-scale and difference-score designs. Leave the `engine` option at its automatic default and `psyrat_run` falls back to whichever native engine implements the design and says so in a console banner; the exported tables from `psyrat_report` carry the same statement in their header. The results are not equivalent. Native HMC is Bayesian but not bit-identical to CmdStan, and the fitlme engine's intervals come from a parametric bootstrap rather than a posterior, so they are not credible intervals. Both native engines require the Statistics and Machine Learning Toolbox. Chapter 5 lists which designs each engine covers. Install CmdStan for the reference path and for the GUI.

Why can the save path not contain spaces?

CmdStan does not accept whitespace in filenames or paths, and a run that reaches it with a space in either fails in a way that is hard to read. The toolbox checks for whitespace three times rather than letting that happen: at the save dialog, at the shared preflight before estimation starts, and again in the estimation setup. Choose a folder with no spaces in any part of its path. The same restriction applies to the folder holding CmdStan build artifacts; a whitespace path there is replaced with a temporary folder and a warning.

What does the weight column mean for a splits analysis?

It is n_i, the number of items or trials averaged into that split's mean score, and it must be a positive whole number on every row. A splits input has one row per split rather than one row per trial, so the toolbox has no other way to know how many observations each score summarizes. The weight enters the model as the sampling variance of a mean: a split of n_i items carries the per-item residual variance divided by n_i. For nonparallel splits it does more than that. Variation in n_i across splits is the only thing that separates the per-item residual from the person-by-split interaction, which is why equal weights are rejected outright in that mode and low variation raises a warning. Chapter 13 works through a splits analysis end to end.

Appendix A. Glossary

Terms as the toolbox uses them. Where a term has a broader meaning elsewhere in the psychometric literature, the definition here is the one that governs the output, the GUI labels, and Chapter 4.

chains. Independent runs of the sampler over the same model and data, started from different initial values. Agreement between chains is what makes R-hat informative, so the toolbox requires at least 3 and uses 4 by default.

coefficient of equivalence. The test-retest coefficient that generalizes over trials while holding the occasion fixed. Trial-related terms enter the error; the occasion main effect does not.

coefficient of stability. The test-retest coefficient that generalizes over occasions while holding the trial sample fixed. Occasion-related terms enter the error. The coefficient of trial equivalence and stability generalizes over both facets at once and cannot exceed either of the other two.

credible interval. The interval between two quantiles of the posterior distribution of a coefficient, at the width the viewer uses: 0.95, or the value entered in the Credible interval control on the viewers that offer one. Reported coefficients are posterior means with such an interval around them, except in the few summary windows that print point estimates only. A credible interval is a statement about the posterior, not a frequentist confidence statement.

cut score (criterion-referenced reliability). A threshold on the measurement scale, entered as Criterion Cutoff, used to ask how reliably scores are classified relative to that value. Cut score coefficients are absolute-error coefficients only; a relative-error version is not defined and is rejected.

dependability. The absolute-error reliability coefficient, selected in the toolbox as gcoeff = 1. Its error term includes the trial main effect divided by the projected number of trials, so it speaks to how well an observed score estimates a participant's own universe score on the measurement scale. Dependability and generalizability are not interchangeable, and neither is a synonym for reliability.

difference score. A score formed by subtracting one condition's score from another's, selected with Estimate internal consistency of difference scores. The two constituents are modeled jointly, so their between-person covariance enters the difference's universe-score variance. The four-event contrast is a difference of two difference scores.

dimension. A continuous between-person covariate selected in the Dimension 1 (dynamic) or Dimension 2 (dynamic) popup, which turns on dynamic (conditional) reliability. Each covariate must carry one value per participant, repeated across that participant's rows; the toolbox standardizes it across participants before estimation, so a dimension value of 0 is the sample mean.

divergent transition. A sampler iteration whose trajectory could not be followed accurately, reported by CmdStan and counted by the toolbox. Divergences never trigger a rerun and never change the convergence verdict, but a nonzero count flags possible bias in the posterior. More warmup, a higher Adapt delta, or stronger priors are the remedies.

D-study. The step that projects estimated variance components to a chosen number of observations, producing the coefficient at n' trials and n' occasions rather than at the numbers the sample happened to contain. Trial-count curves and trial cutoffs are D-study output.

dynamic (conditional) reliability. Reliability reported as a function of one or two standardized between-person dimensions rather than as one number, from a model in which the residual scale varies with the dimension (Rast & Clayson, in press). Selected by naming a Dimension 1 (dynamic) column.

effective sample size. The number of independent draws a correlated chain is worth for a given parameter. The convergence check fails when any monitored parameter's effective sample size falls below ten times the number of chains.

ERN (error-related negativity). The negative-going, response-locked ERP component elicited by errors, and the manual's running example. The tutorial datasets are simulated ERN-like scores.

ERP (event-related potential). An EEG waveform time-locked to an event and averaged over trials. PsyRAT's reliability analyses take the single-trial scores that go into such an average; the Import + Score ERP workflow reads the waveforms to produce those scores.

error variance (absolute and relative). The denominator term that separates the two coefficients. Absolute error is the variance of the difference between an observed score and the universe score, and it includes the trial main effect divided by n'. Relative error concerns only the ordering of participants, so the trial main effect drops out. Absolute error is therefore at least as large as relative error, and dependability is at most generalizability.

estimation engine. The fitter that estimates the variance components: CmdStan (the reference path), native HMC (Hamiltonian Monte Carlo), or native fitlme (restricted maximum likelihood with a bootstrap). Automatic (recommended) uses CmdStan whenever it is installed and falls back to a native engine only when it is not. The engine that ran is recorded in every export header and in the run-config sidecar.

event. A level of the condition facet, taken from the column assigned to Event Type. Difference scores are formed across two events; the four-event contrast is formed across four.

facet. A source of measurement error over which a score is generalized. The trial facet is present in every design; the occasion facet is added by assigning an Occasion ID column. Participants are the object of measurement rather than a facet.

gamma family. The scaled chi-square (Gamma with a log link) observation likelihood, selected with Observation likelihood family, for strictly positive, right-skewed scores such as single-trial time-frequency power. It runs only on CmdStan and covers a subset of the designs.

generalizability. The relative-error reliability coefficient, selected as gcoeff = 2. Its error term excludes the trial main effect, so it speaks to how well the ordering of participants is reproduced. See dependability.

G-study. The step that estimates the variance components of the design from the observed data. In this toolbox the G-study is the Bayesian fit, and its output is a set of posterior draws for each component.

ICC (intraclass correlation coefficient). The proportion of total variance attributable to between-person differences at a single observation. It is the reliability coefficient evaluated at one trial, so it does not depend on a projected trial count and is not comparable with a coefficient reported at n' trials.

location-scale. A model in which the residual standard deviation (the scale) is modeled alongside the mean (the location), per participant or as a function of a dimension. The subject-level and dynamic designs are location-scale models. Under the gamma family, Location-scale is also the name of one Gamma scale parameterization.

modular cut posterior. The result of fitting a model in two stages: the margins first, without a coupling term, and the coupling afterward, conditional on the retained margin draws, with nothing fed back. The gamma-family subject-level concurrent dynamic difference designs (analyses 19 and 20) are fitted this way. Their intervals are not full joint-posterior credible intervals, and the export header says so.

nonparallel splits. Splits holding unequal numbers of items, declared as Nonparallel (unequal n_i). They are fitted with a weighted observed-design model that is identified only by the variation in n_i, report the observed design with no D-study projection, and carry generalizability as a lower bound.

occasion. A level of the occasion facet, taken from the column assigned to Occasion ID. Its presence makes a design test-retest.

parallel splits. Splits holding equal numbers of items, declared as Parallel (equal n_i). They reuse the one-facet and test-retest machinery with the split mean as the score and the split count in place of the trial count.

person variance. The between-person variance component, the variance of participants' universe scores. It is the numerator of the one-facet coefficients.

posterior. The distribution of a parameter given the data, the priors, and the model, represented in PsyRAT by the draws retained from every chain. Reported point estimates are the means of those draws, and credible intervals are pairs of their quantiles.

prior. The distribution assumed for a parameter before the data are seen. The Gaussian-family defaults are weakly informative on the ERP microvolt scale and are not scale-free. Priors are set from a script, never from a screen, and the merged set a run used is recorded in its sidecar.

R-hat (potential scale reduction factor). A convergence diagnostic comparing variation between chains with variation within them; values near 1 indicate the chains describe the same distribution. The convergence check fails when any monitored parameter has R-hat of 1.1 or higher.

reliability. The umbrella term for the psychometric quality of scores. Dependability and generalizability are its two precise generalizability-theory forms in this toolbox, and output that names one of them means that one.

seed. The integer that initializes the sampler's random number generator, 12345 by default. The same data, settings, and seed reproduce the same draws on the same platform and CmdStan version.

SEM (standard error of measurement). The square root of the error variance at the projected number of observations, reported on the measurement scale. The absolute SEM pairs with dependability and includes the trial main effect; the relative SEM pairs with generalizability and excludes it.

sentinel. A value that marks the absence of something rather than measuring it. `(none)` in a Specify Inputs popup means the facet is not used. In the trial-cutoff table, -1 in the Trial Cutoff column means no cutoff could be found at the threshold, and -1 in a coefficient cell means the cutoff lies beyond the observed trial counts. None of these is a count or a column.

split. One row of a splits input, holding the mean of several items rather than a single trial. A splits analysis treats the split as the unit, and output that would otherwise say "trial" says "split".

stratum. One separately modeled cell of the design: a group, an event, or a group-by-event combination. Variance components and coefficients are estimated within each stratum, and dynamic-reliability figures draw one panel per stratum.

subject-level (subject-specific) error variance. A residual variance estimated separately for each participant, rather than one residual shared across the sample. Turning it on yields a reliability estimate for each participant, which is what makes participant-by-participant inclusion decisions possible.

trial cutoff. The smallest number of trials at which the projected coefficient reaches the reliability threshold you set. A cutoff above the largest observed trial count is an extrapolation and is flagged as one. A cutoff of -1 is not a count: it records that no cutoff could be found at that threshold.

universe score. The score a participant would obtain averaged over the universe of admissible observations, which is the quantity a reliability coefficient asks how well an observed score estimates. It appears as the numerator of every coefficient in this toolbox, and it is a property of the score rather than of the construct the score indexes.

warmup. The sampler iterations used for adaptation and discarded before the posterior summaries are computed. Warmup and sampling iterations are set separately; warmup draws never enter a reported estimate.

weight (n_i). The items-per-split column required by a splits analysis, giving the number of items averaged into each split mean. It must be a positive whole number. Variation in n_i across splits is what identifies the per-item residual separately from the person-by-split interaction, so the nonparallel model rejects all-equal weights.

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